INCH-POUND

MIL-STD-1651B
w/CHANGE 1
23 November 2016
SUPERSEDING
MIL-STD-1651B
6 November 2015

# DEPARTMENT OF DEFENSE INTERFACE STANDARD

INSERT ARRANGEMENTS
FOR
SAE-AS50151,
MIL-DTL-22992 (CLASSES C, J AND R),
MIL-DTL-83723 (SERIES II),
AND SAE-AS95234
CIRCULAR ELECTRICAL CONNECTORS

AMSC N/A

FSC 5935

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# FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Comments, suggestions, or questions on this document should be addressed to Commander, DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus, Ohio, 43218-3990 or emailed to CircularConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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SUMMARY OF CHANGE 1 MODIFICATIONS

1. Correction and clarification to Table I, contact size 12-16 and 16-22: In column 5, change socket part number from M39029/45-297 to M39029/45-269. Also, for size 12-16, add the column 5 PINs to column 2, SAE-AS50151, series AS34001 front release connectors. Therefore, pin part number MS39029/44-289 and socket part number MS39029/45-296 have been added to column 3. In addition, contact size 16-22 (front-release SAE-AS95234) also applies to contact size 16-22 in column 3 (front-release AS50151 connectors) and have been added to column 3.

2. Clarification: Add footnote under table I stating size 16S socket contacts are only used with rear-release connector shell sizes 8S, 10S, 10SL, 12S, 14S and 16S.

3. Clarification: For insert arrangements used in “short” connector shells, including insert arrangement 8S-1, 10S-2, 10SL-3, 10SL-4, 12S-3, 12S-4, 14S-1, 14S-2, 14S-4, 14S-5, 14S-6, 14S-7, 14S-9, 16S-1, 16S-3, 16S-4, 16S-5, 16S-6 and 16S-8, update the associated tables to indicate that both 16S (socket) and 16 (pin) contact sizes apply, depending on which connector family and series are specified.

4. Correction: For insert arrangements which apply to SAE-AS95234, but which include a parenthetical note stating “not applicable” to SAE-AS95234, including 10SL-3, 14S-1, 14S-2, 14S-5, 14S-6, 14S-9, 16S-3 and 16S-4.

5. Insert arrangement 18-22 on page 58: Update the figure and the Contact X/Y coordinate table to change the cavity identification letter for contact location “G” to “C”. Rationale: During review of the required contact marking requirements, a NAWC memo dated 20 January 1993 was discovered. The memo indicates that this change was intended for a future revision of MIL-STD-1651.

6. Add new insert arrangement 22-63 on page 118. This new arrangement is necessary for a military application.



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## 1. SCOPE

1.1 Scope. This standard gives the dimensional requirements for insert arrangements used with SAE-AS50151, MIL-DTL-22992 (Classes C, J and R), and MIL-DTL-83723 (Series II) and SAE-AS95234 electrical connectors.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, and 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, and 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-22992 | - | Connectors, Plugs and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specification for. |
| MIL-DTL-83723 | - | Connectors, Electrical (Circular, Environment Resisting), Receptacles and Plugs, General Specification for. |

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE-AS50151 | - | Connectors, Electrical, Circular Threaded, AN Type, General Specification for. |
| SAE-AS39029 | - | Contacts, Electrical Connector, General Specification for. |
| SAE-AS34001 | - | Connectors, Receptacle, Electric, Wall Mounting, Front Release, Crimp Contact, AN Type. |
| SAE-AS34501 | - | Connectors, Receptacle, Electric, Wall Mounting, Rear Release, Crimp Contact, AN Type. |
| SAE-AS95234 | - | Connectors, Electrical, Reverse Bayonet, General Specification for. |

(Copies of these documents are available at http://www.sae.org or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 (inside USA and Canada) or 724-776-4970 (outside USA), or email: CustomerService@sae.org.)

ASME INTERNATIONAL

| | | |
|---|---|---|
| ASME Y14.5 | - | Dimensioning and Tolerancing. |

(Copies of this document are available online at http://www.asme.org or from the ASME International, Three Park Avenue, New York, NY 10016-5990.)



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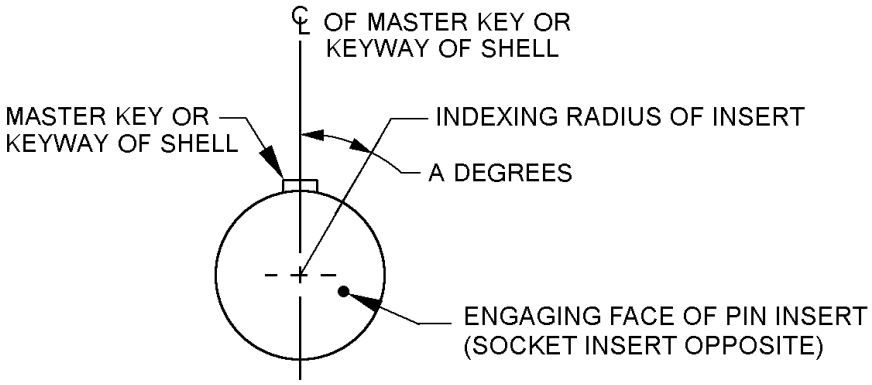
2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

3.1 Definitions. The terms used in this standard are generally accepted by the electrical and electronics industries and are commonly used in electrical connector engineering practice, except for the following terminology:

a) Inactive for new design. For this standard, insert arrangements that are specified as “inactive for new design” shall not be used in new designs or equipment. An inactive arrangement may be used to support existing designs or equipment.

b) Common terminology. An illustration depicting common terminology used in this standard is provided below:




## 4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

## 5. DETAILED REQUIREMENTS

5.1 Dimensions. Dimensions shall be in accordance with the applicable figure of this standard and the following dimensional data:

a. Dimensioning and tolerancing in accordance with ASME Y14.5. Dimensions are true position and are in inches.
b. Metric equivalents (to the nearest .01 mm) are given for information only and are based upon 1 inch = 25.4 mm.
c. Dimensions and markings shown are for engaging face of the pin insert. Socket insert is opposite.
d. The following tolerances apply to insert installed in shell:



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1. The center of each contact cavity in the insert for crimp socket contacts shall be located at true position within .014 dia.
2. The center of engaging end of each pin contact shall be located in true position within .024 dia.

e. Unless otherwise indicated, dimensions are symmetrical about centerline.

The arrow symbol, "↓" indicates the ℄ (center line) of the insert arrangement.

f. Each insert arrangement is shown in the "normal position" (also known as normal polarization or normal "clocking") in the shell.

5.2 Marking. Marking shall be in accordance with SAE-AS50151, MIL-DTL-22992, MIL-DTL-83723 or SAE-AS95234, as applicable, and as shown in the applicable insert arrangements included in this standard.

5.3. Contacts. Unless otherwise specified in the insert arrangement, crimp contact Part or Identifying Numbers (PINs) shall be in accordance with table I. Solder contacts shall be in accordance with SAE-AS50151, MIL-DTL-22992 or MIL-DTL-83723 or SAE-AS95234, as applicable.

TABLE I. Contact Part or Identifying Numbers.

| Crimp contact PINs 1/ | | | | |
|---|---|---|---|---|
| Contact size | Contact type | SAE-AS39029 PIN for SAE-AS50151 2/ | SAE-AS39029 PIN for SAE-AS50151 3/, or MIL-DTL-83723, series II | SAE-AS39029 PIN for SAE-AS95234, front release |
| 0 | pin | M39029/44-293 | M39029/29-216 | M39029/44-293 |
| | socket | M39029/45-300 | M39029/30-222 | M39029/45-300 |
| 4 | pin | M39029/44-292 | M39029/29-215 | M39029/44-292 |
| | socket | M39029/45-299 | M39029/30-221 | M39029/45-299 |
| 8 | pin | M39029/44-291 | M39029/29-214 | M39029/44-291 |
| | socket | M39029/45-298 | M39029/30-220 | M39029/45-298 |
| 12 | pin | M39029/44-290 | M39029/29-213 | M39029/44-290 |
| | socket | M39029/45-297 | M39029/30-219 | M39029/45-297 |
| 12-16 | pin | M39029/44-289 | Not applicable | M39029/44-289 |
| | socket | M39029/45-296 | | M39029/45-296 |
| 16 | pin | M39029/44-288 | M39029/29-212 | M39029/44-288 |
| | socket | M39029/45-295 | M39029/30-218 | M39029/45-295 |
| 16-22 | pin | M39029/44-287 | Not applicable | M39029/44-287 |
| | socket | M39029/45-294 | | M39029/45-294 |
| 16S 4/ | socket | Not applicable | M39029/30-217 | Not applicable |

1/ For connectors with non-removable, fixed contacts with solderable terminations that utilize MIL-STD-1651 insert arrangements, see applicable connector specifications for pin and socket contact requirements.

2/ For SAE-AS50151, series AS34001 (front release) connectors.

3/ For SAE-AS50151, series AS34501 (rear release) connectors.

4/ 16S socket contacts in accordance with SAE-AS39029/30, which have shorter length mating ends than a size 16 socket in accordance with the same document, are only used with rear release SAE-AS50151 connector shell sizes 8S, 10S, 10SL, 12S, 14S and 16S.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This standard covers insert arrangements for use with SAE-AS50151, series II and III, MIL-DTL-22992, classes C, J or R, MIL-DTL-83723, series II, or SAE-AS95234 connectors. Information pertaining to shell size, arrangement number, number of contacts, size of contacts, service rating, contact location, and superseding information is contained in this document.

6.2 Application notes: The following information is provided as a convenience only. Users should review the applicable military or non-government electrical connector specification for additional details. Note: Some insert arrangements are inactive for new design for one or more connector specifications, but are active for new design for other specifications (see figures 1 through 9).

1. Shell size 8:
   - Shell size 8 is identical to 8S and applies only to MIL-DTL-83723, series II.
   - Shell size 8S applies to SAE-AS50151 only.
   - Shell size 8 and 8S are not applicable to MIL-DTL-22992 or SAE-AS95234.
2. Shell size 10, 10S, 10SL and 11:
   - Shell size 10 is identical to 10S and applies only to MIL-DTL-83723, series II.
   - Shell size 10S applies to SAE-AS50151 only.
   - Shell size 10SL applies to SAE-AS50151 and SAE-AS95234 only.
   - Shell size 10, 10S, 10SL and 11 are not applicable to MIL-DTL-22992.
   - Shell size 11 is identical to 10SL and applies to MIL-DTL-83723, series II only.
3. Shell size 12, 12S, and 13:
   - Shell size 12 applies to SAE-AS50151, MIL-DTL-22992 and MIL-DTL-83723, series II.
   - Shell size 12S applies to SAE-AS50151 only.
   - Shell size 13 is identical to 12S and applies only to MIL-DTL-83723, series II.
   - Shell size 12, 12S and 13 do not apply to SAE-AS95234 only.
4. Shell size 14, 14S, and 15:
   - Shell size 14 applies to SAE-AS50151, MIL-DTL-22992 and MIL-DTL-83723, series II.
   - Shell size 14S applies to SAE-AS50151 and SAE-AS95234 only.
   - Shell size 15 is identical to 14S and applies only to MIL-DTL-83723, series II.
5. Shell size 16, 16S, and 17:
   - Shell size 16 applies to SAE-AS50151, MIL-DTL-22992 and MIL-DTL-83723, series II and SAE-AS95234.
   - Shell size 16S applies to SAE-AS50151 and SAE-AS95234 only.
   - Shell size 17 is identical to 16S and applies only to MIL-DTL-83723, series II.
6. Shell size 44 applies to SAE-AS50151, except insert arrangement 44-52, which also applies to MIL-DTL-22992.
7. Shell size 48 applies to SAE-AS50151 only.

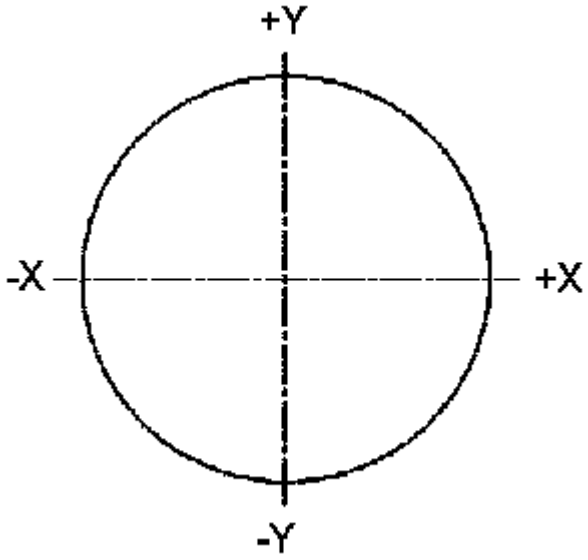
6.3 Cross-reference. See table II for a cross-reference of figures to previous sections. See table III for a cross-reference of figures to their associated insert arrangements for each shell size. For the convenience of the user, table III includes hyper-links to each insert arrangement.



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6.4 Drawing notes. The following information is applicable to all insert arrangements specified by this standard.

a. Dimensions are in inches, unless otherwise specified.
b. Metric equivalents are given for information only.
c. Each insert arrangement is shown using a conventional two-dimensional Cartesian system, with two perpendicular axes. The x-axis is always the horizontal centerline, and the y-axis is always the vertical centerline. The origin, or “0,0” point, is always the intersection of the two axes. The axes in each insert arrangement follow the standard two-dimensional, “positive” orientation, with the positive x-axis pointed to the right and the positive y-axis pointed up, as illustrated in the figure below:




6.5 Subject term (key word) listing.

Alternate insert positions
Insert position
Pin contact
Shell size
Socket contact

6.4 Changes from previous issue. The margins of this standard are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.



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TABLE II. Cross-reference of figures to sections. 1/

| Insert arrangements by shell size | Current figure | Previous section | Applicable connector document 2/ | Series or class |
|---|---|---|---|---|
| Insert arrangement, shell sizes 8, 8S | Figure 1 | Section 100 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II, III |
| Insert arrangement, shell sizes 10, 10S, 10SL, 11 | Figure 2 | Section 200 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | |
| Insert arrangement, shell sizes 12, 12S, 13 | Figure 3 | Section 300 | MIL-DTL-83723 | II |
| | | | MIL-DTL-22992 | C,J,R |
| | | | SAE-AS50151 | II, III |
| Insert arrangement, shell sizes 14, 14S, 15 | Figure 4 | Section 400 | MIL-DTL-83723 | II |
| | | | MIL-DTL-22992 | C,J,R |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell sizes 16, 16S, 17 | Figure 5 | Section 500 | MIL-DTL-83723 | II |
| | | | MIL-DTL-22992 | C,J,R |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 18 | Figure 6 | Section 600 | MIL-DTL-83723 | II |
| | | | MIL-DTL-22992 | C,J,R |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 20 | Figure 7 | Section 700 | MIL-DTL-83723 | II |
| | | | MIL-DTL-22992 | C,J,R |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 22 | Figure 8 | Section 800 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 24 | Figure 9 | Section 900 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II, III |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 28 | Figure 10 | Section 1000 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II |
| | | | SAE-AS95234 | --- |

See footnotes at end of table.



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TABLE II. Cross-reference of figures to sections - Continued. 1/

| Insert arrangements by shell size | Current figure | Previous section | Applicable connector document 2/ | Series or class |
|---|---|---|---|---|
| Insert arrangement, shell size 32 | Figure 11 | Section 1100 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 36 | Figure 12 | Section 1200 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II |
| | | | SAE-AS95234 | --- |
| Insert arrangement, shell size 40 | Figure 13 | Section 1300 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II |
| Insert arrangement, shell size 44 | Figure 14 | Section 1400 | MIL-DTL-83723 | II |
| | | | SAE-AS50151 | II, III |
| Insert arrangement, shell size 48 | Figure 15 | Section 1500 | SAE-AS50151 | II, III |

1/ In previous revisions of this document, insert arrangements were separated by shell sizes into separate sections.

2/ Some insert arrangements may not be applicable to a referenced electrical connector document. See figures 1 through 15, insert arrangements for shell sizes 8 through 48, for specific information on applicable documents associated with specific insert arrangements. In addition, each figure title and the associated application notes (6.2) provide useful information on applicable insert arrangements.

TABLE III. Cross-reference of insert arrangements by shell size.

| Shell sizes 8, 8S, figure 1 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 8-1, 8S-1 | 14 | ---- | --- |

See footnote at end of table.

| Shell sizes 10, 10S, 10SL, 11 figure 2 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 10-2 | 15 | 10SL-4, 11-4 | 17 |
| 10SL-3, 11-3 1/ | 16 | ---- | --- |

See footnote at end of table.

| Shell sizes 12, 12S, 13, figure 3 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 12S-3, 13-3 | 18 | 12-5 | 20 |
| 12S-4, 13-4 | 19 | ---- | ---- |

See footnote at end of table.



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TABLE III. Cross-reference of insert arrangements by shell size -Continued.

| Shell sizes 14, 14S, 15, figure 4 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 14S-1, 15-1 | 21 | 14S-5, 15-5 | 25 |
| 14S-2, 15-2 | 22 | 14S-6, 15-6 | 26 |
| 14-3 | 23 | 14S-7, 15-7 | 27 |
| 14S-4, 15-4 1/ | 24 | 14S-9, 15-9 | 28 |

See footnote at end of table.

| Shell sizes 16, 16S, 17, figure 5 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 16S-1, 17-1 | 29 | 16S-8, 17-8 | 36 |
| 16-2 | 30 | 16-9 | 37 |
| 16S-3, 17-3 | 31 | 16-10 | 38 |
| 16S-4, 17-4 | 32 | 16-11 | 39 |
| 16S-5, 17-5 1/ | 33 | 16-12 | 40 |
| 16S-6, 17-6 1/ | 34 | 16-13 | 41 |
| 16-7 | 35 | ---- | --- |

See footnote at end of table.

| Shell size 18, figure 6 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 18-1 1/ | 42 | 18-11 1/ | 51 |
| 18-3 1/ | 43 | 18-12 1/ | 52 |
| 18-4 1/ | 44 | 18-13 1/ | 53 |
| 18-5 1/ | 45 | 18-14 1/ | 54 |
| 18-6 | 46 | 18-15 1/ | 55 |
| 18-7 | 47 | 18-16 1/ | 56 |
| 18-8 1/ | 48 | 18-20 1/ | 57 |
| 18-9 | 49 | 18-22 1/ | 58 |
| 18-10 1/ | 50 | 18-29 1/ | 59 |

See footnote at end of table.



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TABLE III. Cross-reference of insert arrangements by shell size - Continued.

| Shell size 20, figure 7 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 20-2 | 60 | 20-17 | 73 |
| 20-3 1/ | 61 | 20-18 | 74 |
| 20-4 | 62 | 20-19 1/ | 75 |
| 20-5 1/ | 63 | 20-20 1/ | 76 |
| 20-6 1/ | 64 | 20-21 | 77 |
| 20-7 | 65 | 20-22 | 78 |
| 20-8 | 66 | 20-23 1/ | 79 |
| 20-9 | 67 | 20-24 1/ | 80 |
| 20-11 1/ | 68 | 20-27 | 81 |
| 20-12 1/ | 69 | 20-29 | 82 |
| 20-14 | 70 | 20-31 | 83 |
| 20-15 | 71 | 20-33 | 84 |
| 20-16 | 72 | --- | --- |

See footnote at end of table.

| Shell size 22, figure 8 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 22-1 1/ | 85 | 22-17 | 102 |
| 22-2 | 86 | 22-18 | 103 |
| 22-3 1/ | 87 | 22-19 | 104 |
| 22-4 1/ | 88 | 22-20 1/ | 105 |
| 22-5 | 89 | 22-21 | 106 |
| 22-6 1/ | 90 | 22-22 | 107 |
| 22-7 | 91 | 22-23 | 108 |
| 22-8 1/ | 92 | 22-24 1/ | 109 |
| 22-9 | 93 | 22-25 1/ | 110 |
| 22-10 | 94 | 22-27 | 111 |
| 22-11 | 95 | 22-28 1/ | 112 |
| 22-12 | 96 | 22-29 1/ | 113 |
| 22-13 1/ | 97 | 22-33 1/ | 114 |
| 22-14 | 98 | 22-34 1/ | 115 |
| 22-15 | 100 | 22-36 | 116 |
| 22-16 1/ | 101 | 22-63 | 117 |

See footnote at end of table.



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TABLE III. Cross-reference of insert arrangements by shell size - Continued.

| Shell size 24, figure 9 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 24-1 1/ | 117 | 24-14 1/ | 128 |
| 24-2 | 118 | 24-16 | 129 |
| 24-3 1/ | 119 | 24-17 1/ | 130 |
| 24-4 | 120 | 24-20 | 131 |
| 24-5 | 121 | 24-21 | 132 |
| 24-6 | 122 | 24-22 | 133 |
| 24-7 | 123 | 24-23 1/ | 134 |
| 24-9 1/ | 124 | 24-27 | 135 |
| 24-10 | 125 | 24-28 | 136 |
| 24-11 | 126 | 24-80 1/ | 137 |
| 24-12 | 127 | --- | --- |

See footnote at end of table.

| Shell size 28, figure 10 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 28-1 | 139 | 28-12 | 149 |
| 28-2 | 140 | 28-14 1/ | 151 |
| 28-3 | 141 | 28-15 1/ | 152 |
| 28-4 | 142 | 28-16 1/ | 154 |
| 28-5 | 143 | 28-17 | 156 |
| 28-6 1/ | 144 | 28-18 | 157 |
| 28-8 | 145 | 28-19 | 158 |
| 28-9 | 146 | 28-20 | 159 |
| 28-10 | 147 | 28-21 | 160 |
| 28-11 | 148 | 28-22 | 162 |

See footnote at end of table.

| Shell size 32, figure 11 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 32-1 | 163 | 32-10 1/ | 177 |
| 32-2 | 164 | 32-12 1/ | 178 |
| 32-3 | 165 | 32-13 | 180 |
| 32-4 1/ | 166 | 32-14 1/ | 182 |
| 32-5 1/ | 168 | 32-15 | 183 |
| 32-6 | 169 | 32-17 | 184 |
| 32-7 | 171 | 32-22 | 185 |
| 32-8 1/ | 173 | 32-63 | 187 |
| 32-9 | 175 | 32-73 | 188 |

See footnote at end of table.



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TABLE III. Cross-reference of insert arrangements by shell size - Continued.

| Shell size 36, figure 12 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 36-1 1/ | 191 | 36-10 | 203 |
| 36-3 | 193 | 36-13 1/ | 206 |
| 36-4 1/ | 194 | 36-14 1/ | 208 |
| 36-5 | 195 | 36-15 | 210 |
| 36-6 | 196 | 36-19 1/ | 212 |
| 36-7 | 197 | 36-52 | 214 |
| 36-8 | 199 | 36-66 | 216 |
| 36-9 | 201 | ---- | --- |

See footnote at end of table.

| Shell size 40, figure 13 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 40-1 | 219 | 40-7 | 231 |
| 40-2 | 221 | 40-9 | 233 |
| 40-3 | 223 | 40-10 | 235 |
| 40-4 | 225 | 40-11 | 237 |
| 40-5 | 227 | 40-56 | 239 |
| 40-6 | 229 | 40-62 | 243 |

See footnote at end of table.

| Shell size 44, figure 14 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 44-1 | 246 | 44-52 | 248 |

See footnote at end of table.

| Shell size 48, figure 15 | | | |
|---|---|---|---|
| Insert arrangement | Page | Insert arrangement | Page |
| 48-62 | 252 | ---- | --- |

1/ Inactive for new design.



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TABLE IV. Insert arrangements removed from MIL-STD-1651 by revision B.

| Insert arrangement | Status in MIL-STD-1651 | Replacement |
|---|---|---|
| 12S, 13-1 | Cancelled after 6 November 2015 | --- |
| 12S, 13-2 | Cancelled after 6 November 2015 | --- |
| 14S, 15-10 | Cancelled after 6 November 2015 | --- |
| 14S, 15-11 | Cancelled after 6 November 2015 | --- |
| 14S, 15-12 | Cancelled after 6 November 2015 | --- |
| 14S, 15-13 | Cancelled after 6 November 2015 | --- |
| 16S, 17-14 | Cancelled after 6 November 2015 | --- |
| 16S, 17-15 | Cancelled after 6 November 2015 | 16S, 17-4 |
| 16S, 17-16 | Cancelled after 6 November 2015 | 16S, 17-4 |
| 16S, 17-17 | Cancelled after 6 November 2015 | 16-14 |
| 18-17 | Cancelled after 6 November 2015 | 18-9 |
| 18-18 | Cancelled after 6 November 2015 | 18-9 |
| 18-19 | Cancelled after 6 November 2015 | --- |
| 18-23 | Cancelled after 6 November 2015 | 18-1 |
| 18-24 | Cancelled after 6 November 2015 | 18-1 |
| 18-25 | Cancelled after 6 November 2015 | 18-3 |
| 18-26 | Cancelled after 6 November 2015 | 18-3 |
| 18-27 | Cancelled after 6 November 2015 | 18-5 |
| 18-28 | Cancelled after 6 November 2015 | 18-5 |
| 18-30 | Cancelled after 6 November 2015 | 18-20 |
| 18-31 | Cancelled after 6 November 2015 | 18-20 |
| 20-10 | Cancelled after 6 November 2015 | --- |
| 20-13 | Cancelled after 6 November 2015 | --- |
| 20-25 | Cancelled after 6 November 2015 | 20-11 |
| 20-26 | Cancelled after 6 November 2015 | --- |
| 20-30 | Cancelled after 6 November 2015 | 20-11 |
| 20-32 | Cancelled after 6 November 2015 | 20-7 |
| 22-30 | Cancelled after 6 November 2015 | 22-14 |
| 22-32 | Cancelled after 6 November 2015 | 22-5 |
| 22-35 | Cancelled after 6 November 2015 | --- |
| 24-15 | Cancelled after 6 November 2015 | 24-5 |
| 24-18 | Cancelled after 6 November 2015 | --- |
| 24-24 | Cancelled after 6 November 2015 | 24-5 |
| 24-25 | Cancelled after 6 November 2015 | 24-6 |
| 24-26 | Cancelled after 6 November 2015 | 24-6 |
| 28-7 | Cancelled after 6 November 2015 | --- |
| 28-13 | Cancelled after 6 November 2015 | 28-12 |
| 28-23 | Cancelled after 6 November 2015 | --- |
| 28-24 | Cancelled after 6 November 2015 | --- |
| 32-16 | Cancelled after 6 November 2015 | 32-6 |
| 32-18 | Cancelled after 6 November 2015 | 32-4 |
| 32-19 | Cancelled after 6 November 2015 | 32-1 |
| 32-20 | Cancelled after 6 November 2015 | 32-6 |
| 32-101 | Cancelled after 6 November 2015 | --- |
| 32-102 | Cancelled after 6 November 2015 | --- |
| 36-2 | Cancelled after 6 November 2015 | --- |
| 36-11 | Cancelled after 6 November 2015 | 36-10 |
| 36-12 | Cancelled after 6 November 2015 | 36-10 |



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TABLE IV. Insert arrangements removed from MIL-STD-1651 by Revision B.

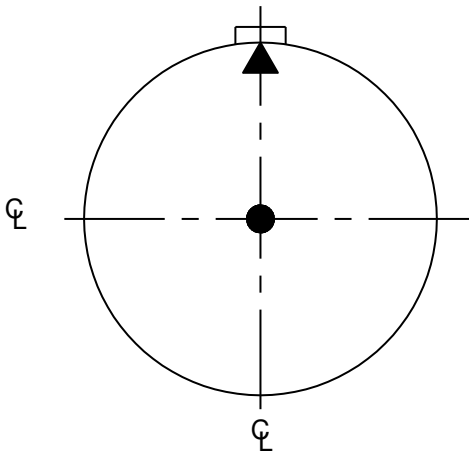
| Insert arrangement | Status in MIL-STD-1651 | Replacement |
|---|---|---|
| 36-16 | Cancelled after 6 November 2015 | 36-7 |
| 36-17 | Cancelled after 6 November 2015 | 36-7 |
| 36-18 | Cancelled after 6 November 2015 | 36-9 |
| 36-20 | Cancelled after 6 November 2015 | --- |
| 36-21 | Cancelled after 6 November 2015 | 36-9 |
| 40-12 | Cancelled after 6 November 2015 | --- |
| 40-13 | Cancelled after 6 November 2015 | 40-2 |
| 40-14 | Cancelled after 6 November 2015 | --- |
| 40-15 | Cancelled after 6 November 2015 | --- |
| 44-2 | Cancelled after 6 November 2015 | --- |
| 44-3 | Cancelled after 6 November 2015 | --- |
| 44-4 | Cancelled after 6 November 2015 | --- |
| 44-5 | Cancelled after 6 November 2015 | 44-1 |
| 44-6 | Cancelled after 6 November 2015 | 44-1 |
| 48-1 | Cancelled after 6 November 2015 | --- |
| 48-2 | Cancelled after 6 November 2015 | --- |
| 48-3 | Cancelled after 6 November 2015 | --- |
| 48-4 | Cancelled after 6 November 2015 | --- |
| 48-5 | Cancelled after 6 November 2015 | --- |

TABLE V. Insert arrangements removed from MIL-STD-1651 by Revision A.

| Insert Arrangement | Status in MIL-STD-1651 |
|---|---|
| 10, 10S-5 | Not to be used |
| 10, 10S-6 | Cancelled after 17 December 1958 |
| 12S, 13-6 | Cancelled after 8 March 1945 |
| 14S, 15-8 | Not to be used |
| 18-2 | Cancelled after 19 May 1945 |
| 18-21 | Cancelled after 25 November 1949 |
| 20-1 | Cancelled after 1 June 1945 |
| 20-28 | Cancelled after 1 June 1945 |
| 22-26 | Not to be used |
| 22-31 | Not to be used |
| 22-37 | Cancelled after 26 June 1945 |
| 24-8 | Cancelled after 26 June 1945 |
| 24-13 | Cancelled after 26 June 1945 |
| 24-19 | Cancelled after 26 June 1945 |
| 32-11 | Cancelled after 26 June 1945 |
| 40-8 | Not to be used |



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8-1 and 8S-1

(Not applicable to MIL-DTL-22992 or SAE-AS95234.)

| Contacts (Insert arrangements 8-1 and 8S-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

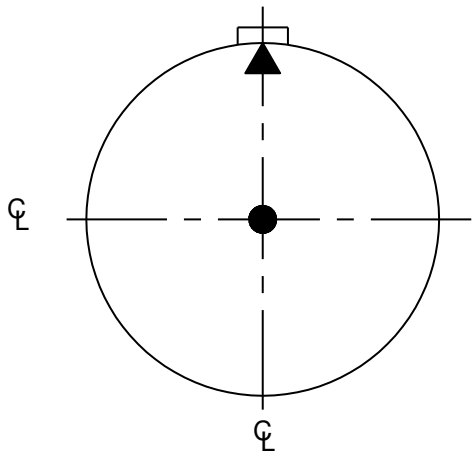
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 8S | -1 | 1 | 16 (pin)<br>16S (socket) | A | All | MS33680-1 |
| 8 | -1 | 1 | 16 | A | All | |

Note: This insert arrangement has no alternate insert positions.

FIGURE 1. Insert arrangement, electrical connector, shell size 8, MIL-DTL-83723, series II only, and shell size 8S, SAE-AS50151 only.



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10-2 and 10S-2

(Not applicable to MIL-DTL-22992 or SAE-AS95234.)

| Contacts (Insert arrangements 10-2 and 10S-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

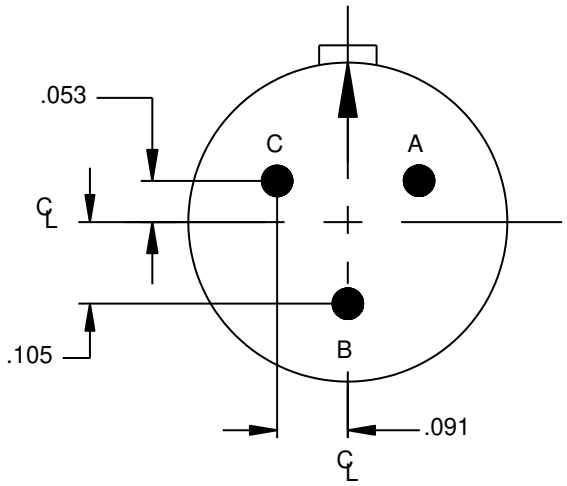
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 10S | -2 | 1 | 16S (socket)<br>16 (pin) | A | All | MS33680-2 |
| 10 | -2 | 1 | 16 | A | All | |

Note: This insert arrangement has no alternate insert positions.

FIGURE 2. Insert arrangement, electrical connector, shell sizes 10 and 11, MIL-DTL-83723, series II only, and shell sizes 10S and 10SL, SAE-AS50151 and SAE- AS95234 only – Continued.



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10SL-3 and 11-3

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangements 10SL-3 and 11-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.091 (2.31) | +.053 (1.35) |
| B | +.000 (0.00) | -.105 (2.66) |
| C | -.091 (2.31) | +.053 (1.35) |

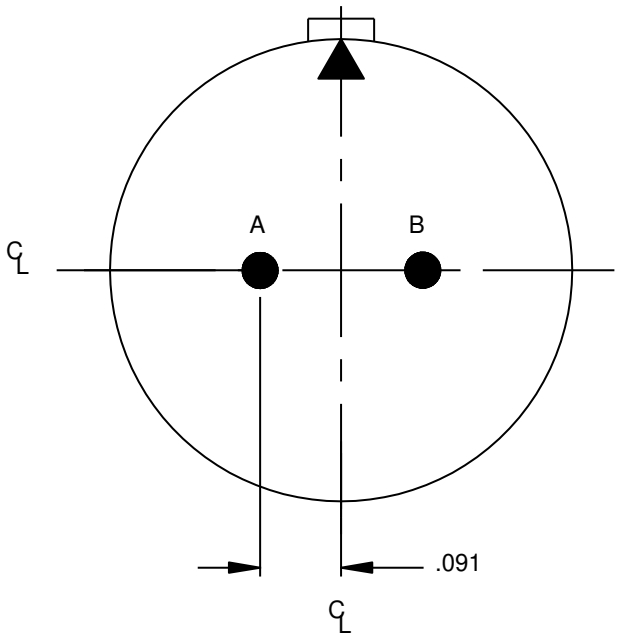
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 10SL | -3 | 3 | 16S (socket)<br>16 (socket)<br>16 (pin) | INST | All | MS33680-3 |
| 11 | -3 | 3 | 16 | A | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 2. Insert arrangement, electrical connector, shell sizes 10 and 11, MIL-DTL-83723, series II only, and shell sizes 10S and 10SL, SAE-AS50151 and SAE- AS95234 only – Continued.



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10SL-4 and 11-4

(Inactive for new design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangements 10SL-4 and 11-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.091 (2.31) | +.000 (0.00) |
| B | +.091 (2.31) | +.000 (0.00) |

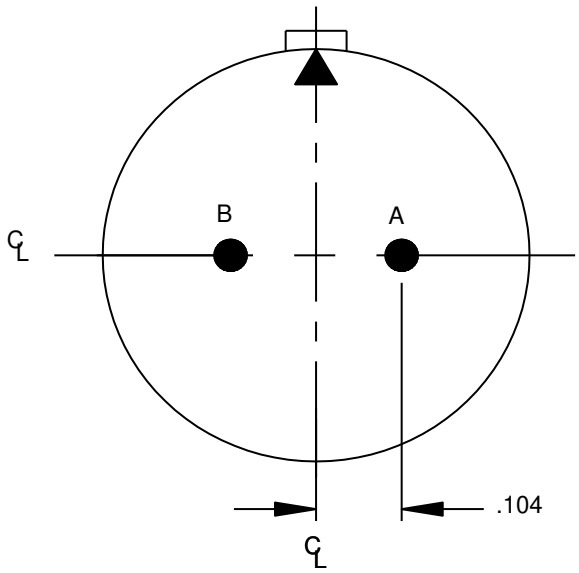
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 10SL | -4 | 2 | 16S (socket)<br>16 (socket)<br>16 (pin) | A | All | MS33680-4 |
| 11 | -4 | 2 | 16 | A | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 2. Insert arrangement, electrical connector, shell sizes 10 and 11, MIL-DTL-83723, series II only, and shell sizes 10S and 10SL, SAE-AS50151 and SAE- AS95234 only – Continued.



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12S-3 and 13-3

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 12S-3 and 13-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.104 (2.64) | +.000 (0.00) |
| B | -.104 (2.64) | -.000 (0.00) |

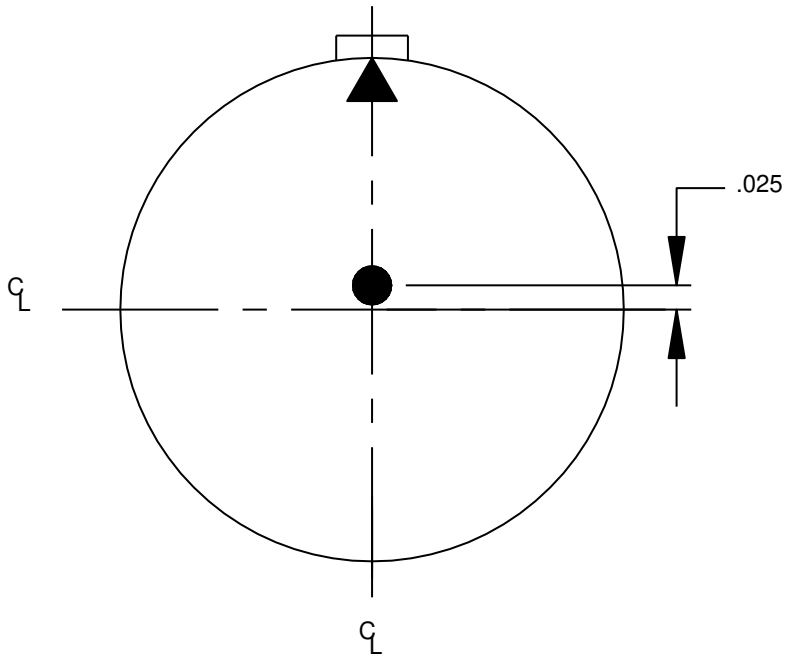
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 12S | -3 | 2 | 16S (socket)<br>16 (pin) | A | All | MS33681-3 |
| 13 | -3 | 2 | 16 | A | All | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 3. Insert arrangement, electrical connector, shell size 12, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, shell sizes 12S, SAE-AS50151 only, and shell size 13, MIL-DTL-83723, series II only.



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12S-4 and 13-4

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 12S-4 and 13-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.025 (0.64) |

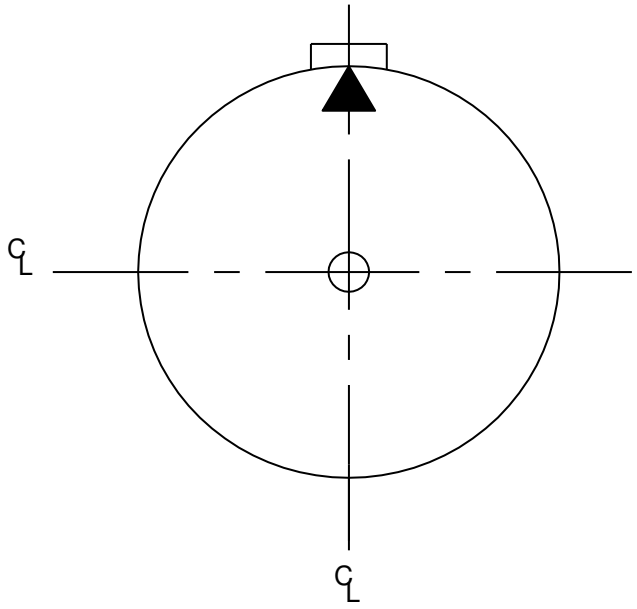
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 12S | -4 | 1 | 16S (socket)<br>16 (pin) | D | All | MS33681-4 |
| 13 | -4 | 1 | 16 | D | All | |

Note: This insert arrangement has no alternate insert positions.

FIGURE 3. Insert arrangement, electrical connector, shell size 12, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, shell sizes 12S, SAE-AS50151 only, and shell size 13, MIL-DTL-83723, series II only – Continued.



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12-5

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 12-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

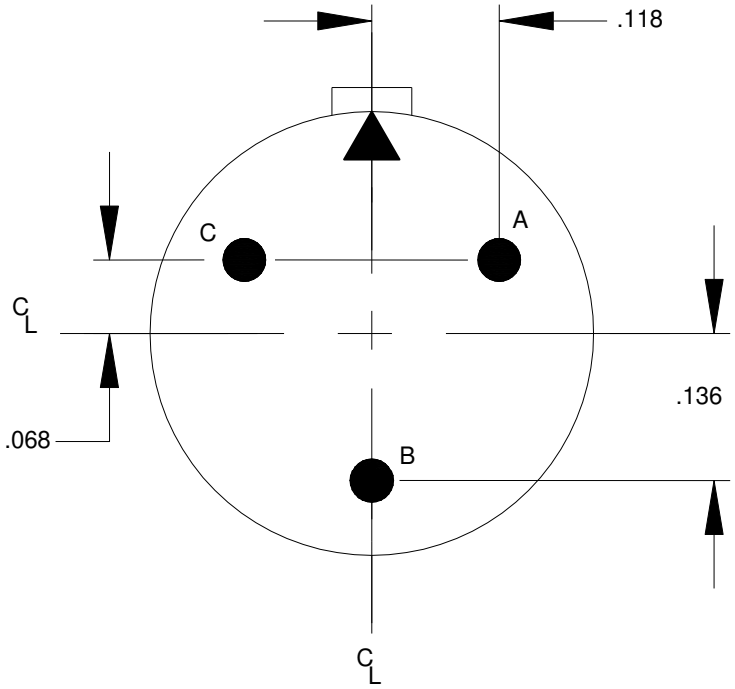
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 12 | -5 | 1 | 12 | D | All | MS33681-5 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 3. Insert arrangement, electrical connector, shell size 12, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, shell sizes 12S, SAE-AS50151 only, and shell size 13, MIL-DTL-83723, series II only – Continued.



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14S-1 and 15-1

(Insert arrangement 14S-1 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14S-1 and 15-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.118 (3.00) | +.068 (1.73) |
| B | +.000 (0.00) | -.136 (3.45) |
| C | -.118 (3.00) | +.068 (1.73) |

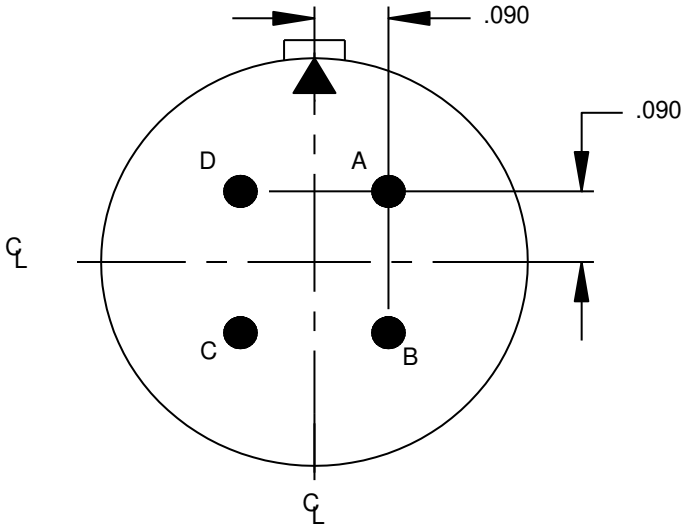
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -1 | 3 | 16S (socket)<br>16 (socket)<br>16 (pin) | A | All | MS33682-1 |
| 15 | -1 | 3 | 16 | A | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only.



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14S-2 and 15-2

(Insert arrangement 14S-2 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14S-2 and 15-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.090 (2.29) | +.090 (2.29) |
| B | +.090 (2.29) | -.090 (2.29) |
| C | -.090 (2.29) | -.090 (2.29) |
| D | -.090 (2.29) | +.090 (2.29) |

| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -2 | 4 | 16S (socket)<br>16 (socket)<br>16 (pin) | INST | All | MS33682-2 |
| 15 | -2 | 4 | 16 | INST | All | |

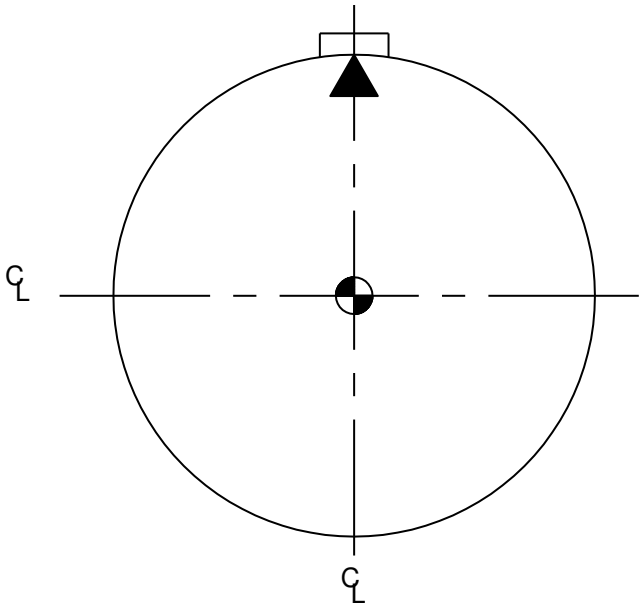
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 120 | 240 | --- |

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14-3

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

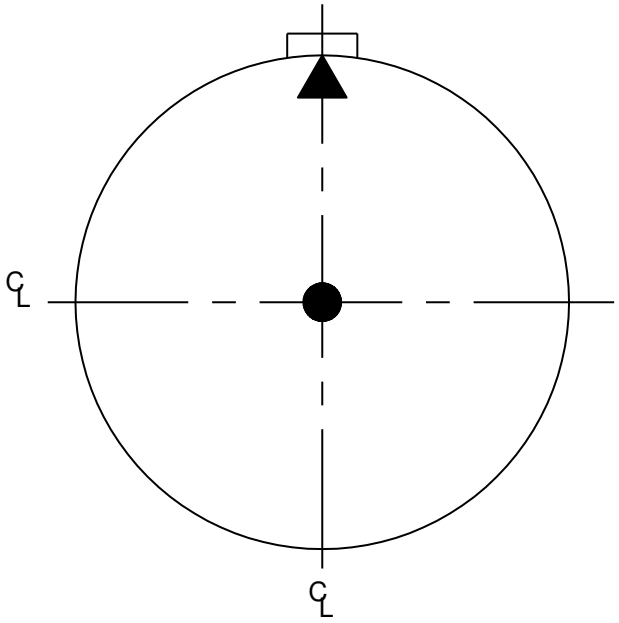
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14 | -3 | 1 | 8 | A | All | MS33682-3 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14S-4 and 15-4

(Inactive for new design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangements 14S-4, 15-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

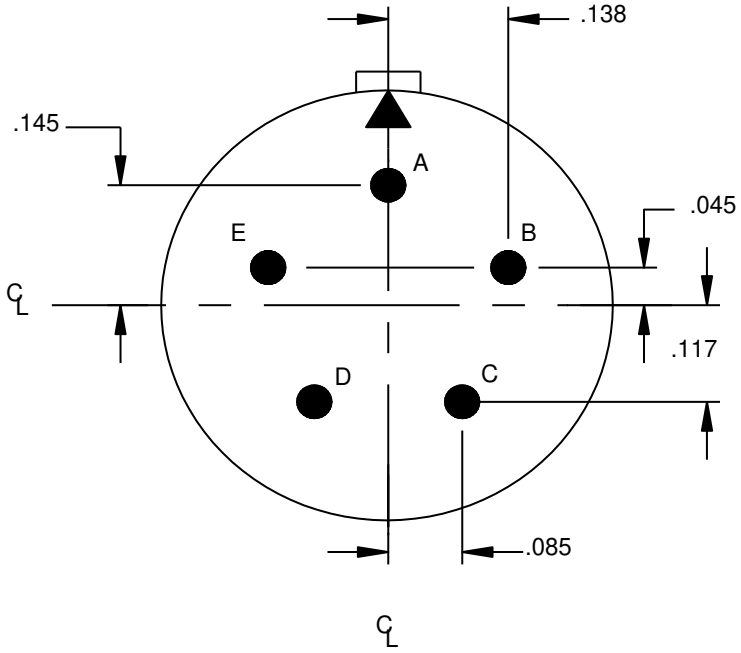
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -4 | 1 | 16S (socket)<br>16 (socket)<br>16 (pin) | D | All | MS33682-4 |
| 15 | -4 | 1 | 16 | D | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14S-5 and 15-5

(Insert arrangement 14S-5 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14S-5 and 15-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.145 (3.68) |
| B | +.138 (3.51) | +.045 (1.14) |
| C | +.085 (2.16) | -.117 (2.97) |
| D | -.085 (2.16) | -.117 (2.97) |
| E | -.138 (3.51) | +.045 (1.14) |

| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -5 | 5 | 16S (socket)<br>16 (socket)<br>16 (pin) | INST | All | MS33682-5 |
| 15 | -5 | 5 | 16 | INST | All | |

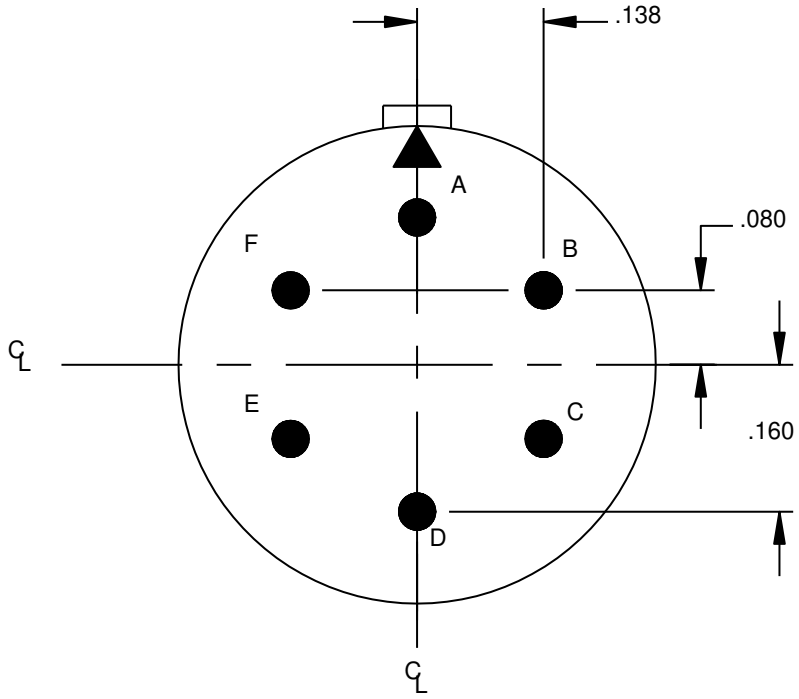
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 110 | --- | --- |

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14S-6 and 15-6

(Insert arrangement 14S-6 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14S-6 and 15-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.160 (4.06) |
| B | +.138 (3.51) | +.080 (2.03) |
| C | +.138 (3.51) | -.080 (2.03) |
| D | +.000 (0.00) | -.160 (4.06) |
| E | -.138 (3.51) | -.080 (2.03) |
| F | -.138 (3.51) | +.080 (2.03) |

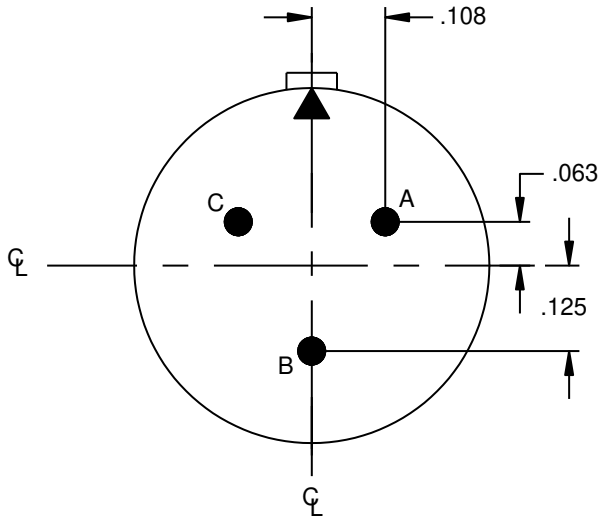
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -6 | 6 | 16S (socket)<br>16 (socket)<br>16 (pin) | INST | All | MS33682-6 |
| 15 | -6 | 6 | 16 | INST | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14S-7 and 15-7

| Contacts (Insert Arrangements 14S-7 and 15-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.108 (2.74) | +.063 (1.60) |
| B | +.000 (0.00) | -.125 (3.18) |
| C | -.108 (2.74) | +.063 (1.60) |

| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -7 | 3 | 16S (socket)<br>16 (socket)<br>16 (pin) | A | All | MS33682-7 |
| 15 | -7 | 3 | 16 | A | All | |

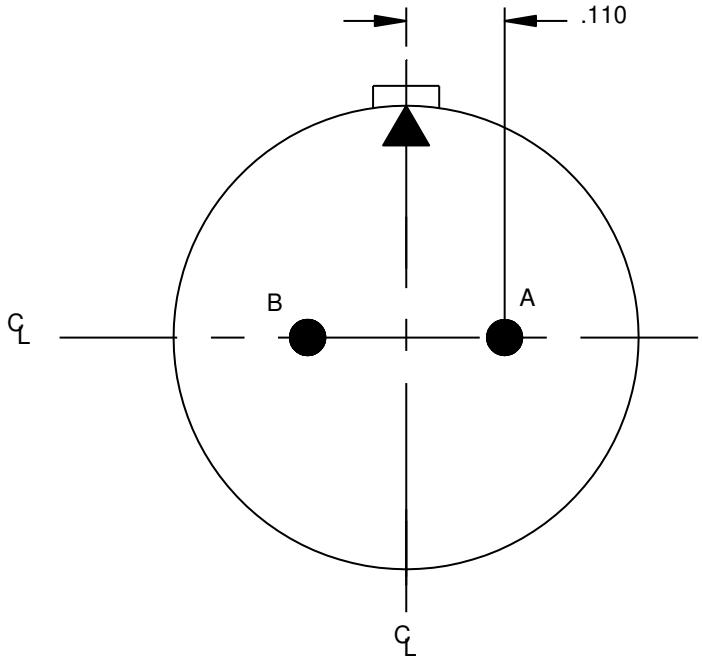
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | | |
|---|---|---|---|---|
| A degrees (A°) | | | | |
| W | X | Y | Z | |
| 90 | 180 | 270 | --- | |

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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14S-9 and 15-9

(Insert arrangement 14S-9 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 14S-9 and 15-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.110 (2.79) | +.000 (0.00) |
| B | -.110 (2.79) | -.000 (0.00) |

| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 14S | -9 | 2 | 16S (socket)<br>16 (socket)<br>16 (pin) | A | All | MS33682-9 |
| 15 | -9 | 2 | 16 | A | All | |

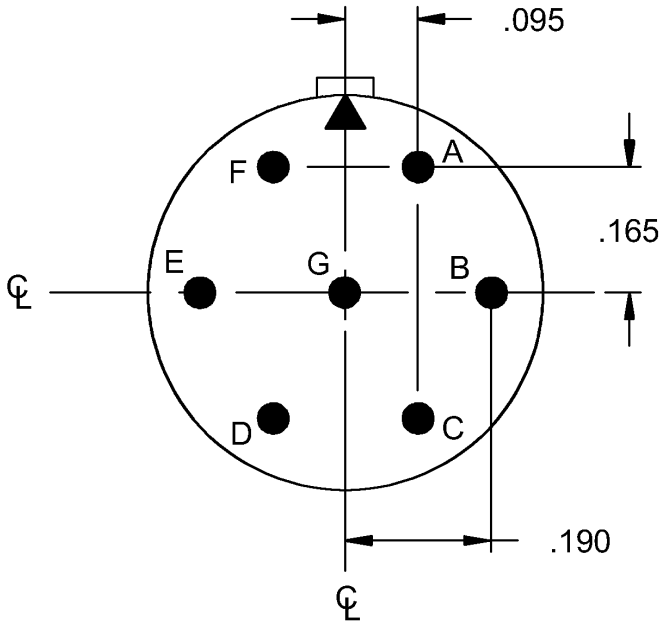
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 4. Insert arrangement, electrical connector, shell size 14, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151, shell size 14S, SAE-AS50151 and SAE-AS95234 only, and shell size 15, MIL-DTL-83723 only – Continued.



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16S-1 and 17-1

| Contacts (Insert arrangements 16S-1 and 17-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.095 (2.41) | +.165 (4.19) |
| B | +.190 (4.83) | +.000 (0.00) |
| C | +.095 (2.41) | -.165 (4.19) |
| D | -.095 (2.41) | -.165 (4.19) |
| E | -.190 (4.83) | +.000 (0.00) |
| F | -.095 (2.41) | +.165 (4.19) |
| G | +.000 (0.00) | +.000 (0.00) |

| Shell sizes | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -1 | 7 | 16S (socket)<br>16 (socket)<br>16 (pin) | A | All | MS33683-1 |
| 17 | -1 | 7 | 16 | A | All | |

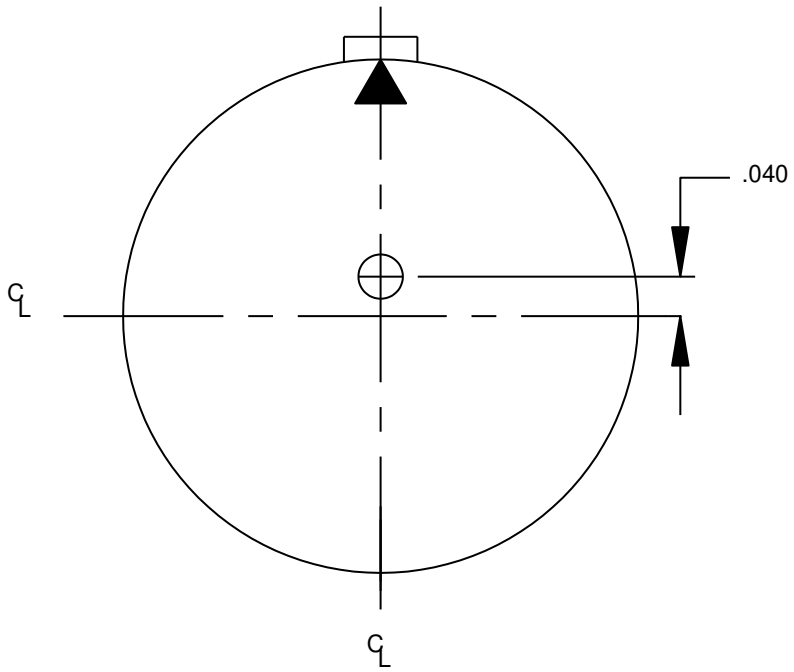
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only.



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16-2

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 16-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.040 (10.16) |

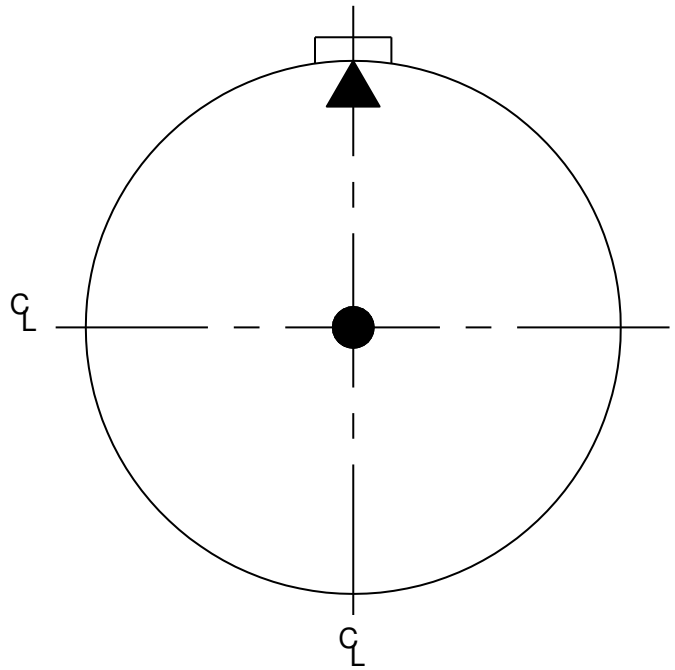
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -2 | 1 | 12 | E | All | MS33683-2 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16S-3 and 17-3

(Insert arrangement 16S-3 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 16S-3 and 17-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

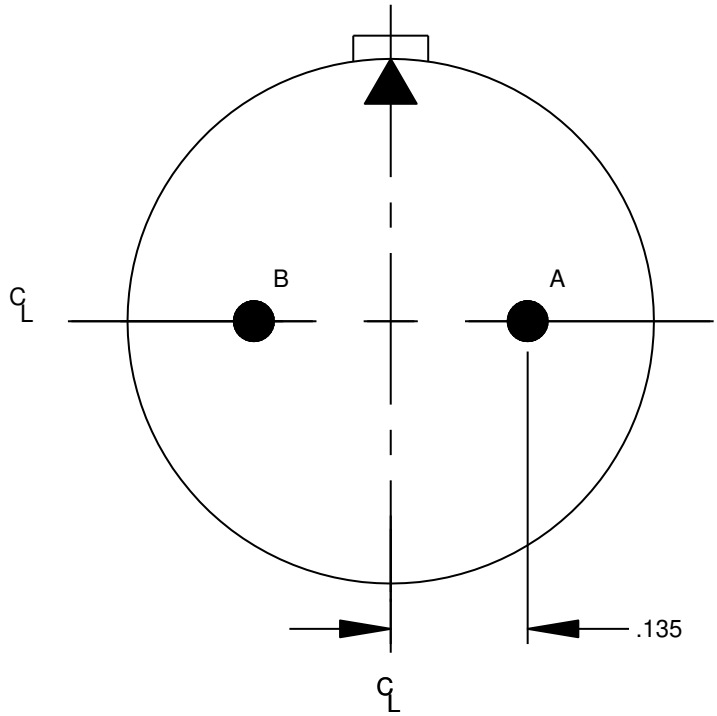
| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -3 | 1 | 16S (socket)<br>16 (socket)<br>16 (pin) | B | All | MS33683-3 |
| 17 | -3 | 1 | 16 | B | All | |

Note: This insert arrangement has no alternate insert positions. 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16S-4 and 17-4

(Insert arrangement 16S-4 applicable to SAE-AS95234.)

| Contacts (Insert arrangements 16S-4 and 17-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.135 (3.43) | +.000 (0.00) |
| B | -.135 (3.43) | +.000 (0.00) |

| Shell size | Arrangement number | Number of contacts | Size contacts (see note) | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -4 | 2 | 16S (socket)<br>16 (socket)<br>16 (pin) | D | All | MS33683-4 |
| 17 | -4 | 2 | 16 | D | All | |

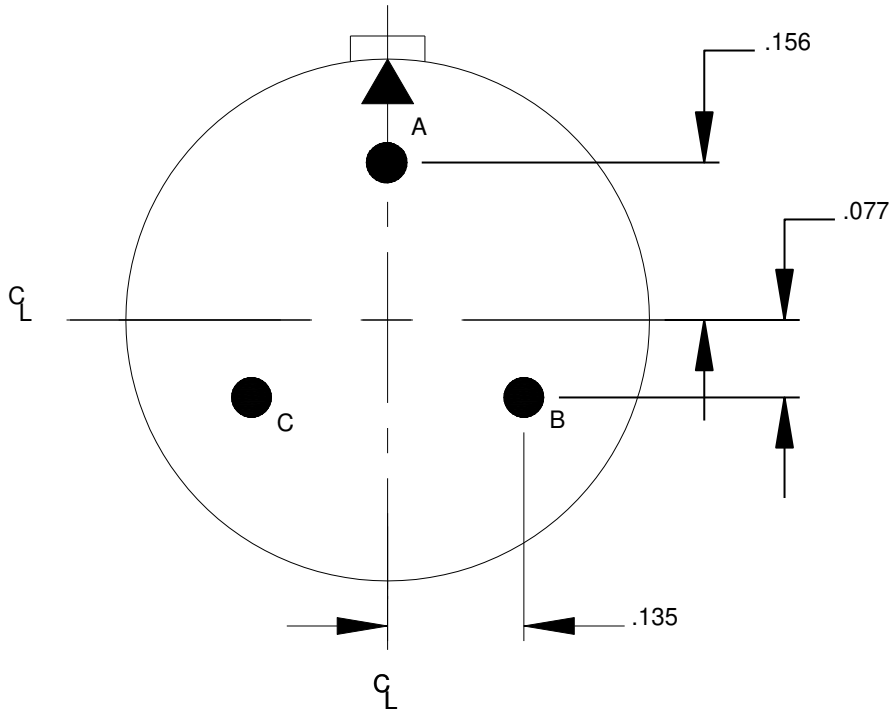
Note: 16S sockets (M39029/30-217) are applicable to rear release, SAE-AS50151 connectors only. Size 16 sockets (M39029/45-295) are applicable to SAE-AS95234 front release connectors.

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16S-5 and 17-5

(Inactive for New Design after 13 May 1954.)

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 16S-5 and 17-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.156 (3.96) |
| B | +.135 (3.43) | -.077 (1.96) |
| C | -.135 (3.43) | -.077 (1.96) |

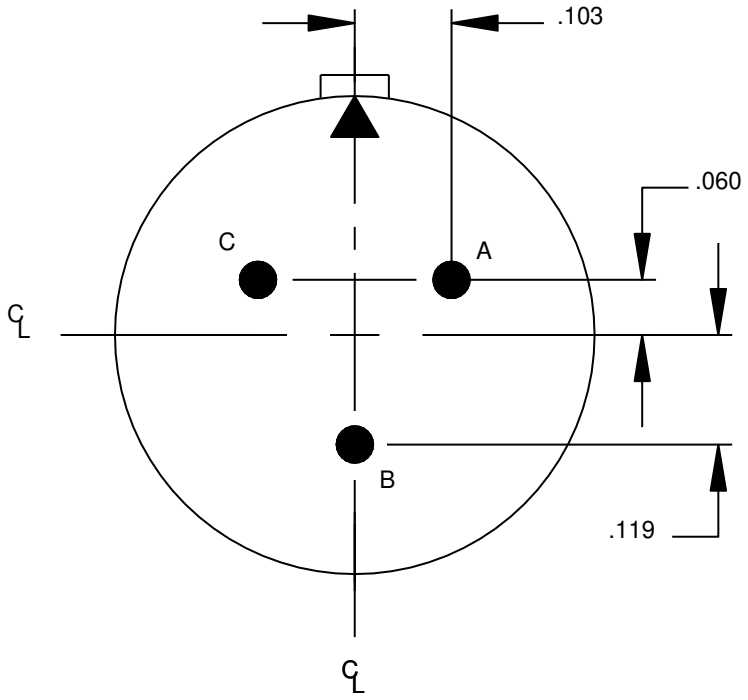
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -5 | 3 | 16S (socket)<br>16 (pin) | A | All | MS33683-5 |
| 17 | -5 | 3 | 16 | A | All | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16S-6 and 17-6

(Inactive for New Design after 30 July 1951.)

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 16S-6 and 17-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.103 (2.62) | +.060 (1.52) |
| B | +.000 (0.00) | -.119 (3.02) |
| C | -.103 (2.62) | +.060 (1.52) |

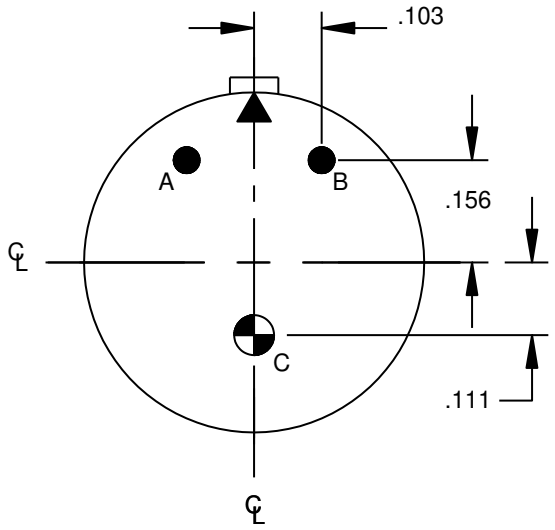
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -6 | 3 | 16S (socket)<br>16 (pin) | A | All | MS33683-6 |
| 17 | -6 | 3 | 16 | A | All | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-7

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 16-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.103 (2.62) | +.156 (3.96) |
| B | +.103 (2.62) | +.156 (3.96) |
| C | +.000 (0.00) | -.111 (2.82) |

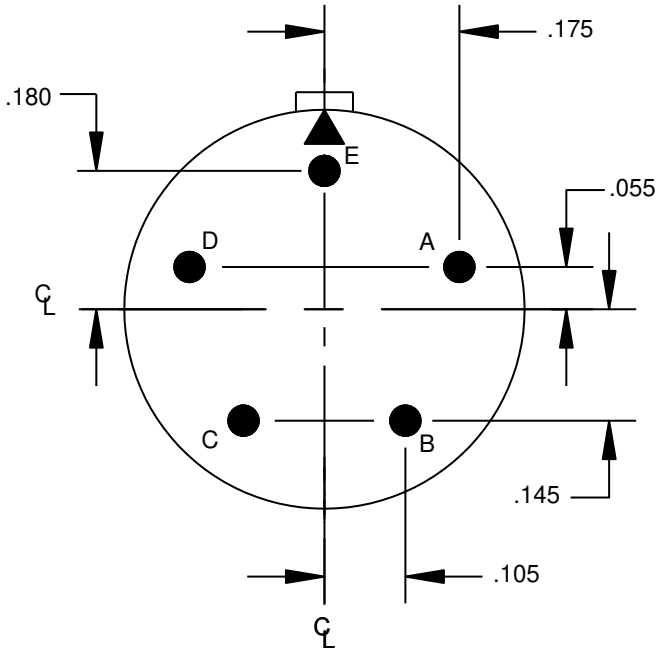
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -7 | 2 | 16 | A | A,B | MS33683-7 |
| | | 1 | 8 | | C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16S-8 and 17-8

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangements 16S-8 and 17-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.175 (4.45) | +.055 (1.40) |
| B | +.105 (2.67) | -.145 (3.68) |
| C | -.105 (2.67) | -.145 (3.68) |
| D | -.175 (4.45) | +.055 (1.40) |
| E | +.000 (0.00) | +.180 (4.47) |

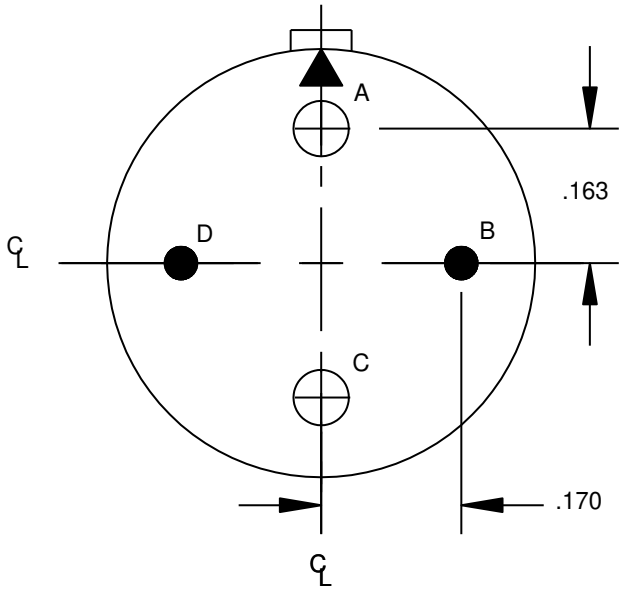
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16S | -8 | 5 | 16S (socket)<br>16 (pin) | A | All | MS33683-8 |
| 17 | -8 | 5 | 16 | A | All | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 170 | 265 | --- |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-9

| Contacts (Insert arrangement 16-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.163 (4.14) |
| B | +.170 (4.32) | +.000 (0.00) |
| C | +.000 (0.00) | -.163 (4.14) |
| D | -.170 (4.32) | +.000 (0.00) |

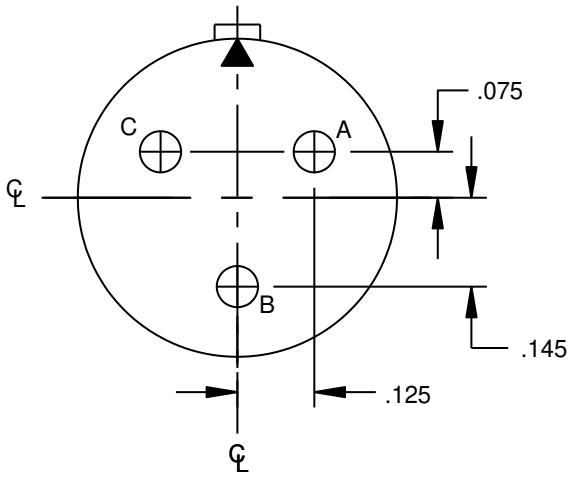
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -9 | 2 | 16 | A | All | MS33683-9 |
| | | 2 | 12 | | | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-10

| Contacts (Insert arrangement 16-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.125 (3.18) | +.075 (1.91) |
| B | +.000 (0.00) | -.145 (3.68) |
| C | -.125 (3.18) | +.075 (1.91) |

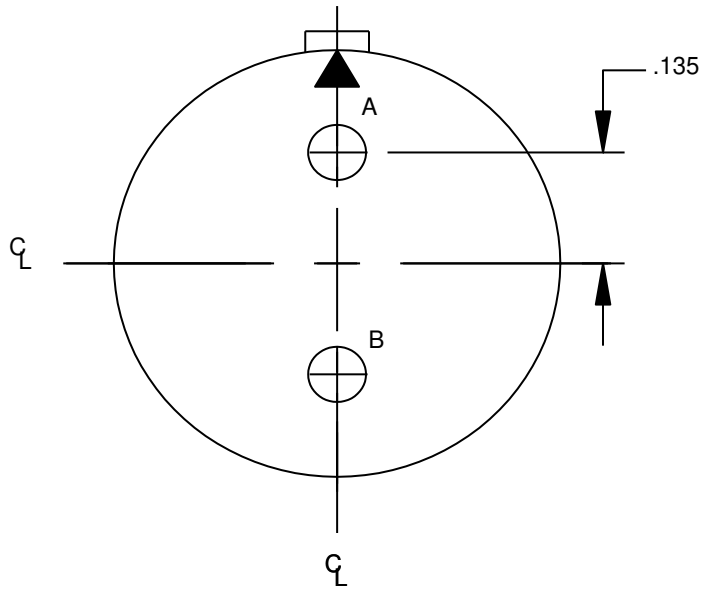
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -10 | 3 | 12 | A | All | MS33683-10 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-11

| Contacts (Insert arrangement 16-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.135 (3.43) |
| B | +.000 (0.00) | -.135 (3.43) |

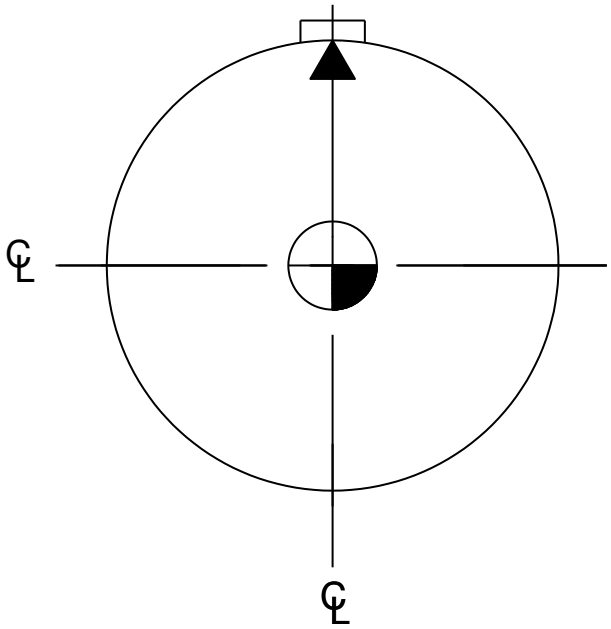
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -11 | 2 | 12 | A | All | MS33683-11 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-12

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 16-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

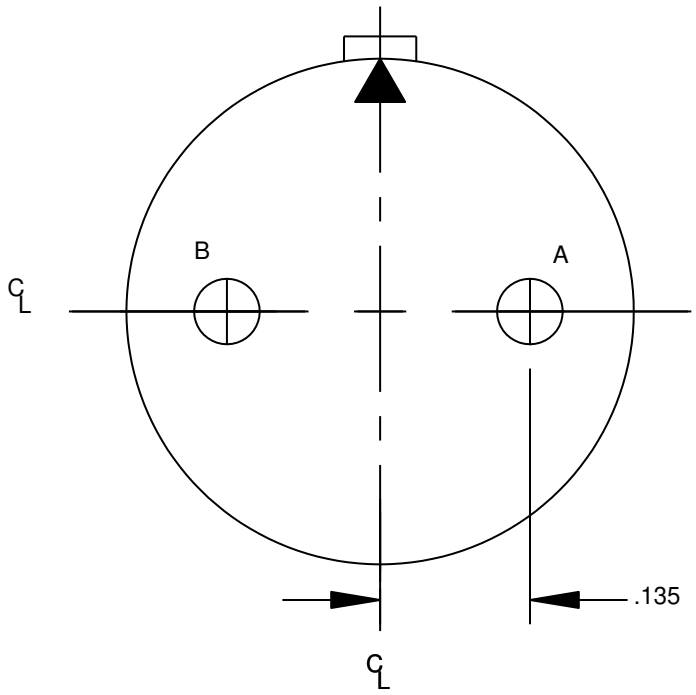
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -12 | 1 | 4 | A | All | MS33683-12 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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16-13

CONTACT A – IRON
CONTACT B- COPPER-NICKEL ALLOY

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 16-13) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.135 (3.43) | +.000 (0.00) |
| B | -.135 (3.43) | +.000 (0.00) |

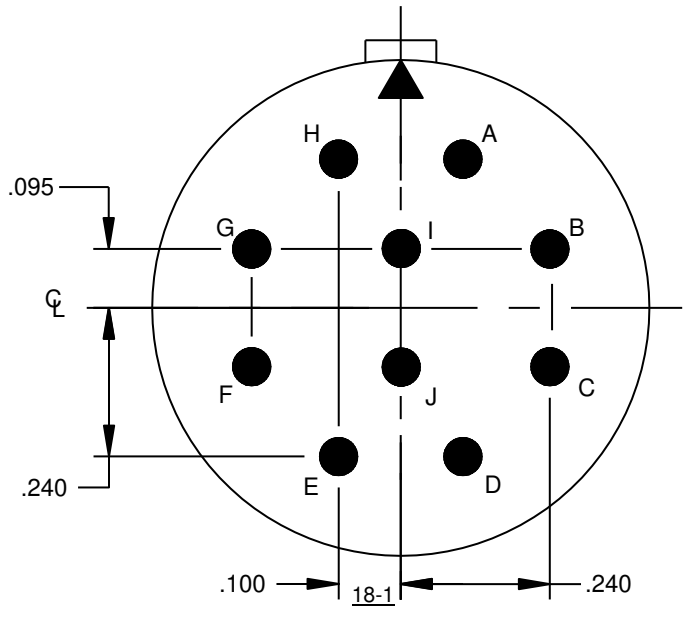
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 16 | -13 | 2 | 12 | A | All | MS33683-13 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 5. Insert arrangement, electrical connector, shell size 16, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234, shell size 16S, SAE-AS50151 and SAE-AS95234 only, and shell size 17, MIL-DTL-83723 only – Continued.



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(Inactive for new design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.100 (2.54) | +.240 (6.10) |
| B | +.240 (6.10) | +.095 (2.41) |
| C | +.240 (6.10) | -.095 (2.41) |
| D | +.100 (2.54) | -.240 (6.10) |
| E | -.100 (2.54) | -.240 (6.10) |
| F | -.240 (6.10) | -.095 (2.41) |
| G | -.240 (6.10) | +.095 (2.41) |
| H | -.100 (2.54) | +.240 (6.10) |
| I | +.000 (0.00) | +.095 (2.41) |
| J | +.000 (0.00) | -.095 (2.41) |

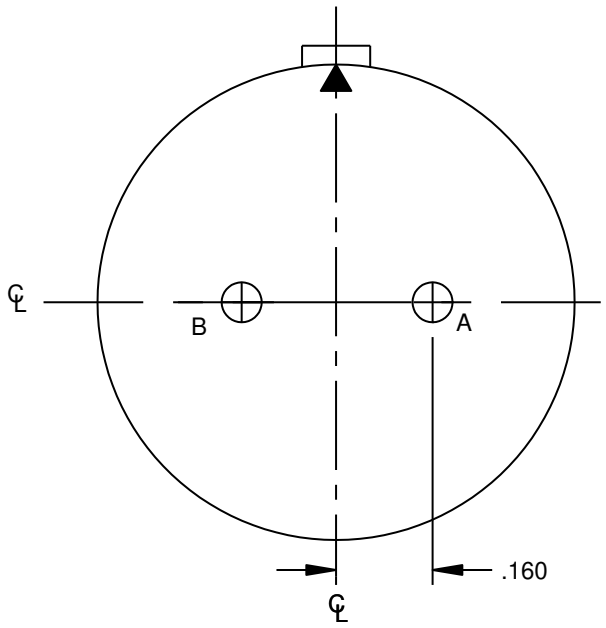
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -1 | 4 | 16 | A | B, C, F, G | MS33684-1 |
| | | 6 | | INST | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234.



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18-3

(Inactive for New Design after 13 May 1954 for SAE-AS50151
and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 18-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.160 (4.06) | +.000 (0.00) |
| B | -.160 (4.06) | +.000 (0.00) |

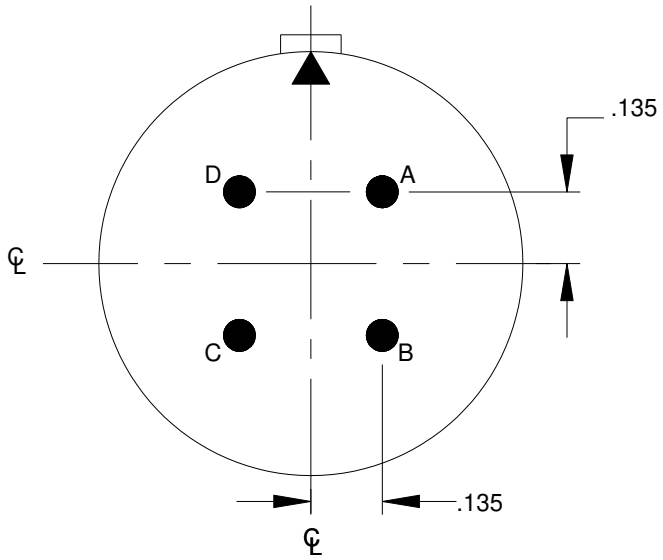
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -3 | 2 | 12 | D | All | MS33684-3 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-4

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.135 (3.43) | +.135 (3.43) |
| B | +.135 (3.43) | -.135 (3.43) |
| C | -.135 (3.43) | -.135 (3.43) |
| D | -.135 (3.43) | +.135 (3.43) |

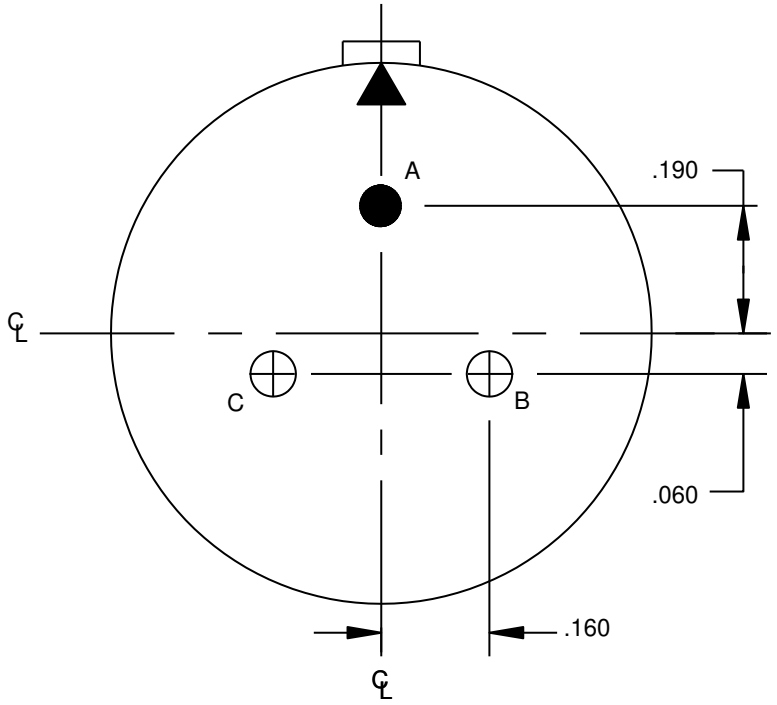
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -4 | 4 | 16 | D | All | MS33684-4 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-5

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.190 (4.83) |
| B | +.160 (4.06) | -.060 (1.52) |
| C | -.160 (4.06) | -.060 (1.52) |

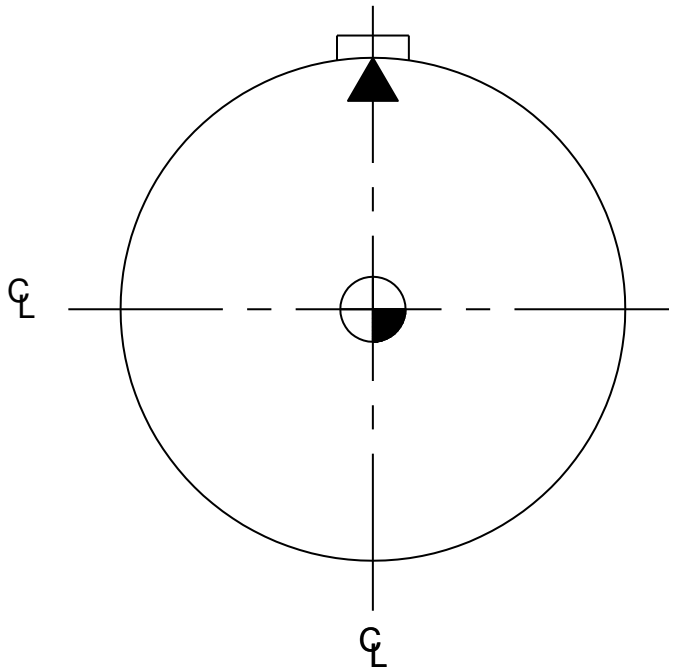
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -5 | 1 | 16 | D | All | MS33684-5 |
| | | 2 | 12 | | | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-6

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 18-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

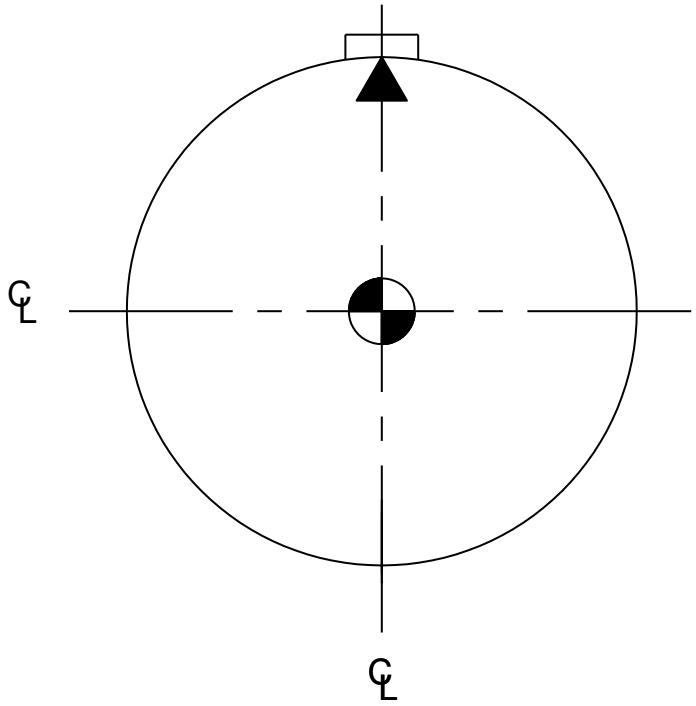
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -6 | 1 | 4 | D | All | MS33684-6 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-7

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 18-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable. | +.000 (0.00) | +.000 (0.00) |

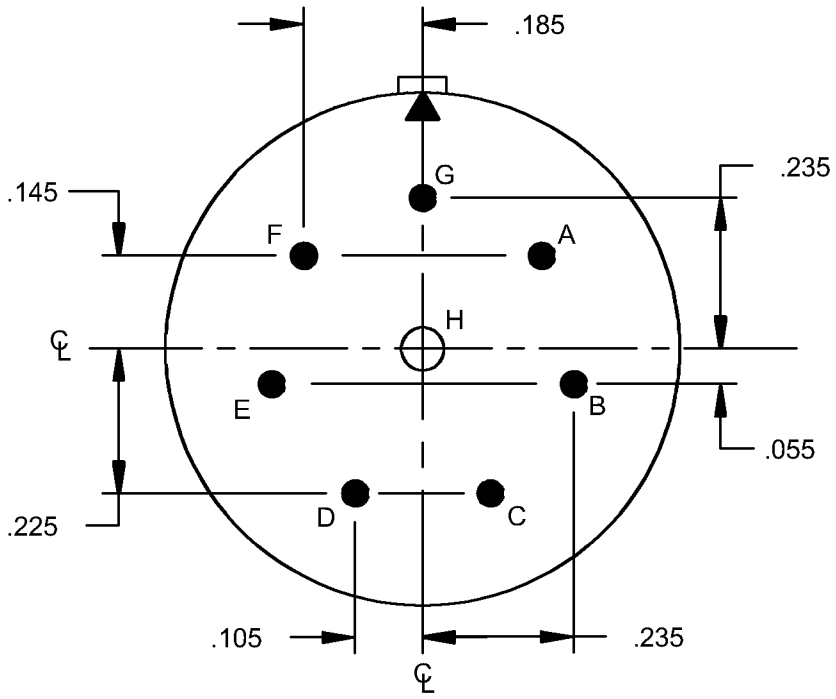
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -7 | 1 | 8 | B | All | MS33684-7 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-8

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.185 (4.70) | +.145 (3.68) |
| B | +.235 (5.97) | -.055 (1.40) |
| C | +.105 (2.67) | -.225 (5.72) |
| D | -.105 (2.67) | -.225 (5.72) |
| E | -.235 (5.97) | -.055 (1.40) |
| F | -.185 (4.70) | +.145 (3.68) |
| G | +.000 (0.00) | +.235 (5.97) |
| H | +.000 (0.00) | +.000 (0.00) |

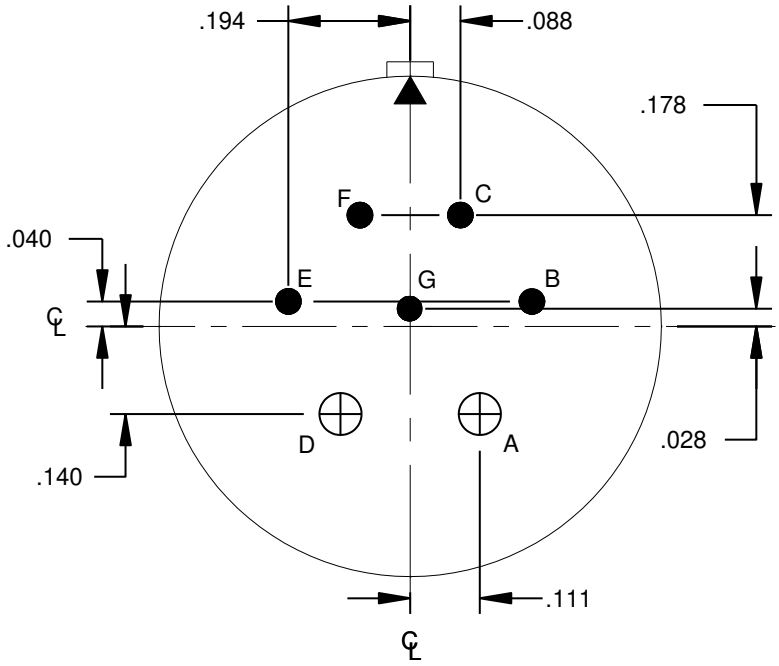
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -8 | 7 | 16 | A | All | MS33684-8 |
| | | 1 | 12 | | | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | --- | --- | 290 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-9

| Contacts (Insert arrangement 18-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.111 (2.82) | -.140 (3.56) |
| B | +.194 (4.93) | +.040 (1.02) |
| C | +.088 (2.24) | +.178 (4.52) |
| D | -.111 (2.82) | -.140 (3.56) |
| E | -.194 (4.93) | +.040 (1.02) |
| F | -.088 (2.24) | +.178 (4.52) |
| G | +.000 (0.00) | +.028 (0.71) |

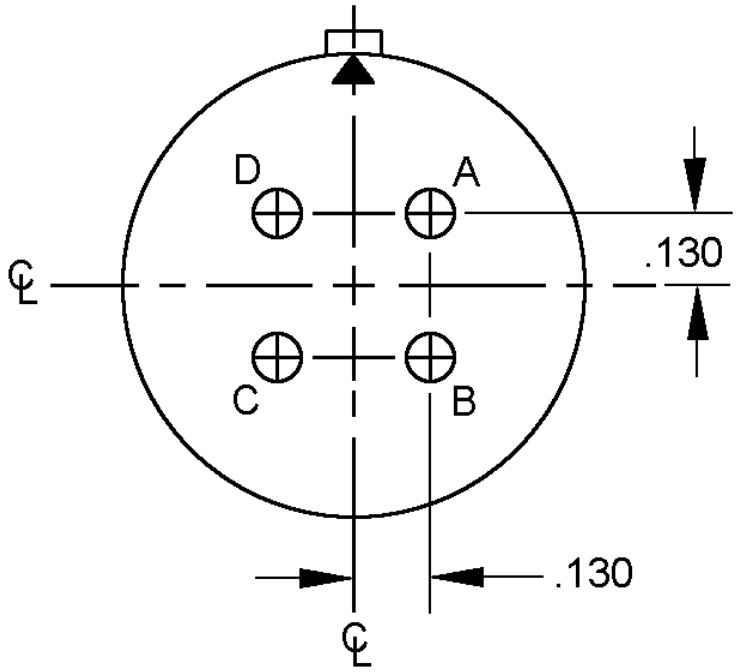
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -9 | 5 | 16 | INST | B,C,E,F,G | MS33684-9 |
| | | 2 | 12 | | A,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-10

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 18-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.130 (3.30) | +.130 (3.30) |
| B | +.130 (3.30) | -.130 (3.30) |
| C | -.130 (3.30) | -.130 (3.30) |
| D | -.130 (3.30) | +.130 (3.30) |

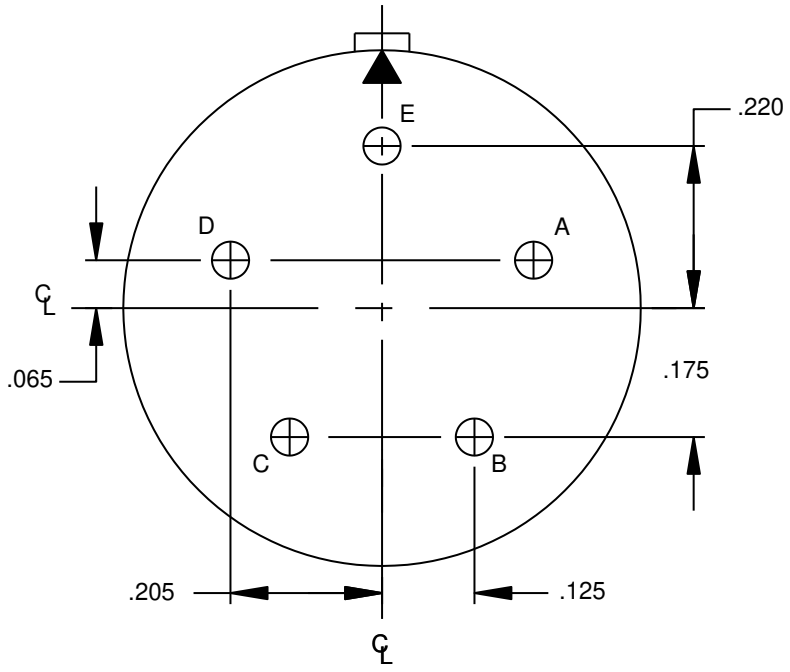
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -10 | 4 | 12 | A | All | MS33684-10 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 120 | 240 | --- |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-11

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.205 (5.21) | +.065 (1.65) |
| B | +.125 (3.18) | -.175 (4.45) |
| C | -.125 (3.18) | -.175 (4.45) |
| D | -.205 (5.21) | +.065 (1.65) |
| E | +.000 (0.00) | +.220 (5.59) |

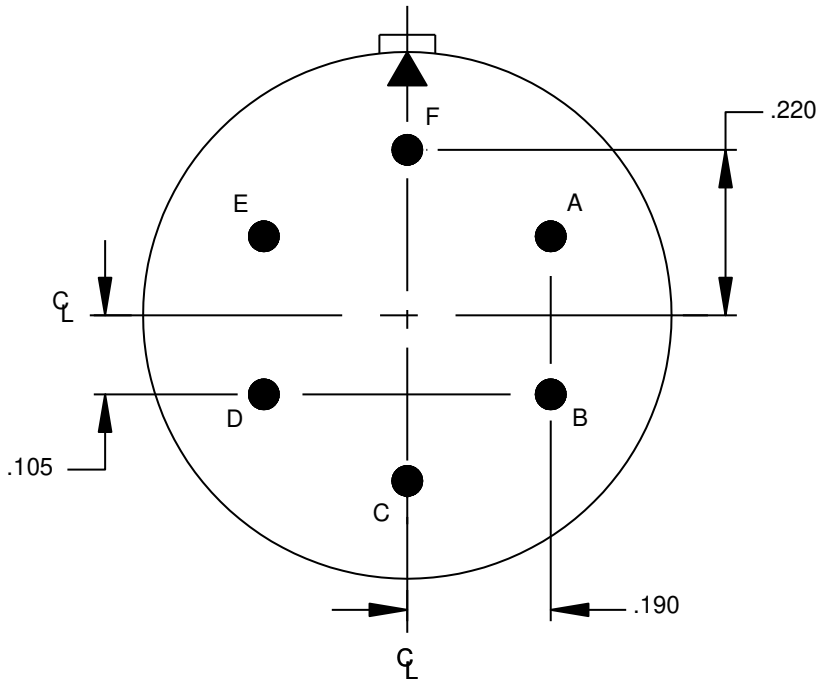
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -11 | 5 | 12 | A | All | MS33684-11 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 170 | 265 | --- |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-12

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.190 (4.83) | +.105 (2.67) |
| B | +.190 (4.83) | -.105 (2.67) |
| C | +.000 (0.00) | -.220 (5.59) |
| D | -.190 (4.83) | -.105 (2.67) |
| E | -.190 (4.83) | +.105 (2.67) |
| F | +.000 (0.00) | +.220 (5.59) |

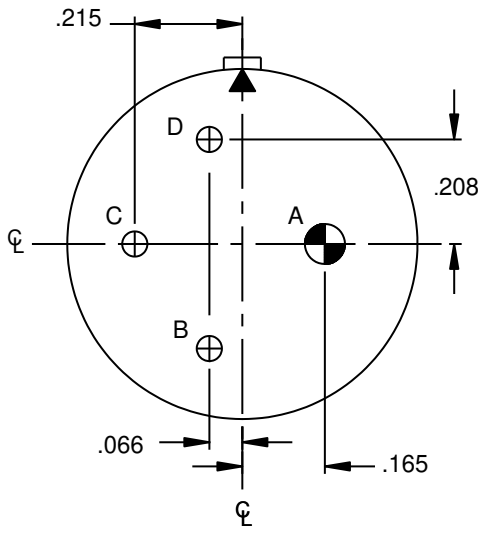
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -12 | 6 | 16 | A | All | MS33684-12 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-13

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-13) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.165 (4.19) | +.000 (0.00) |
| B | - .066 (1.68) | -.208 (5.28) |
| C | - .215 (5.46) | +.000 (0.00) |
| D | - .066 (1.68) | +.208 (5.28) |

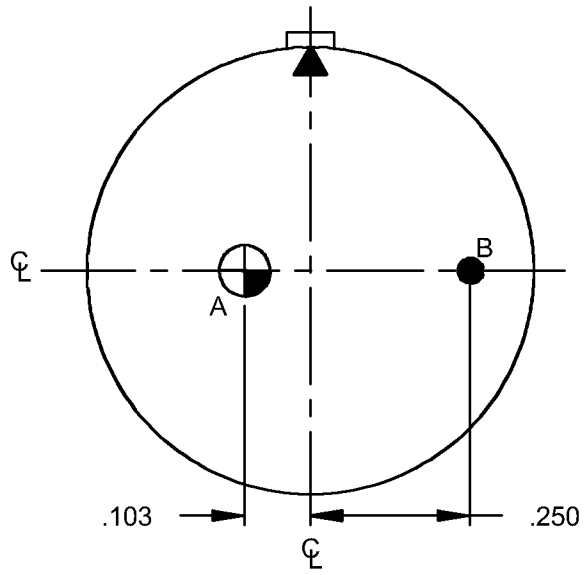
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -13 | 3 | 12 | A | All others | MS33684-13 |
| | | 1 | 8 | | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-14

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-14) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | - .103 (2.62) | +.000 (0.00) |
| B | +.250 (6.35) | +.000 (0.00) |

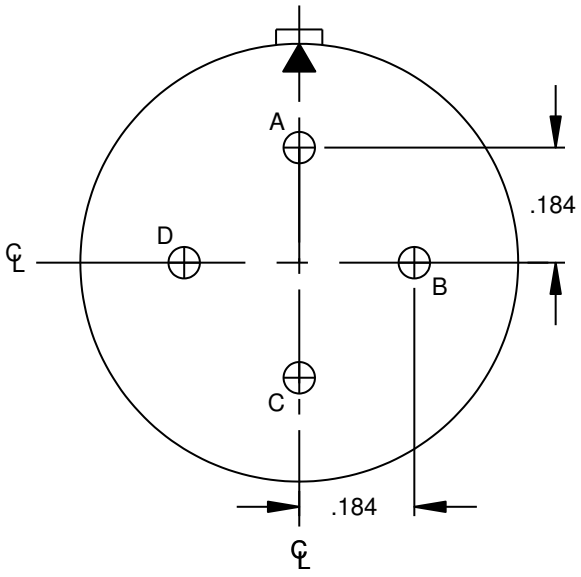
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -14 | 1 | 16 | A | All | MS33684-14 |
| | | 1 | 4 | | | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-15

CONTACTS A AND C – IRON
CONTACTS B AND D – COPPER-NICKEL ALLOY

(Inactive for New Design of Air Force airborne equipment after 28 May 1971 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 18-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.184 (4.67) |
| B | +.184 (4.67) | +.000 (0.00) |
| C | +.000 (0.00) | -.184 (4.67) |
| D | -.184 (4.67) | +.000 (0.00) |

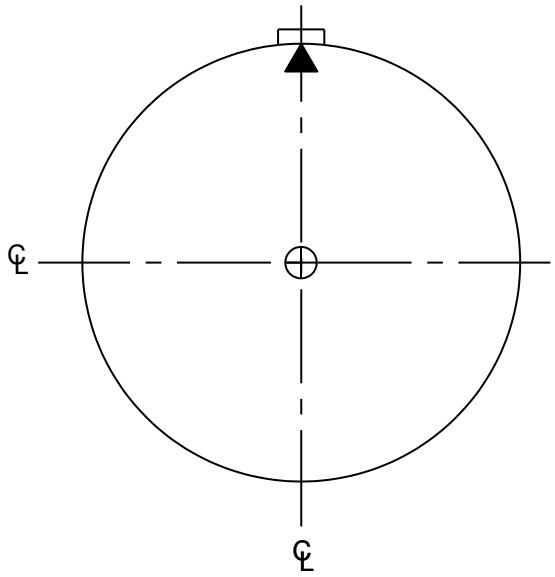
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -15 | 4 | 12 | A | All | MS33684-15 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 120 | 240 | --- |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-16

(Inactive for New Design of Air Force airborne equipment after 28 May 1971.)

(Not applicable to SAE-AS95234.)

| Contacts (Insert arrangement 18-16) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

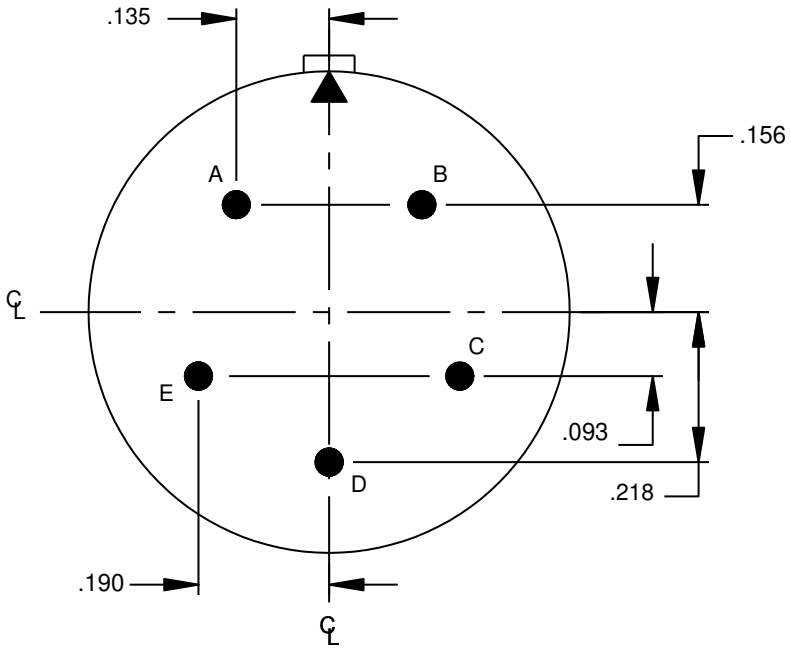
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -16 | 1 | 12 | C | All | MS33684-16 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-20

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 18-20) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.135 (3.43) | +.156 (3.96) |
| B | +.135 (3.43) | +.156 (3.96) |
| C | +.190 (4.83) | -.093 (2.36) |
| D | -.000 (0.00) | -.218 (5.54) |
| E | -.190 (4.83) | -.093 (4.83) |

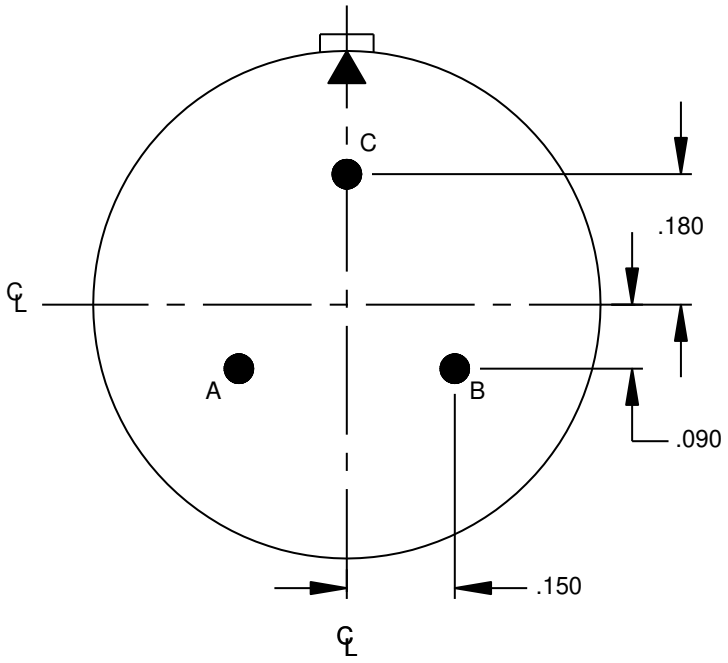
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -20 | 5 | 16 | A | All | MS33684-20 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-22

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 18-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.150 (3.81) | -.090 (2.29) |
| B | +.150 (3.81) | -.090 (2.29) |
| C | +.000 (0.00) | +.180 (4.57) |

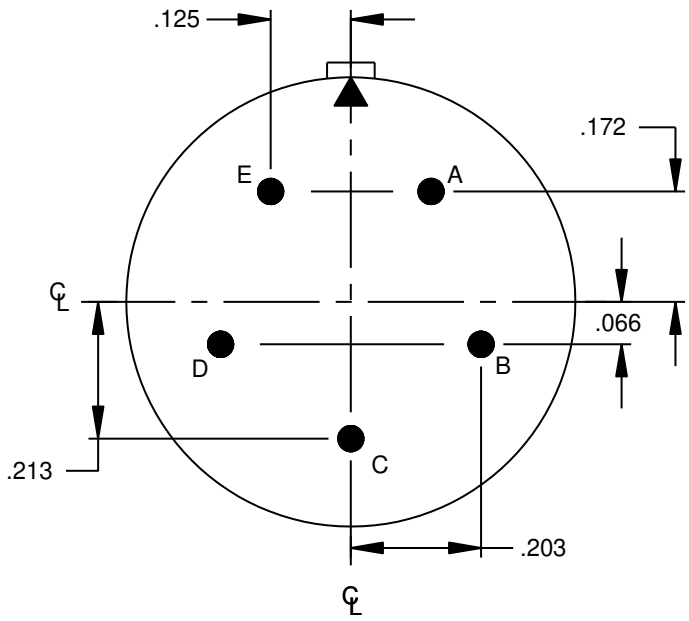
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -22 | 3 | 16 | D | All | MS33684-22 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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18-29

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 18-29) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.125 (3.18) | +.172 (4.37) |
| B | +.203 (5.16) | -.066 (1.68) |
| C | +.000 (0.00) | -.213 (5.41) |
| D | -.203 (5.16) | -.066 (1.68) |
| E | -.125 (3.18) | +.172 (4.37) |

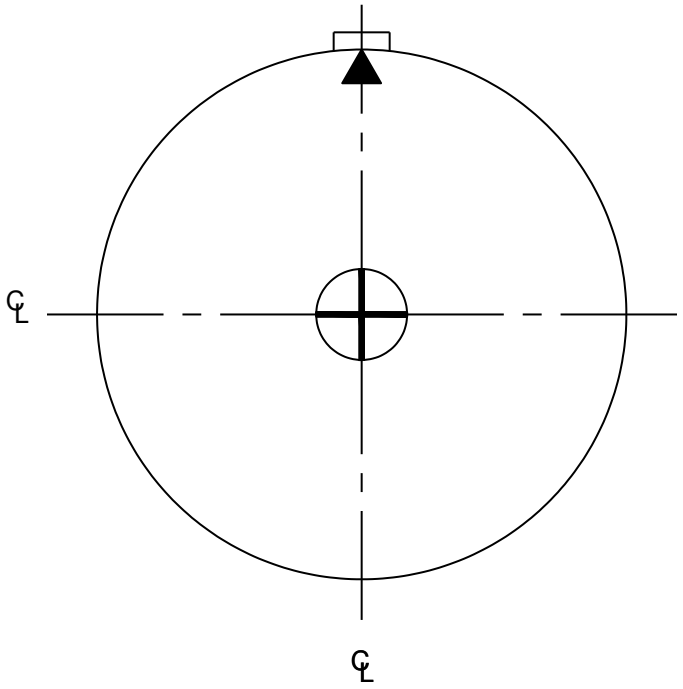
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 18 | -29 | 5 | 16 | A | All | MS33684-29 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 6. Insert arrangement, electrical connector, shell size 18, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-2

| Contacts (Insert arrangement 20-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| Not applicable | +.000 (0.00) | +.000 (0.00) |

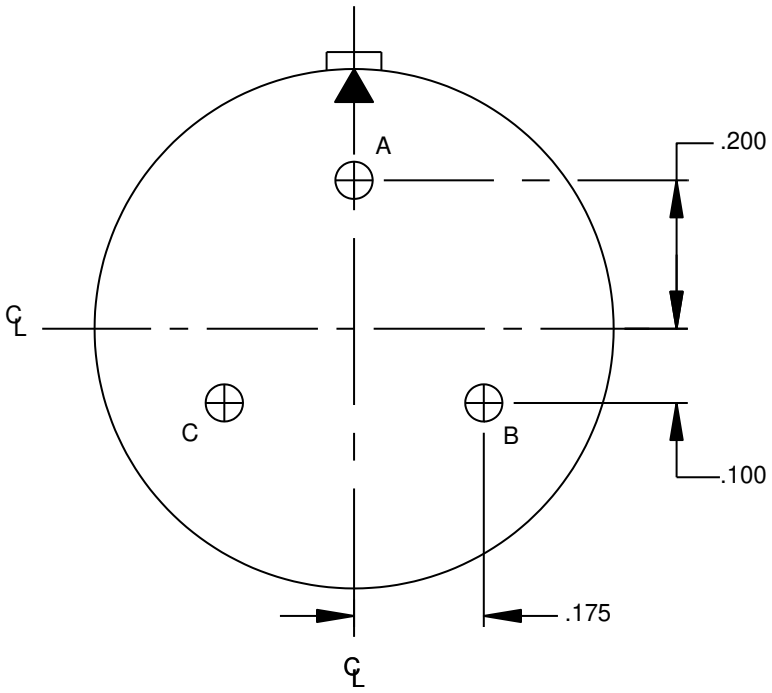
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -2 | 1 | 0 | D | All | MS33685-2 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234.



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20-3

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.200 (5.08) |
| B | +.175 (4.45) | -.100 (2.54) |
| C | -.175 (4.45) | -.100 (2.54) |

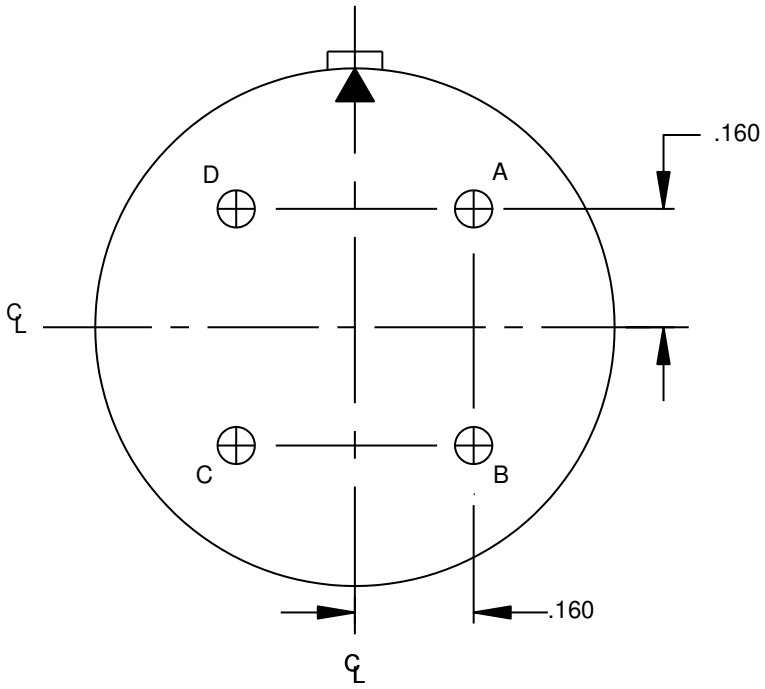
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -3 | 3 | 12 | D | All | MS33685-3 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-4

| Contacts (Insert arrangement 20-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.160 (4.06) | +.160 (4.06) |
| B | +.160 (4.06) | -.160 (4.06) |
| C | -.160 (4.06) | -.160 (4.06) |
| D | -.160 (4.06) | +.160 (4.06) |

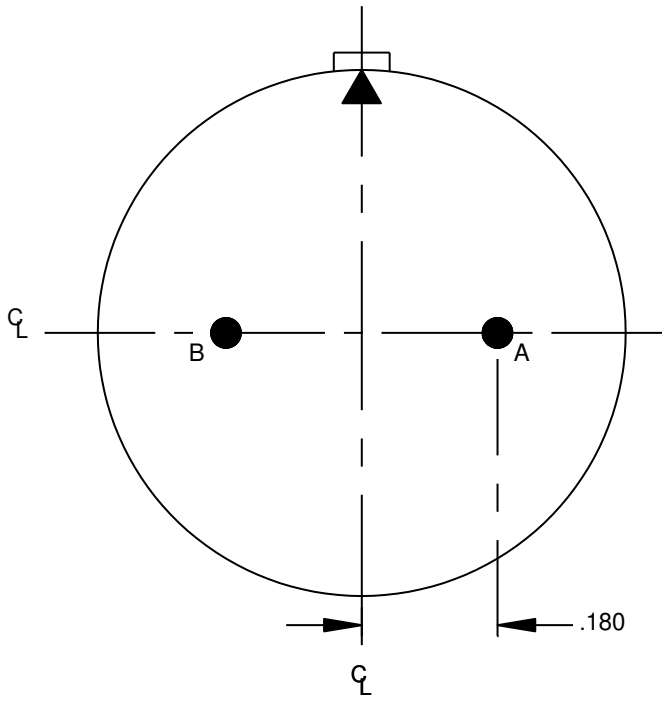
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -4 | 4 | 12 | D | All | MS33685-4 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 45 | 110 | 250 | --- |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-5

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.180 (4.57) | +.000 (0.00) |
| B | -.180 (4.57) | +.000 (0.00) |

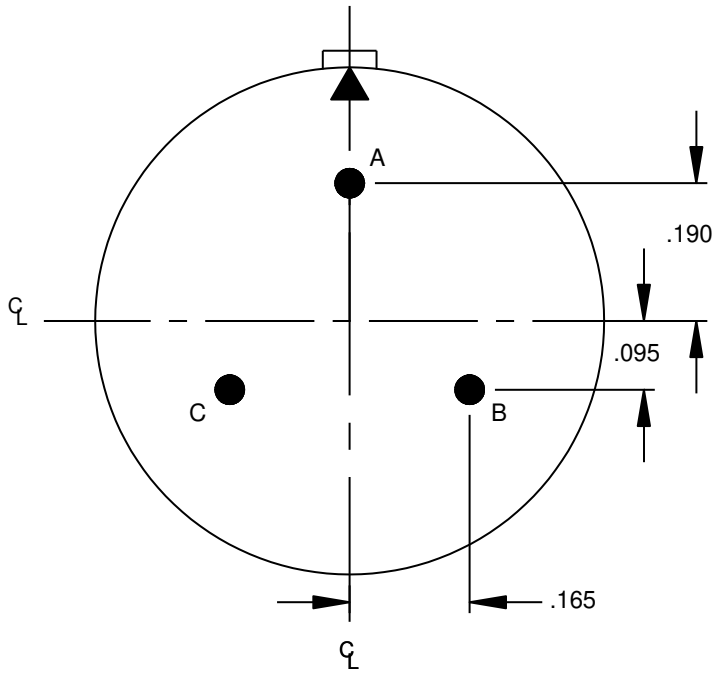
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -5 | 2 | 16 | E | All | MS33685-5 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-6

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.190 (4.83) |
| B | +.165 (4.19) | -.095 (2.41) |
| C | -.165 (4.19) | -.095 (2.41) |

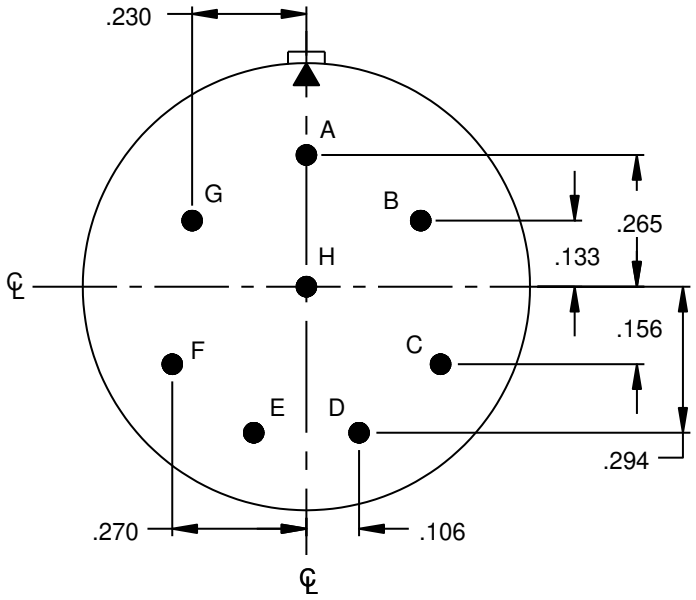
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -6 | 3 | 16 | D | All | MS33685-6 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-7

| Contacts (Insert arrangement 20-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.265 (6.73) |
| B | +.230 (5.84) | +.133 (3.38) |
| C | +.270 (6.86) | -.156 (3.96) |
| D | +.106 (2.69) | -.294 (7.47) |
| E | -.106 (2.69) | -.294 (7.47) |
| F | -.270 (6.86) | -.156 (3.96) |
| G | -.230 (5.84) | +.133 (3.38) |
| H | +.000 (0.00) | +.000 (0.00) |

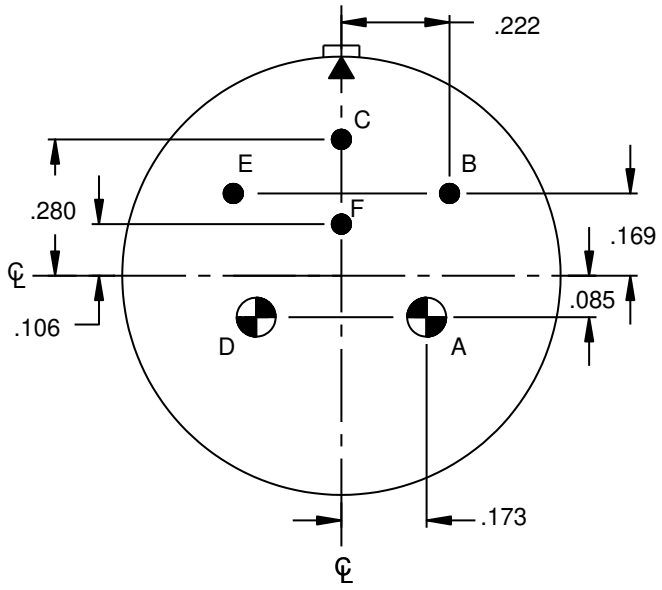
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -7 | 4 | 16 | D | A, B, G, H | MS33685-7 |
| | | 4 | | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-8

| Contacts (Insert arrangement 20-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.173 (4.39) | -.085 (2.16) |
| B | +.222 (5.64) | +.169 (4.29) |
| C | +.000 (0.00) | +.280 (7.11) |
| D | -.173 (4.39) | -.085 (2.16) |
| E | -.222 (5.64) | +.169 (4.29) |
| F | +.000 (0.00) | +.106 (2.69) |

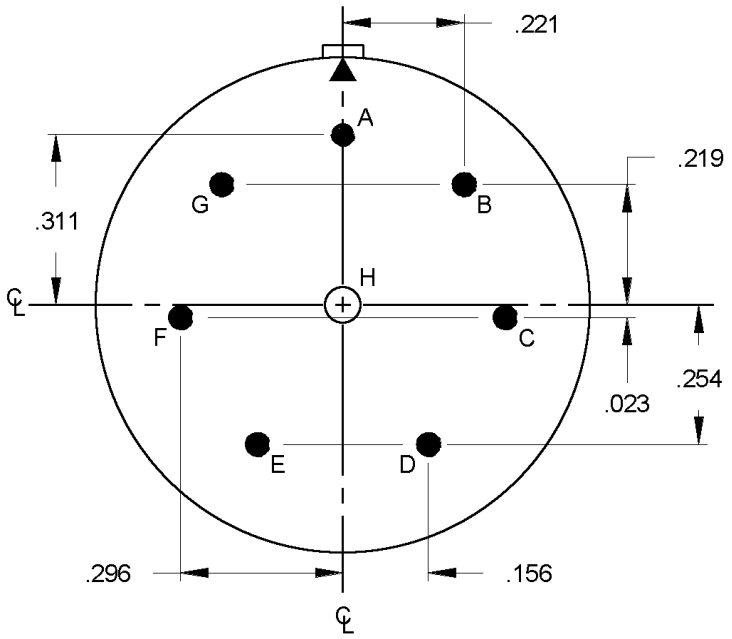
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -8 | 4 | 16 | INST | All others | MS33685-8 |
| | | 2 | 8 | | A,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-9

| Contacts (Insert arrangement 20-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.311 (7.90) |
| B | +.221 (5.61) | +.219 (5.56) |
| C | +.296 (7.52) | -.023 (0.58) |
| D | +.156 (3.96) | -.254 (6.45) |
| E | -.156 (3.96) | -.254 (6.45) |
| F | -.296 (7.52) | -.023 (0.58) |
| G | -.221 (5.61) | +.219 (5.56) |
| H | +.000 (0.00) | +.000 (0.00) |

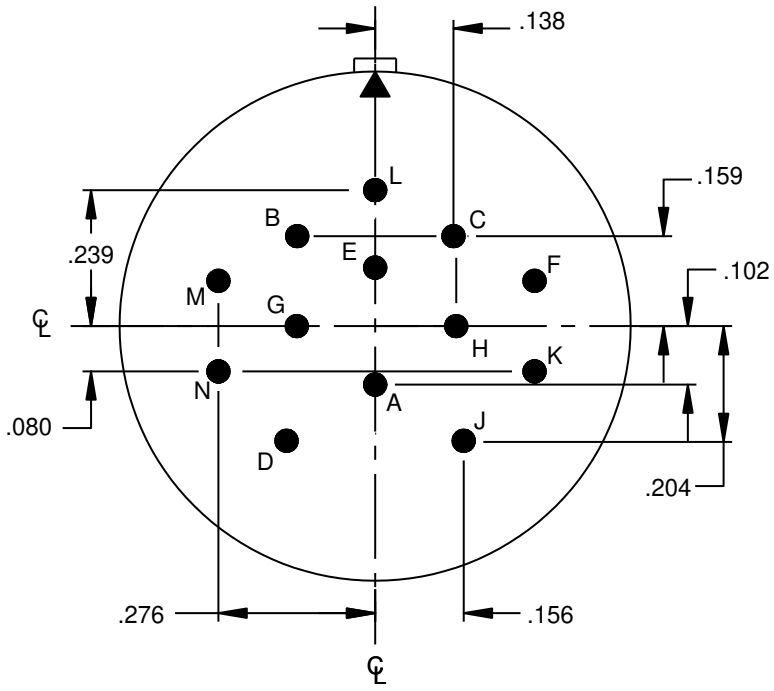
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -9 | 1 | 12 | D | H | MS33685-9 |
| | | 7 | 16 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-11

(Inactive for New Design after 1 June 1945 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | -.102 (2.59) |
| B | -.138 (3.51) | +.159 (4.04) |
| C | +.138 (3.51) | +.159 (4.04) |
| D | -.156 (3.96) | -.204 (5.18) |
| E | +.000 (0.00) | +.080 (2.03) |
| F | +.276 (7.01) | +.080 (2.03) |
| G | -.138 (3.51) | +.000 (0.00) |
| H | +.138 (3.51) | +.000 (0.00) |
| J | +.156 (3.96) | -.204 (5.18) |
| K | +.276 (7.01) | -.080 (2.03) |
| L | +.000 (0.00) | +.239 (6.07) |
| M | -.276 (7.01) | +.080 (2.03) |
| N | -.276 (7.01) | -.080 (2.03) |

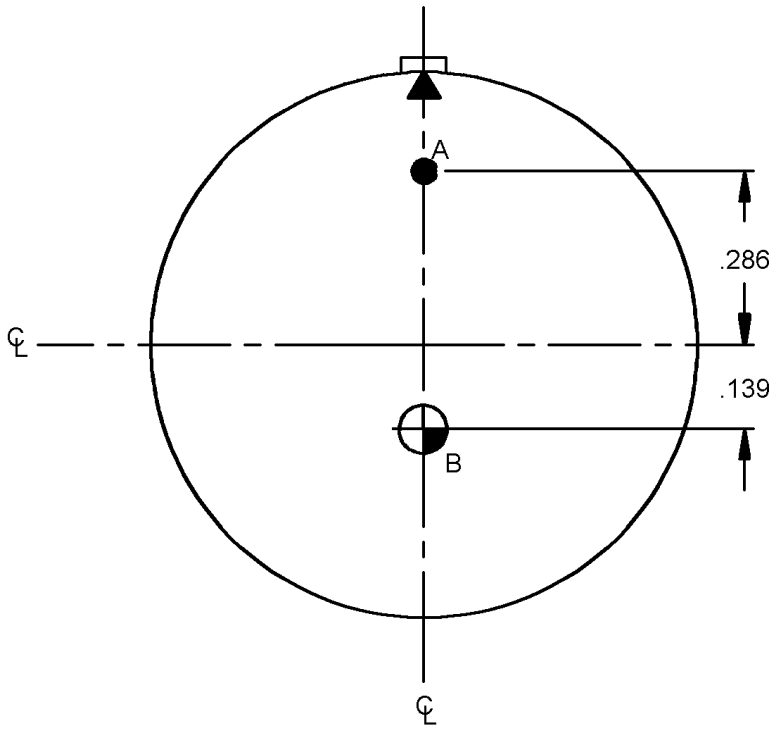
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -11 | 13 | 16 | INST | All | MS33685-11 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-12

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.286 (7.26) |
| B | +.000 (0.00) | -.139 (3.53) |

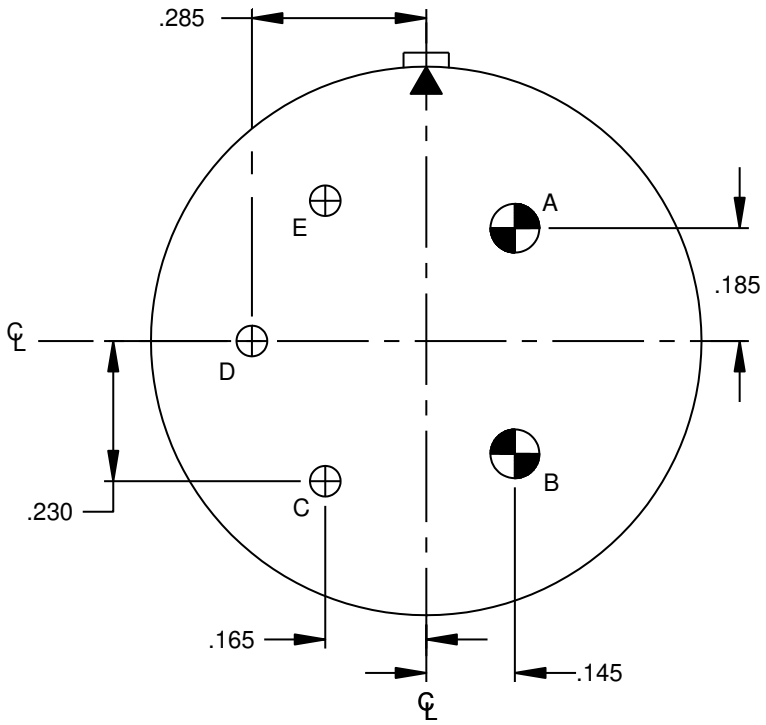
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -12 | 1 | 16 | A | All | MS33685-12 |
| | | 1 | 4 | | | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-14

| Contacts (Insert arrangement 20-14) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.145 (3.68) | +.185 (4.70) |
| B | +.145 (3.68) | -.185 (4.70) |
| C | -.165 (4.19) | -.230 (5.84) |
| D | -.285 (7.24) | +.000 (0.00) |
| E | -.165 (4.19) | +.230 (5.84) |

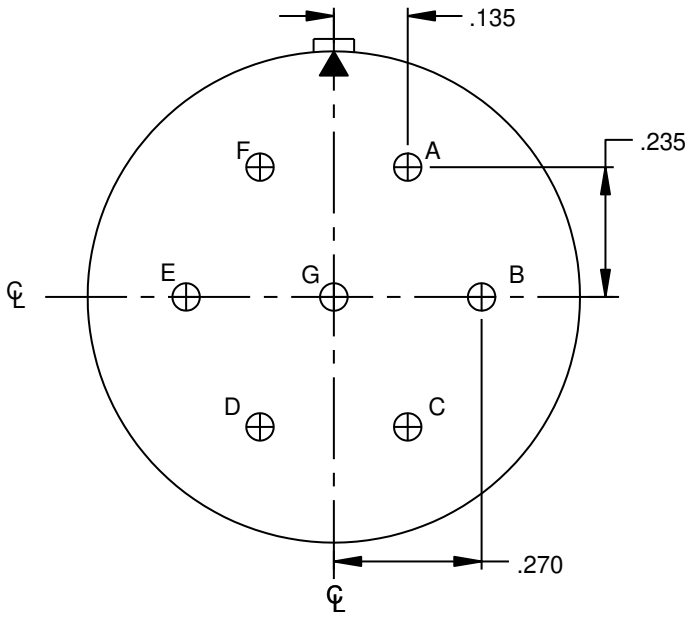
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -14 | 3 | 12 | A | C,D,E | MS33685-14 |
| | | 2 | 8 | | A,B | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-15

| Contacts (Insert arrangement 20-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.135 (3.43) | +.235 (5.97) |
| B | +.270 (6.86) | +.000 (0.00) |
| C | +.135 (3.43) | -.235 (5.97) |
| D | -.135 (3.43) | -.235 (5.97) |
| E | -.270 (6.86) | +.000 (0.00) |
| F | -.135 (3.43) | +.235 (5.97) |
| G | +.000 (0.00) | +.000 (0.00) |

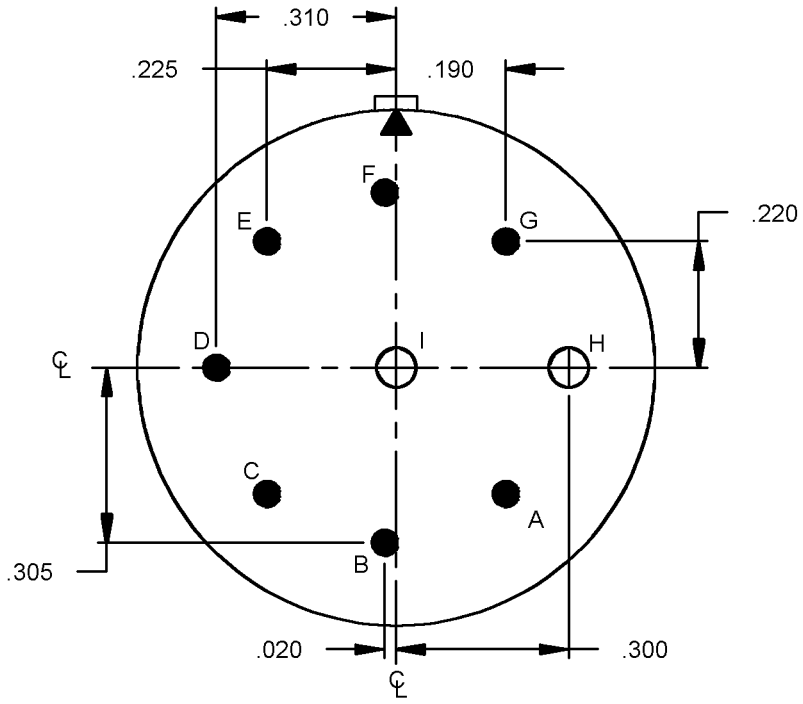
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -15 | 7 | 12 | A | All | MS33685-15 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-16

| Contacts (Insert arrangement 20-16) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.190 (4.83) | -.220 (5.59) |
| B | -.020 (0.51) | -.305 (7.75) |
| C | -.225 (5.72) | -.220 (5.59) |
| D | -.310 (7.87) | +.000 (0.00) |
| E | -.225 (5.72) | +.220 (5.59) |
| F | -.020 (0.51) | +.305 (7.75) |
| G | +.190 (4.83) | +.220 (5.59) |
| H | +.300 (7.62) | +.000 (0.00) |
| I | +.000 (0.00) | +.000 (0.00) |

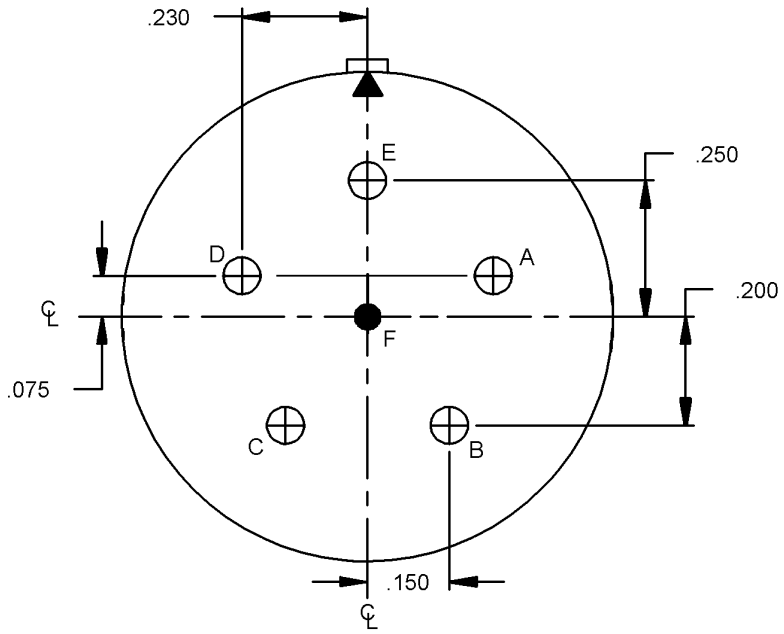
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -16 | 7 | 16 | A | A,B,C,D,E,F,G | MS33685-16 |
| | | 2 | 12 | | H, I | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-17

| Contacts (Insert arrangement 20-17) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.230 (5.84) | +.075 (1.91) |
| B | +.150 (3.81) | -.200 (5.08) |
| C | -.150 (3.81) | -.200 (5.08) |
| D | -.230 (5.84) | +.075 (1.91) |
| E | +.000 (0.00) | +.250 (6.35) |
| F | +.000 (0.00) | +.000 (0.00) |

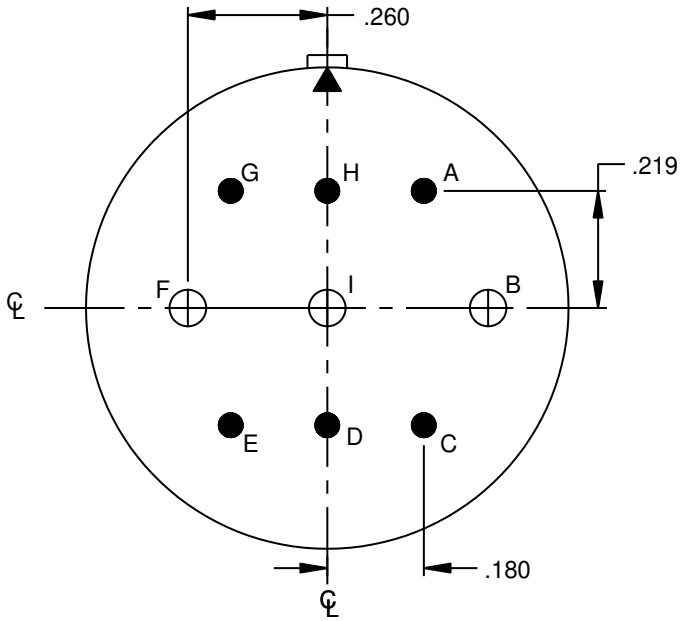
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -17 | 1 | 16 | A | F | MS33685-17 |
| | | 5 | 12 | | A,B,C,D,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-18

| Contacts (Insert arrangement 20-18) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.180 (4.57) | +.219 (5.56) |
| B | +.260 (6.60) | +.000 (0.00) |
| C | +.180 (4.57) | -.219 (5.56) |
| D | +.000 (0.00) | -.219 (5.56) |
| E | -.180 (4.57) | -.219 (5.56) |
| F | -.260 (6.60) | +.000 (0.00) |
| G | -.180 (4.57) | +.219 (5.56) |
| H | +.000 (0.00) | +.219 (5.56) |
| I | +.000 (0.00) | +.000 (0.00) |

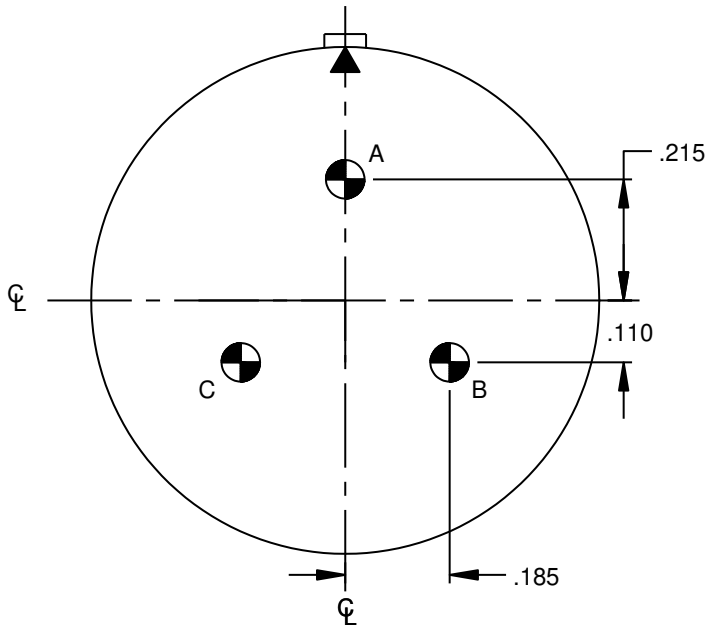
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -18 | 6 | 16 | A | A,C,D,E,G,H | MS33685-18 |
| | | 3 | 12 | | B,F,I | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-19

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-19) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.215 (5.46) |
| B | +.185 (4.70) | -.110 (2.79) |
| C | -.185 (4.70) | -.110 (2.79) |

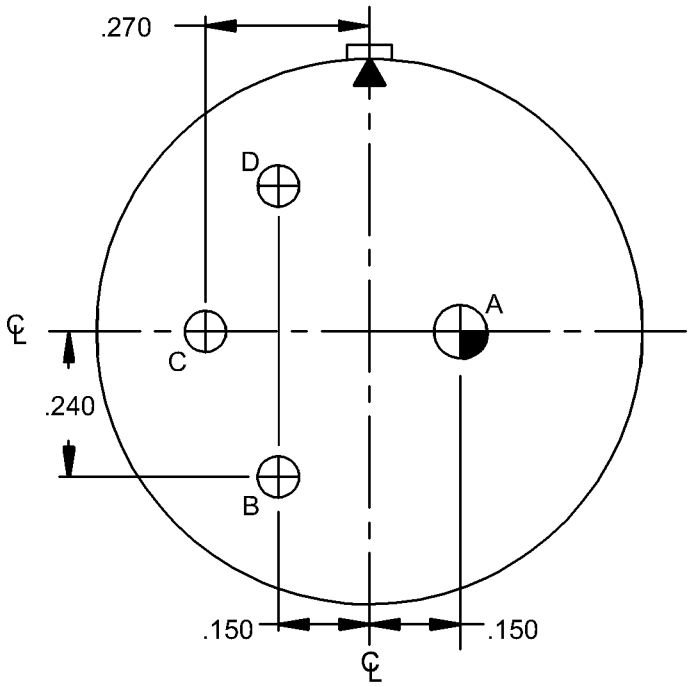
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -19 | 3 | 8 | A | All | MS33685-19 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-20

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-20) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.150 (3.81) | +.000 (0.00) |
| B | -.150 (3.81) | -.240 (6.10) |
| C | -.270 (6.86) | +.000 (0.00) |
| D | -.150 (3.81) | +.240 (6.10) |

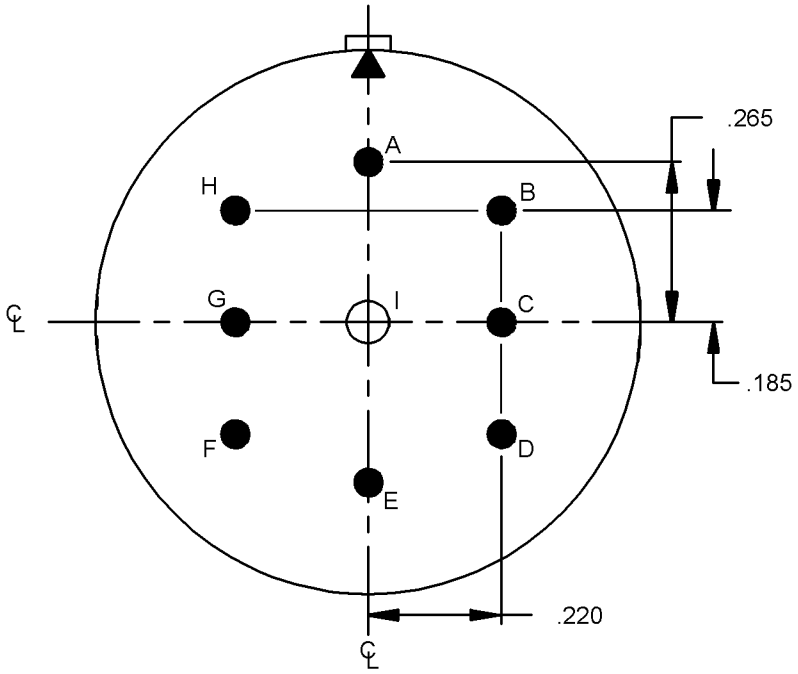
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -20 | 3 | 12 | A | B,C,D | MS33685-20 |
| | | 1 | 4 | | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-21

| Contacts (Insert arrangement 20-21) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.265 (6.73) |
| B | +.220 (5.59) | +.185 (4.70) |
| C | +.220 (5.59) | +.000 (0.00) |
| D | +.220 (5.59) | -.185 (4.70) |
| E | +.000 (0.00) | -.265 (6.73) |
| F | -.220 (5.59) | -.185 (4.70) |
| G | -.220 (5.59) | + 000 (0.00) |
| H | -.220 (5.59) | +.185 (4.70) |
| I | +.000 (0.00) | +.000 (0.00) |

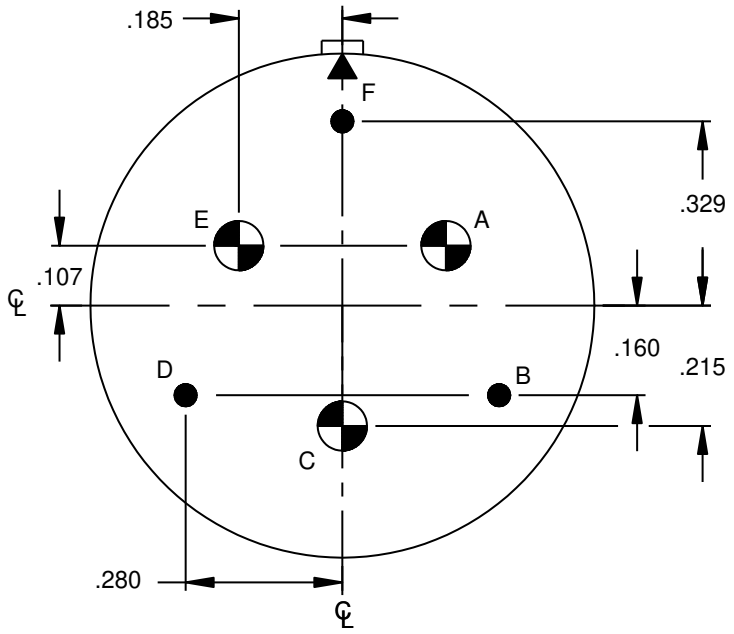
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -21 | 8 | 16 | A | All others | MS33685-21 |
| | | 1 | 12 | | I | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-22

| Contacts (Insert arrangement 20-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.185 (4.70) | +.107 (2.72) |
| B | +.280 (7.11) | -.160 (4.06) |
| C | +.000 (0.00) | -.215 (5.46) |
| D | -.280 (7.11) | -.160 (4.06) |
| E | -.185 (4.70) | +.107 (2.72) |
| F | +.000 (0.00) | +.329 (8.36) |

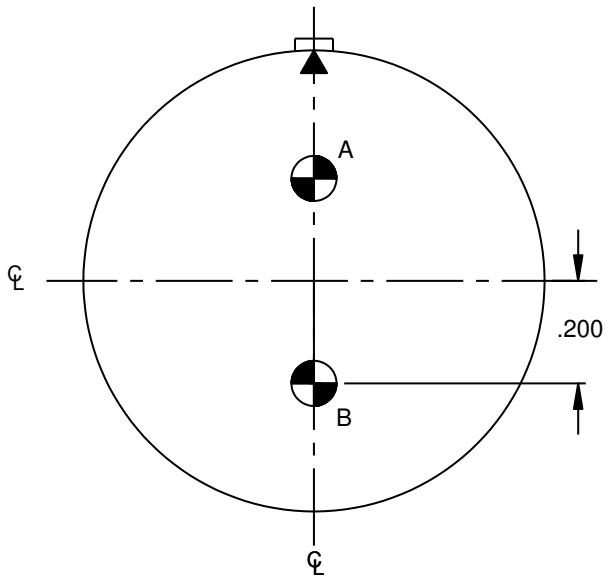
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -22 | 3 | 16 | A | B,D,F | MS33685-22 |
| | | 3 | 8 | | A,C,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-23

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 20-23) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.200 (5.08) |
| B | +.000 (0.00) | -.200 (5.08) |

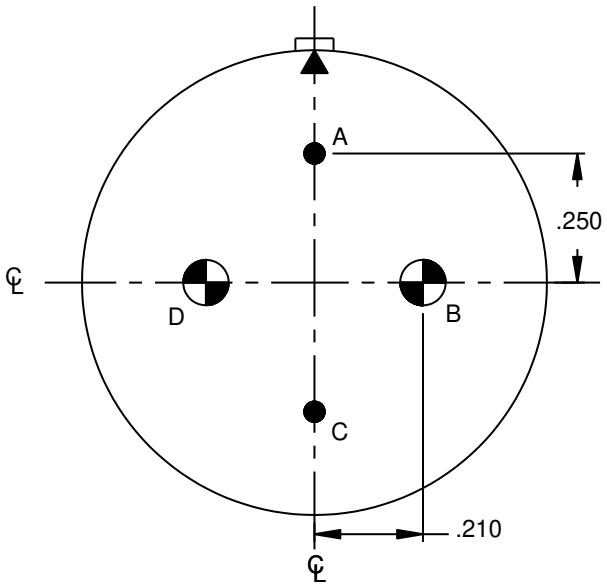
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -23 | 2 | 8 | A | All | MS33685-23 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-24

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 20-24) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.250 (6.35) |
| B | +.210 (5.33) | +.000 (0.00) |
| C | +.000 (0.00) | -.250 (6.35) |
| D | -.210 (5.33) | +.000 (0.00) |

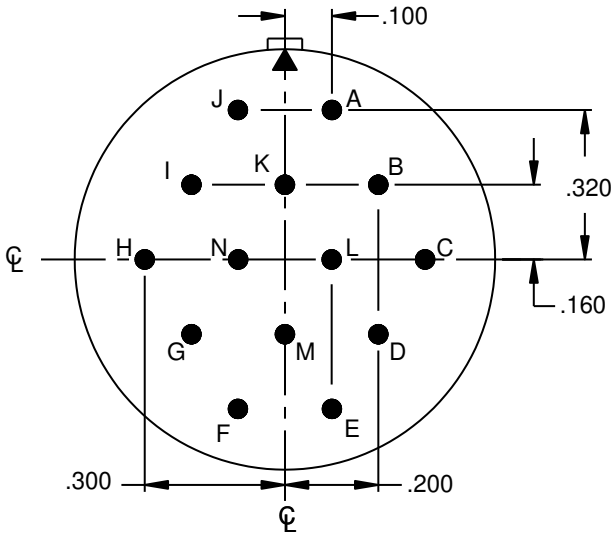
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -24 | 2 | 16 | A | A,C | MS33685-24 |
| | | 2 | 8 | | B,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-27

| Contacts (Insert arrangement 20-27) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.100 (2.54) | +.320 (8.13) |
| B | +.200 (5.08) | +.160 (4.06) |
| C | +.300 (7.62) | +.000 (0.00) |
| D | +.200 (5.08) | -.160 (4.06) |
| E | +.100 (2.54) | -.320 (8.13) |
| F | -.100 (2.54) | -.320 (8.13) |
| G | -.200 (5.08) | -.160 (4.06) |
| H | -.300 (7.62) | +.000 (0.00) |
| I | -.200 (5.08) | +.160 (4.06) |
| J | -.100 (2.54) | +.320 (8.13) |
| K | +.000 (0.00) | +.160 (4.06) |
| L | +.100 (2.54) | +.000 (0.00) |
| M | +.000 (0.00) | -.160 (4.06) |
| N | -.100 (2.54) | +.000 (0.00) |

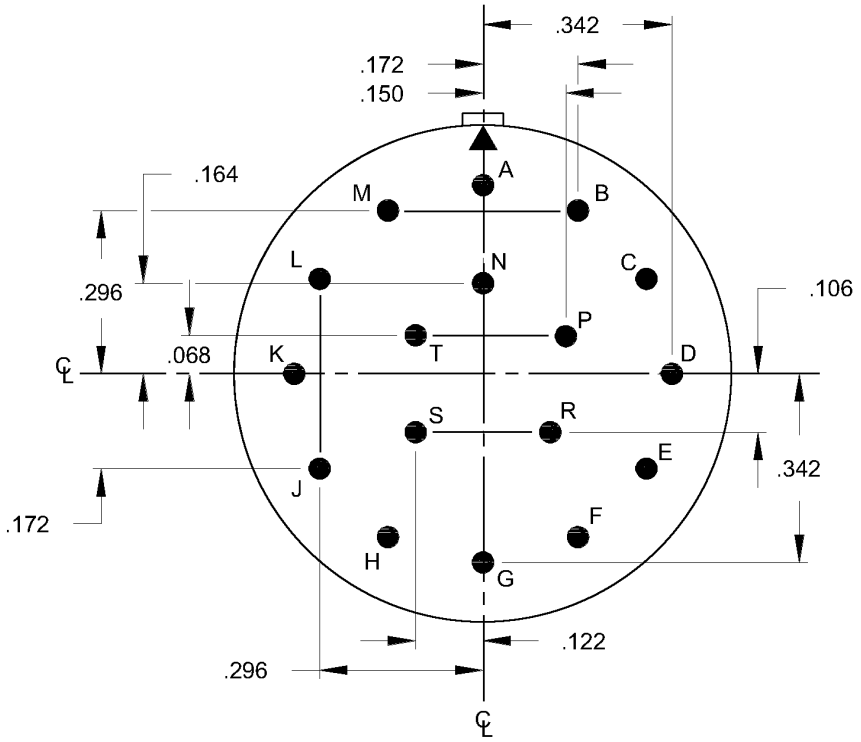
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -27 | 14 | 16 | A | All | MS33685-27 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-29

| Contacts (Insert arrangement 20-29) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.342 (8.69) |
| B | +.172 (4.37) | +.296 (7.52) |
| C | +.296 (7.52) | +.172 (4.37) |
| D | +.342 (8.69) | +.000 (0.00) |
| E | +.296 (7.52) | -.172 (4.37) |
| F | +.172 (4.37) | -.296 (7.52) |
| G | +.000 (0.00) | -.342 (8.69) |
| H | -.172 (4.37) | -.296 (7.52) |
| J | -.296 (7.52) | -.172 (4.37) |

| Contacts (Insert arrangement 20-29) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| K | -.342 (8.69) | +.000 (0.00) |
| L | -.296 (7.52) | +.172 (4.37) |
| M | -.172 (4.37) | +.296 (7.52) |
| N | +.000 (0.00) | +.164 (4.17) |
| P | +.150 (3.81) | +.068 (1.73) |
| R | +.122 (3.10) | -.106 (2.69) |
| S | -.122 (3.10) | -.106 (2.69) |
| T | -.150 (3.81) | +.068 (1.73) |
| --- | --- | --- |

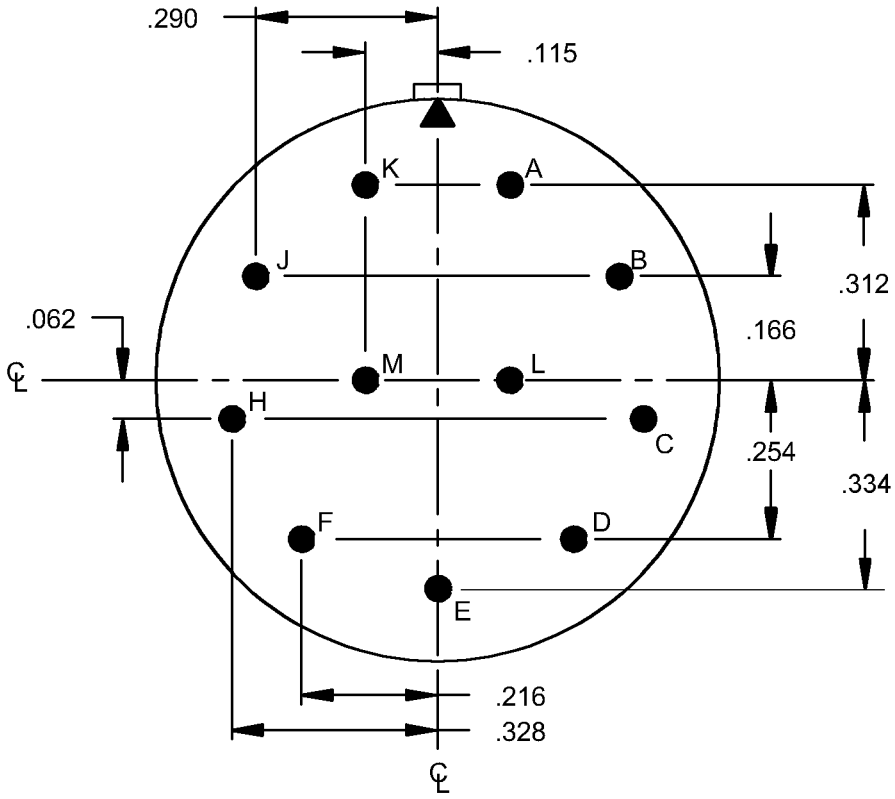
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -29 | 17 | 16 | A | All | MS33685-29 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-31

| Contacts (Insert arrangement 20-31) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.115 (2.92) | +.312 (7.92) |
| B | +.290 (7.37) | +.166 (4.22) |
| C | +.328 (8.33) | -.062 (1.57) |
| D | +.216 (5.49) | -.254 (6.45) |
| E | +.000 (0.00) | -.334 (8.48) |
| F | -.216 (5.49) | -.254 (6.45) |
| H | -.328 (8.33) | -.062 (1.57) |
| J | -.290 (7.37) | +.166 (4.22) |
| K | -.115 (2.92) | +.312 (7.92) |
| L | +.115 (2.92) | +.000 (0.00) |
| M | -.115 (2.92) | +.000 (0.00) |

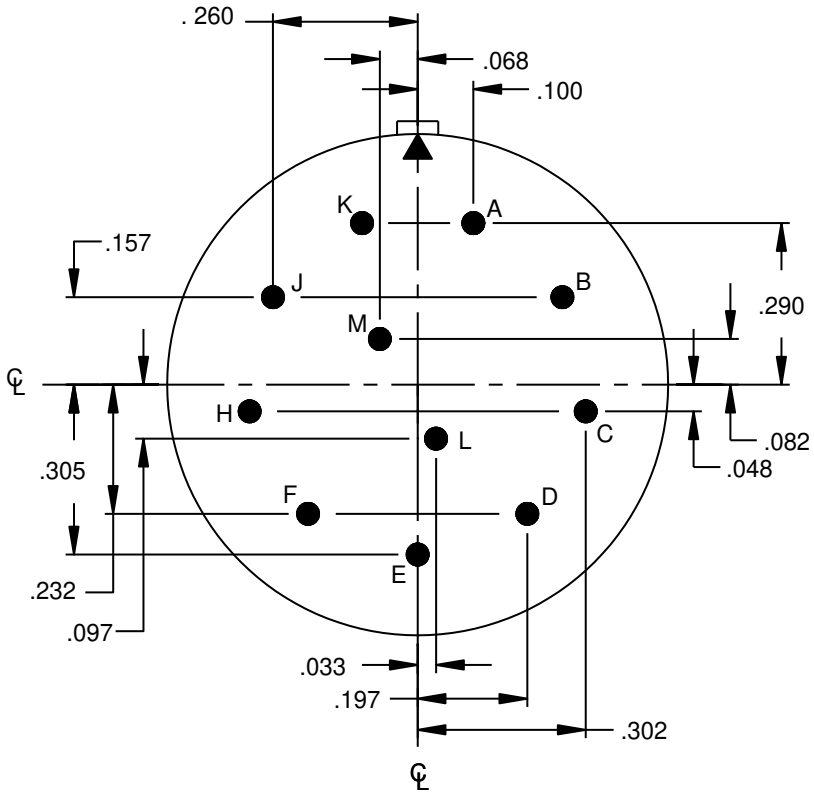
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -31 | 11 | 16 | A | All | MS33685-31 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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20-33

| Contacts (Insert arrangement 20-33) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.100 (2.54) | +.290 (7.37) |
| B | +.260 (6.60) | +.157 (3.99) |
| C | +.302 (7.67) | -.048 (1.22) |
| D | +.197 (5.00) | -.232 (5.89) |
| E | +.000 (0.00) | -.305 (7.75) |
| F | -.197 (5.00) | -.232 (5.89) |
| H | -.302 (7.67) | -.048 (1.22) |
| J | -.260 (6.60) | +.157 (3.99) |
| K | -.100 (2.54) | +.290 (7.37) |
| L | +.033 (0.84) | -.097 (2.46) |
| M | -.068 (1.73) | +.082 (2.08) |

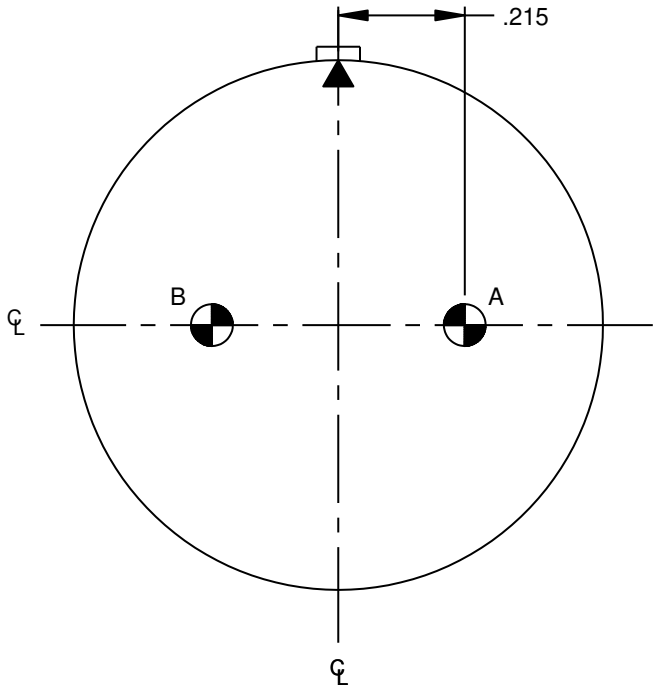
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 20 | -33 | 11 | 16 | A | All | MS33685-33 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 7. Insert arrangement, electrical connector, shell size 20, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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22-1

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and
MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.215 (5.46) | +.000 (0.00) |
| B | -.215 (5.46) | +.000 (0.00) |

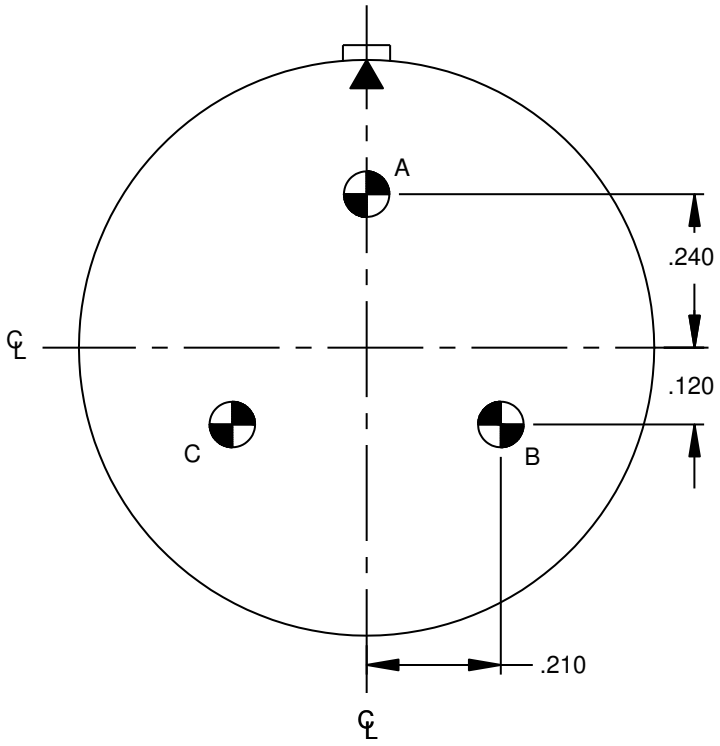
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -1 | 2 | 8 | D | All | MS33686-1 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234.



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22-2

| Contacts (Insert arrangement 22-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.240 (6.10) |
| B | +.210 (5.33) | -.120 (3.05) |
| C | -.210 (5.33) | -.120 (3.05) |

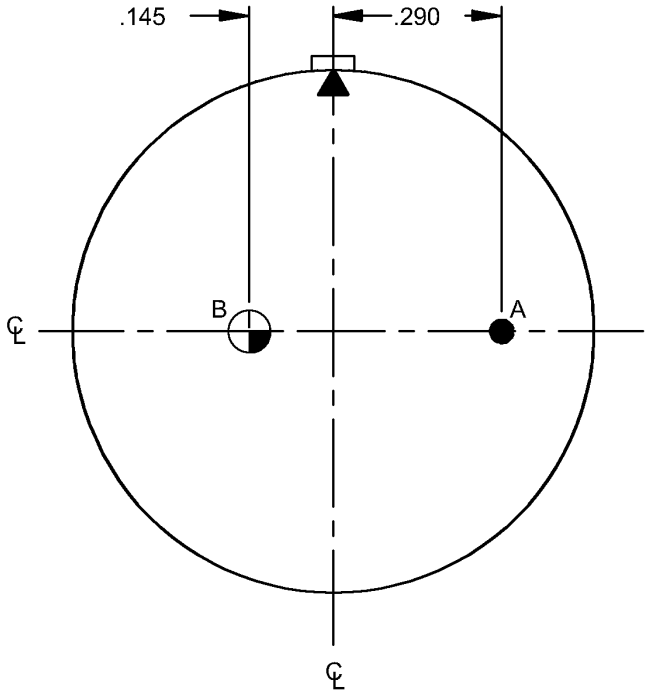
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -2 | 3 | 8 | D | All | MS33686-2 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-3

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and
MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.290 (7.37) | +.000 (0.00) |
| B | -.145 (3.68) | +.000 (0.00) |

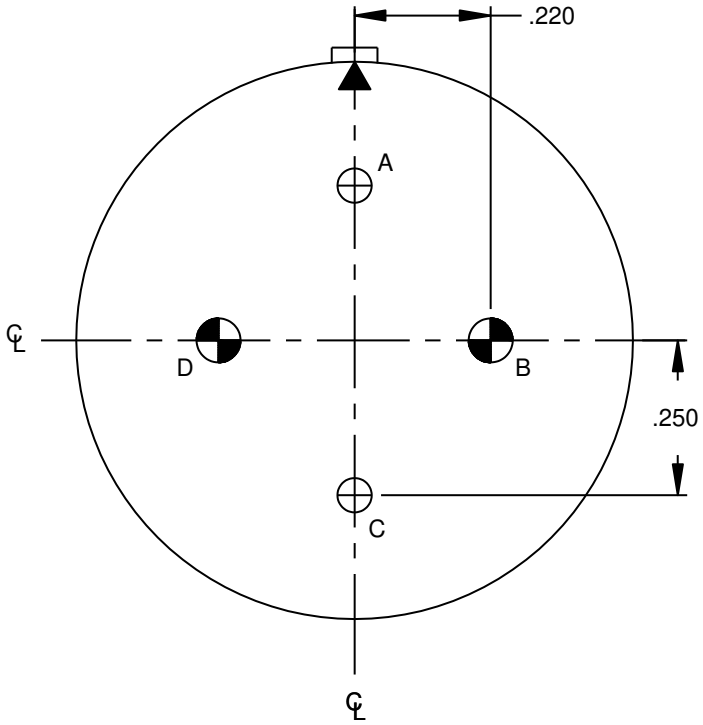
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -3 | 1 | 16 | D | A | MS33686-3 |
| | | 1 | 4 | | B | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-4

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 19 Dec 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.250 (6.35) |
| B | +.220 (5.59) | +.000 (0.00) |
| C | +.000 (0.00) | -.250 (6.35) |
| D | -.220 (5.59) | +.000 (0.00) |

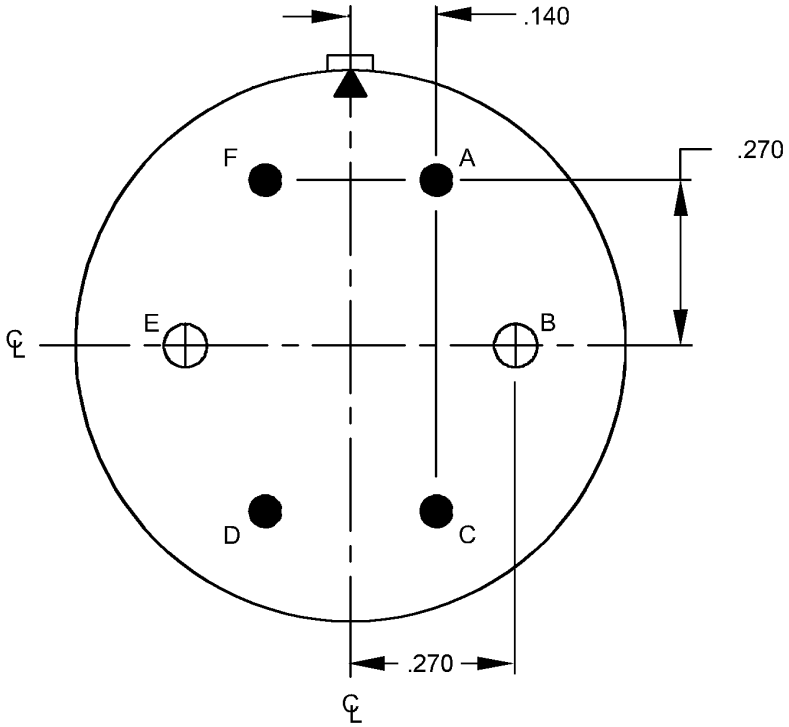
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -4 | 2 | 12 | A | A, C | MS33686-4 |
| | | 2 | 8 | | B, D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-5

| Contacts (Insert arrangement 22-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.140 (3.56) | +.270 (6.86) |
| B | +.270 (6.86) | +.000 (0.00) |
| C | +.140 (3.56) | -.270 (6.86) |
| D | -.140 (3.56) | -.270 (6.86) |
| E | -.270 (6.86) | +.000 (0.00) |
| F | -.140 (3.56) | +.270 (6.86) |

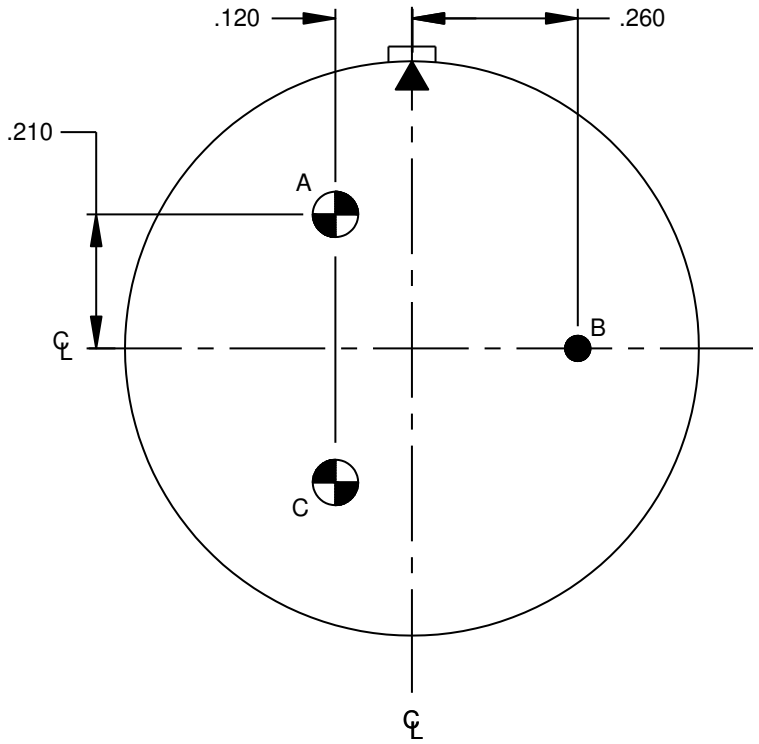
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -5 | 4 | 16 | D | A,C,D,F | MS33686-5 |
| | | 2 | 12 | | B,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-6

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.120 (3.05) | +.210 (5.53) |
| B | +.260 (6.60) | +.000 (0.00) |
| C | -.120 (3.05) | -.210 (5.53) |

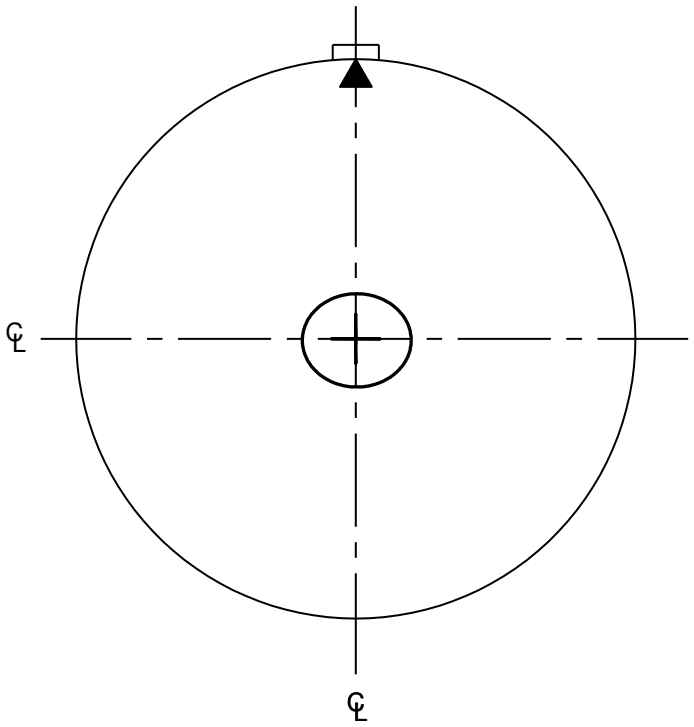
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -6 | 1 | 16 | D | B | MS33686-6 |
| | | 2 | 8 | | A, C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-7

| Contacts (Insert arrangement 22-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| --- | +.000 (0.00) | +.000 (0.00) |

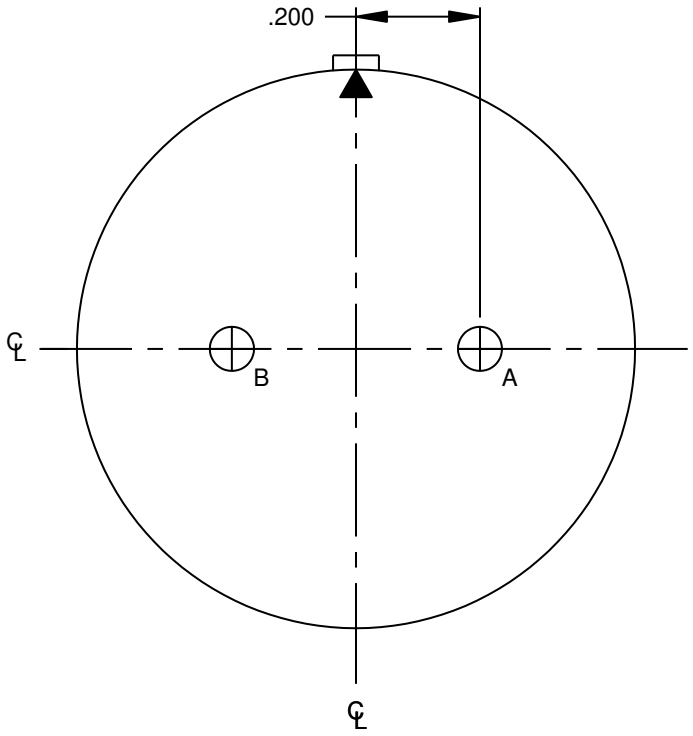
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -7 | 1 | 0 | E | All | MS33686-7 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-8

(Inactive for new design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 22-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.200 (5.08) | +.000 (0.00) |
| B | -.200 (5.08) | +.000 (0.00) |

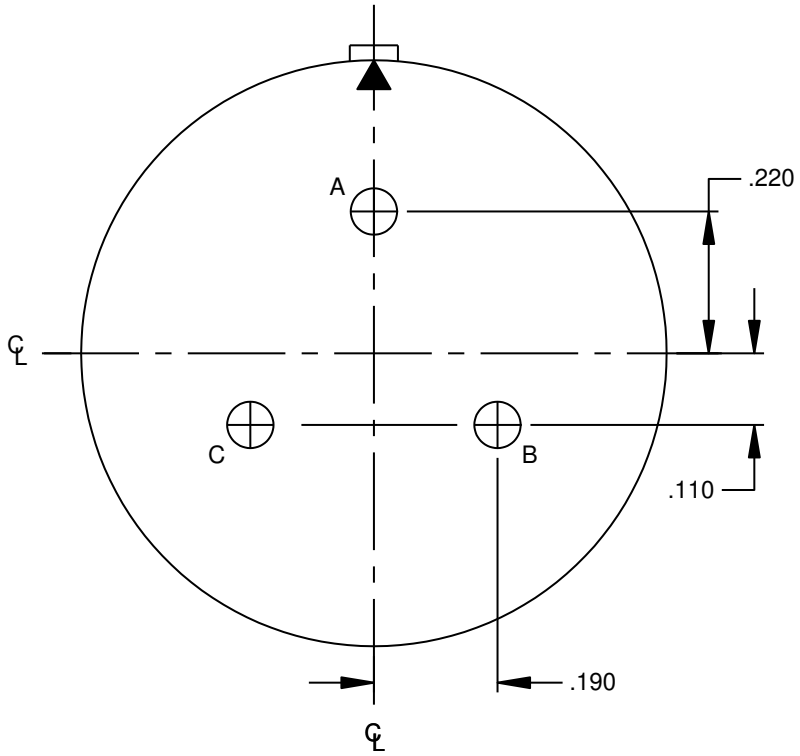
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -8 | 2 | 12 | E | All | MS33686-8 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-9

| Contacts (Insert arrangement 22-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.220 (5.59) |
| B | +.190 (4.83) | -.110 (2.79) |
| C | -.190 (4.83) | -.110 (2.79) |

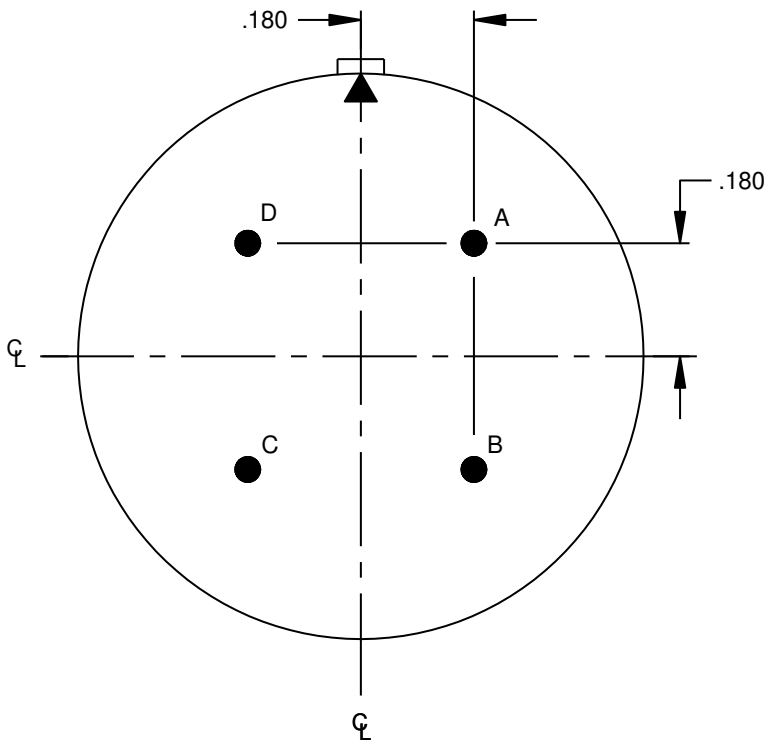
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -9 | 3 | 12 | E | All | MS33686-9 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-10

| Contacts (Insert arrangement 22-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.180 (4.57) | +.180 (4.57) |
| B | +.180 (4.57) | -.180 (4.57) |
| C | -.180 (4.57) | -.180 (4.57) |
| D | -.180 (4.57) | +.180 (4.57) |

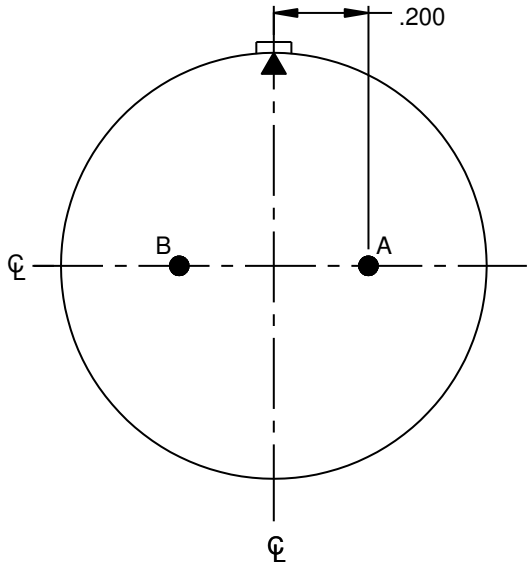
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -10 | 4 | 16 | E | All | MS33686-10 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-11

| Contacts (Insert arrangement 22-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.200 (5.08) | +.000 (0.00) |
| B | -.200 (5.08) | +.000 (0.00) |

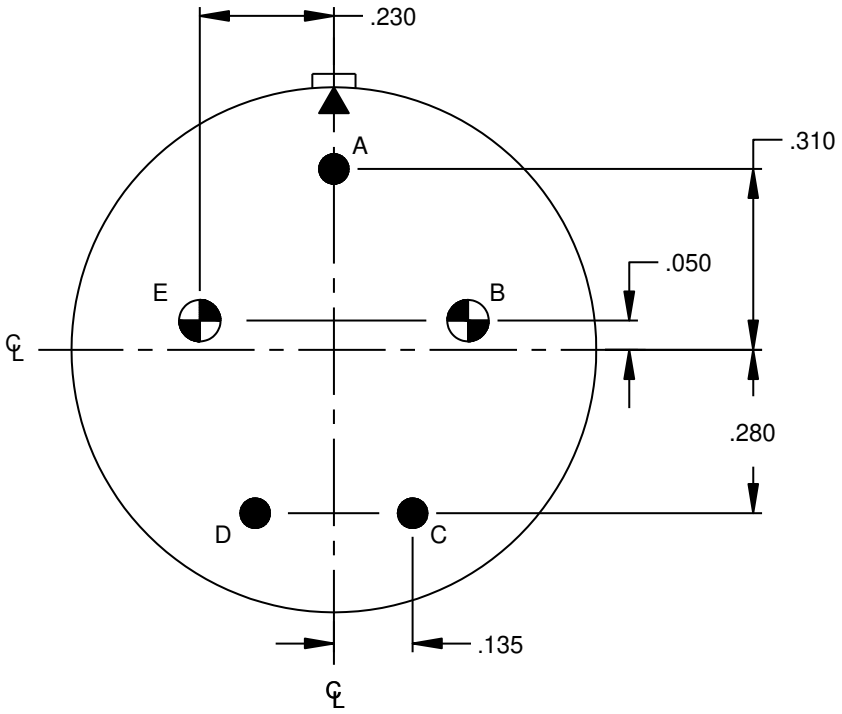
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -11 | 2 | 16 | B | All | MS33686-11 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-12

| Contacts (Insert arrangement 22-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.310 (7.87) |
| B | +.230 (5.84) | +.050 (1.27) |
| C | +.135 (3.43) | -.280 (7.11) |
| D | -.135 (3.43) | -.280 (7.11) |
| E | -.230 (5.84) | +.050 (1.27) |

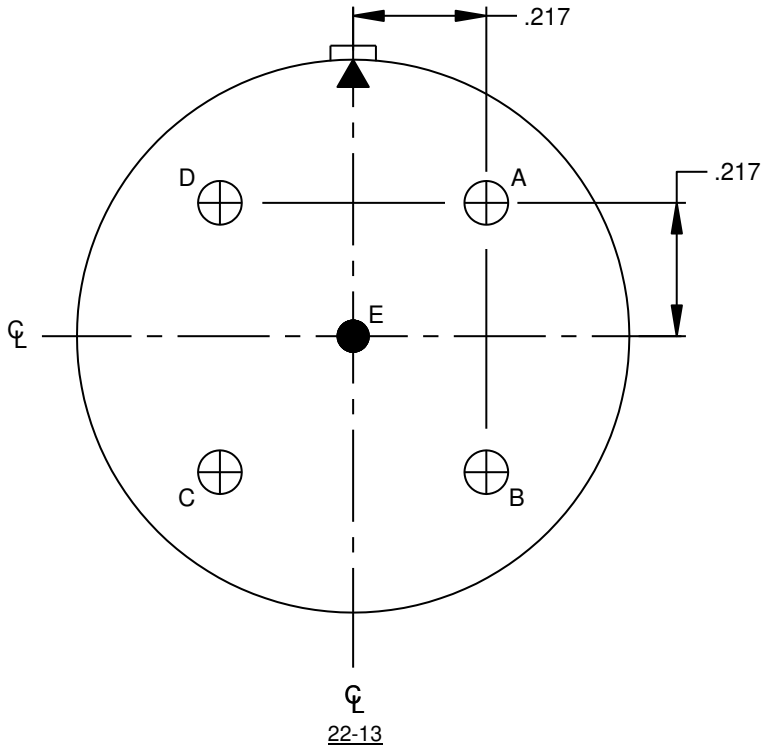
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -12 | 3 | 16 | D | A,C,D | MS33686-12 |
| | | 2 | 8 | | B,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-13

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-13) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.217 (5.51) | +.217 (5.51) |
| B | +.217 (5.51) | -.217 (5.51) |
| C | -.217 (5.51) | -.217 (5.51) |
| D | -.217 (5.51) | +.217 (5.51) |
| E | +.000 (0.00) | +.000 (0.00) |

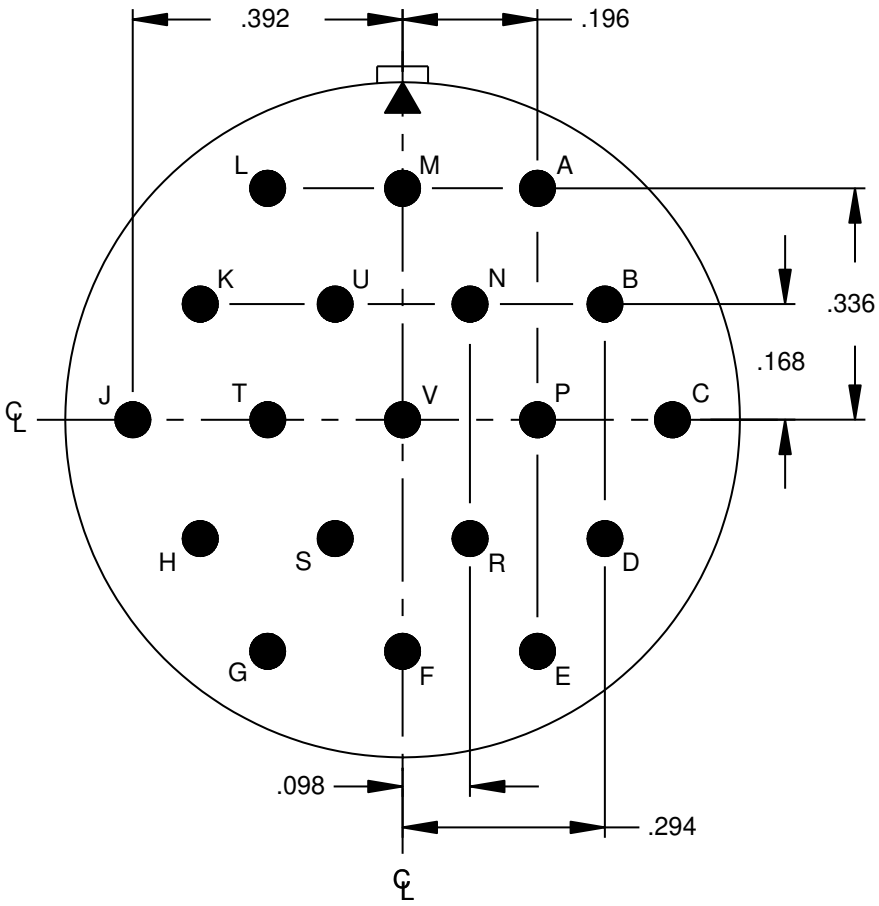
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -13 | 1 | 16 | D | E | MS33686-13 |
| | | 4 | 12 | A | A,B,C,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-14

(Alternate positions X and Y canceled after 26 June 1968 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-14) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.196 (4.98) | +.336 (8.53) |
| B | +.294 (7.47) | +.168 (4.27) |
| C | +.392 (9.96) | +.000 (0.00) |
| D | +.294 (7.47) | -.168 (4.27) |
| E | +.196 (4.98) | -.336 (8.53) |
| F | +.000 (0.00) | -.336 (8.53) |
| G | -.196 (4.98) | -.336 (8.53) |
| H | -.294 (7.47) | -.168 (4.27) |
| J | -.392 (9.96) | +.000 (0.00) |
| K | -.294 (7.47) | +.168 (4.27) |

| Contacts (Insert arrangement 22-14) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | -.196 (4.98) | +.336 (8.53) |
| M | +.000 (0.00) | +.336 (8.53) |
| N | +.098 (2.49) | +.168 (4.27) |
| P | +.196 (4.98) | +.000 (0.00) |
| R | +.098 (2.49) | -.168 (4.27) |
| S | -.098 (2.49) | -.168 (4.27) |
| T | -.196 (4.98) | +.000 (0.00) |
| U | -.098 (2.49) | +.168 (4.27) |
| V | +.000 (0.00) | +.000 (0.00) |
| --- | --- | --- |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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| Insert arrangement 22-14 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 22 | -14 | 19 | 16 | A | All | MS33686-14 |

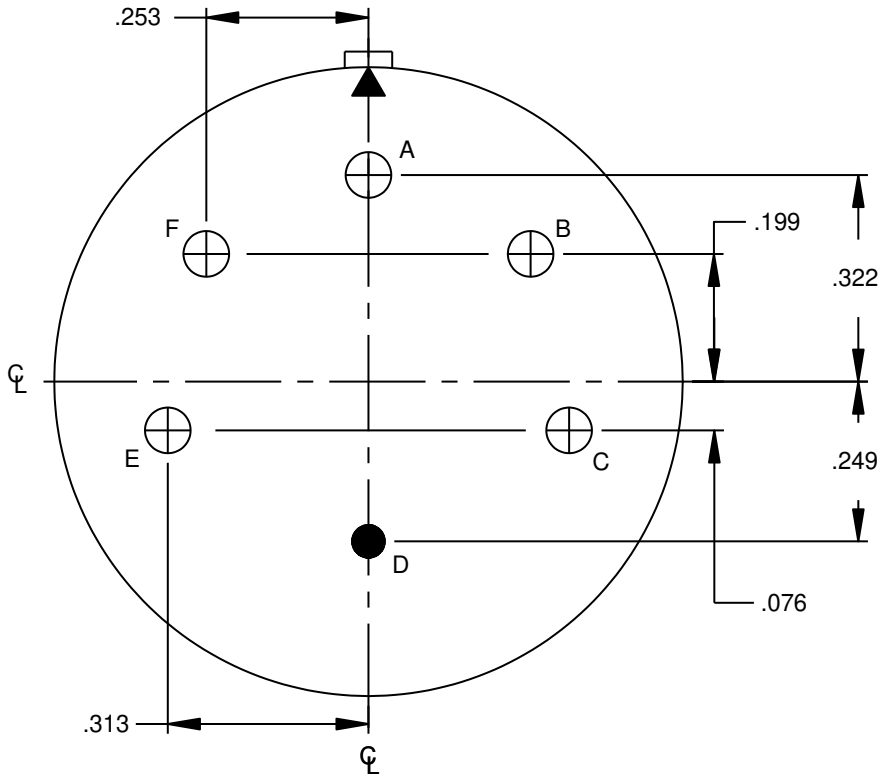
| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

(Note: Alternate positions X and Y cancelled after 26 June 1968.)

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-15

| Contacts (Insert arrangement 22-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.322 (8.18) |
| B | +.253 (6.43) | +.199 (5.05) |
| C | +.313 (7.95) | -.076 (1.93) |
| D | +.000 (0.00) | -.249 (6.32) |
| E | -.313 (7.95) | -.076 (1.93) |
| F | -.253 (6.43) | +.199 (5.05) |

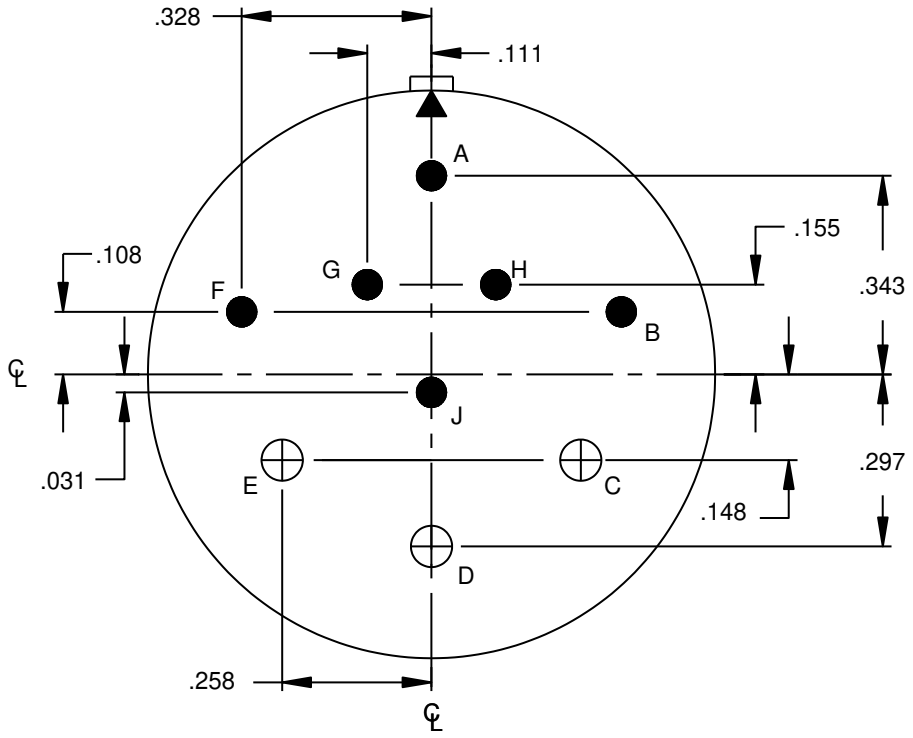
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -15 | 1 | 16 | D | D | MS33686-15 |
| | | 5 | 12 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-16

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-16) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.343 (8.71) |
| B | +.328 (8.33) | +.108 (2.74) |
| C | +.258 (6.55) | -.148 (3.76) |
| D | +.000 (0.00) | -.297 (7.54) |
| E | -.258 (6.55) | -.148 (3.76) |

| Contacts (Insert arrangement 22-16) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| F | -.328 (8.33) | +.108 (2.74) |
| G | -.111 (2.82) | +.155 (3.94) |
| H | +.111 (2.82) | +.155 (3.94) |
| J | +.000 (0.00) | -.031 (0.79) |
| --- | --- | --- |

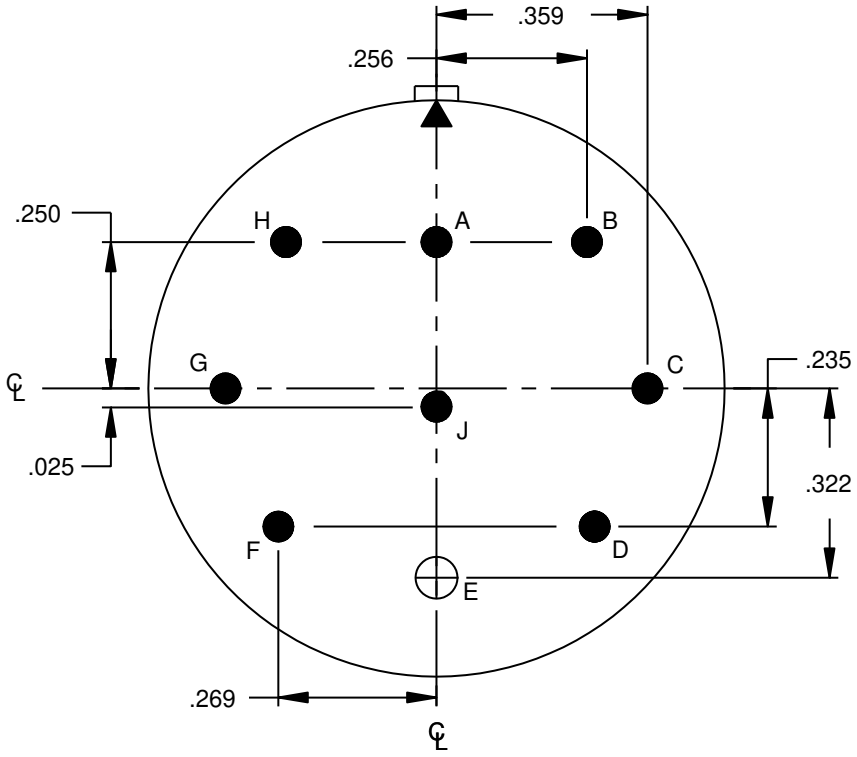
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -16 | 6 | 16 | A | All others | MS33686-16 |
| | | 3 | 12 | | C,D,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-17

| Contacts (Insert arrangement 22-17) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.250 (6.35) |
| B | +.256 (6.50) | +.250 (6.35) |
| C | +.359 (9.12) | +.000 (0.00) |
| D | +.269 (6.83) | -.235 (5.97) |
| E | +.000 (0.00) | -.322 (8.18) |
| F | -.269 (6.83) | -.235 (5.97) |
| G | -.359 (9.12) | +.000 (0.00) |
| H | -.256 (6.50) | +.250 (6.35) |
| J | +.000 (0.00) | -.025 (0.64) |

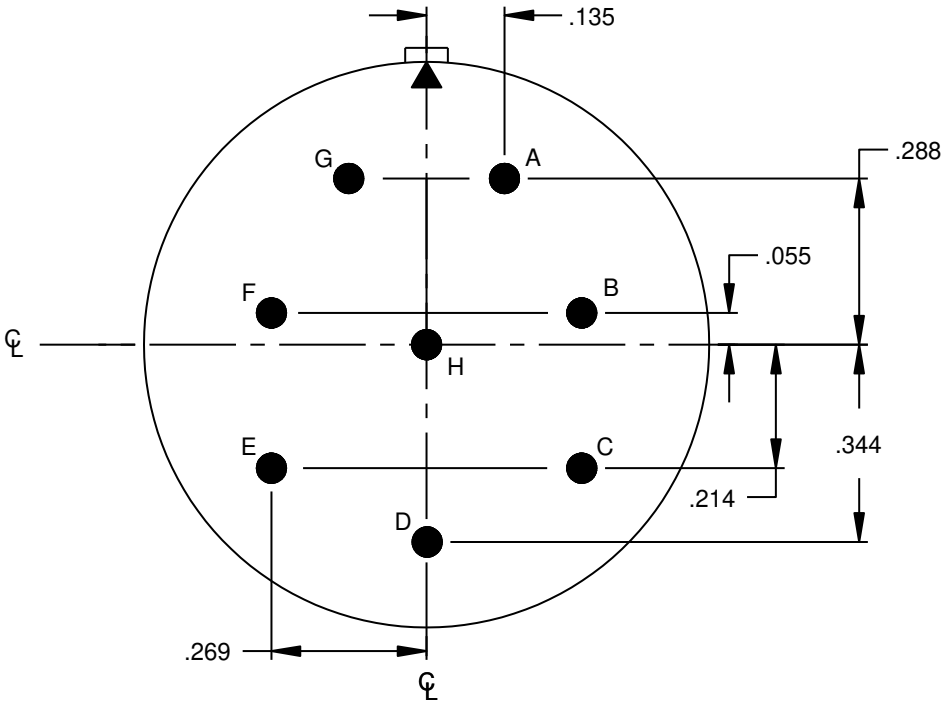
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -17 | 1 | 16 | D | A | MS33686-17 |
| | | 7 | 16 | A | B,C,D,F,G,H,J | |
| | | 1 | 12 | A | E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-18

| Contacts (Insert arrangement 22-18) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.135 (3.43) | +.288 (7.32) |
| B | +.269 (6.83) | +.055 (1.40) |
| C | +.269 (6.83) | -.214 (5.44) |
| D | +.000 (0.00) | -.344 (8.74) |
| E | -.269 (6.83) | -.214 (5.44) |
| F | -.269 (6.83) | +.055 (1.40) |
| G | -.135 (3.43) | +.288 (7.32) |
| H | +.000 (0.00) | +.000 (0.00) |

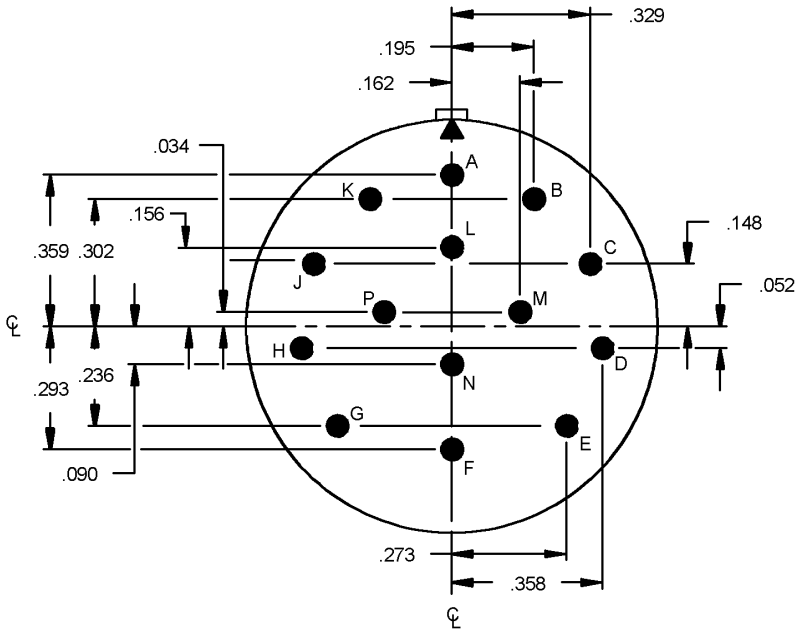
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -18 | 3 | 16 | A | C,D,E | MS33686-18 |
| | | 5 | | D | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-19

| Contacts (Insert arrangement 22-19) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.359 (9.12) |
| B | +.195 (4.95) | +.302 (7.67) |
| C | +.329 (8.36) | +.148 (3.76) |
| D | +.358 (9.09) | -.052 (1.32) |
| E | +.273 (6.93) | -.236 (5.99) |
| F | +.000 (0.00) | -.293 (7.44) |
| G | -.273 (6.93) | -.236 (5.99) |
| H | -.358 (9.09) | -.052 (1.32) |
| J | -.329 (8.36) | +.148 (3.76) |
| K | -.195 (4.95) | +.302 (7.67) |
| L | +.000 (0.00) | +.156 (3.96) |
| M | +.162 (4.11) | +.034 (0.86) |
| N | +.000 (0.00) | -.090 (2.29) |
| P | -.162 (4.11) | +.034 (0.86) |

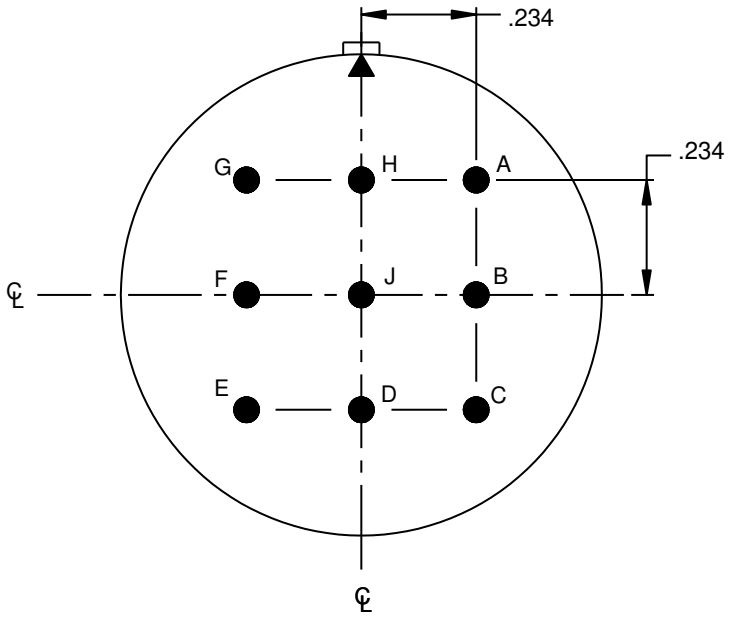
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -19 | 14 | 16 | A | All | MS33686-19 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-20

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-20) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.234 (5.94) | +.234 (5.94) |
| B | +.234 (5.94) | +.000 (0.00) |
| C | +.234 (5.94) | -.234 (5.94) |
| D | +.000 (0.00) | -.234 (5.94) |
| E | -.234 (5.94) | -.234 (5.94) |
| F | -.234 (5.94) | +.000 (0.00) |
| G | -.234 (5.94) | +.234 (5.94) |
| H | +.000 (0.00) | +.234 (5.94) |
| J | +.000 (0.00) | +.000 (0.00) |

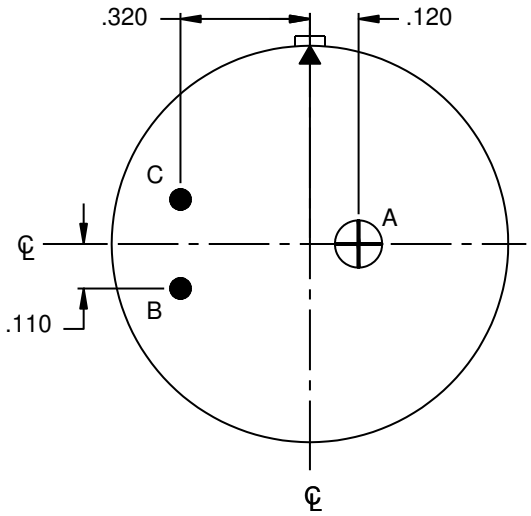
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -20 | 9 | 16 | A | All | MS33686-20 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-21

| Contacts (Insert arrangement 22-21) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.120 (3.05) | +.000 (0.00) |
| B | -.320 (8.13) | -.110 (2.79) |
| C | -.320 (8.13) | +.110 (2.79) |

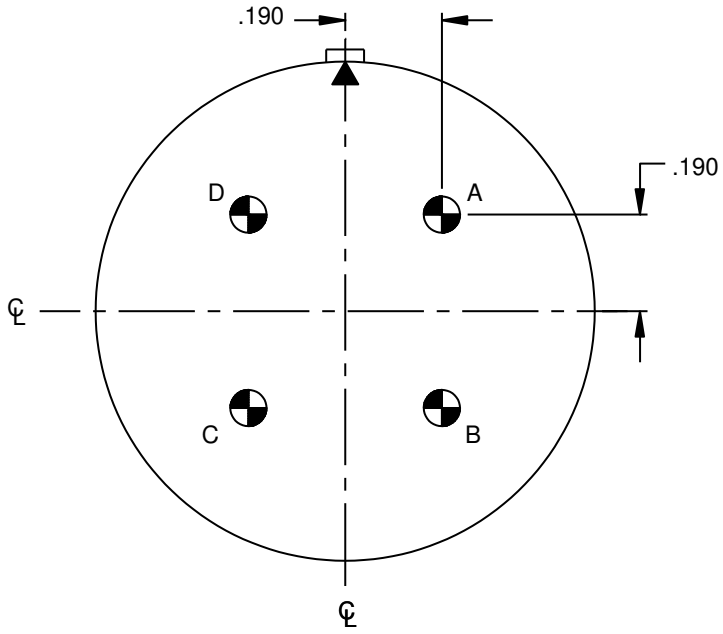
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -21 | 2 | 16 | A | B, C | MS33686-21 |
| | | 1 | 0 | | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-22

| Contacts (Insert arrangement 22-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.190 (4.83) | +.190 (4.83) |
| B | +.190 (4.83) | -.190 (4.83) |
| C | -.190 (4.83) | -.190 (4.83) |
| D | -.190 (4.83) | +.190 (4.83) |

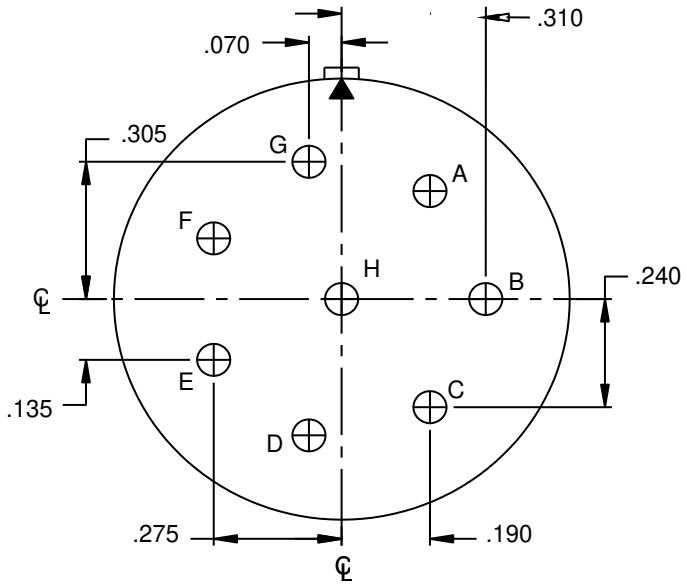
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -22 | 4 | 8 | A | All | MS33686-22 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 110 | 250 | --- |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-23

| Contacts (Insert arrangement 22-23) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.190 (4.83) | +.240 (6.10) |
| B | +.310 (7.87) | +.000 (0.00) |
| C | +.190 (4.83) | -.240 (6.10) |
| D | -.070 (1.78) | -.305 (7.75) |
| E | -.275 (6.99) | -.135 (3.43) |
| F | -.275 (6.99) | +.135 (3.43) |
| G | -.070 (1.78) | +.305 (7.75) |
| H | +.000 (0.00) | +.000 (0.00) |

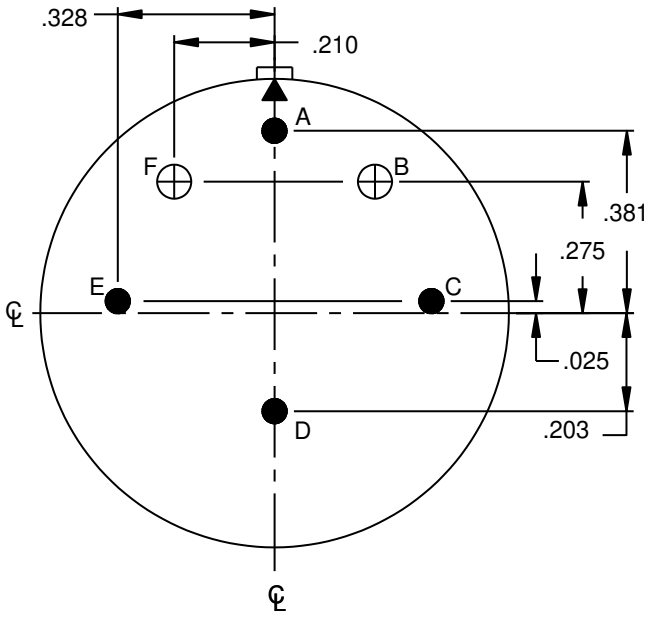
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -23 | 1 | 12 | D | H | MS33686-23 |
| | | 7 | 12 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | --- | 250 | --- |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-24

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-24) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.381 (9.68) |
| B | +.210 (5.33) | +.275 (6.99) |
| C | +.328 (8.33) | +.025 (0.64) |
| D | +.000 (0.00) | -.203 (5.16) |
| E | -.328 (8.33) | +.025 (0.64) |
| F | -.210 (5.33) | +.275 (6.99) |

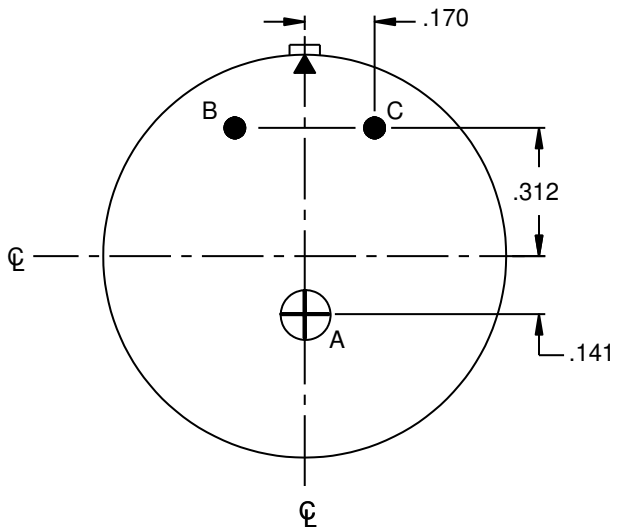
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -24 | 3 | 16 | D | C,D,E | MS33686-24 |
| | | 1 | 16 | A | A | |
| | | 2 | 12 | A | B,F | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-25

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 31 July 1951 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-25) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | -.141 (3.58) |
| B | -.170 (4.32) | +.312 (7.92) |
| C | +.170 (4.32) | +.312 (7.92) |

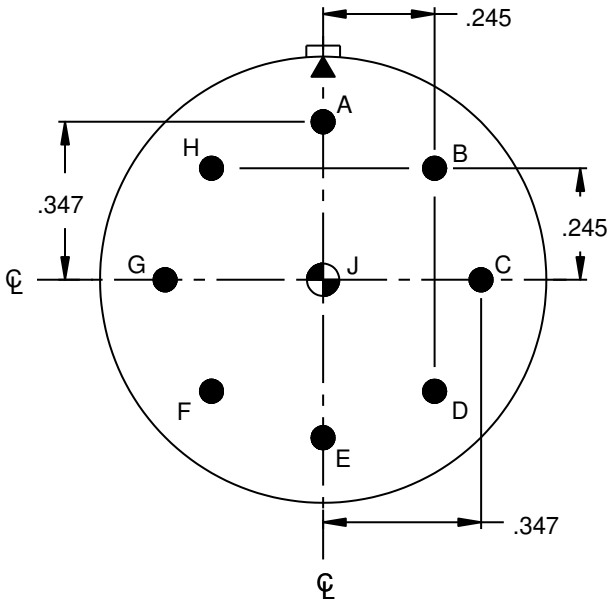
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -25 | 2 | 16 | A | B, C | MS33686-25 |
| | | 1 | 0 | | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-27

| Contacts (Insert arrangement 22-27) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.347 (8.81) |
| B | +.245 (6.22) | +.245 (6.22) |
| C | +.347 (8.81) | +.000 (0.00) |
| D | +.245 (6.22) | -.245 (6.22) |
| E | +.000 (0.00) | -.347 (8.81) |
| F | -.245 (6.22) | -.245 (6.22) |
| G | -.347 (8.81) | +.000 (0.00) |
| H | -.245 (6.22) | +.245 (6.22) |
| J | +.000 (0.00) | +.000 (0.00) |

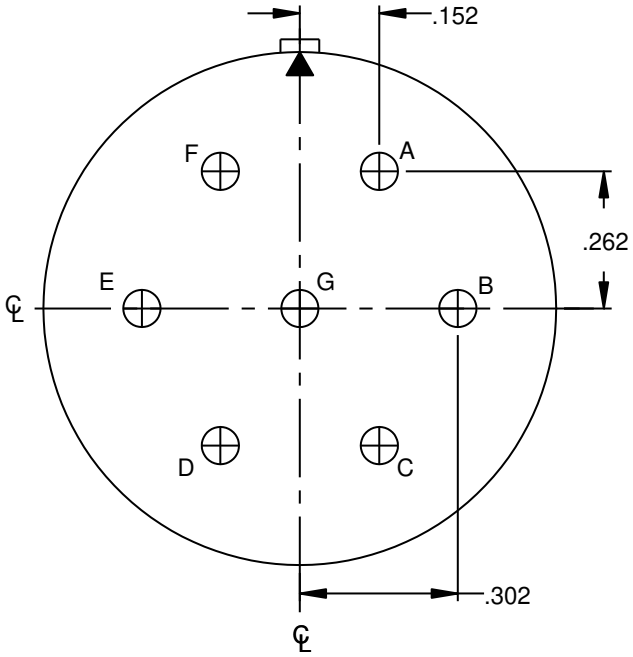
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -27 | 1 | 8 | D | J | MS33686-27 |
| | | 8 | 16 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-28

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-28) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.152 (3.86) | +.262 (6.65) |
| B | +.302 (7.67) | +.000 (0.00) |
| C | +.152 (3.86) | -.262 (6.65) |
| D | -.152 (3.86) | -.262 (6.65) |
| E | -.302 (7.67) | +.000 (0.00) |
| F | -.152 (3.86) | +.262 (6.65) |
| G | +.000 (0.00) | +.000 (0.00) |

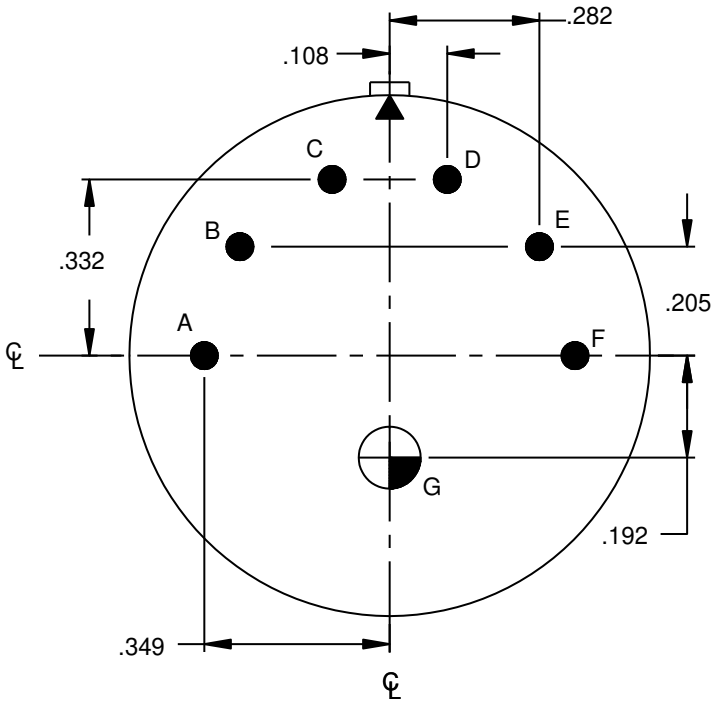
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -28 | 7 | 12 | A | All | MS33686-28 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-29

(Not applicable to MIL-DTL-22992.)
(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-29) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.349 (8.86) | +.000 (0.00) |
| B | -.282 (7.16) | +.205 (5.21) |
| C | -.108 (2.74) | +.332 (8.43) |
| D | +.108 (2.74) | +.332 (8.43) |
| E | +.282 (7.16) | +.205 (5.21) |
| F | +.349 (8.86) | +.000 (0.00) |
| G | +.000 (0.00) | -.192 (4.88) |

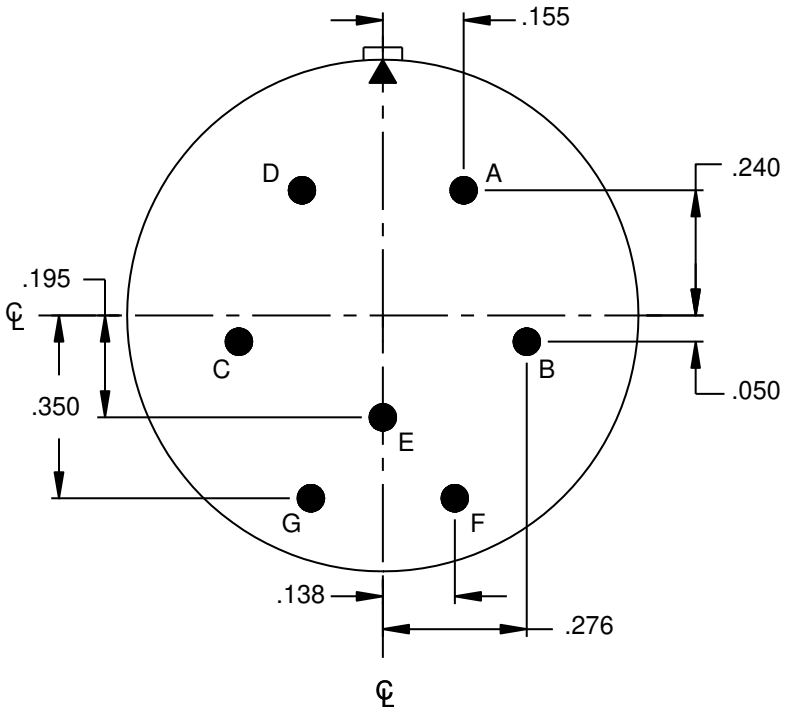
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -29 | 6 | 16 | A | All others | MS33686-29 |
| | | 1 | 4 | A | G | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-33

(Not applicable to MIL-DTL-22992.)
(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-33) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.155 (3.94) | +.240 (6.10) |
| B | +.276 (7.01) | -.050 (0.00) |
| C | -.276 (7.01) | -.050 (0.00) |
| D | -.155 (3.94) | +.240 (6.10) |
| E | +.000 (0.00) | -.195 (4.95) |
| F | +.138 (3.51) | -.350 (8.89) |
| G | -.138 (3.51) | -.350 (8.89) |

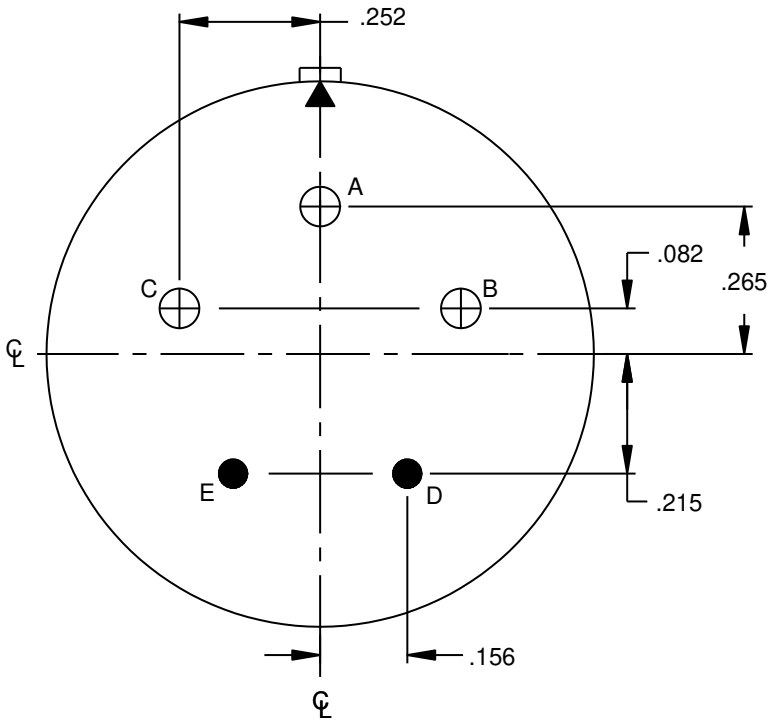
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -33 | 3 | 16 | A | E,F,G | MS33686-33 |
| | | 4 | 16 | D | A,B,C,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-34

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 22-34) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.265 (6.73) |
| B | +.252 (6.40) | +.082 (2.08) |
| C | -.252 (6.40) | +.082 (2.08) |
| D | +.156 (3.96) | -.215 (5.46) |
| E | -.156 (3.96) | -.215 (5.46) |

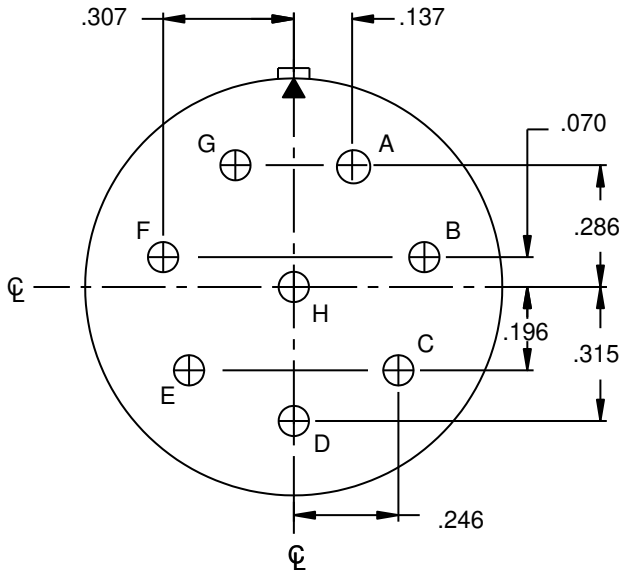
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -34 | 2 | 16 | D | D,E | MS33686-34 |
| | | 3 | 12 | D | A,B,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-36

CONTACTS A, C, E AND G – IRON
CONTACTS B, D, F AND H – COPPER-NICKEL ALLOY

| Contacts (Insert arrangement 22-36) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.137 (3.48) | +.286 (7.26) |
| B | +.307 (7.80) | +.070 (1.78) |
| C | +.246 (6.25) | -.196 (4.98) |
| D | +.000 (0.00) | -.315 (8.00) |
| E | -.246 (6.25) | -.196 (4.98) |
| F | -.307 (7.80) | +.070 (1.78) |
| G | -.137 (3.48) | +.286 (7.26) |
| H | +.000 (0.00) | +.000 (0.00) |

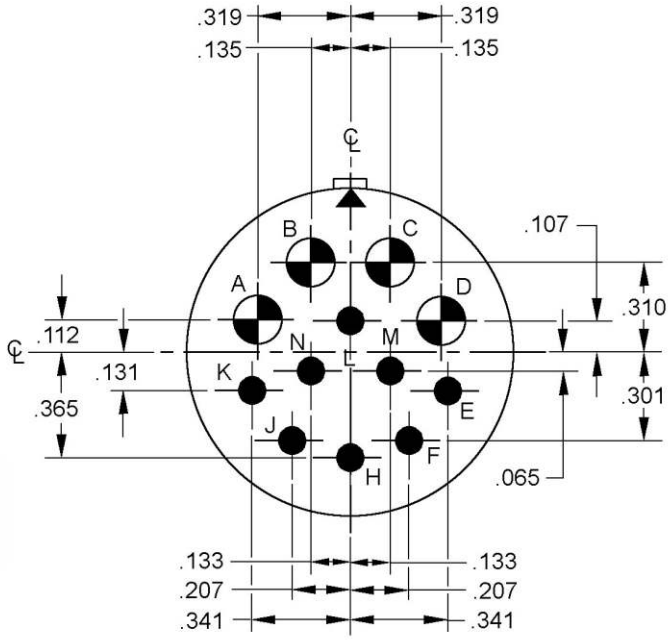
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 22 | -36 | 1 | 12 | D | H | MS33686-36 |
| | | 7 | 12 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | --- | 270 | --- |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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22-63

| Contacts (Insert arrangement 22-63) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.319 (8.10) | +.112 (2.84) |
| B | -.135 (3.43) | +.310 (7.87) |
| C | +.135 (3.43) | +.310 (7.87) |
| D | +.319 (8.10) | +.112 (2.84) |
| E | +.341 (8.66) | -.131 (3.33) |
| F | +.207 (5.26) | -.301 (7.65) |
| H | +.000 (0.00) | -.365 (9.27) |
| J | -.207 (5.26) | -.301 (7.65) |
| K | -.341 (8.66) | -.131 (3.33) |
| L | +.000 (0.00) | +.107 (2.72) |
| M | +.133 (3.38) | -.065 (1.65) |
| N | -.133 (3.38) | -.065 (1.65) |

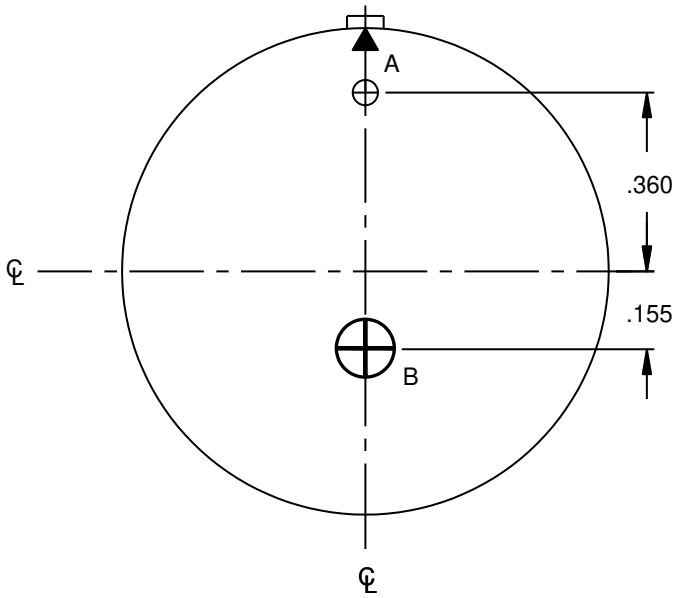
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location |
|---|---|---|---|---|---|
| 22 | -63 | 4 | 12 | A | A,B,C,D |
| | | 8 | 16 | | All others |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 20 | --- | --- | --- |

FIGURE 8. Insert arrangement, electrical connector, shell size 22, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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24-1

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 24-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.360 (9.14) |
| B | +.000 (0.00) | -.155 (3.94) |

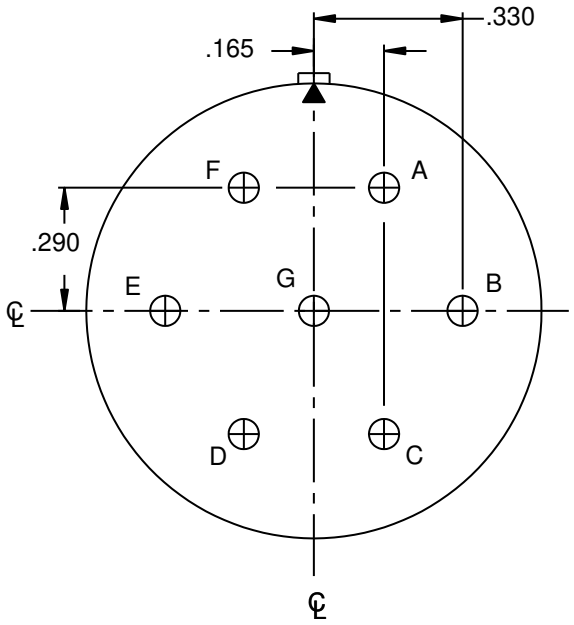
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -1 | 1 | 12 | D | A | MS33687-1 |
| | | 1 | 0 | D | B | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234.



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24-2

| Contacts (Insert arrangement 24-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.165 (4.19) | +.290 (7.37) |
| B | +.330 (8.38) | +.000 (0.00) |
| C | +.165 (4.19) | -.290 (7.37) |
| D | -.165 (4.19) | -.290 (7.37) |
| E | -.330 (8.38) | +.000 (0.00) |
| F | -.165 (4.19) | +.290 (7.37) |
| G | +.000 (0.00) | +.000 (0.00) |

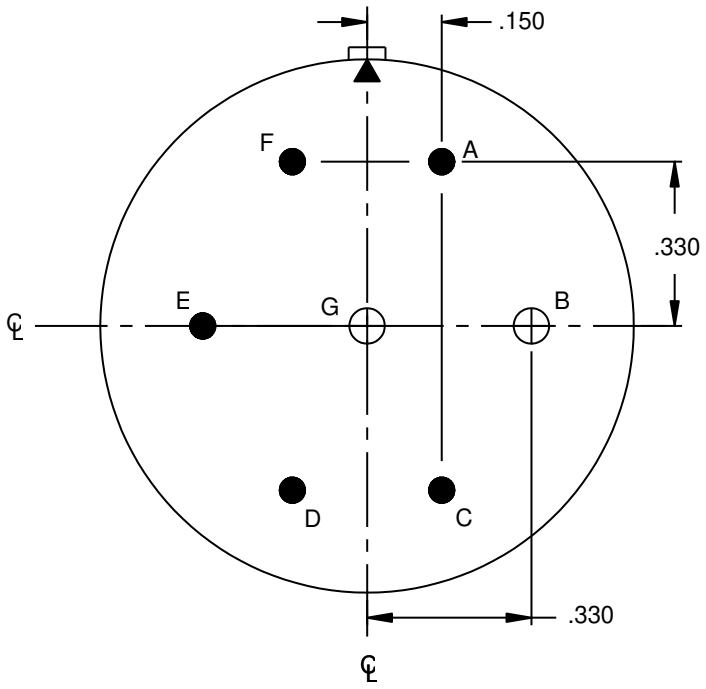
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -2 | 7 | 12 | D | All | MS33687-2 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-3

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 24-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.150 (3.81) | +.330 (8.38) |
| B | +.330 (8.38) | +.000 (0.00) |
| C | +.150 (3.81) | -.330 (8.38) |
| D | -.150 (3.81) | -.330 (8.38) |
| E | -.330 (8.38) | +.000 (0.00) |
| F | -.150 (3.81) | +.330 (8.38) |
| G | +.000 (0.00) | +.000 (0.00) |

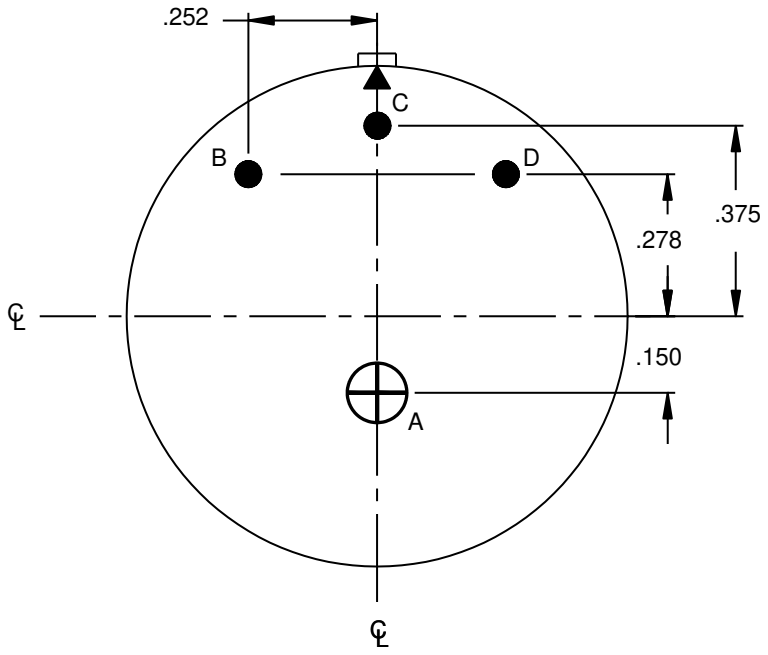
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -3 | 5 | 16 | D | A,C,D,E,F | MS33687-3 |
| | | 2 | 12 | D | B,G | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-4

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | -.150 (3.81) |
| B | -.252 (6.40) | +.278 (7.06) |
| C | +.000 (0.00) | +.375 (9.53) |
| D | +.252 (6.40) | +.278 (7.06) |

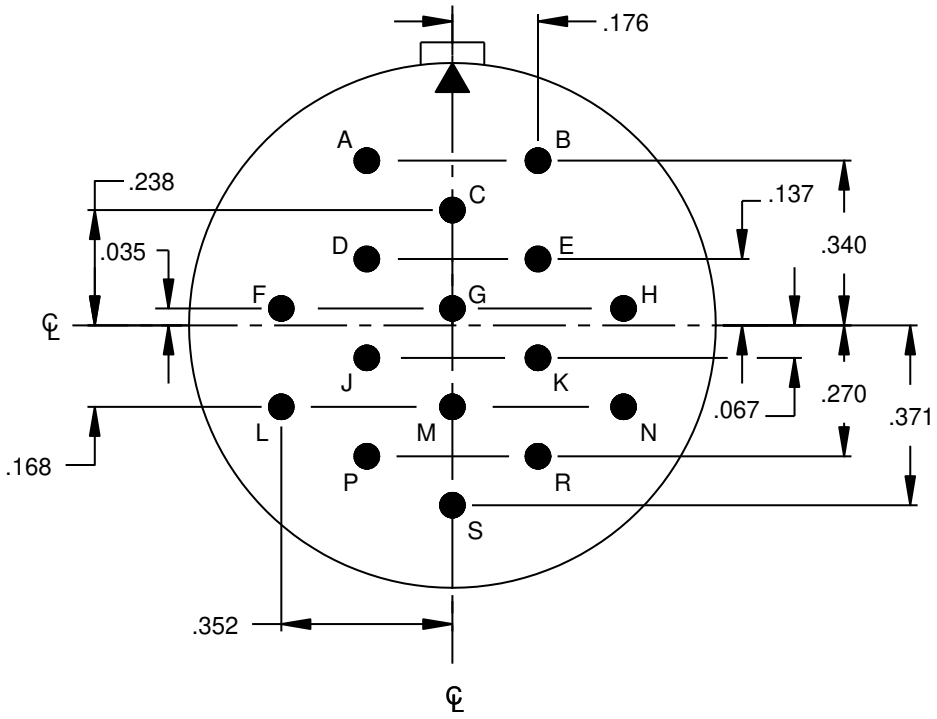
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -4 | 3 | 16 | D | B,C,D | MS33687-4 |
| | | 1 | 0 | D | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-5

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-5) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.176 (4.47) | +.340 (8.64) |
| B | +.176 (4.47) | +.340 (8.64) |
| C | +.000 (0.00) | +.238 (6.05) |
| D | -.176 (4.47) | +.137 (3.48) |
| E | +.176 (4.47) | +.137 (3.48) |
| F | -.352 (8.94) | +.035 (0.89) |
| G | +.000 (0.00) | +.035 (0.89) |
| H | +.352 (8.94) | +.035 (0.89) |

| Contacts (Insert arrangement 24-5) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| J | -.176 (4.47) | -.067 (1.70) |
| K | +.176 (4.47) | -.067 (1.70) |
| L | -.352 (8.94) | -.168 (4.27) |
| M | +.000 (0.00) | -.168 (4.27) |
| N | +.352 (8.94) | -.168 (4.27) |
| P | -.176 (4.47) | -.270 (6.86) |
| R | +.176 (4.47) | -.270 (6.86) |
| S | +.000 (0.00) | -.371 (9.42) |

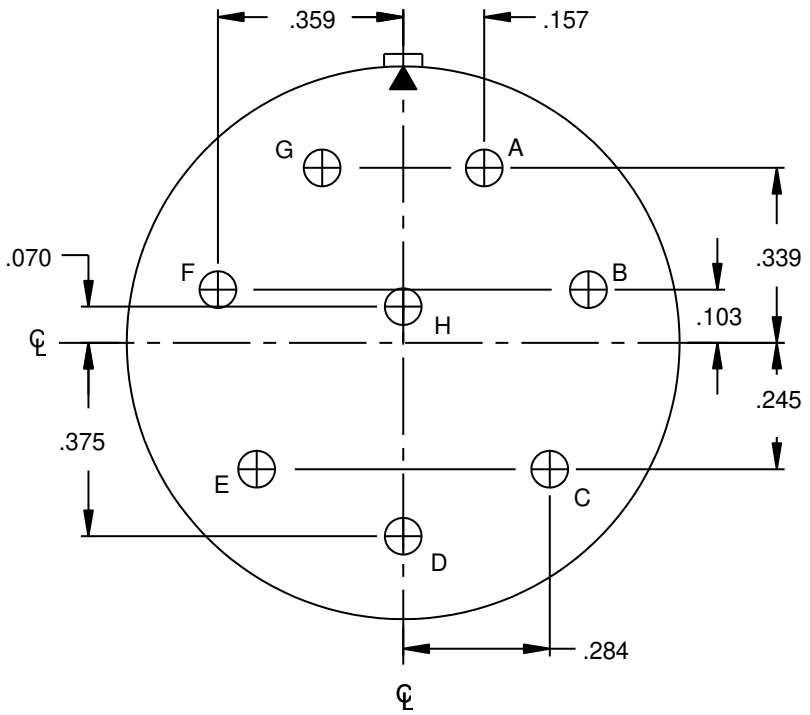
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -5 | 16 | 16 | A | All | MS33687-5 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-6

| Contacts (Insert arrangement 24-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.157 (3.99) | +.339 (8.61) |
| B | +.359 (9.12) | +.103 (2.62) |
| C | +.284 (7.21) | -.245 (6.22) |
| D | +.000 (0.00) | -.375 (9.53) |
| E | -.284 (7.21) | -.245 (6.22) |
| F | -.359 (9.12) | +.103 (2.62) |
| G | -.157 (3.99) | +.339 (8.61) |
| H | +.000 (0.00) | +.070 (1.78) |

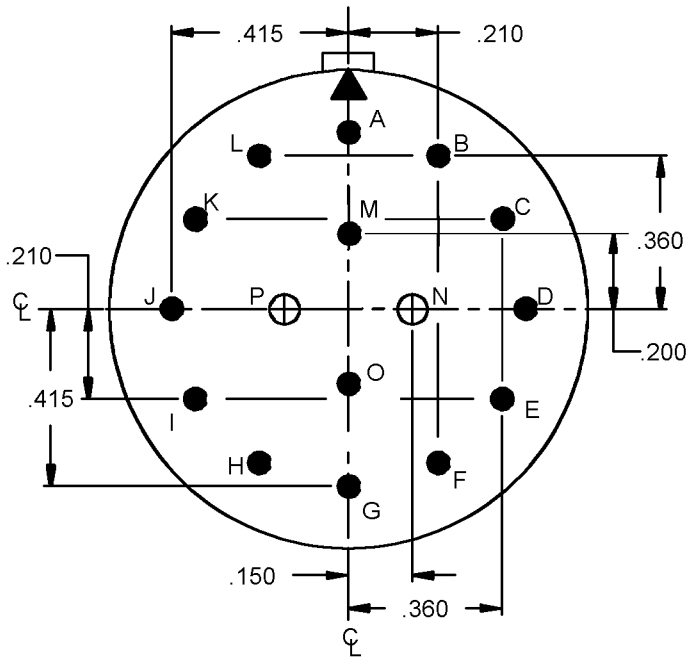
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -6 | 3 | 12 | D | A,G,H | MS33687-6 |
| | | 5 | 12 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-7

| Contacts (Insert arrangement 24-7) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.415 (10.54) |
| B | +.210 (5.33) | +.360 (9.14) |
| C | +.360 (9.14) | +.210 (5.33) |
| D | +.415 (10.54) | +.000 (0.00) |
| E | +.360 (9.14) | -.210 (5.33) |
| F | +.210 (5.33) | -.360 (9.14) |
| G | +.000 (0.00) | -.415 (10.54) |
| H | -.210 (5.33) | -.360 (9.14) |

| Contacts (Insert arrangement 24-7) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| I | -.360 (9.14) | -.210 (5.33) |
| J | -.415 (10.54) | +.000 (0.00) |
| K | -.360 (9.14) | +.210 (5.33) |
| L | -.210 (5.33) | +.360 (9.14) |
| M | +.000 (0.00) | +.200 (5.08) |
| N | +.150 (3.81) | +.000 (0.00) |
| O | +.000 (0.00) | -.200 (5.08) |
| P | -.150 (3.81) | +.000 (0.00) |

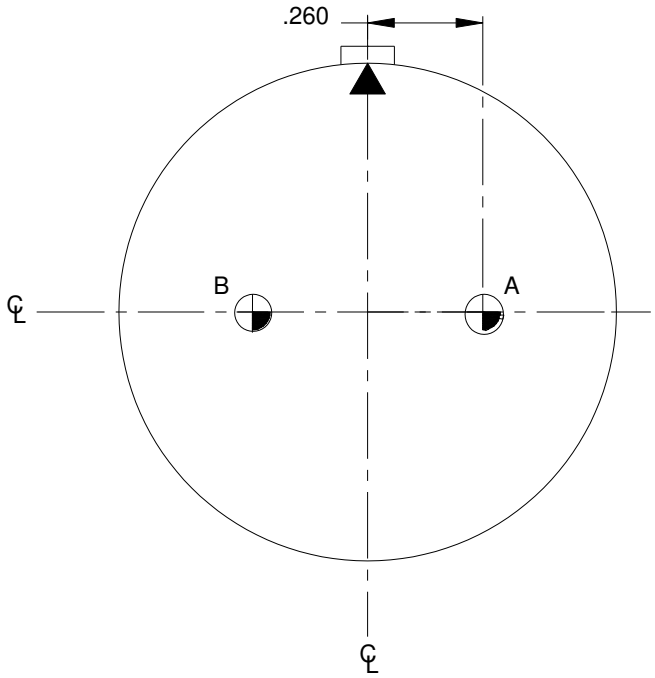
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -7 | 14 | 16 | A | All others | MS33687-7 |
| | | 2 | 12 | | P,N | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-9

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.260 (6.60) | +.000 (0.00) |
| B | -.260 (6.60) | +.000 (0.00) |

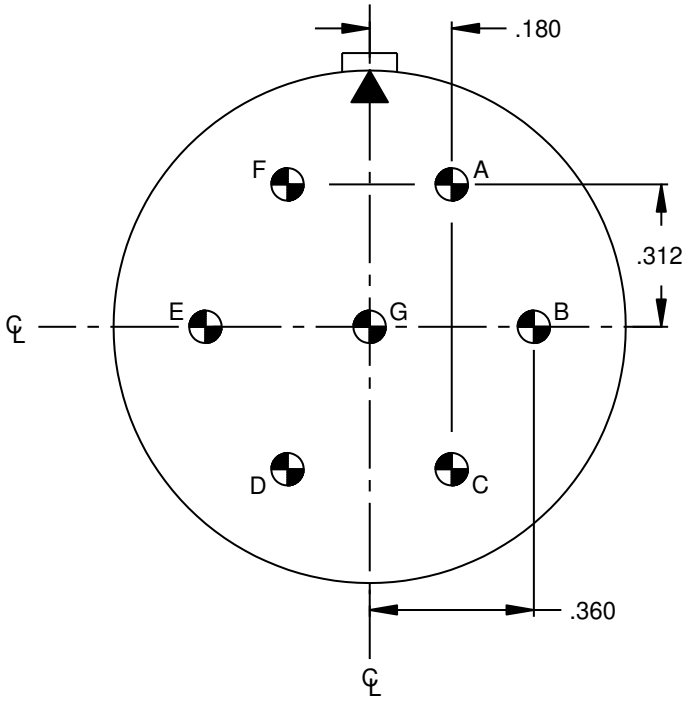
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -9 | 2 | 4 | A | All | MS33687-9 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-10

| Contacts (Insert arrangement 24-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.180 (4.57) | +.312 (7.92) |
| B | +.360 (9.14) | +.000 (0.00) |
| C | +.180 (4.57) | -.312 (7.92) |
| D | -.180 (4.57) | -.312 (7.92) |
| E | -.360 (9.14) | +.000 (0.00) |
| F | -.180 (4.57) | +.312 (7.92) |
| G | +.000 (0.00) | +.000 (0.00) |

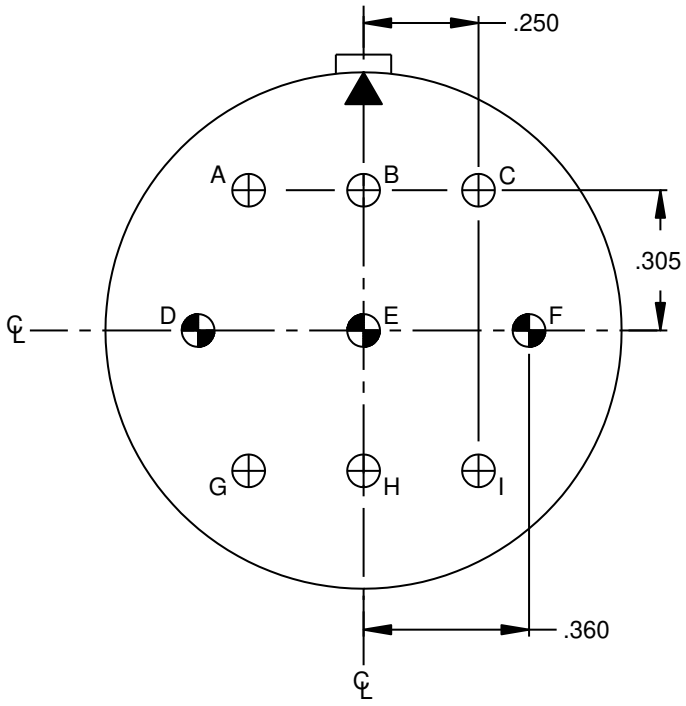
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -10 | 7 | 8 | A | All | MS33687-10 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-11

| Contacts (Insert arrangement 24-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.250 (6.35) | +.305 (7.75) |
| B | +.000 (0.00) | +.305 (7.75) |
| C | +.250 (6.35) | +.305 (7.75) |
| D | -.360 (9.14) | +.000 (0.00) |
| E | +.000 (0.00) | +.000 (0.00) |
| F | +.360 (9.14) | +.000 (0.00) |
| G | -.250 (6.35) | -.305 (7.75) |
| H | +.000 (0.00) | -.305 (7.75) |
| I | +.250 (6.35) | -.305 (7.75) |

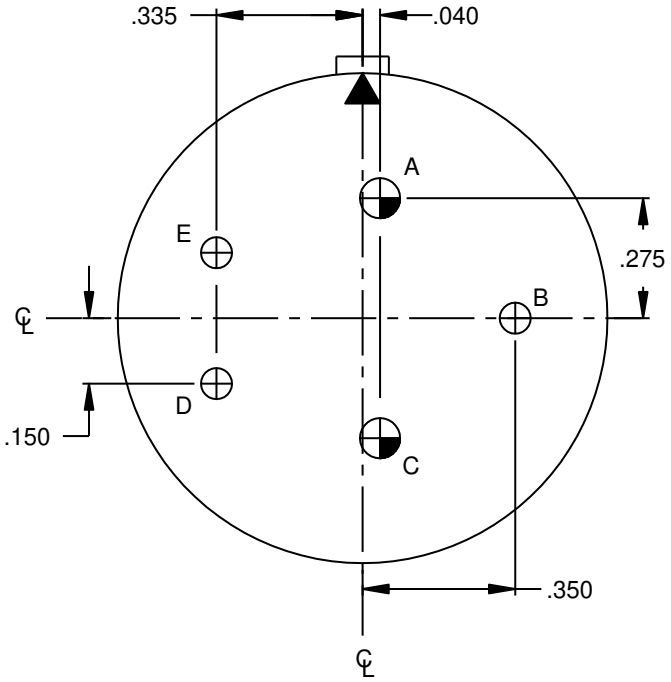
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -11 | 6 | 12 | A | All others | MS33687-11 |
| | | 3 | 8 | | D,E,F | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-12

| Contacts (Insert arrangement 24-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.040 (1.02) | +.275 (6.99) |
| B | +.350 (8.89) | +.000 (0.00) |
| C | +.040 (1.02) | -.275 (6.99) |
| D | -.335 (8.51) | -.150 (3.81) |
| E | -.335 (8.51) | +.150 (3.81) |

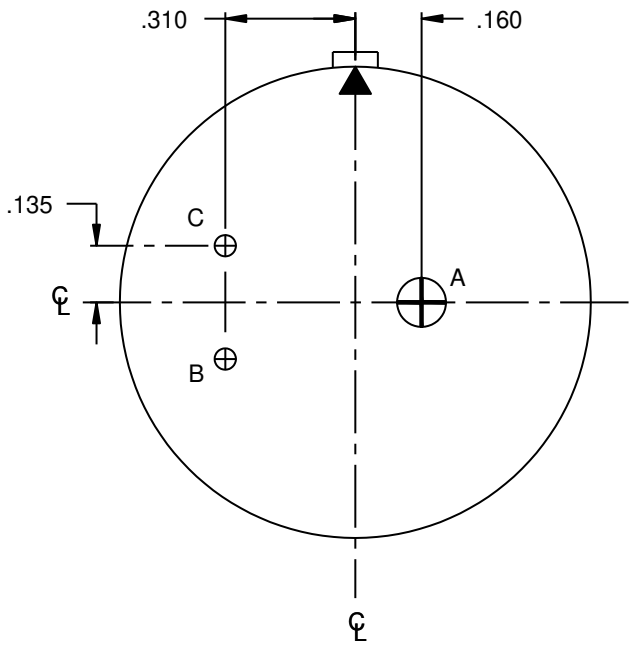
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -12 | 3 | 12 | A | All others | MS33687-12 |
| | | 2 | 4 | | A,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-14

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-14) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.160 (4.06) | +.000 (0.00) |
| B | -.310 (7.87) | -.135 (3.43) |
| C | -.310 (7.87) | +.135 (3.43) |

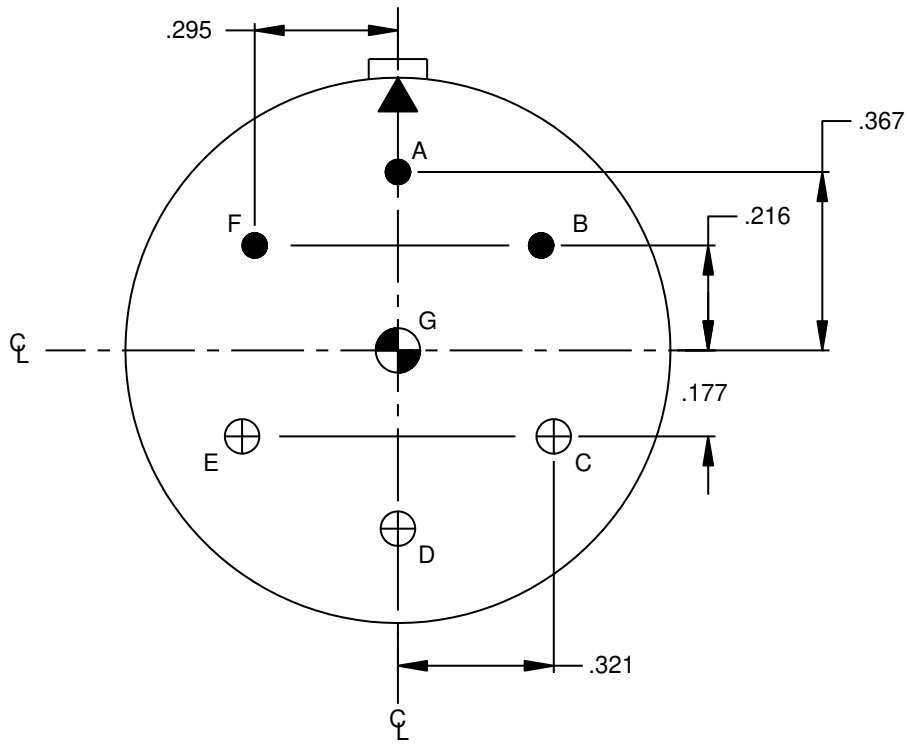
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -14 | 2 | 12 | A | B,C | MS33687-14 |
| | | 1 | 0 | | A | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-16

| Contacts (Insert arrangement 24-16) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.367 (9.32) |
| B | +.295 (7.49) | +.216 (5.49) |
| C | +.321 (8.15) | -.177 (4.50) |
| D | +.000 (0.00) | -.367 (9.32) |
| E | -.321 (8.15) | -.177 (4.50) |
| F | -.295 (7.49) | +.216 (5.49) |
| G | +.000 (0.00) | +.000 (0.00) |

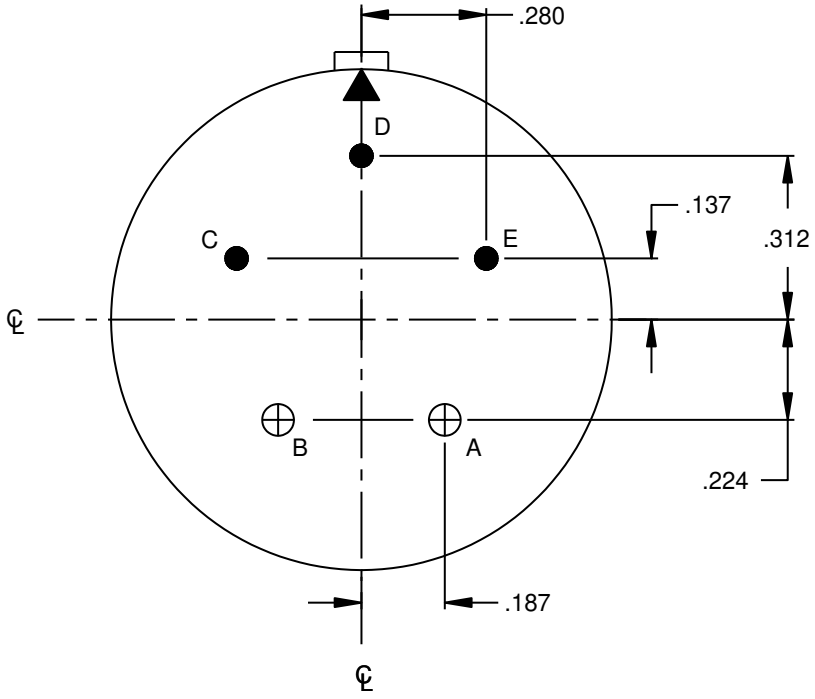
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -16 | 3 | 16 | D | A,B,F | MS33687-16 |
| | | 1 | 8 | D | G | |
| | | 3 | 12 | A | C,D,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-17

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-17) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.187 (4.75) | -.224 (5.69) |
| B | -.187 (4.75) | -.224 (5.69) |
| C | -.280 (7.11) | +.137 (3.48) |
| D | +.000 (0.00) | +.312 (7.92) |
| E | +.280 (7.11) | +.137 (3.48) |

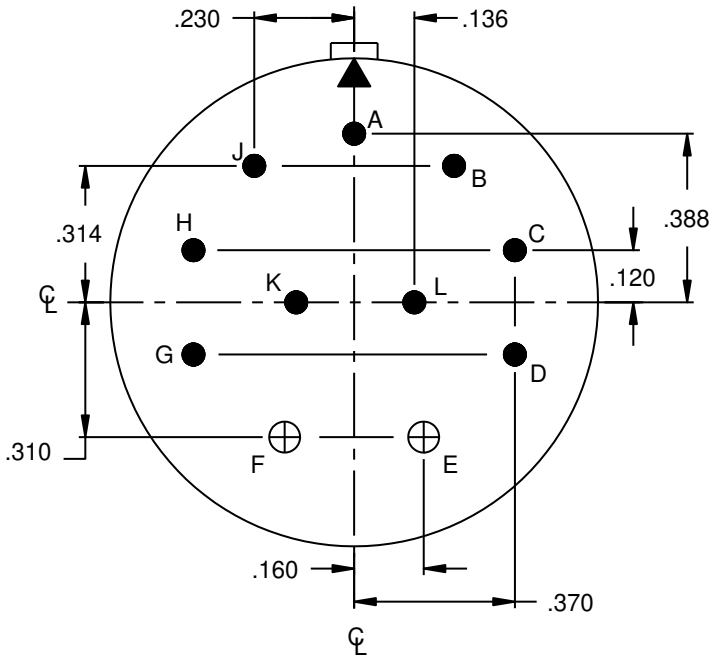
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -17 | 3 | 16 | D | C,D,E | MS33687-17 |
| | | 2 | 12 | | A,B | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-20

| Contacts (Insert arrangement 24-20) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.388 (9.86) |
| B | +.230 (5.84) | +.314 (7.98) |
| C | +.370 (9.40) | +.120 (3.05) |
| D | +.370 (9.40) | -.120 (3.05) |
| E | +.160 (4.06) | -.310 (7.87) |
| F | -.160 (4.06) | -.310 (7.87) |
| G | -.370 (9.40) | -.120 (3.05) |
| H | -.370 (9.40) | +.120 (3.05) |
| J | -.230 (5.84) | +.314 (7.98) |
| K | -.136 (3.45) | +.000 (0.00) |
| L | +.136 (3.45) | +.000 (0.00) |

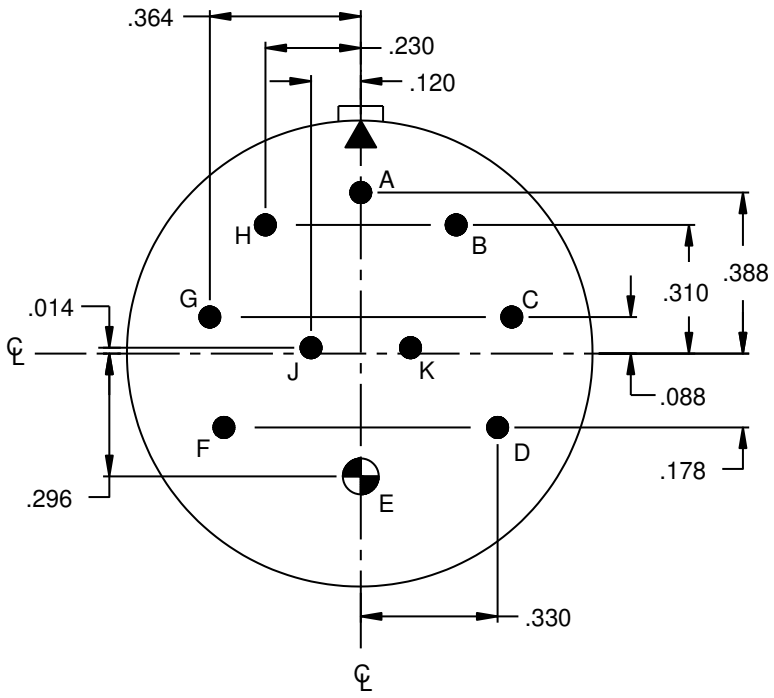
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -20 | 9 | 16 | D | All others | MS33687-20 |
| | | 2 | 12 | | F,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-21

| Contacts (Insert arrangement 24-21) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.388 (9.86) |
| B | +.230 (5.84) | +.310 (7.87) |
| C | +.364 (9.25) | +.088 (2.24) |
| D | +.330 (8.38) | -.178 (4.52) |
| E | +.000 (0.00) | -.296 (7.52) |
| F | -.330 (8.38) | -.178 (4.52) |
| G | -.364 (9.25) | +.088 (2.24) |
| H | -.230 (5.84) | +.310 (7.87) |
| J | -.120 (3.05) | +.014 (0.36) |
| K | +.120 (3.05) | +.014 (0.36) |

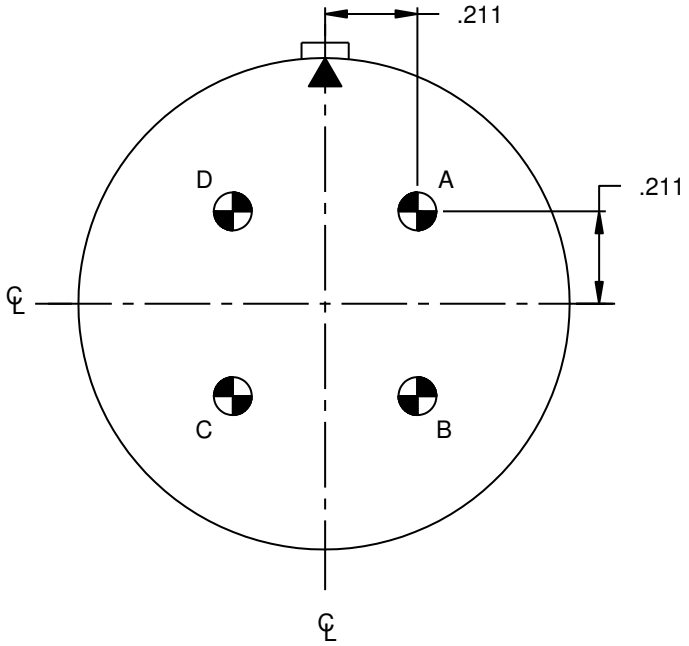
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -21 | 9 | 16 | D | All others | MS33687-21 |
| | | 1 | 8 | | E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-22

| Contacts (Insert arrangement 24-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.211 (5.33) | +.211 (5.33) |
| B | +.211 (5.33) | -.211 (5.33) |
| C | -.211 (5.33) | -.211 (5.33) |
| D | -.211 (5.33) | +.211 (5.33) |

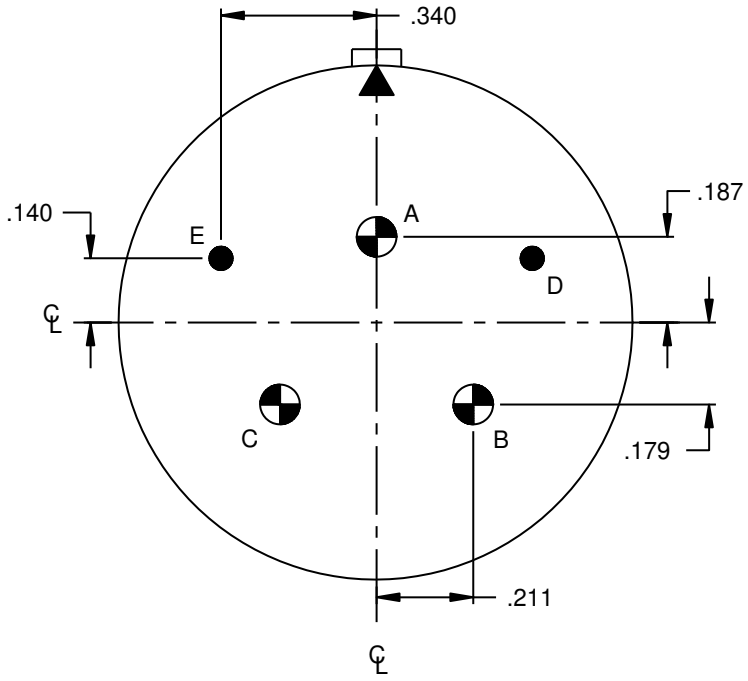
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -22 | 4 | 8 | A | All | MS33687-22 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 45 | 110 | 250 | --- |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-23

(Inactive for New Design after 11 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 24-23) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.187 (4.75) |
| B | +.211 (5.36) | -.179 (4.55) |
| C | -.211 (5.36) | -.179 (4.55) |
| D | +.340 (8.64) | +.140 (3.56) |
| E | -.340 (8.64) | +.140 (3.56) |

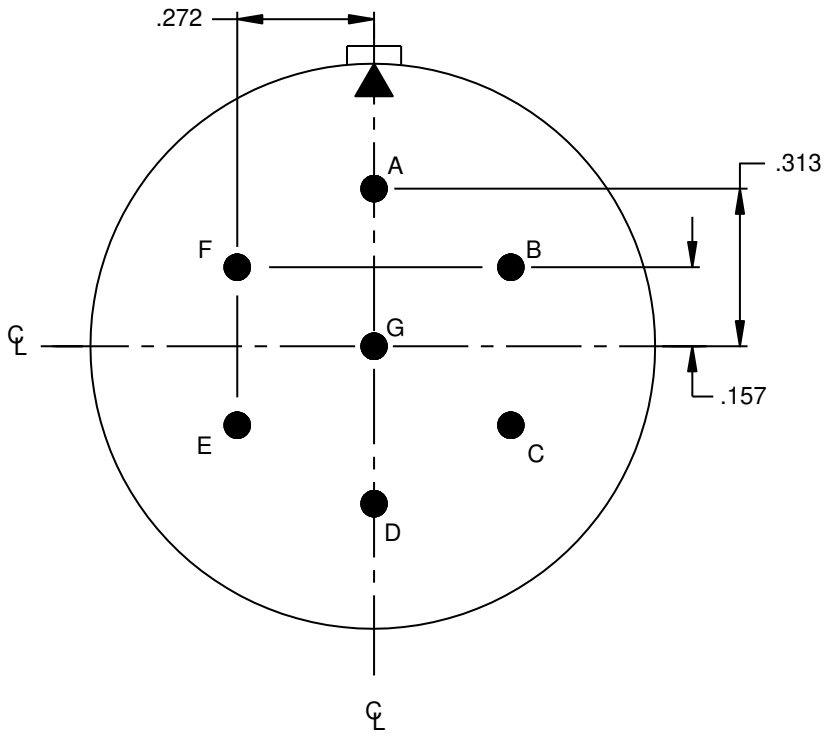
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -23 | 2 | 16 | D | D,E | MS33687-23 |
| | | 3 | 8 | | A,B,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-27

| Contacts (Insert arrangement 24-27) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.313 (7.95) |
| B | +.272 (6.91) | +.157 (3.99) |
| C | +.272 (6.91) | -.157 (3.99) |
| D | +.000 (0.00) | -.313 (7.95) |
| E | -.272 (6.91) | -.157 (3.99) |
| F | -.272 (6.91) | +.157 (3.99) |
| G | +.000 (0.00) | +.000 (0.00) |

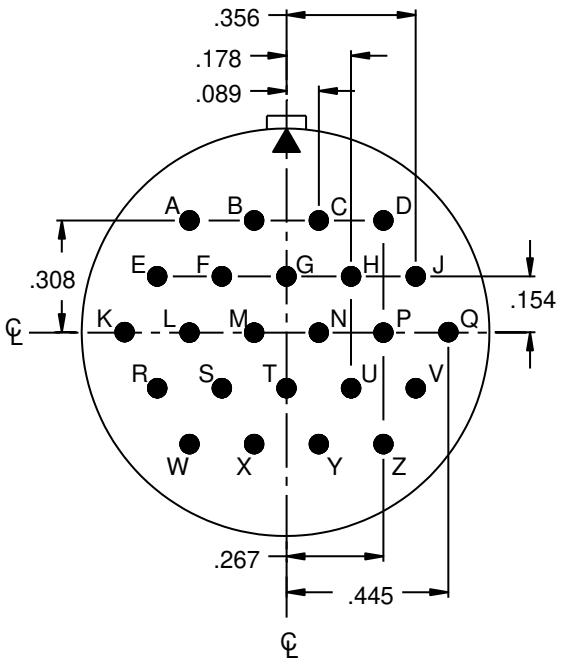
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -27 | 7 | 16 | E | All | MS33687-27 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | --- | --- | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-28

| Contacts (Insert arrangement 24-28) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.267 (6.78) | +.308 (7.82) |
| B | -.089 (2.26) | +.308 (7.82) |
| C | +.089 (2.26) | +.308 (7.82) |
| D | +.267 (6.78) | +.308 (7.82) |
| E | -.356 (9.04) | +.154 (3.91) |
| F | -.178 (4.52) | +.154 (3.91) |
| G | +.000 (0.00) | +.154 (3.91) |
| H | +.178 (4.52) | +.154 (3.91) |
| J | +.356 (9.04) | +.154 (3.91) |
| K | -.445 (11.30) | +.000 (0.00) |
| L | -.267 (6.78) | +.000 (0.00) |
| M | -.089 (2.26) | +.000 (0.00) |

| Contacts (Insert arrangement 24-28) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | +.089 (2.26) | +.000 (0.00) |
| P | +.267 (6.78) | +.000 (0.00) |
| Q | +.445 (11.30) | +.000 (0.00) |
| R | -.356 (9.04) | -.154 (3.91) |
| S | -.178 (4.52) | -.154 (3.91) |
| T | +.000 (0.00) | -.154 (3.91) |
| U | +.178 (4.52) | -.154 (3.91) |
| V | +.356 (9.04) | -.154 (3.91) |
| W | -.267 (6.78) | -.308 (7.82) |
| X | -.089 (2.26) | -.308 (7.82) |
| Y | +.089 (2.26) | -.308 (7.82) |
| Z | +.267 (6.78) | -.308 (7.82) |

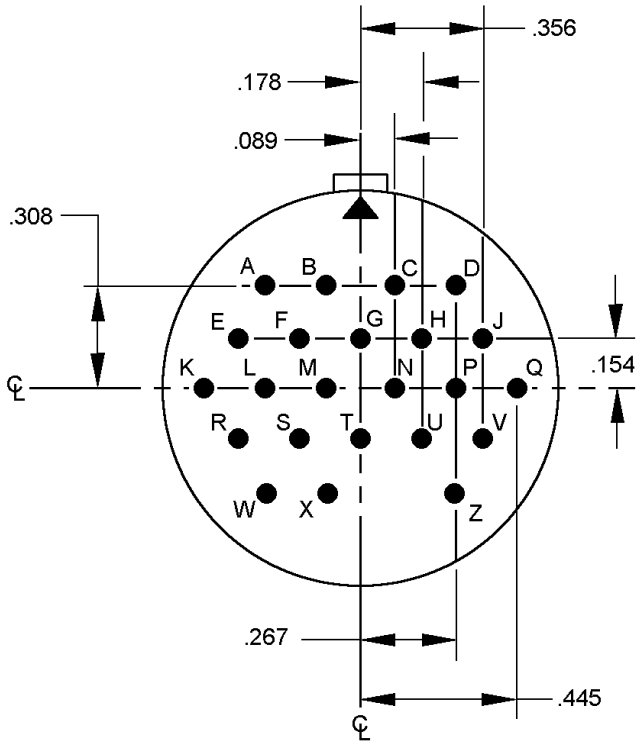
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -28 | 24 | 16 | INST | All | MS33687-28 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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24-80

(Inactive for New Design of Air Force airborne equipment for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 24-80) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.267 (6.78) | +.308 (7.82) |
| B | -.089 (2.26) | +.308 (7.82) |
| C | +.089 (2.26) | +.308 (7.82) |
| D | +.267 (6.78) | +.308 (7.82) |
| E | -.356 (9.04) | +.154 (3.91) |
| F | -.178 (4.52) | +.154 (3.91) |
| G | +.000 (0.00) | +.154 (3.91) |
| H | +.178 (4.52) | +.154 (3.91) |
| J | +.356 (9.04) | +.154 (3.91) |
| K | -.445 (11.30) | +.000 (0.00) |
| L | -.267 (6.78) | +.000 (0.00) |
| M | -.089 (2.26) | +.000 (0.00) |

| Contacts (Insert arrangement 24-80) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | +.089 (2.26) | +.000 (0.00) |
| P | +.267 (6.78) | +.000 (0.00) |
| Q | +.445 (11.30) | +.000 (0.00) |
| R | -.356 (9.04) | -.154 (3.91) |
| S | -.178 (4.52) | -.154 (3.91) |
| T | +.000 (0.00) | -.154 (3.91) |
| U | +.178 (4.52) | -.154 (3.91) |
| V | +.356 (9.04) | -.154 (3.91) |
| W | -.267 (6.78) | -.308 (7.82) |
| X | -.089 (2.26) | -.308 (7.82) |
| Z | +.267 (6.78) | -.308 (7.82) |
| --- | --- | --- |

| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 24 | -80 | 23 | 16 | INST | All | MS33687-80 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



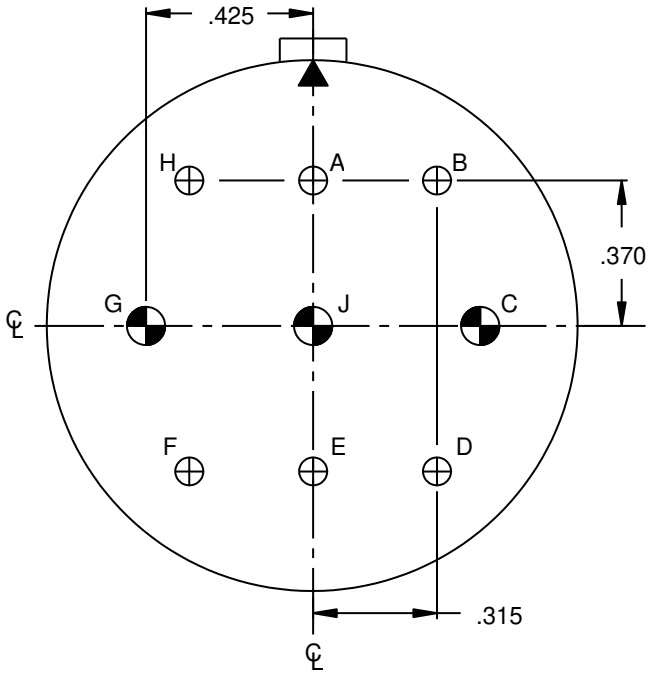
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Insert arrangement 24-80 | | | |
|---|---|---|---|
| Alternate insert positions | | | |
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 145 | 240 | 300 |

FIGURE 9. Insert arrangement, electrical connector, shell size 24, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-1

| Contacts (Insert arrangement 28-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.370 (9.40) |
| B | +.315 (8.00) | +.370 (9.40) |
| C | +.425 (10.80) | +.000 (0.00) |
| D | +.315 (8.00) | -.370 (9.40) |
| E | +.000 (0.00) | -.370 (9.40) |
| F | -.315 (8.00) | -.370 (9.40) |
| G | -.425 (10.80) | +.000 (0.00) |
| H | -.315 (8.00) | +.370 (9.40) |
| J | +.000 (0.00) | +.000 (0.00) |

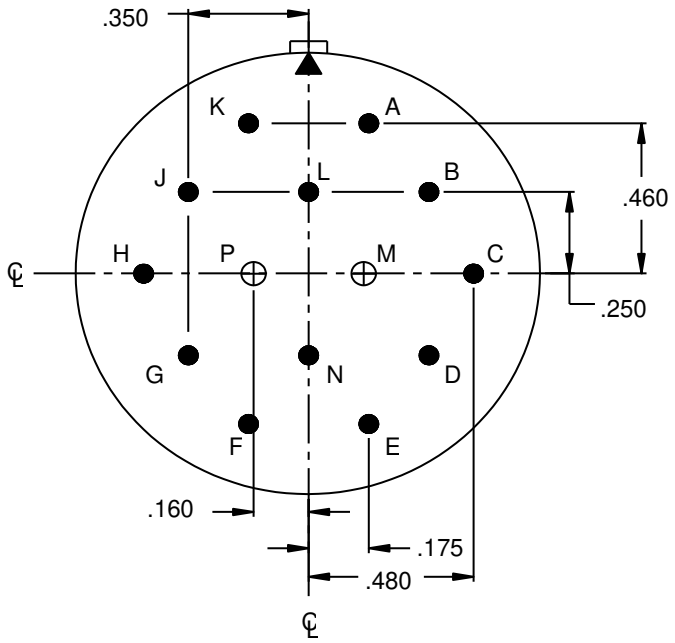
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -1 | 2 | 12 | D | A,E | MS33688-1 |
| | | 4 | 12 | A | B,D,F,H | |
| | | 1 | 8 | D | J | |
| | | 2 | 8 | A | G,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234.



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Check the source to verify that this is the current version before use.




28-2

| Contacts (Insert arrangement 28-2) | | |
|---|---|---|
| Contact position | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.175 (4.45) | +.460 (11.68) |
| B | +.350 (8.89) | +.250 (6.35) |
| C | +.480 (12.19) | +.000 (0.00) |
| D | +.350 (8.89) | -.250 (6.35) |
| E | +.175 (4.45) | -.460 (11.68) |
| F | -.175 (4.45) | -.460 (11.68) |
| G | -.350 (8.89) | -.250 (6.35) |

| Contacts (Insert arrangement 28-2) | | |
|---|---|---|
| Contact position | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| H | -.480 (12.19) | +.000 (0.00) |
| J | -.350 (8.89) | +.250 (6.35) |
| K | -.175 (4.45) | +.460 (11.68) |
| L | +.000 (0.00) | +.250 (6.35) |
| M | +.160 (4.06) | +.000 (0.00) |
| N | +.000 (0.00) | -.250 (6.35) |
| P | -.160 (4.06) | +.000 (0.00) |

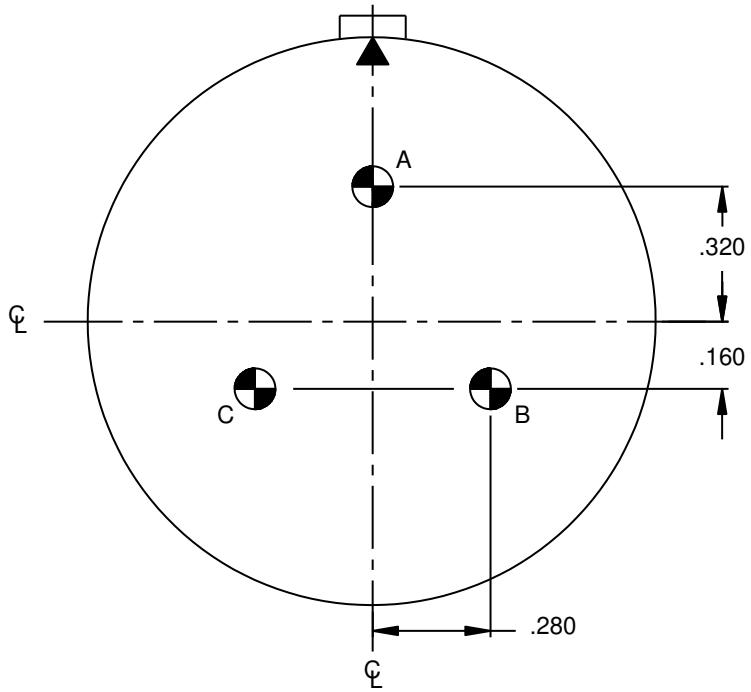
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -2 | 12 | 16 | D | All others | MS33688-2 |
| | | 2 | 12 | | M,P | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-3

| Contacts (Insert arrangement 28-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.320 (8.13) |
| B | +.280 (7.11) | -.160 (4.06) |
| C | -.280 (7.11) | -.160 (4.06) |

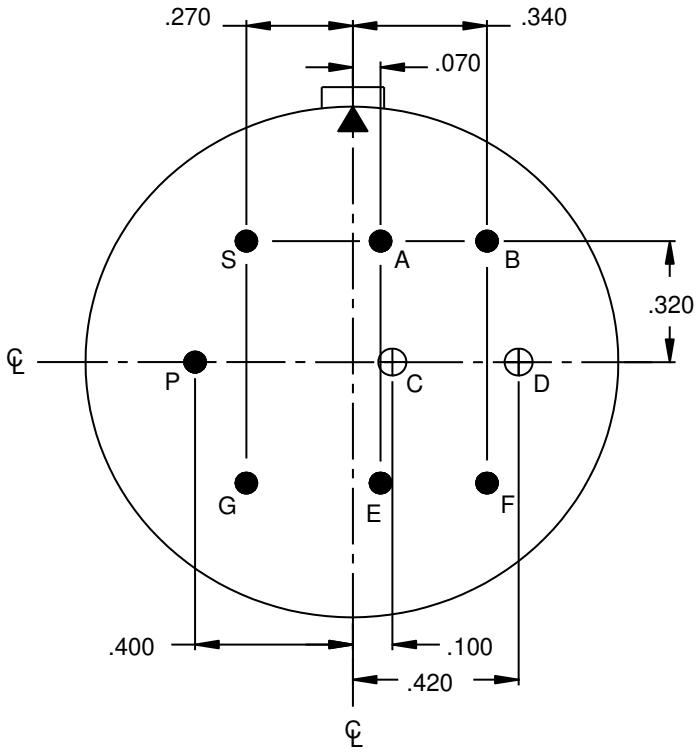
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -3 | 3 | 8 | E | All | MS33688-3 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-4

| Contacts (Insert arrangement 28-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.070 (1.78) | +.320 (8.13) |
| B | +.340 (8.64) | +.320 (8.13) |
| C | +.100 (2.54) | +.000 (0.00) |
| D | +.420 (10.67) | +.000 (0.00) |
| E | +.070 (1.78) | -.320 (8.13) |
| F | +.340 (8.64) | -.320 (8.13) |
| G | -.270 (6.86) | -.320 (8.13) |
| P | -.400 (10.16) | +.000 (0.00) |
| S | -.270 (6.86) | +.320 (8.13) |

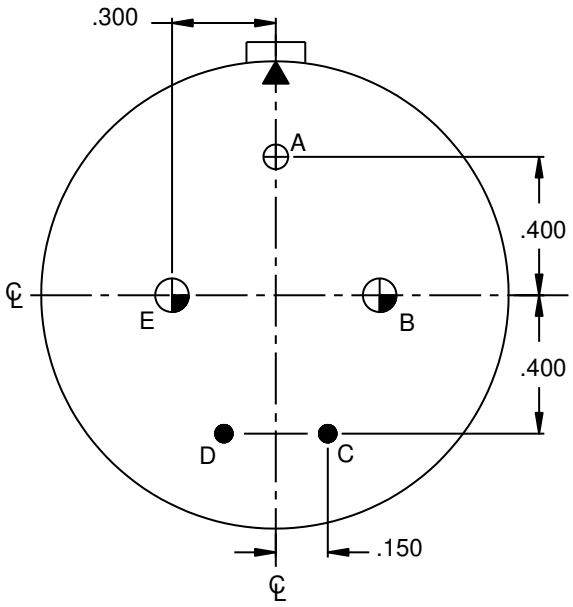
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -4 | 3 | 16 | E | G,P,S | MS33688-4 |
| | | 4 | 16 | D | A,B,E,F | |
| | | 2 | 12 | D | C,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-5

| Contacts (Insert arrangement 28-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.400 (10.16) |
| B | +.300 (7.62) | +.000 (0.00) |
| C | +.150 (3.81) | -.400 (10.16) |
| D | -.150 (3.81) | -.400 (10.16) |
| E | -.300 (7.62) | +.000 (0.00) |

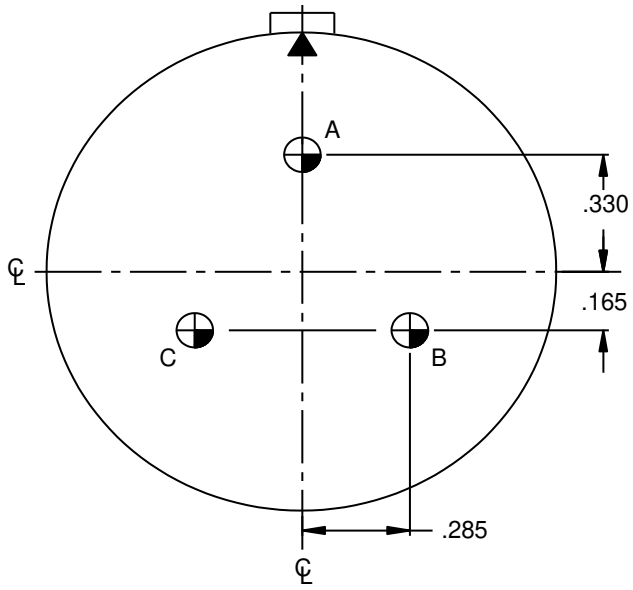
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -5 | 2 | 16 | D | C,D | MS33688-5 |
| | | 1 | 12 | D | A | |
| | | 2 | 4 | D | B,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-6

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 28-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.330 (8.38) |
| B | +.285 (7.24) | -.165 (4.19) |
| C | -.285 (7.24) | -.165 (4.19) |

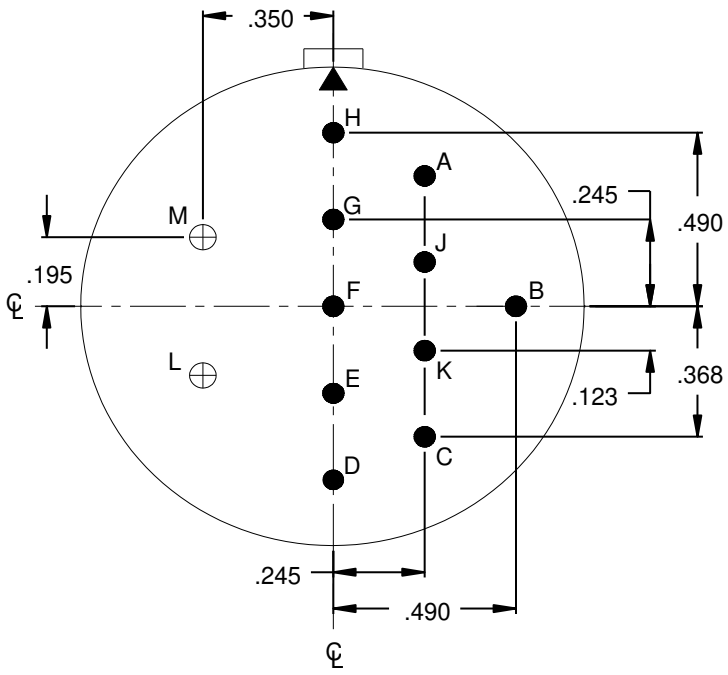
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -6 | 3 | 4 | D | All | MS33688-6 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-8

| Contacts (Insert arrangement 28-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.245 (6.22) | +.368 (9.35) |
| B | +.490 (12.45) | +.000 (0.00) |
| C | +.245 (6.22) | -.368 (9.35) |
| D | +.000 (0.00) | -.490 (12.45) |
| E | +.000 (0.00) | -.245 (6.22) |
| F | +.000 (0.00) | +.000 (0.00) |
| G | +.000 (0.00) | +.245 (6.22) |
| H | +.000 (0.00) | +.490 (12.45) |
| J | +.245 (6.22) | +.123 (3.12) |
| K | +.245 (6.22) | -.123 (3.12) |
| L | -.350 (8.89) | -.195 (4.95) |
| M | -.350 (8.89) | +.195 (4.95) |

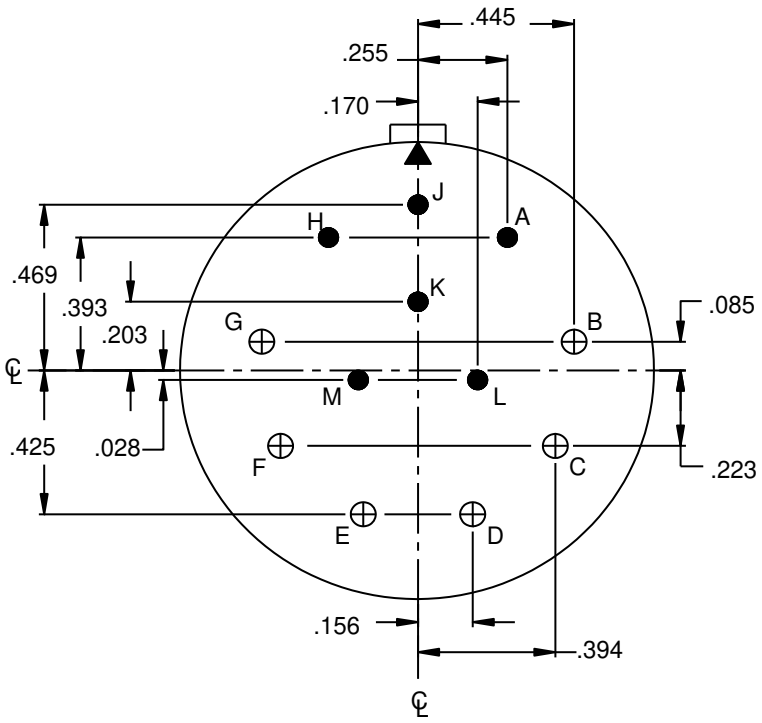
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -8 | 2 | 12 | E | L,M | MS33688-8 |
| | | 1 | 16 | D | B | |
| | | 9 | 16 | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-9

| Contacts (Insert arrangement 28-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.225 (6.48) | +.393 (9.98) |
| B | +.445 (11.30) | +.085 (2.16) |
| C | +.394 (10.01) | -.223 (5.66) |
| D | +.156 (3.96) | -.425 (10.80) |
| E | -.156 (3.96) | -.425 (10.80) |
| F | -.394 (10.01) | -.223 (5.66) |
| G | -.445 (11.30) | +.085 (2.16) |
| H | -.225 (6.48) | +.393 (9.98) |
| J | +.000 (0.00) | +.469 (11.91) |
| K | +.000 (0.00) | +.203 (5.16) |
| L | +.170 (4.32) | -.028 (0.71) |
| M | -.170 (4.32) | -.028 (0.71) |

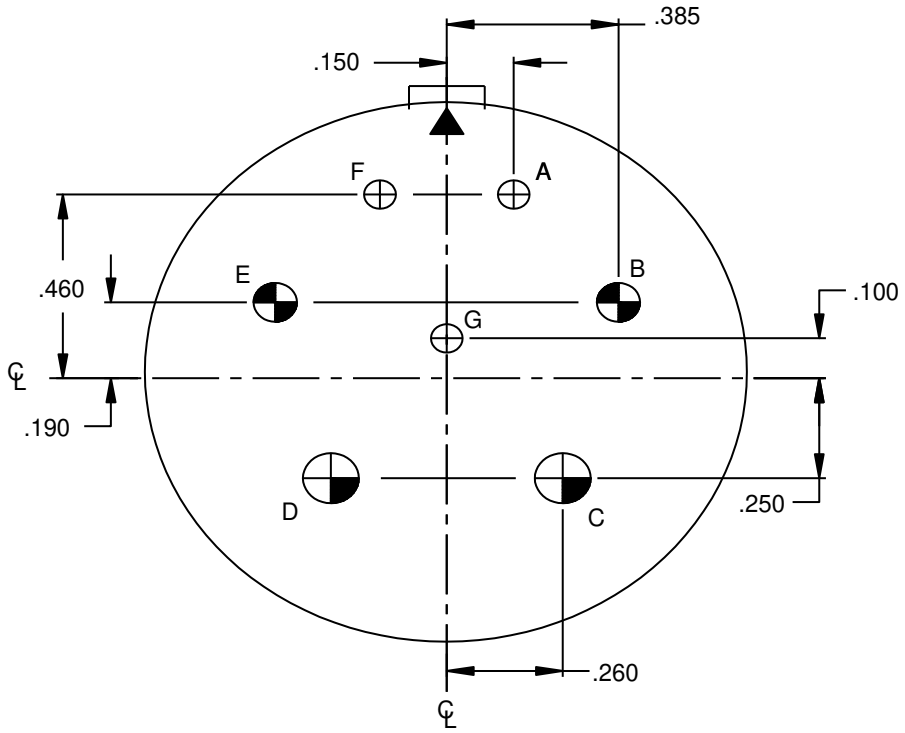
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -9 | 6 | 16 | D | A,H,J,K,L,M | MS33688-9 |
| | | 6 | 12 | | B,C,D,E,F,G | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-10

| Contacts (Insert arrangement 28-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.150 (3.81) | +.460 (11.68) |
| B | +.385 (9.78) | +.190 (4.83) |
| C | +.260 (6.60) | -.250 (6.35) |
| D | -.260 (6.60) | -.250 (6.35) |
| E | -.385 (9.78) | +.190 (4.83) |
| F | -.150 (3.81) | +.460 (11.46) |
| G | +.000 (0.00) | +.100 (2.54) |

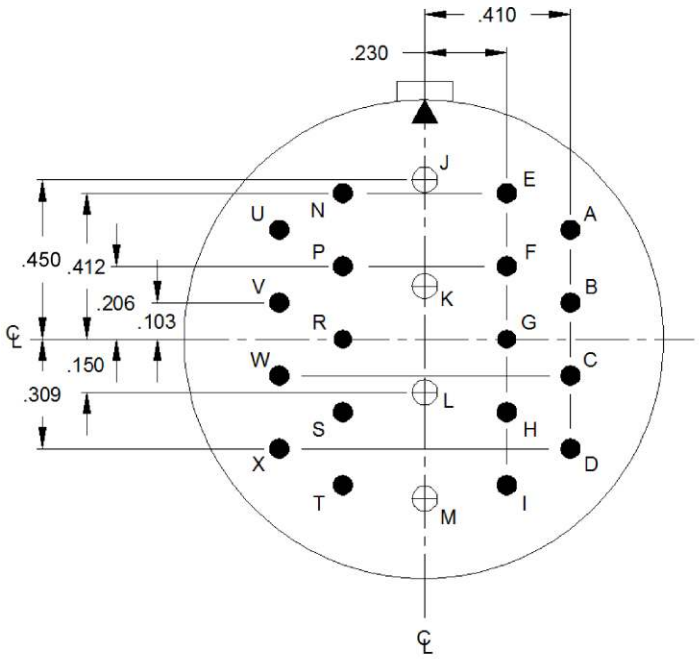
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -10 | 1 | 12 | D | G | MS33688-10 |
| | | 2 | 12 | A | A,F | |
| | | 2 | 8 | A | B,E | |
| | | 2 | 4 | A | C,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-11

| Contacts (Insert arrangement 28-11) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.410 (10.41) | +.309 (7.85) |
| B | +.410 (10.41) | +.103 (2.62) |
| C | +.410 (10.41) | -.103 (2.62) |
| D | +.410 (10.41) | -.309 (7.85) |
| E | +.230 (5.84) | +.412 (10.46) |
| F | +.230 (5.84) | +.206 (5.23) |
| G | +.230 (5.84) | +.000 (0.00) |
| H | +.230 (5.84) | -.206 (5.23) |
| I | +.230 (5.84) | -.412 (10.46) |
| J | +.000 (0.00) | +.450 (11.43) |
| K | +.000 (0.00) | +.150 (3.81) |

| Contacts (Insert arrangement 28-11) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | +.000 (0.00) | -.150 (3.81) |
| M | +.000 (0.00) | -.450 (11.43) |
| N | -.230 (5.84) | +.412 (10.46) |
| P | -.230 (5.84) | +.206 (5.23) |
| R | -.230 (5.84) | +.000 (0.00) |
| S | -.230 (5.84) | -.206 (5.23) |
| T | -.230 (5.84) | -.412 (10.46) |
| U | -.410 (10.41) | +.309 (7.85) |
| V | -.410 (10.41) | +.103 (2.62) |
| W | -.410 (10.41) | -.103 (2.62) |
| X | -.410 (10.41) | -.309 (7.85) |

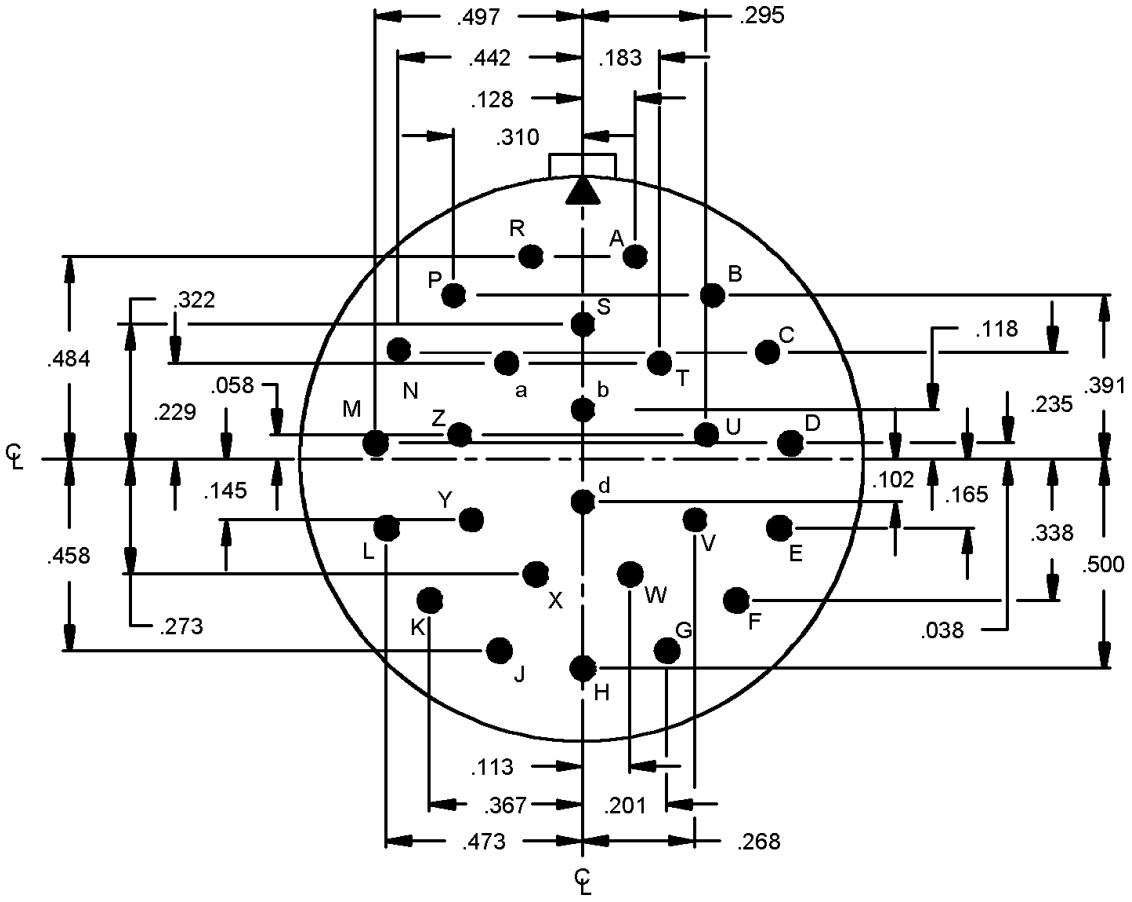
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -11 | 18 | 16 | A | All others | MS33688-11 |
| | | 4 | 12 | A | J,K,L,M | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-12

| Contacts (Insert arrangement 28-12) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.128 (3.25) | +.484 (12.29) |
| B | +.310 (7.87) | +.391 (9.93) |
| C | +.442 (11.23) | +.235 (5.97) |
| D | +.497 (12.62) | +.038 (0.97) |
| E | +.473 (12.01) | -.165 (4.19) |
| F | +.367 (9.32) | -.338 (8.59) |
| G | +.201 (5.11) | -.458 (11.63) |
| H | +.000 (0.00) | -.500 (12.70) |
| J | -.201 (5.11) | -.458 (11.63) |
| K | -.367 (9.32) | -.338 (8.59) |
| L | -.473 (12.01) | -.165 (4.19) |
| M | -.497 (12.62) | +.038 (0.97) |
| N | -.442 (11.23) | +.235 (5.97) |

| Contacts (Insert arrangement 28-12) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | -.310 (7.87) | +.391 (9.93) |
| R | -.128 (3.25) | +.484 (12.29) |
| S | +.000 (0.00) | +.322 (8.18) |
| T | +.183 (4.65) | +.229 (5.82) |
| U | +.295 (7.49) | +.058 (1.47) |
| V | +.268 (6.81) | -.145 (3.68) |
| W | +.113 (2.87) | -.273 (6.93) |
| X | -.113 (2.87) | -.273 (6.93) |
| Y | -.268 (6.81) | -.145 (3.68) |
| Z | -.295 (7.49) | +.058 (147) |
| a | -.183 (4.65) | +.229 (5.82) |
| b | +.000 (0.00) | +.118 (3.00) |
| d | +.000 (0.00) | -.102 (2.59) |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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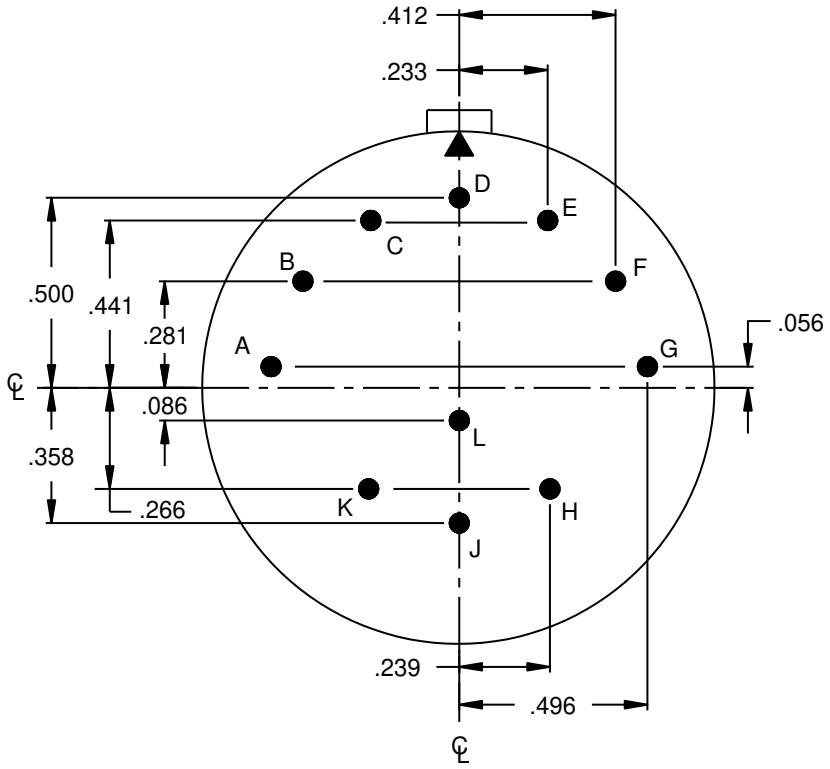
| Insert arrangement 28-12 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 28 | -12 | 26 | 16 | A | All | MS33688-12 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-14

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 28-14) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.496 (12.60) | +.056 (1.42) |
| B | -.412 (10.46) | +.281 (7.14) |
| C | -.233 (5.92) | +.441 (11.20) |
| D | +.000 (0.00) | +.500 (12.70) |
| E | +.233 (5.92) | +.441 (11.20) |
| F | +.412 (10.46) | +.281 (7.14) |

| Contacts (Insert arrangement 28-14) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| G | +.496 (12.60) | +.056 (1.42) |
| H | +.239 (6.07) | -.266 (6.76) |
| J | +.000 (0.00) | -.358 (9.09) |
| K | -.239 (6.07) | -.266 (6.76) |
| L | +.000 (0.00) | -.086 (2.18) |
| --- | --- | --- |

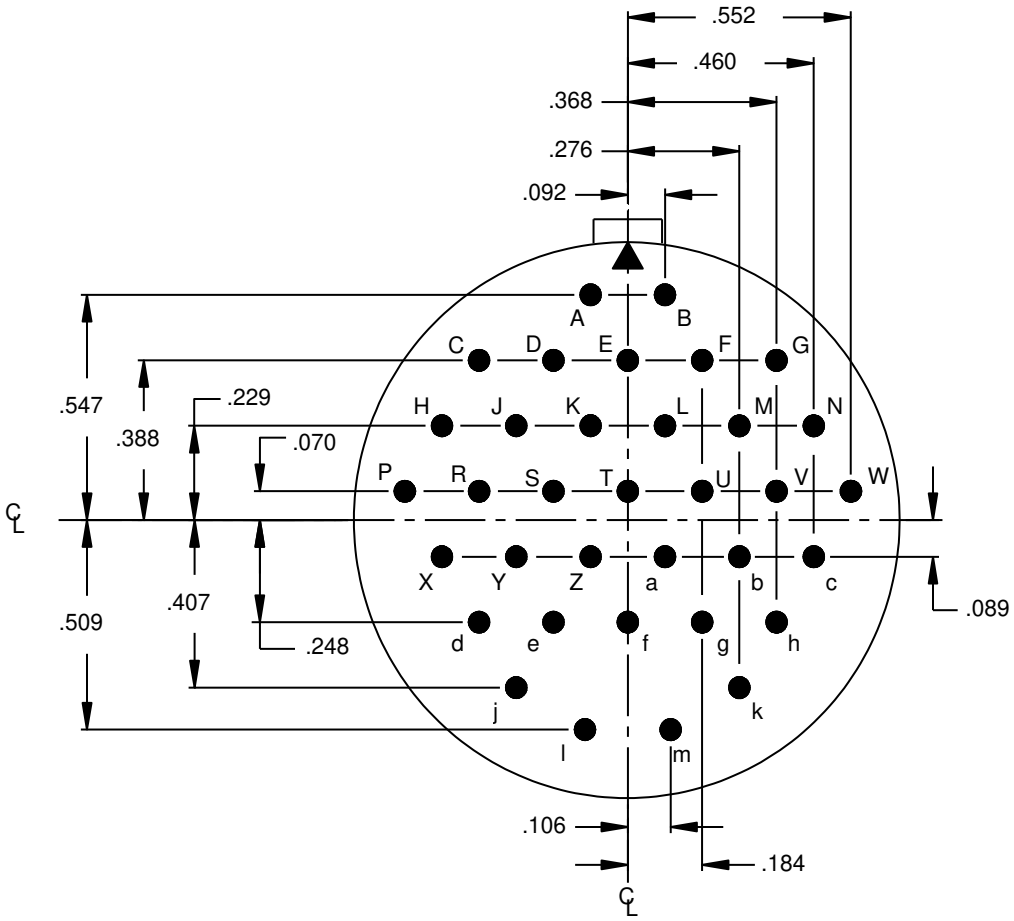
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -14 | 11 | 16 | D | All | MS33688-14 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-15

(Inactive for New Design of Air Force Airborne Equipment after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 28-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.092 (2.34) | +.547 (13.89) |
| B | +.092 (2.34) | +.547 (13.89) |
| C | -.368 (9.35) | +.388 (9.86) |
| D | -.184 (4.67) | +.388 (9.86) |
| E | +.000 (0.00) | +.388 (9.86) |
| F | +.184 (4.67) | +.388 (9.86) |
| G | +.368 (9.35) | +.388 (9.86) |
| H | -.460 (11.68) | +.229 (5.82) |
| J | -.276 (7.01) | +.229 (5.82) |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 28-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| K | -.092 (2.34) | +.229 (5.82) |
| L | +.092 (2.34) | +.229 (5.82) |
| M | +.276 (7.01) | +.229 (5.82) |
| N | +.460 (11.68) | +.229 (5.82) |
| P | -.552 (14.02) | +.070 (1.78) |
| R | -.368 (9.35) | +.070 (1.78) |
| S | -.184 (4.67) | +.070 (1.78) |
| T | +.000 (0.00) | +.070 (1.78) |
| U | +.184 (4.67) | +.070 (1.78) |
| V | +.368 (9.35) | +.070 (1.78) |
| W | +.552 (14.02) | +.070 (1.78) |
| X | -.460 (11.68) | -.089 (2.26) |
| Y | +.276 (7.01) | -.089 (2.26) |
| Z | -.092 (2.34) | -.089 (2.26) |
| a | +.092 (2.34) | -.089 (2.26) |
| b | +.276 (7.01) | -.089 (2.26) |
| c | +.460 (11.68) | -.089 (2.26) |
| d | -.368 (9.35) | -.248 (6.30) |
| e | -.184 (4.67) | -.248 (6.30) |
| f | +.000 (0.00) | -.248 (6.30) |
| g | +.184 (4.67) | -.248 (6.30) |
| h | +.368 (9.35) | -.248 (6.30) |
| j | -.276 (7.01) | -.407 (10.34) |
| k | +.276 (7.01) | -.407 (10.34) |
| l | -.106 (2.69) | -.509 (12.93) |
| m | +.106 (2.69) | -.509 (12.93) |

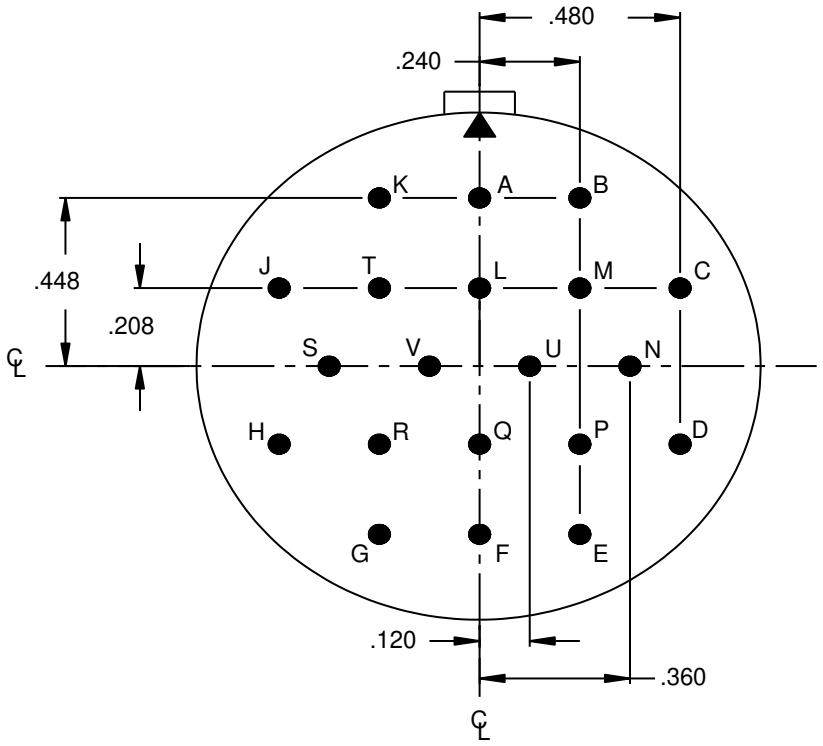
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -15 | 35 | 16 | A | All | MS33688-15 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-16

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 28-16) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.448 (11.38) |
| B | +.240 (6.10) | +.448 (11.38) |
| C | +.480 (12.19) | +.208 (5.28) |
| D | +.480 (12.19) | -.208 (5.28) |
| E | +.240 (6.10) | -.448 (11.38) |
| F | +.000 (0.00) | -.448 (11.38) |
| G | -.240 (6.10) | -.448 (11.38) |
| H | -.480 (12.19) | -.208 (5.28) |
| J | -.480 (12.19) | +.208 (5.28) |
| K | -.240 (6.10) | +.448 (11.38) |

| Contacts (Insert arrangement 28-16) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | +.000 (0.00) | +.208 (5.28) |
| M | +.240 (6.10) | +.208 (5.28) |
| N | +.360 (9.14) | +.000 (0.00) |
| P | +.240 (6.10) | -.208 (5.28) |
| Q | +.000 (0.00) | -.208 (5.28) |
| R | -.240 (6.10) | -.208 (5.28) |
| S | -.360 (9.14) | +.000 (0.00) |
| T | -.240 (6.10) | +.208 (5.28) |
| V | -.120 (3.05) | +.000 (0.00) |
| U | +.120 (3.05) | +.000 (0.00) |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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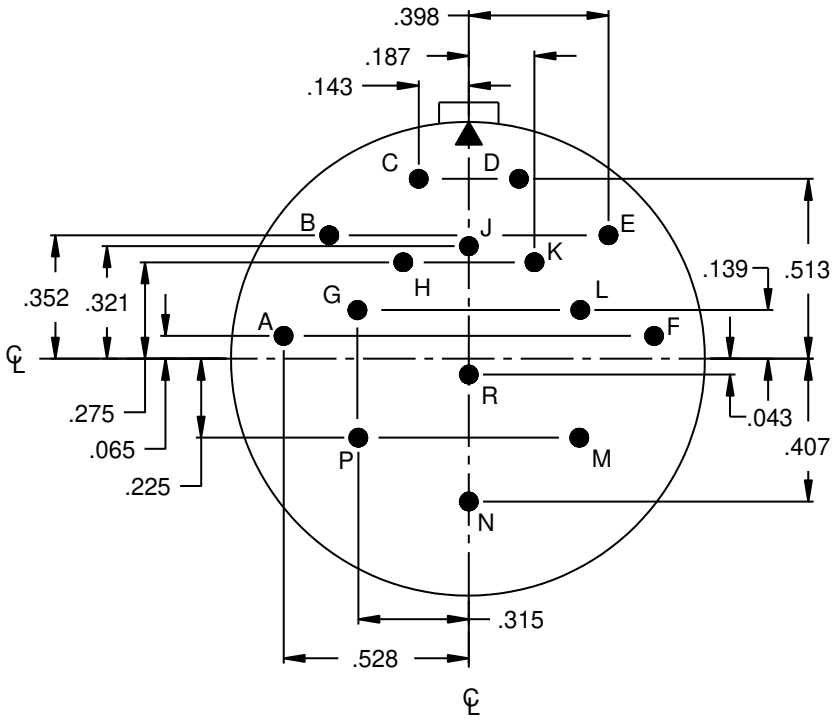
| Insert arrangement 28-16 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 28 | -16 | 20 | 16 | A | All | MS33688-16 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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28-17

| Contacts (Insert arrangement 28-17) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.528 (13.41) | +.065 (1.65) |
| B | -.398 (10.11) | +.352 (8.94) |
| C | -.143 (3.63) | +.513 (13.03) |
| D | +.143 (3.63) | +.513 (13.03) |
| E | +.398 (10.11) | +.352 (8.94) |
| F | +.528 (13.41) | +.065 (1.65) |
| G | -.315 (8.00) | +.139 (3.53) |
| H | -.187 (4.75) | +.275 (6.99) |

| Contacts (Insert arrangement 28-17) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| J | +.000 (0.00) | +.321 (8.15) |
| K | +.187 (4.75) | +.275 (6.99) |
| L | +.315 (8.00) | +.139 (3.53) |
| M | +.315 (8.00) | -.225 (5.72) |
| N | +.000 (0.00) | -.407 (10.34) |
| P | -.315 (8.00) | -.225 (5.72) |
| R | +.000 (0.00) | -.043 (1.09) |
| --- | --- | --- |

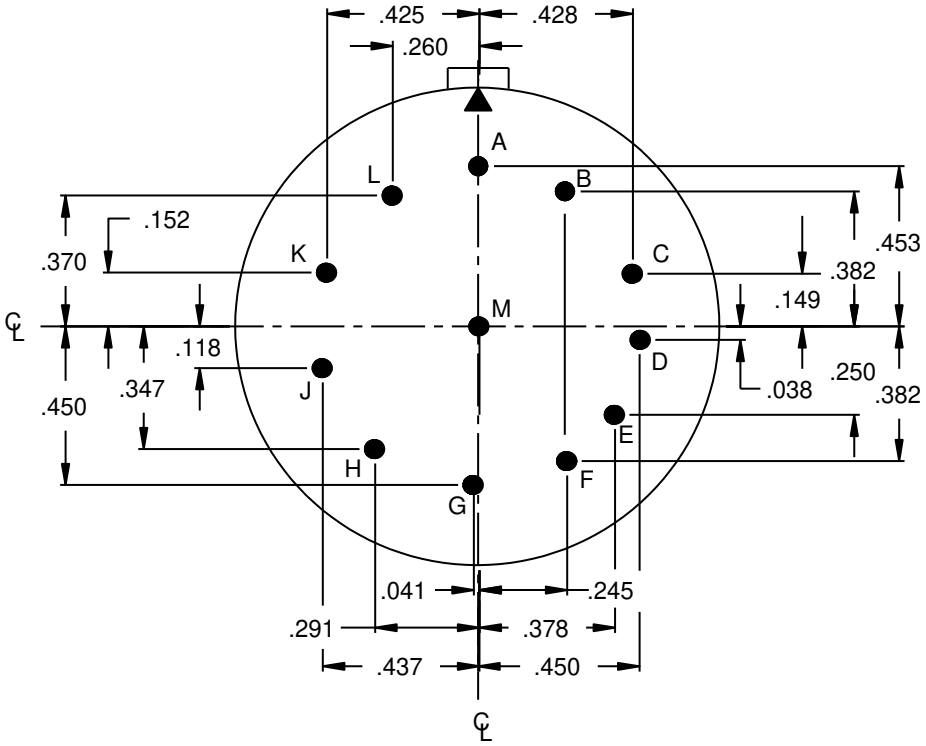
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -17 | 1 | 16 | B | R | MS33688-17 |
| | | 3 | | D | M,N,P | |
| | | 11 | | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-18

| Contacts (Insert arrangement 28-18) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.453 (11.51) |
| B | +.245 (6.22) | +.382 (9.70) |
| C | +.428 (10.87) | +.149 (3.78) |
| D | +.450 (11.43) | -.038 (0.00) |
| E | +.378 (9.60) | -.250 (6.35) |
| F | +.245 (6.22) | -.382 (9.70) |

| Contacts (Insert arrangement 28-18) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| G | -.041 (1.04) | -.450 (11.43) |
| H | -.291 (7.39) | -.347 (8.81) |
| J | -.437 (11.10) | -.118 (3.00) |
| K | -.425 (10.80) | +.152 (3.86) |
| L | -.260 (0.00) | +.370 (9.40) |
| M | +.000 (0.00) | +.000 (0.00) |

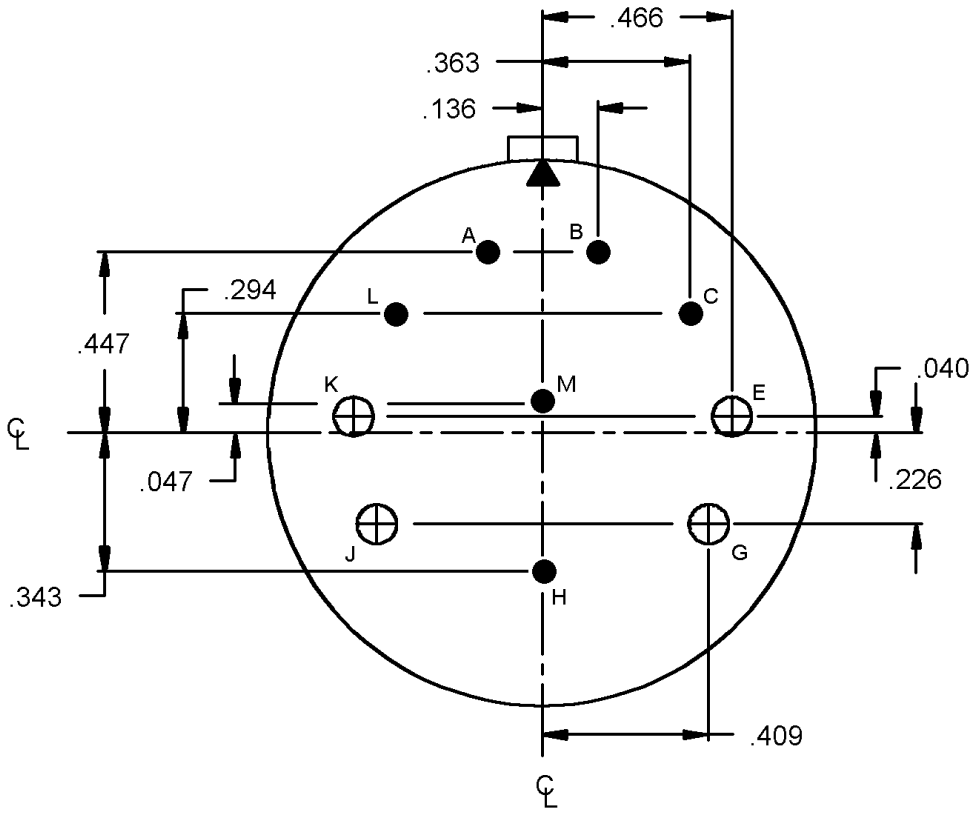
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -18 | 1 | 16 | C | M | MS33688-18 |
| | | 2 | | A | A,B | |
| | | 4 | | INST | C,D,E,F | |
| | | 5 | | D | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-19

| Contacts (Insert arrangement 28-19) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.136 (3.45) | +.447 (11.35) |
| B | +.136 (3.45) | +.447 (11.35) |
| C | +.363 (9.22) | +.294 (7.47) |
| E | +.466 (11.84) | +.040 (1.02) |
| G | +.409 (10.39) | -.226 (5.74) |

| Contacts (Insert arrangement 28-19) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| H | +.000 (0.00) | -.343 (8.71) |
| J | -.409 (10.39) | -.226 (5.74) |
| K | -.466 (11.84) | +.040 (1.02) |
| L | -.363 (9.22) | +.294 (7.47) |
| M | +.000 (0.00) | +.047 (1.19) |

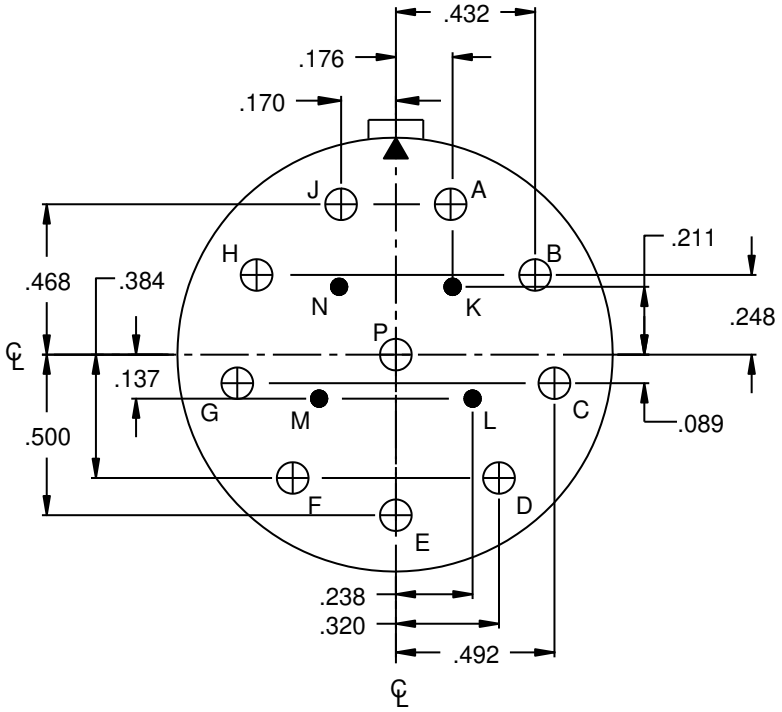
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -19 | 2 | 16 | B | H,M | MS33688-19 |
| | | 2 | 16 | D | A,B | |
| | | 2 | 16 | A | C.L | |
| | | 4 | 12 | | E,G,J,K | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-20

| Contacts (Insert arrangement 28-20) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.170 (4.32) | +.468 (11.89) |
| B | +.432 (10.97) | +.248 (6.30) |
| C | +.492 (12.50) | -.089 (2.26) |
| D | +.320 (8.13) | -.384 (9.75) |
| E | +.000 (0.00) | -.500 (12.70) |
| F | -.320 (8.13) | -.384 (9.75) |
| G | -.492 (12.50) | -.089 (2.26) |

| Contacts (Insert arrangement 28-20) | | |
|---|---|---|
| Contact position ID | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| H | -.432 (10.97) | +.248 (6.30) |
| J | -.170 (4.32) | +.468 (11.89) |
| K | +.176 (4.47) | +.211 (5.36) |
| L | +.238 (6.05) | -.137 (3.48) |
| M | -.238 (6.05) | -.137 (3.48) |
| N | -.176 (4.47) | +.211 (5.36) |
| P | +.000 (0.00) | +.000 (0.00) |

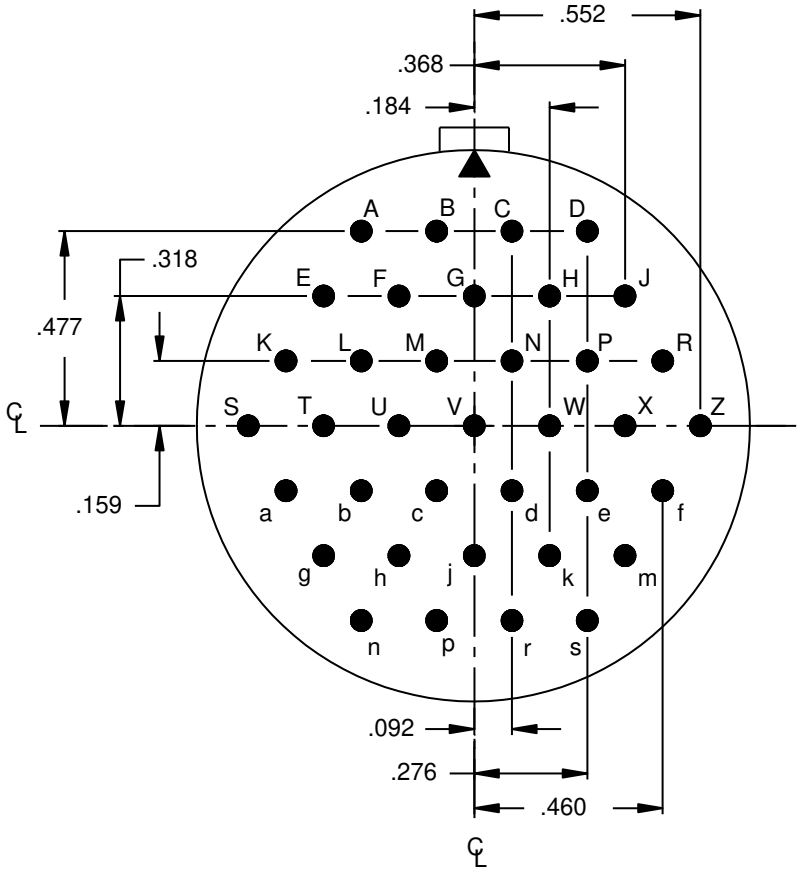
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -20 | 4 | 16 | A | K,L,M,N | MS33688-20 |
| | | 10 | 12 | | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-21

| Contacts (Insert arrangement 28-21) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.276 (7.01) | +.477 (12.12) |
| B | -.092 (2.34) | +.477 (12.12) |
| C | +.092 (2.34) | +.477 (12.12) |
| D | +.276 (7.01) | +.477 (12.12) |
| E | -.368 (9.35) | +.318 (8.08) |
| F | -.184 (4.67) | +.318 (8.08) |
| G | +.000 (0.00) | +.318 (8.08) |
| H | +.184 (4.67) | +.318 (8.08) |
| J | +.368 (9.35) | +.318 (8.08) |
| K | -.460 (11.68) | +.159 (4.04) |
| L | -.276 (7.01) | +.159 (4.04) |
| M | -.092 (2.34) | +.159 (4.04) |
| N | +.092 (2.34) | +.159 (4.04) |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 28-21) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | +.276 (7.01) | +.159 (4.04) |
| R | +.460 (11.68) | +.159 (4.04) |
| S | -.552 (14.02) | +.000 (0.00) |
| T | -.368 (9.35) | +.000 (0.00) |
| U | -.184 (4.67) | +.000 (0.00) |
| V | +.000 (0.00) | +.000 (0.00) |
| W | +.184 (4.67) | +.000 (0.00) |
| X | +.368 (9.35) | +.000 (0.00) |
| Z | +.552 (14.02) | +.000 (0.00) |
| a | -.460 (11.68) | -.159 (4.04) |
| b | -.276 (7.01) | -.159 (4.04) |
| c | -.092 (0.00) | -.159 (4.04) |
| d | +.092 (0.00) | -.159 (4.04) |
| e | +.276 (7.01) | -.159 (4.04) |
| f | +.460 (11.68) | -.159 (4.04) |
| g | -.368 (9.35) | -.318 (8.08) |
| h | -.184 (4.67) | -.318 (8.08) |
| j | +.000 (0.00) | -.318 (8.08) |
| k | +.184 (4.67) | -.318 (8.08) |
| m | +.368 (9.35) | -.318 (8.08) |
| n | -.276 (7.01) | -.477 (12.12) |
| p | -.092 (0.00) | -.477 (12.12) |
| r | +.092 (0.00) | -.477 (12.12) |
| s | +.276 (7.01) | -.477 (12.12) |

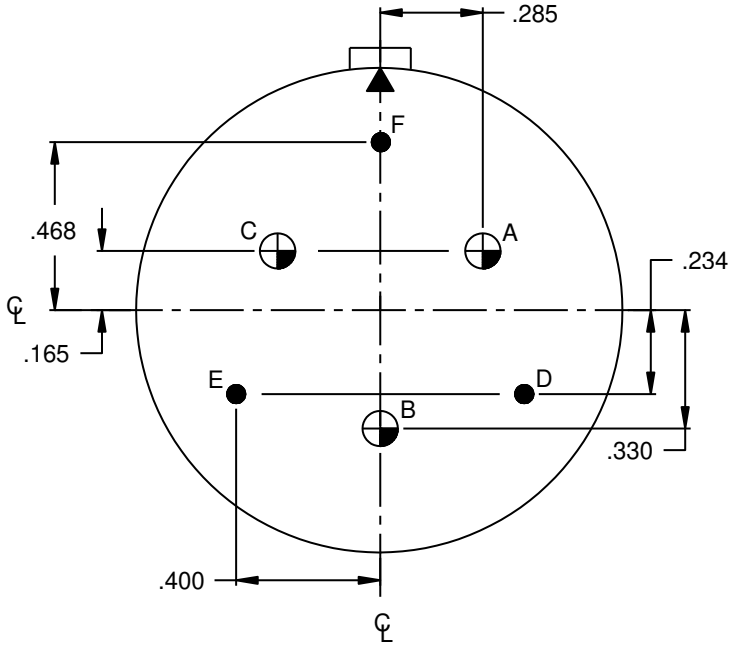
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -21 | 37 | 16 | A | All | MS33688-21 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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28-22

| Contacts (Insert arrangement 28-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.285 (7.24) | +.165 (4.19) |
| B | +.000 (0.00) | -.330 (8.38) |
| C | -.285 (7.24) | +.165 (4.19) |
| D | +.400 (10.16) | -.234 (5.94) |
| E | -.400 (10.16) | -.234 (5.94) |
| F | +.000 (0.00) | +.468 (0.00) |

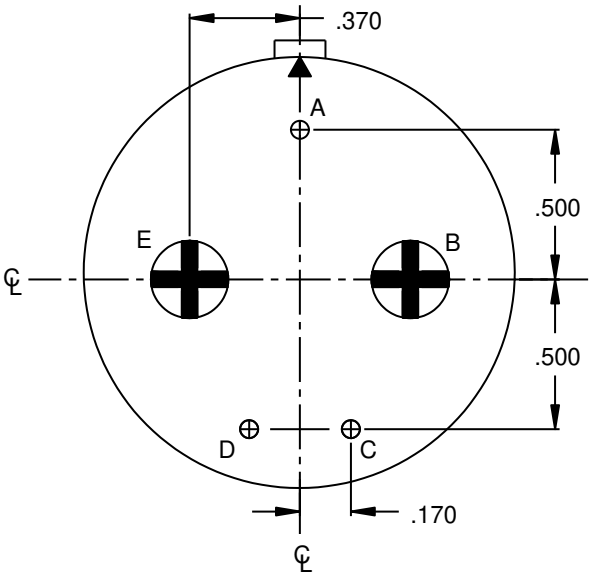
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 28 | -22 | 3 | 16 | D | E,F,D | MS33688-22 |
| | | 3 | 4 | | A,B,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 10. Insert arrangement, electrical connector, shell size 28, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, SAE-AS50151 and SAE-AS95234 – Continued.



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32-1

| Contacts (Insert arrangement 32-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.500 (12.70) |
| B | +.370 (9.40) | +.000 (0.00) |
| C | +.170 (4.32) | -.500 (12.70) |
| D | -.170 (4.32) | -.500 (12.70) |
| E | -.370 (9.40) | +.000 (0.00) |

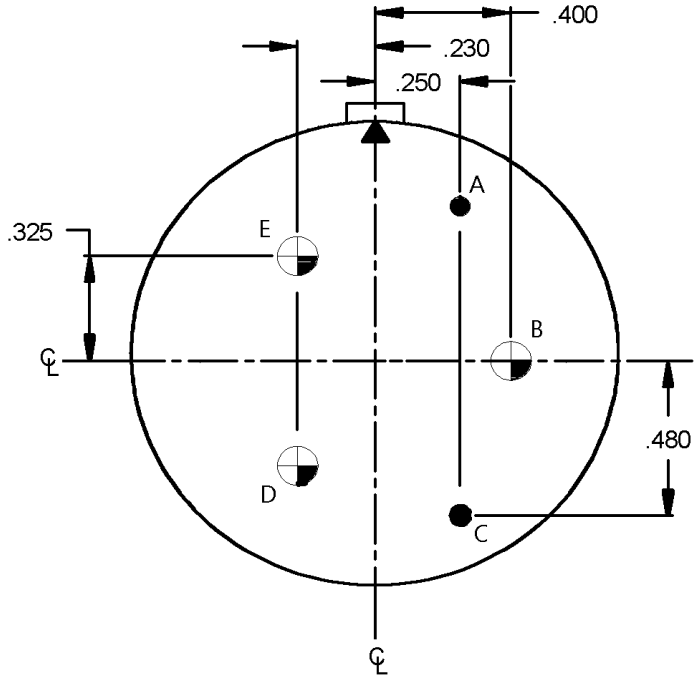
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -1 | 1 | 12 | E | A | MS33689-1 |
| | | 2 | 12 | D | C,D | |
| | | 2 | 0 | D | B,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234.



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32-2

| Contacts (Insert arrangement 32-2) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.250 (6.35) | +.480 (12.19) |
| B | +.400 (10.16) | +.000 (0.00) |
| C | +.250 (6.35) | -.480 (12.19) |
| D | -.230 (5.84) | -.325 (8.26) |
| E | -.230 (5.84) | +.325 (8.26) |

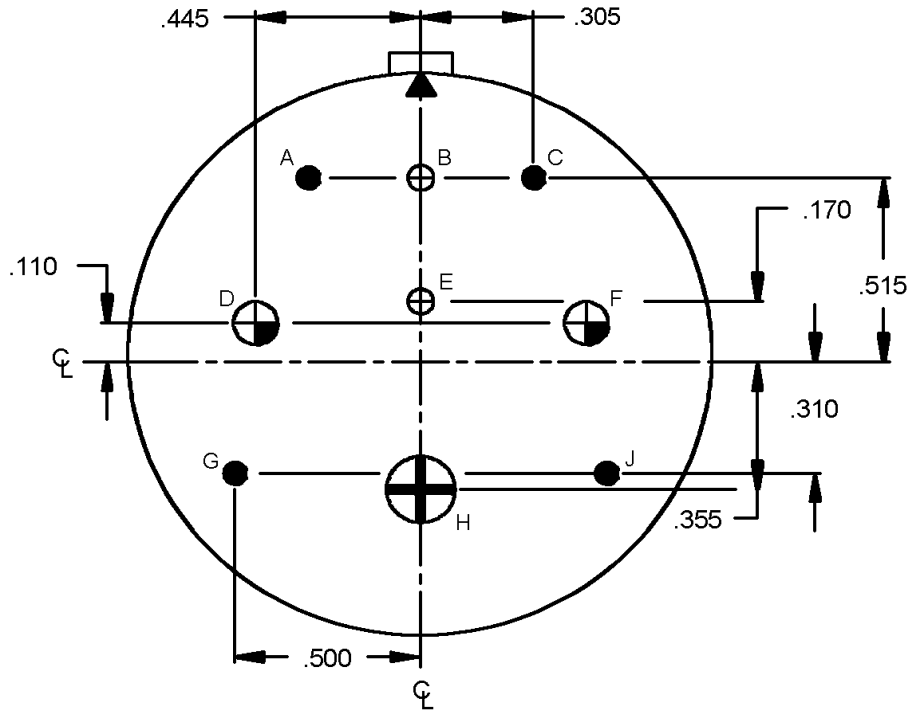
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -2 | 2 | 16 | E | A,C | MS33689-2 |
| | | 3 | 4 | | B,D,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-3

| Contacts (Insert arrangement 32-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.305 (7.75) | +.515 (13.08) |
| B | +.000 (0.00) | +.515 (13.08) |
| C | +.305 (7.75) | +.515 (13.08) |
| D | -.445 (11.30) | +.110 (2.79) |
| E | +.000 (0.00) | +.170 (4.32) |
| F | +.445 (11.30) | +.110 (2.79) |
| G | -.500 (12.70) | -.310 (7.87) |
| H | +.000 (0.00) | -.355 (9.02) |
| J | +.500 (12.70) | -.310 (7.87) |

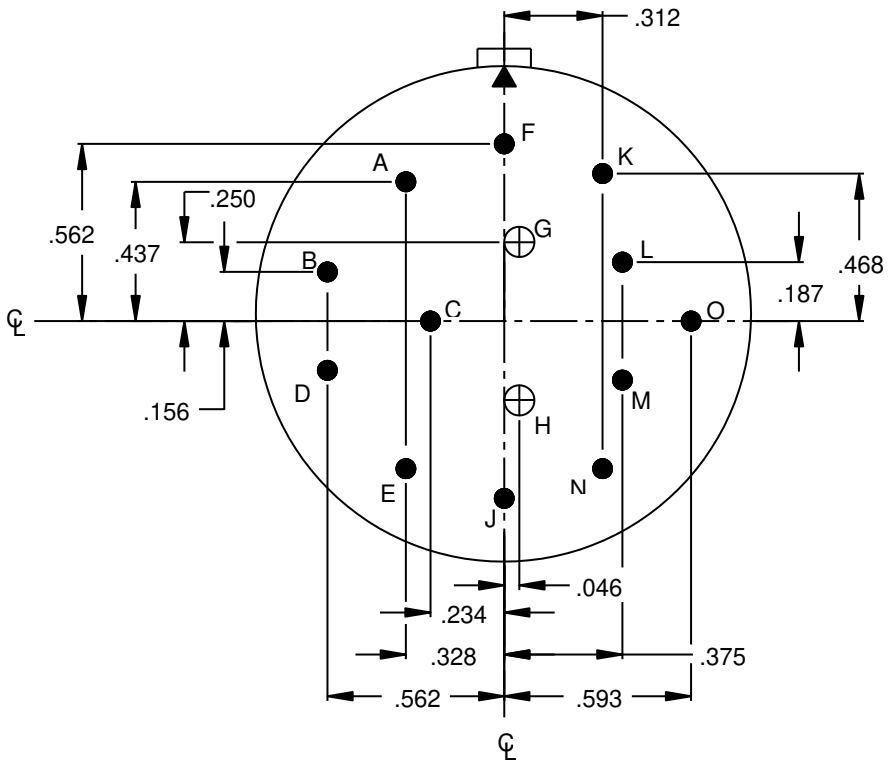
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -3 | 4 | 16 | D | A,C,J,G | MS33689-3 |
| | | 2 | 12 | | B,E | |
| | | 2 | 4 | | D,F | |
| | | 1 | 0 | | H | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-4

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-4) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.328 (8.33) | +.437 (11.10) |
| B | -.562 (14.27) | +.156 (3.96) |
| C | -.234 (5.94) | +.000 (0.00) |
| D | -.562 (14.27) | -.156 (3.96) |
| E | -.328 (8.33) | -.437 (11.10) |
| F | +.000 (0.00) | +.562 (14.27) |
| G | +.046 (1.17) | +.250 (6.35) |

| Contacts (Insert arrangement 32-4) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| H | +.046 (1.17) | -.250 (6.35) |
| J | +.000 (0.00) | -.562 (14.27) |
| K | +.312 (7.92) | +.468 (11.89) |
| L | +.375 (9.53) | +.187 (4.75) |
| M | +.375 (9.53) | -.187 (4.75) |
| N | +.312 (7.92) | -.468 (11.89) |
| O | +.593 (15.06) | +.000 (0.00) |

| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -4 | 2 | 12 | D | G,H | MS33689-4 |
| | | 4 | 16 | A | F,J,K,N | |
| | | 8 | 16 | D | All others | |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



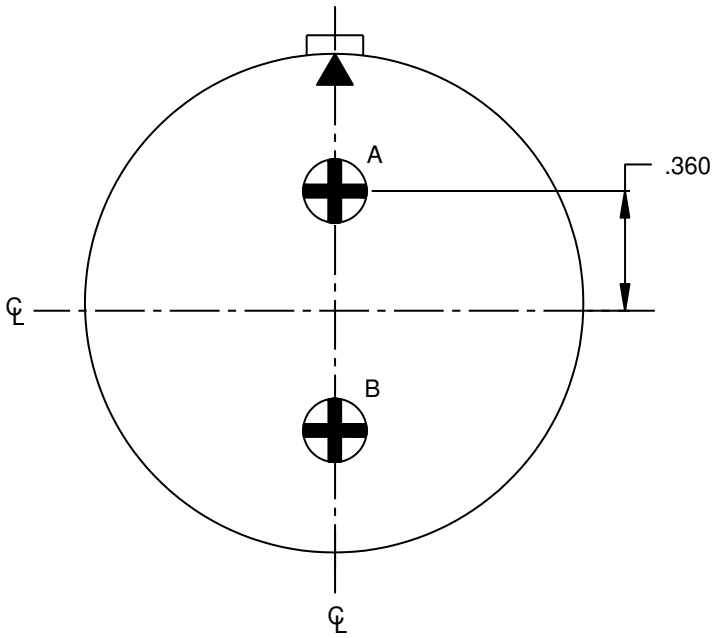
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| Insert arrangement 32-4 | | | |
|---|---|---|---|
| Alternate insert positions | | | |
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-5

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.360 (9.14) |
| B | +.000 (0.00) | -.360 (9.14) |

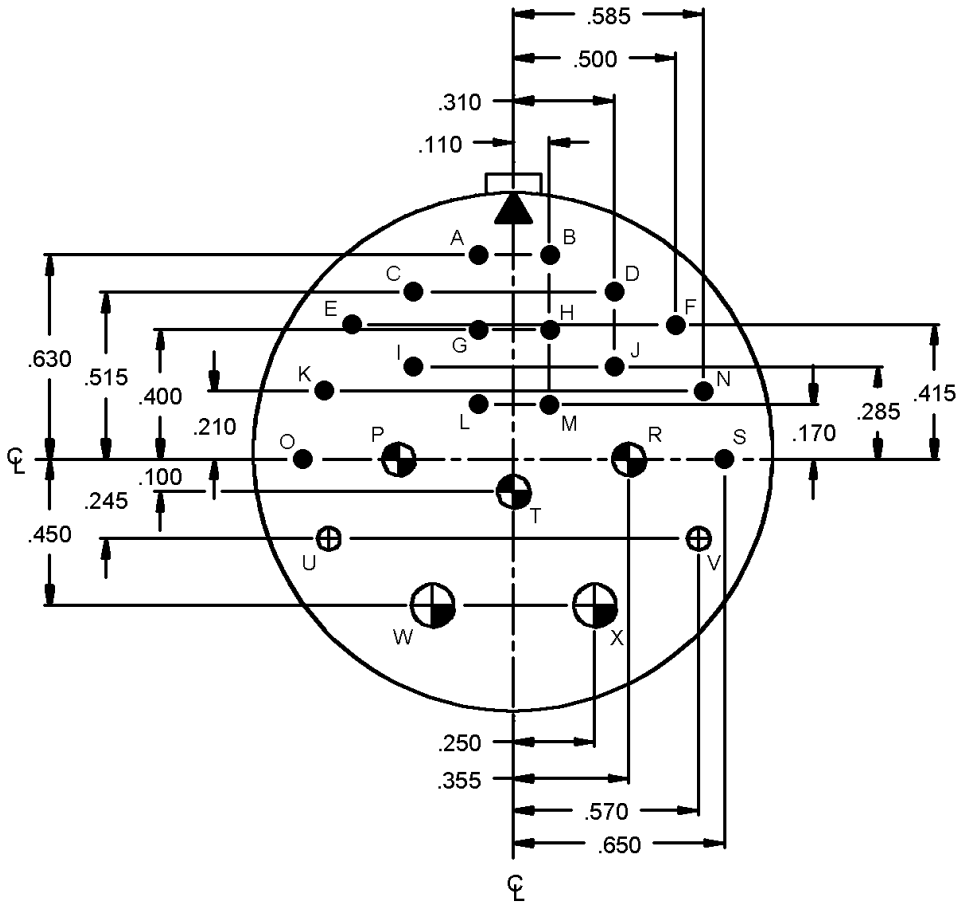
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -5 | 2 | 0 | E | All | MS33689-5 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 325 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-6

| Contacts (Insert arrangement 32-6) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.110 (2.79) | +.630 (16.00) |
| B | +.110 (2.79) | +.630 (16.00) |
| C | -.310 (7.87) | +.515 (13.08) |
| D | +.310 (7.87) | +.515 (13.08) |
| E | -.500 (12.70) | +.415 (10.54) |
| F | +.500 (12.70) | +.415 (10.54) |
| G | -.110 (2.79) | +.400 (10.16) |
| H | +.110 (2.79) | +.400 (10.16) |
| I | -.310 (7.87) | +.285 (7.24) |
| J | +.310 (7.87) | +.285 (7.24) |
| K | -.585 (14.86) | +.210 (5.33) |
| L | -.110 (2.79) | +.170 (4.32) |

| Contacts (Insert arrangement 32-6) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| M | +.110 (2.79) | +.170 (4.32) |
| N | +.585 (14.86) | +.210 (5.33) |
| O | -.650 (16.51) | +.000 (0.00) |
| P | -.355 (9.02) | +.000 (0.00) |
| R | +.355 (9.02) | +.000 (0.00) |
| S | +.650 (16.51) | +.000 (0.00) |
| T | +.000 (0.00) | -.100 (2.54) |
| U | -.570 (14.48) | -.245 (6.22) |
| V | +.570 (14.48) | -.245 (6.22) |
| W | -.250 (6.35) | -.450 (11.43) |
| X | +.250 (6.35) | -.450 (11.43) |
| --- | --- | --- |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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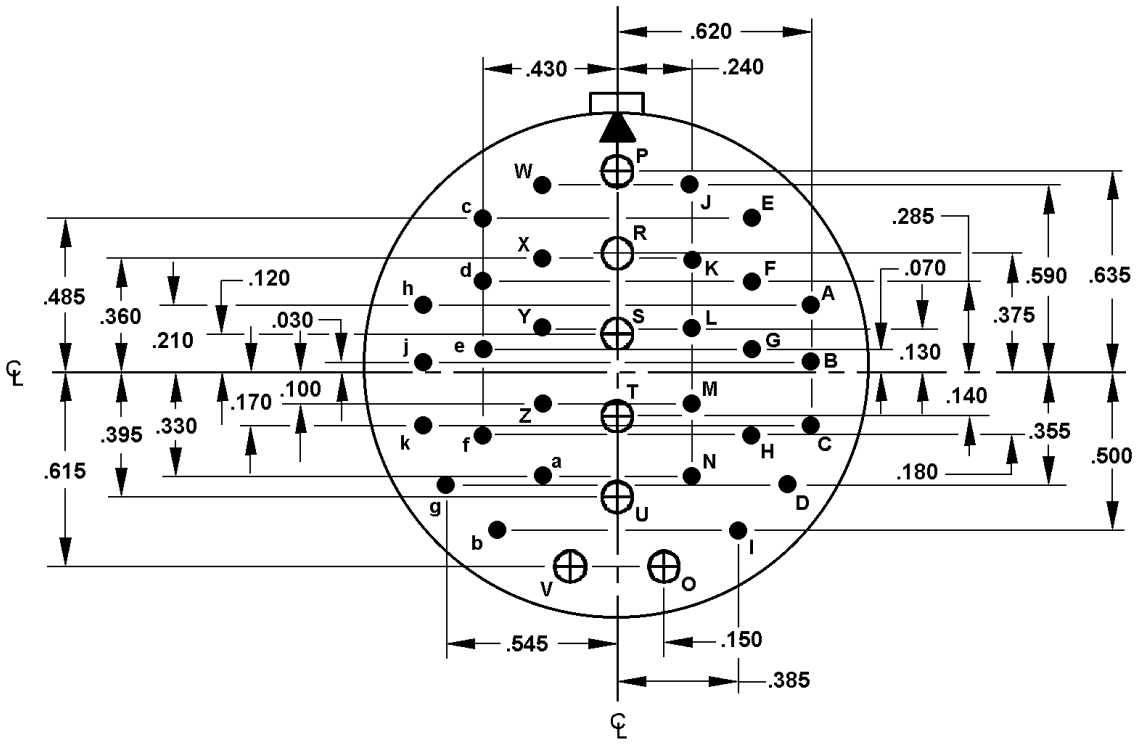
| Insert arrangement 32-6 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 32 | -6 | 16 | 16 | A | All others | MS33689-6 |
| | | 2 | 12 | | U,V | |
| | | 3 | 8 | | P,R,T | |
| | | 2 | 4 | | W,X | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-7

| Contacts (Insert arrangement 32-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.620 (15.75) | +.210 (5.33) |
| B | +.620 (15.75) | +.030 (0.76) |
| C | +.620 (15.75) | -.170 (4.32) |
| D | +.545 (13.84) | -.355 (9.02) |
| E | +.430 (10.92) | +.485 (12.32) |
| F | +.430 (10.92) | +.285 (7.24) |
| G | +.430 (10.92) | +.070 (1.78) |
| H | +.430 (10.92) | -.180 (4.57) |
| I | +.385 (9.78) | -.500 (12.70) |
| J | +.240 (6.10) | +.590 (14.99) |
| K | +.240 (6.10) | +.360 (9.14) |
| L | +.240 (6.10) | +.130 (3.30) |
| M | +.240 (6.10) | -.100 (2.54) |
| N | +.240 (6.10) | -.330 (8.38) |
| O | +.150 (3.81) | -.615 (15.62) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert Arrangement 32-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | +.000 (0.00) | +.635 (16.13) |
| R | +.000 (0.00) | +.375 (9.35) |
| S | +.000 (0.00) | +.120 (3.05) |
| T | +.000 (0.00) | -.140 (3.56) |
| U | +.000 (0.00) | -.395 (10.03) |
| V | -.150 (3.81) | -.615 (15.62) |
| W | -.240 (6.10) | +.590 (14.99) |
| X | -.240 (6.10) | +.360 (9.14) |
| Y | -.240 (6.10) | +.130 (3.30) |
| Z | -.240 (6.10) | -.100 (2.54) |
| a | -.240 (6.10) | -.330 (8.38) |
| b | -.385 (9.78) | -.500 (12.70) |
| c | -.430 (10.92) | +.485 (12.32) |
| d | -.430 (10.92) | +.285 (7.24) |
| e | -.430 (10.92) | +.070 (0.76) |
| f | -.430 (10.92) | -.180 (4.57) |
| g | -.545 (13.84) | -.355 (9.02) |
| h | -.620 (15.75) | +.210 (5.33) |
| j | -.620 (15.75) | +.030 (0.76) |
| k | -.620 (15.75) | -.170 (4.32) |

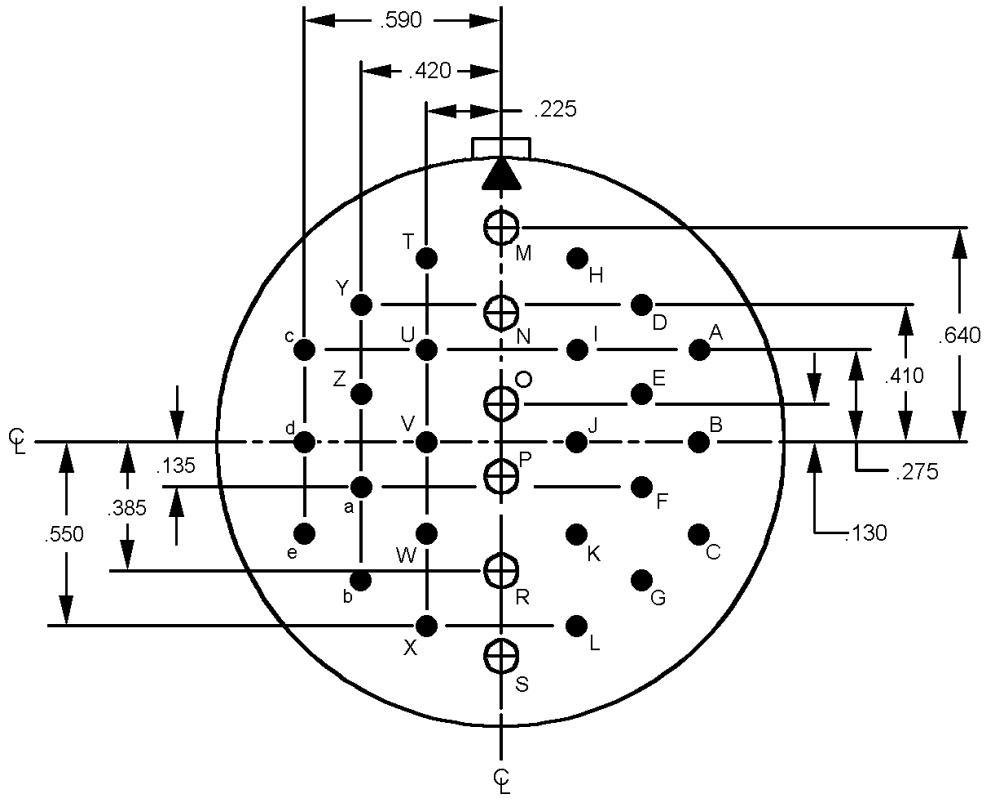
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -7 | 4 | 16 | INST | A,B,h,j | MS33689-7 |
| | | 24 | 16 | | All others | |
| | | 7 | 12 | A | O,P,S,R,T,U,V | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 125 | 235 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-8

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.590 (14.99) | +.275 (6.99) |
| B | +.590 (14.99) | +.000 (0.00) |
| C | +.590 (14.99) | -.275 (6.99) |
| D | +.420 (10.67) | +.410 (10.41) |
| E | +.420 (10.67) | +.135 (3.43) |
| F | +.420 (10.67) | -.135 (3.43) |
| G | +.420 (10.67) | -.410 (10.41) |
| H | +.225 (5.72) | +.550 (13.97) |
| I | +.225 (5.72) | +.275 (6.99) |
| J | +.225 (5.72) | +.000 (0.00) |
| K | +.225 (5.72) | -.275 (6.99) |
| L | +.225 (5.72) | -.550 (13.97) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 32-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | X-axis (mm) |
| M | +.000 (0.00) | +.640 (16.26) |
| N | +.000 (0.00) | +.385 (9.78) |
| O | +.000 (0.00) | +.130 (3.30) |
| P | +.000 (0.00) | -.130 (3.30) |
| R | +.000 (0.00) | -.385 (9.78) |
| S | +.000 (0.00) | -.640 (16.26) |
| T | -.225 (5.72) | +.550 (13.97) |
| U | -.225 (5.72) | +.275 (6.99) |
| V | -.225 (5.72) | +.000 (0.00) |
| W | -.225 (5.72) | -.275 (6.99) |
| X | -.225 (5.72) | -.550 (13.97) |
| Y | -.420 (10.67) | +.410 (10.41) |
| Z | -.420 (10.67) | +.135 (3.43) |
| a | -.420 (10.67) | -.135 (3.43) |
| b | -.420 (10.67) | -.410 (10.41) |
| c | -.590 (14.99) | +.275 (6.99) |
| d | -.590 (14.99) | +.000 (0.00) |
| e | -.590 (14.99) | -.275 (6.99) |

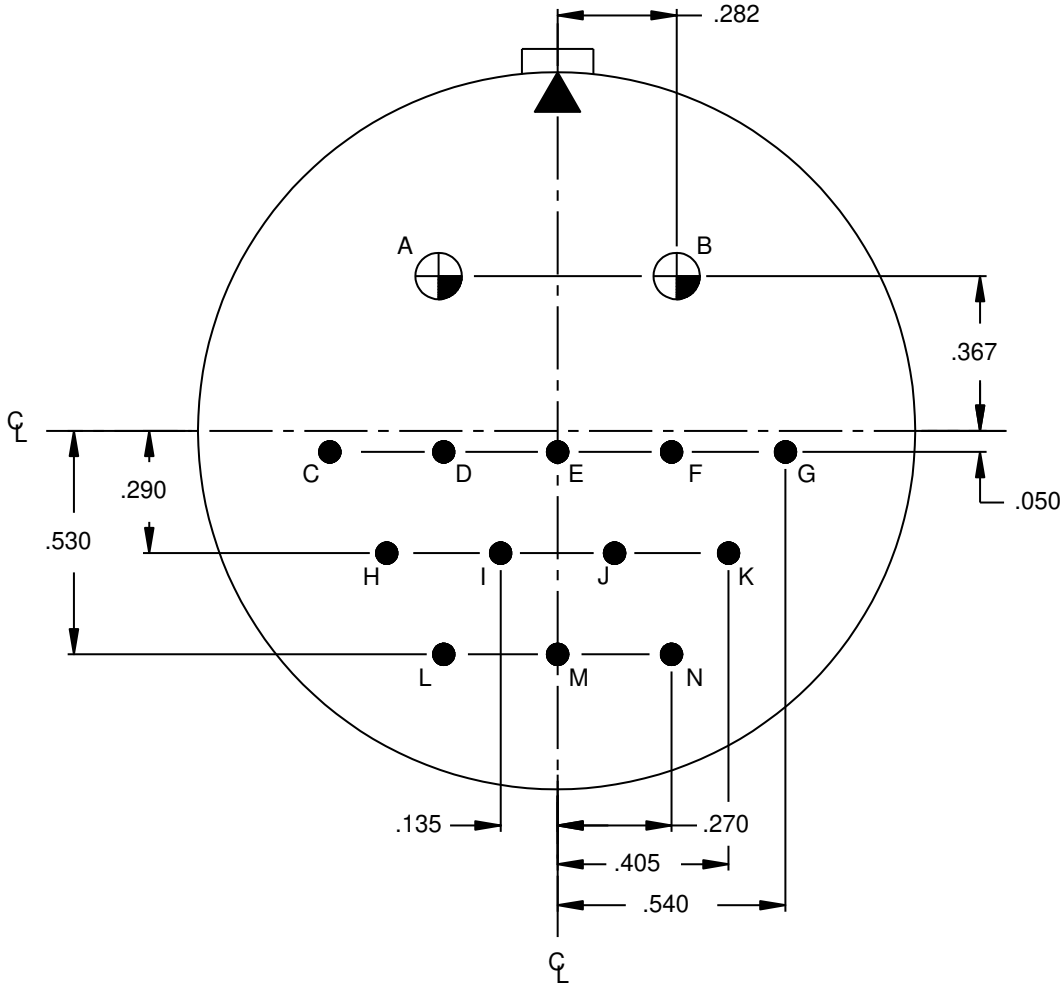
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -8 | 24 | 16 | A | All others | MS33689-8 |
| | | 6 | 12 | | M,N,O,P,R,S | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 125 | 235 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-9

| Contacts (Insert arrangement 32-9) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.282 (7.16) | +.367 (9.32) |
| B | +.282 (7.16) | +.367 (9.32) |
| C | -.540 (13.72) | -.050 (1.27) |
| D | -.270 (6.86) | -.050 (1.27) |
| E | +.000 (0.00) | -.050 (1.27) |
| F | +.270 (6.86) | -.050 (1.27) |
| G | +.540 (13.72) | -.050 (1.27) |

| Contacts (Insert arrangement 32-9) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| H | -.405 (10.29) | -.290 (7.37) |
| I | -.135 (3.43) | -.290 (7.37) |
| J | +.135 (3.43) | -.290 (7.37) |
| K | +.405 (10.29) | -.290 (7.37) |
| L | -.270 (6.86) | -.530 (13.46) |
| M | +.000 (0.00) | -.530 (13.46) |
| N | +.270 (6.86) | -.530 (13.46) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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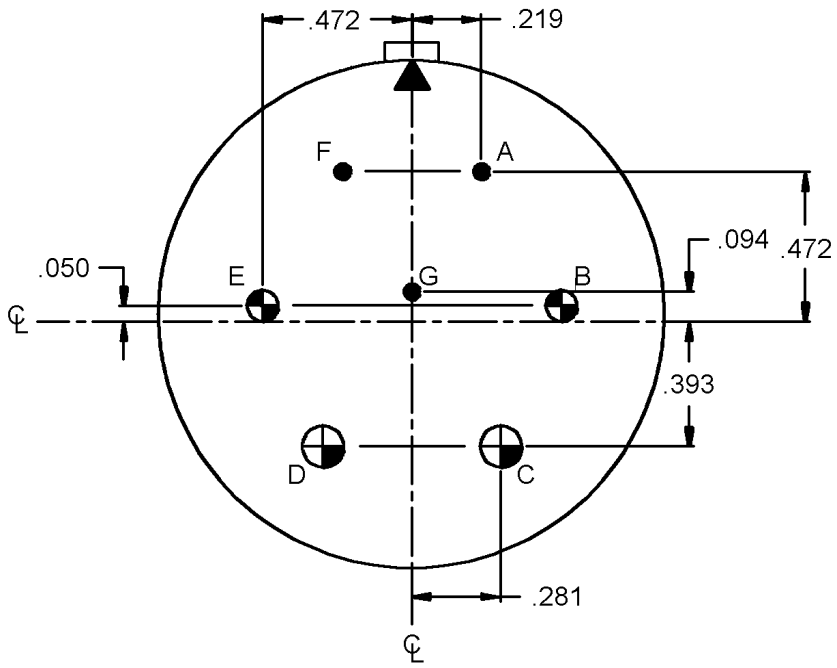
| Insert arrangement 32-9 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 32 | -9 | 12 | 16 | D | All others | MS33689-9 |
| | | 2 | 4 | | A,B | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-10

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.219 (5.56) | +.472 (11.99) |
| B | +.472 (11.99) | +.050 (1.27) |
| C | +.281 (7.14) | -.393 (9.98) |
| D | -.281 (7.14) | -.393 (9.98) |
| E | -.472 (11.99) | +.050 (1.27) |
| F | -.219 (5.56) | +.472 (11.99) |
| G | +.000 (0.00) | +.094 (2.39) |

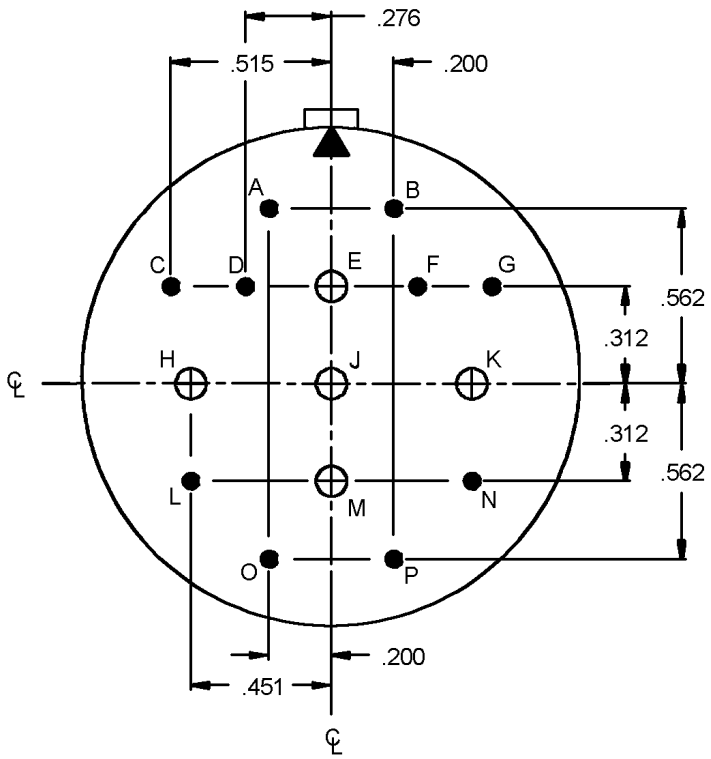
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -10 | 2 | 16 | E | A,F | MS33689-10 |
| | | 1 | 16 | B | G | |
| | | 2 | 8 | D | B,E | |
| | | 2 | 4 | A | C,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-12

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-12) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.200 (5.08) | +.562 (14.27) |
| B | +.200 (5.08) | +.562 (14.27) |
| C | -.515 (13.08) | +.312 (7.92) |
| D | -.276 (7.01) | +.312 (7.92) |
| E | +.000 (0.00) | +.312 (7.92) |
| F | +.276 (7.01) | +.312 (7.92) |
| G | +.515 (13.08) | +.312 (7.92) |
| H | -.451 (11.46) | +.000 (0.00) |
| J | +.000 (0.00) | +.000 (0.00) |
| K | +.451 (11.46) | +.000 (0.00) |
| L | -.451 (11.46) | -.312 (7.92) |
| M | +.000 (0.00) | -.312 (7.92) |
| N | +.451 (11.46) | -.312 (7.92) |
| O | -.200 (5.08) | -.562 (14.27) |
| P | +.200 (5.08) | -.562 (14.27) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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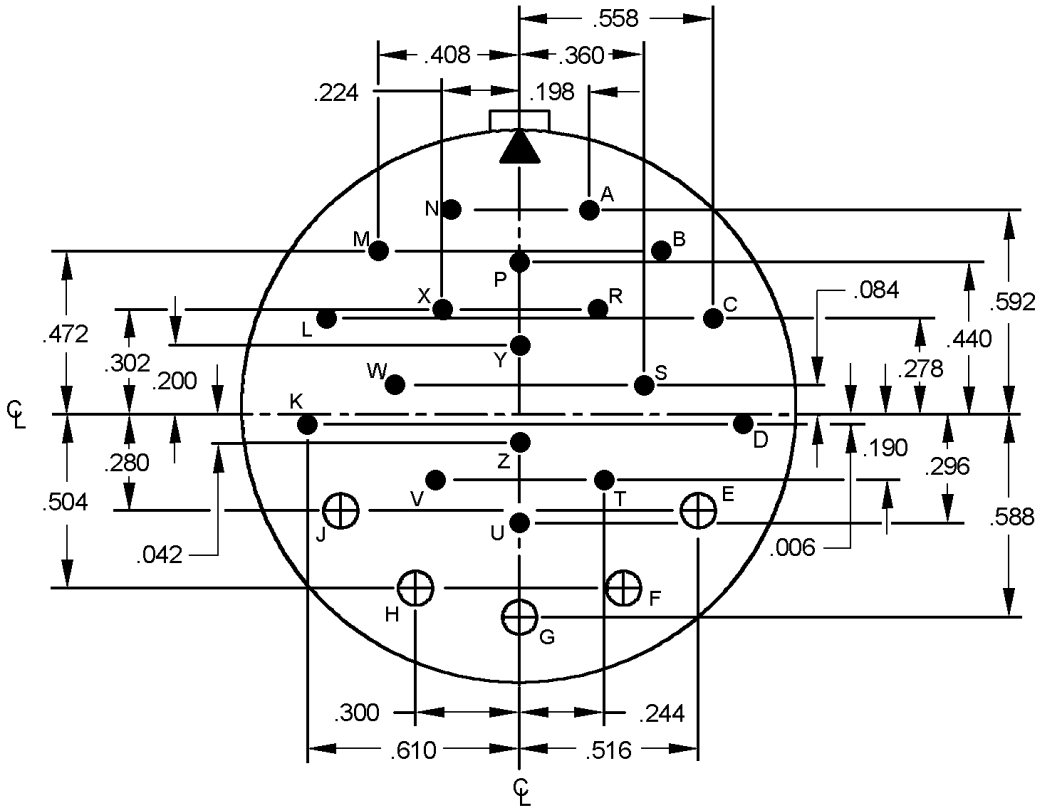
| Insert arrangement 32-12 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 32 | -12 | 1 | 12 | A | E | MS33689-12 |
| | | 4 | 12 | D | H,J,K,M | |
| | | 4 | 16 | A | C,D,F,G | |
| | | 6 | 16 | D | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 32-13) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.198 (5.03) | +.592 (15.04) |
| B | +.408 (10.36) | +.472 (11.99) |
| C | +.558 (14.17) | +.278 (7.06) |
| D | +.610 (15.49) | -.006 (0.15) |
| E | +.516 (13.11) | -.280 (7.11) |
| F | +.300 (7.62) | -.504 (12.80) |
| G | +.000 (0.00) | -.588 (14.94) |
| H | -.300 (7.62) | -.504 (12.80) |
| J | -.516 (13.11) | -.280 (7.11) |
| K | -.610 (15.49) | -.006 (0.15) |
| L | -.558 (14.17) | +.278 (7.06) |
| M | -.408 (10.36) | +.472 (11.99) |

| Contacts (Insert arrangement 32-13) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | -.198 (5.03) | +.592 (15.04) |
| P | +.000 (0.00) | +.440 (11.18) |
| R | +.224 (6.20) | +.302 (7.67) |
| S | +.360 (9.14) | +.084 (2.13) |
| T | +.244 (6.20) | -.190 (4.83) |
| U | +.000 (0.00) | -.296 (7.52) |
| V | -.244 (6.20) | -.190 (4.83) |
| W | -.360 (9.14) | +.084 (2.13) |
| X | -.224 (6.20) | +.302 (7.67) |
| Y | +.000 (0.00) | +.200 (5.08) |
| Z | +.000 (0.00) | -.042 (1.07) |
| --- | --- | --- |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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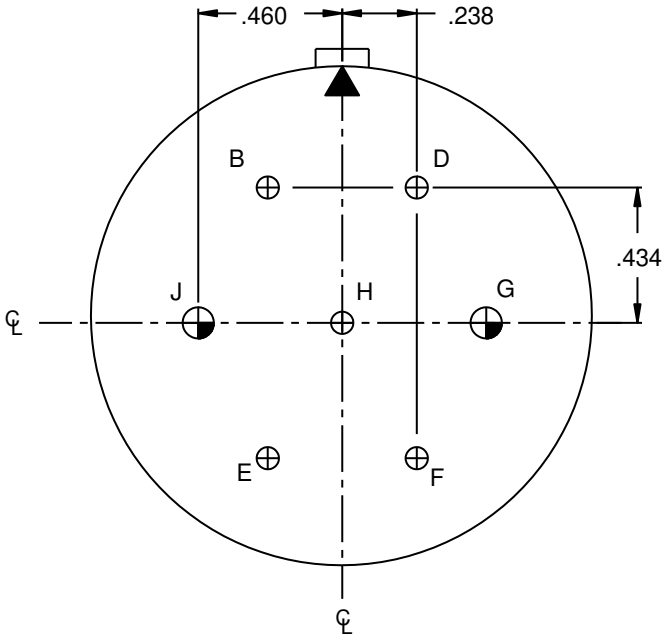
| Insert arrangement 32-13 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 32 | -13 | 18 | 16 | D | All others | MS33689-13 |
| | | 5 | 12 | | E,F,G,H,J | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-14

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-14) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| B | -.238 (6.05) | +.434 (11.02) |
| D | +.238 (6.05) | +.434 (11.02) |
| G | +.460 (11.68) | +.000 (0.00) |
| H | +.000 (0.00) | +.000 (0.00) |
| J | -.460 (11.68) | +.000 (0.00) |
| F | +.238 (6.05) | -.434 (11.02) |
| E | -.238 (6.05) | -.434 (11.02) |

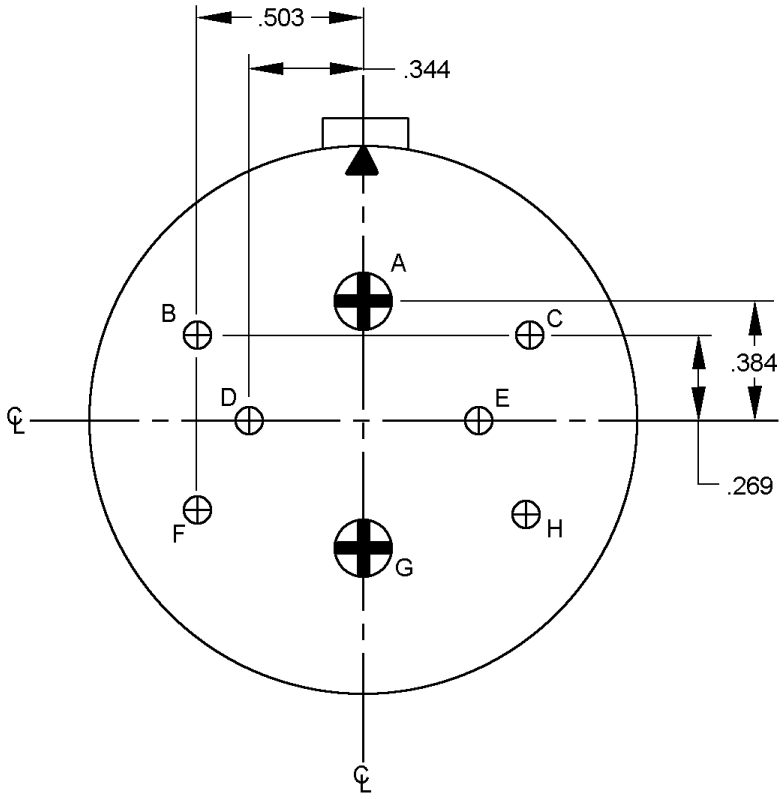
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -14 | 5 | 12 | D | B,D,H,F,Z | MS33689-14 |
| | | 2 | 4 | | G,J | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-15

| Contacts (Insert arrangement 32-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.384 (9.75) |
| B | -.503 (12.78) | +.269 (6.83) |
| C | +.503 (12.78) | +.269 (6.83) |
| D | -.344 (8.74) | +.000 (0.00) |
| E | +.344 (8.74) | +.000 (0.00) |
| F | -.503 (12.78) | -.269 (6.83) |
| G | +.000 (0.00) | -.384 (9.75) |
| H | +.503 (12.78) | -.269 (6.83) |

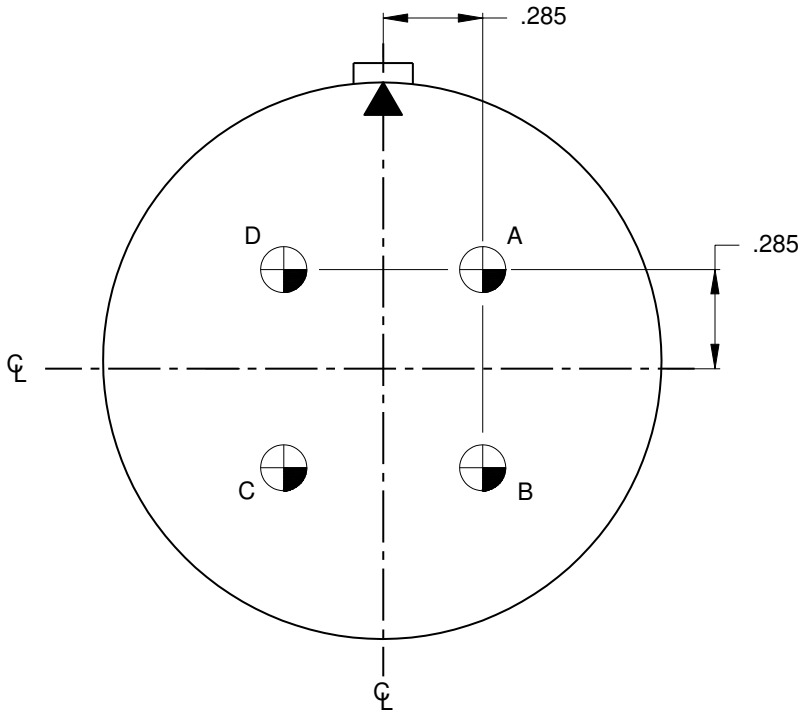
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -15 | 6 | 12 | D | B,C,D,E,F,H | MS33689-15 |
| | | 2 | 0 | | A,G | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-17

| Contacts (Insert arrangement 32-17) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.285 (7.24) | +.285 (7.24) |
| B | +.285 (7.24) | -.285 (7.24) |
| C | -.285 (7.24) | -.285 (7.24) |
| D | -.285 (7.24) | +.285 (7.24) |

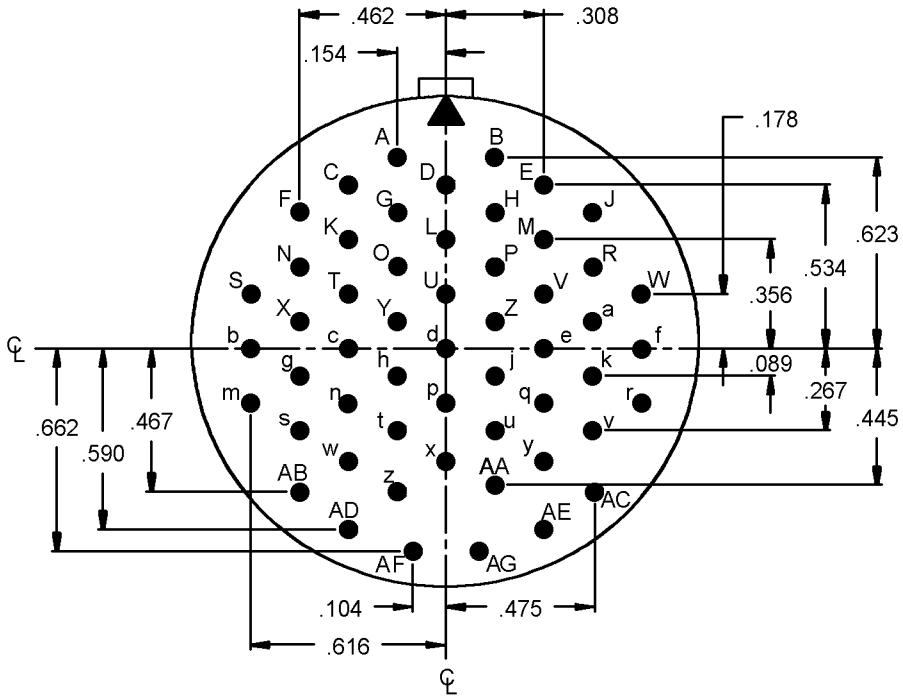
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -17 | 2 | 4 | D | All | MS33689-17 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 45 | 110 | 250 | --- |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-22

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 32-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.154 (3.91) | +.623 (15.82) |
| B | +.154 (3.91) | +.623 (15.82) |
| C | -.308 (7.82) | +.534 (13.56) |
| D | +.000 (0.00) | +.534 (13.56) |
| E | +.308 (7.82) | +.534 (13.56) |
| F | -.462 (11.73) | +.467 (11.86) |
| G | -.154 (3.91) | +.445 (11.30) |
| H | +.154 (3.91) | +.445 (11.30) |
| J | +.462 (11.73) | +.467 (11.86) |
| K | -.308 (7.82) | +.356 (9.04) |
| L | +.000 (0.00) | +.356 (9.04) |
| M | +.308 (7.82) | +.356 (9.04) |
| N | -.462 (11.73) | +.267 (6.78) |
| O | -.154 (3.91) | +.267 (6.78) |
| P | +.154 (3.91) | +.267 (6.78) |
| R | +.462 (11.73) | +.267 (6.78) |
| S | -.616 (15.65) | +.178 (4.52) |
| T | -.308 (7.82) | +.178 (4.52) |
| U | +.000 (0.00) | +.178 (4.52) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 32-22) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| V | +.308 (7.82) | +.178 (4.52) |
| W | +.616 (15.65) | +.178 (4.52) |
| X | -.462 (11.73) | +.089 (2.26) |
| Y | -.154 (3.91) | +.089 (2.26) |
| Z | +.154 (3.91) | +.089 (2.26) |
| a | +.462 (11.73) | +.089 (2.26) |
| b | -.616 (15.65) | +.000 (0.00) |
| c | -.308 (7.82) | +.000 (0.00) |
| d | +.000 (0.00) | +.000 (0.00) |
| e | +.308 (7.82) | +.000 (0.00) |
| f | +.616 (15.65) | +.000 (0.00) |
| g | -.462 (11.73) | -.089 (2.26) |
| h | -.154 (3.91) | -.089 (2.26) |
| j | +.154 (3.91) | -.089 (2.26) |
| k | +.462 (11.73) | -.089 (2.26) |
| m | -.616 (15.65) | -.178 (4.52) |
| n | -.308 (7.82) | -.178 (4.52) |
| p | +.000 (0.00) | -.178 (4.52) |
| q | +.308 (7.82) | -.178 (4.52) |
| r | +.616 (15.65) | -.178 (4.52) |
| s | -.462 (11.73) | -.267 (6.78) |
| t | -.154 (3.91) | -.267 (6.78) |
| u | +.154 (3.91) | -.267 (6.78) |
| v | +.462 (11.73) | -.267 (6.78) |
| w | -.308 (7.82) | -.356 (9.04) |
| x | +.000 (0.00) | -.356 (9.04) |
| y | +.308 (7.82) | -.356 (9.04) |
| z | -.154 (3.91) | -.445 (11.30) |
| AA | +.154 (3.91) | -.445 (11.30) |
| AB | -.475 (12.07) | -.467 (11.86) |
| AC | +.475 (12.07) | -.467 (11.86) |
| AD | -.308 (7.82) | -.590 (14.99) |
| AE | +.308 (7.82) | -.590 (14.99) |
| AF | +.104 (2.64) | -.662 (16.81) |
| AG | -.104 (2.64) | -.662 (16.91) |

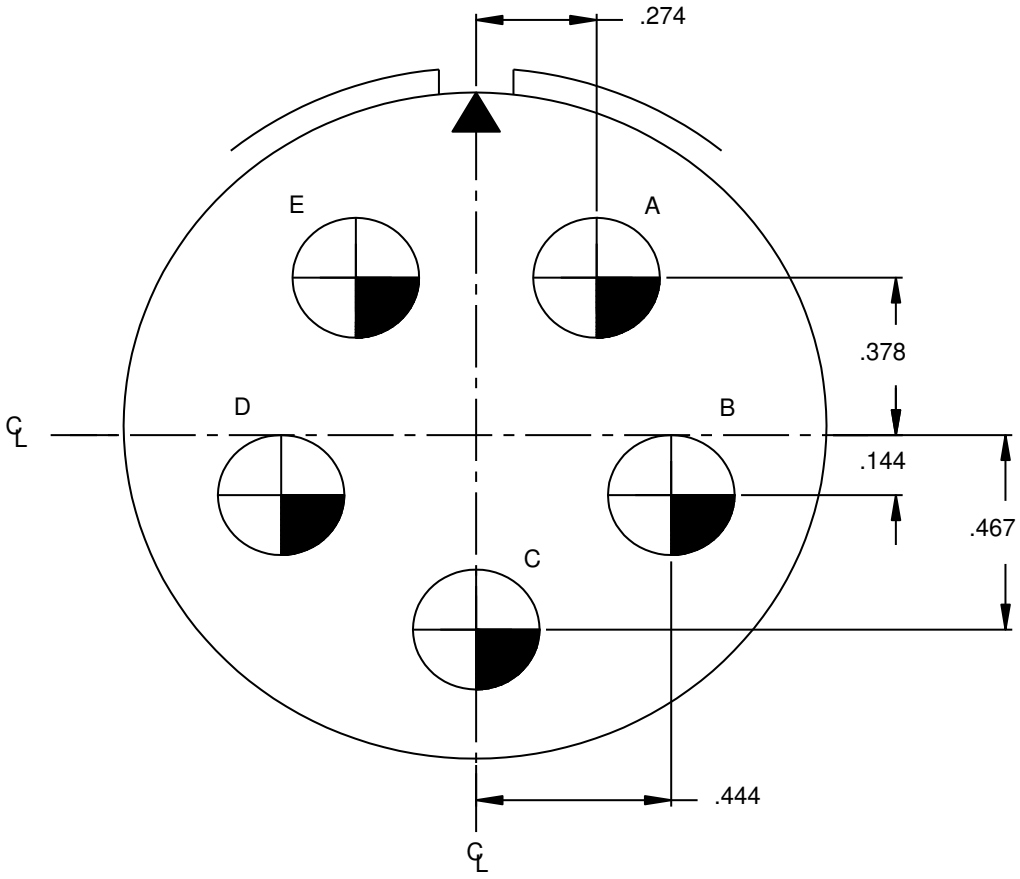
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -22 | 54 | 16 | A | All | MS33689-22 |

| Alternate insert positions | | | | | |
|---|---|---|---|---|---|
| A degrees (A°) | | | | | |
| U | V | W | X | Y | Z |
| 100 | 260 | 80 | 110 | 250 | 280 |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-63

(Not applicable to MIL-DTL-22992 or MIL-DTL-83723.)

| Contacts (Insert arrangement 32-63) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.274 (6.96) | +.378 (14.94) |
| B | +.444 (11.28) | -.144 (3.66) |
| C | +.000 (0.00) | -.467 (11.86) |
| D | -.444 (11.28) | -.144 (3.66) |
| E | -.274 (6.96) | +.378 (14.94) |

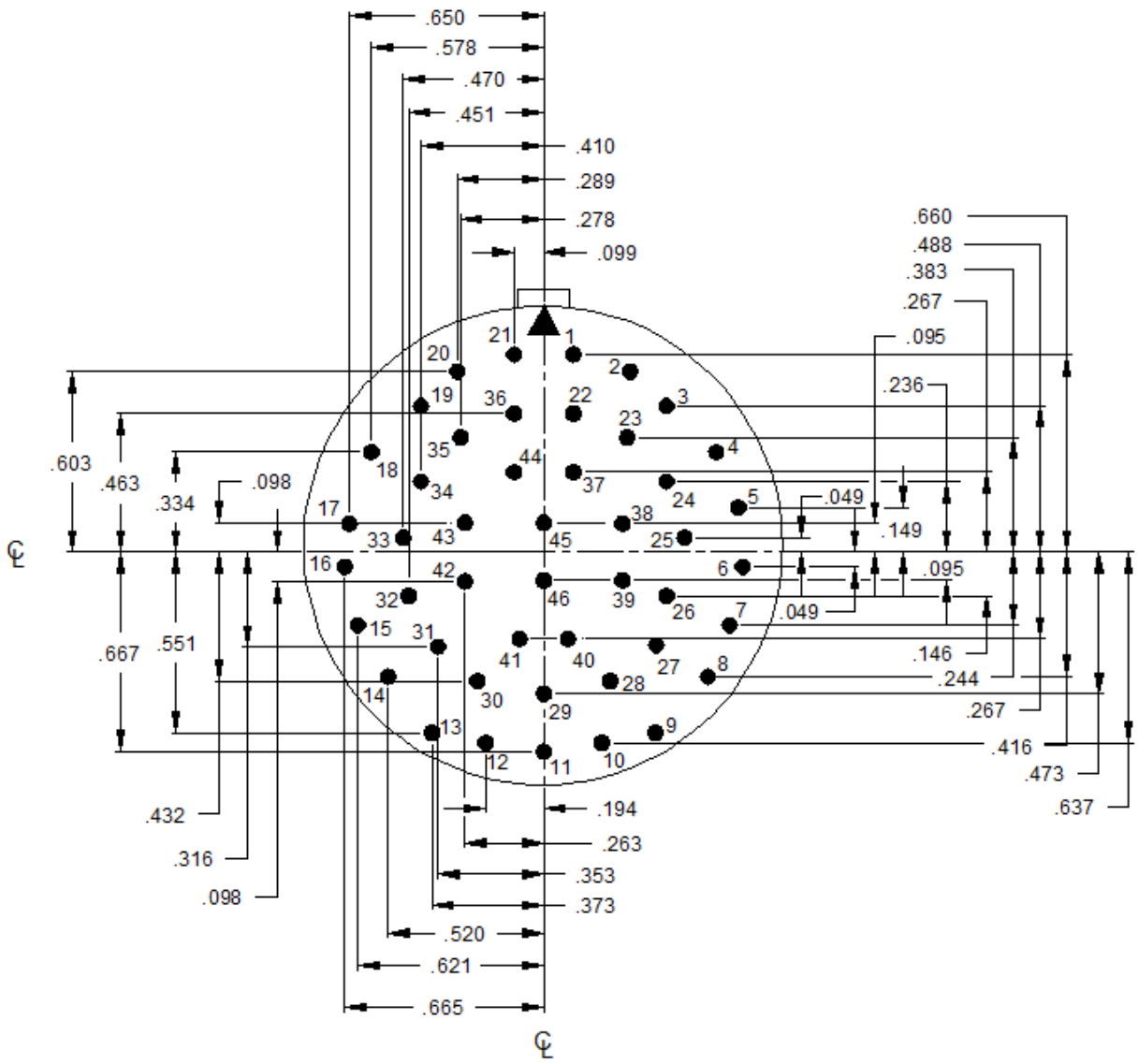
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 32 | -63 | 5 | 4 | D | All | MS33689-63 |

Note: This insert arrangement has no alternate insert positions.

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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32-73

(Not applicable for design of Air Force Airborne Equipment for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 32-73) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 1 | +.099 (2.51) | +.660 (16.76) |
| 2 | +.289 (7.34) | +.603 (15.32) |
| 3 | +.451 (11.46) | +.488 (12.40) |
| 4 | +.578 (14.68) | +.334 (8.48) |
| 5 | +.650 (16.51) | +.149 (3.78) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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| Contacts (Insert arrangement 32-73) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 6 | +.665 (16.89) | -.049 (1.24) |
| 7 | +.621 (15.77) | -.244 (6.20) |
| 8 | +.520 (13.21) | -.416 (10.57) |
| 9 | +.373 (9.47) | -.551 (14.00) |
| 10 | +.194 (4.93) | -.637 (16.18) |
| 11 | +.000 (0.00) | -.667 (16.94) |
| 12 | -.194 (4.93) | -.637 (16.18) |
| 13 | -.373 (9.47) | -.551 (14.00) |
| 14 | -.520 (13.21) | -.416 (10.57) |
| 15 | -.621 (15.77) | -.244 (6.20) |
| 16 | -.665 (16.89) | -.049 (1.24) |
| 17 | -.650 (16.51) | +.149 (3.78) |
| 18 | -.578 (14.68) | +.334 (8.48) |
| 19 | -.451 (11.46) | +.488 (12.40) |
| 20 | -.289 (7.34) | +.603 (15.32) |
| 21 | -.099 (2.51) | +.660 (16.76) |
| 22 | +.099 (2.51) | +.463 (11.76) |
| 23 | +.278 (7.06) | +.383 (9.73) |
| 24 | +.410 (10.41) | +.236 (5.99) |
| 25 | +.470 (11.94) | +.049 (1.24) |
| 26 | +.451 (11.46) | -.146 (3.71) |
| 27 | +.353 (8.97) | -.316 (8.03) |
| 28 | +.194 (4.93) | -.432 (10.97) |
| 29 | +.000 (0.00) | -.473 (12.01) |
| 30 | -.194 (4.93) | -.432 (10.97) |
| 31 | -.353 (8.97) | -.316 (8.03) |
| 32 | -.451 (11.46) | -.146 (3.71) |
| 33 | -.470 (11.94) | +.049 (1.24) |
| 34 | -.410 (10.41) | +.236 (5.99) |
| 35 | -.278 (7.06) | +.383 (9.73) |
| 36 | -.099 (2.51) | +.463 (11.76) |
| 37 | +.099 (2.51) | +.267 (6.78) |
| 38 | +.263 (6.68) | +.095 (2.41) |
| 39 | +.263 (6.68) | -.095 (2.41) |
| 40 | +.099 (2.51) | -.267 (6.78) |
| 41 | -.099 (2.51) | -.267 (6.78) |
| 42 | -.263 (6.68) | -.098 (2.49) |
| 43 | -.263 (6.68) | +.098 (2.49) |
| 44 | -.099 (2.51) | +.267 (6.78) |
| 45 | +.000 (0.00) | +.095 (2.41) |
| 46 | +.000 (0.00) | -.095 (2.41) |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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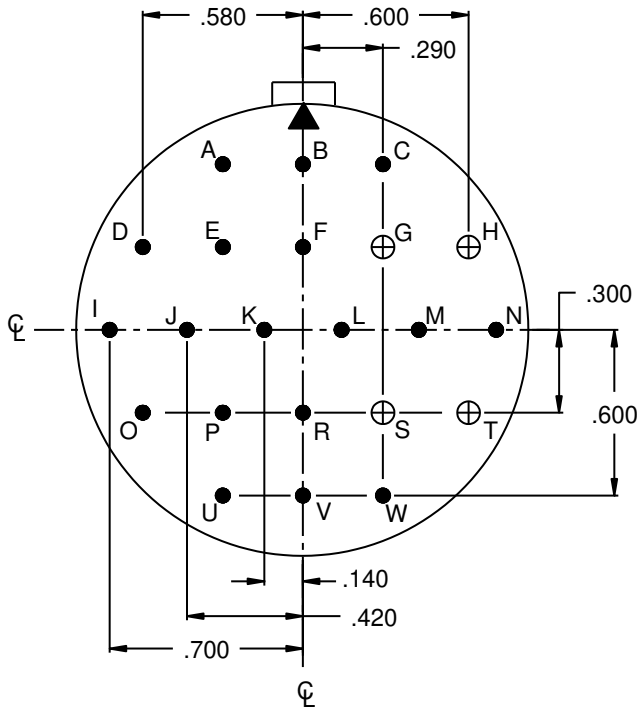
| Insert arrangement 32-73 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 32 | -73 | 46 | 16 | A | All | MS33689-73 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 36 | --- | --- | --- |

FIGURE 11. Insert arrangement, electrical connector, shell size 32, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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36-1

(Not applicable to MIL-DTL-22992.)

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

| Contacts (Insert arrangement 36-1) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.290 (7.37) | +.600 (15.24) |
| B | +.000 (0.00) | +.600 (15.24) |
| C | +.290 (7.37) | +.600 (15.24) |
| D | -.580 (14.73) | +.300 (7.62) |
| E | -.290 (7.37) | +.300 (7.62) |
| F | +.000 (0.00) | +.300 (7.62) |
| G | +.290 (7.37) | +.300 (7.62) |
| H | +.600 (15.24) | +.300 (7.62) |
| I | -.700 (17.78) | +.000 (0.00) |
| J | -.420 (10.67) | +.000 (0.00) |
| K | -.140 (3.56) | +.000 (0.00) |

| Contacts (Insert arrangement 36-1) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | +.140 (3.56) | +.000 (0.00) |
| M | +.420 (10.67) | +.000 (0.00) |
| N | +.700 (17.78) | +.000 (0.00) |
| O | -.580 (14.73) | -.300 (7.62) |
| P | -.290 (7.37) | -.300 (7.62) |
| R | +.000 (0.00) | -.300 (7.62) |
| S | +.290 (7.37) | -.300 (7.62) |
| T | +.600 (15.24) | -.300 (7.62) |
| U | -.290 (7.37) | -.600 (15.24) |
| V | +.000 (0.00) | -.600 (15.24) |
| W | +.290 (7.37) | -.600 (15.24) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234.



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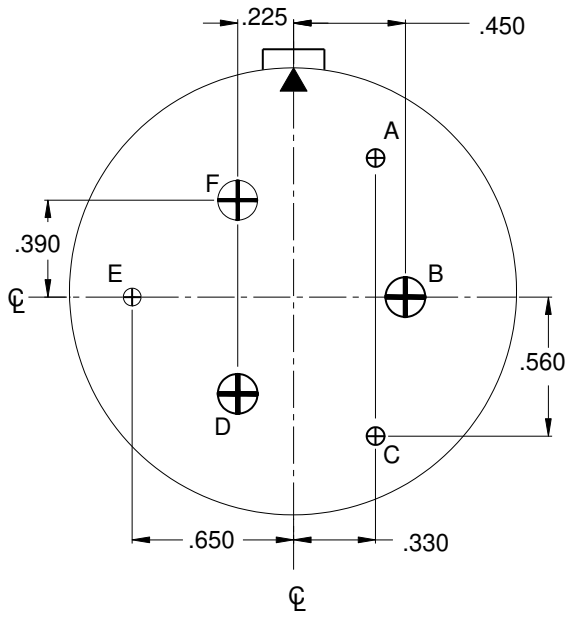
| Insert arrangement 36-1 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -1 | 18 | 16 | D | All others | MS33690-1 |
| | | 4 | 12 | D | G,H,S,T | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 – Continued.



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36-3

| Contacts (Insert arrangement 36-3) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.330 (8.38) | +.560 (14.22) |
| B | +.450 (11.43) | +.000 (0.00) |
| C | +.330 (8.38) | -.560 (14.22) |
| D | -.225 (5.72) | -.390 (9.91) |
| E | -.650 (16.76) | +.000 (0.00) |
| F | -.225 (5.72) | +.390 (9.91) |

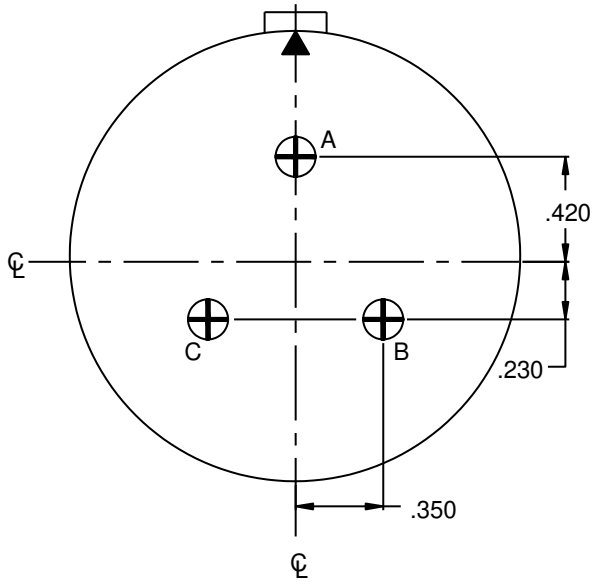
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -3 | 3 | 12 | D | A,C,E | MS33690-3 |
| | | 3 | 0 | D | B,D,F | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-4

(Inactive for New Design after 13 May 1954 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 36-4) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.420 (10.67) |
| B | +.350 (8.89) | -.230 (5.84) |
| C | -.350 (8.89) | -.230 (5.84) |

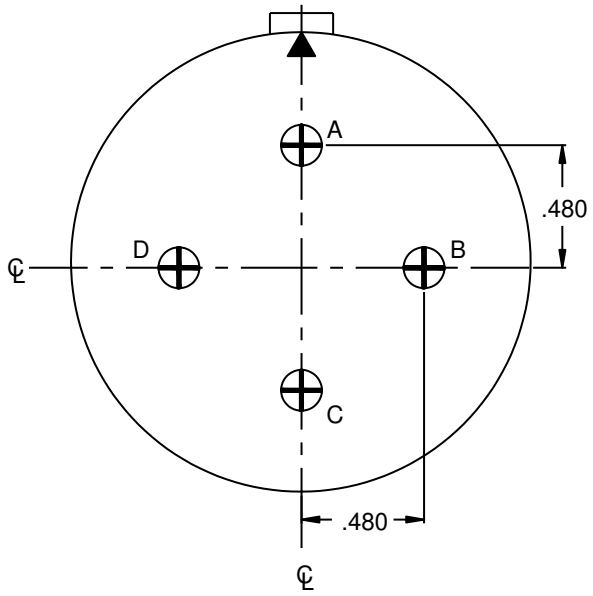
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -4 | 1 | 0 | D | A | MS33690-4 |
| | | 2 | | A | B,C | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 70 | 145 | 215 | 290 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-5

| Contacts (Insert arrangement 36-5) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.480 (12.19) |
| B | +.480 (12.19) | +.000 (0.00) |
| C | +.000 (0.00) | -.480 (12.19) |
| D | -.480 (12.19) | +.000 (0.00) |

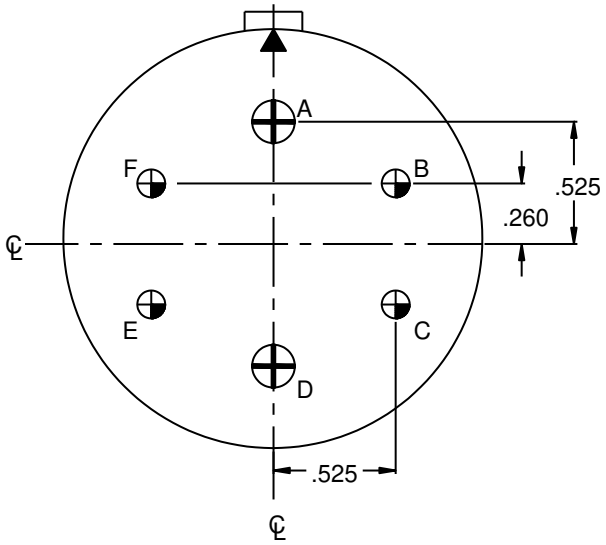
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -5 | 4 | 0 | A | All | MS33690-5 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| --- | 120 | 240 | --- |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-6

| Contacts (Insert arrangement 36-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.525 (13.34) |
| B | +.525 (13.34) | +.260 (6.60) |
| C | +.525 (13.34) | -.260 (6.60) |
| D | +.000 (0.00) | -.525 (13.34) |
| E | -.525 (13.34) | -.260 (6.60) |
| F | -.525 (13.34) | +.260 (6.60) |

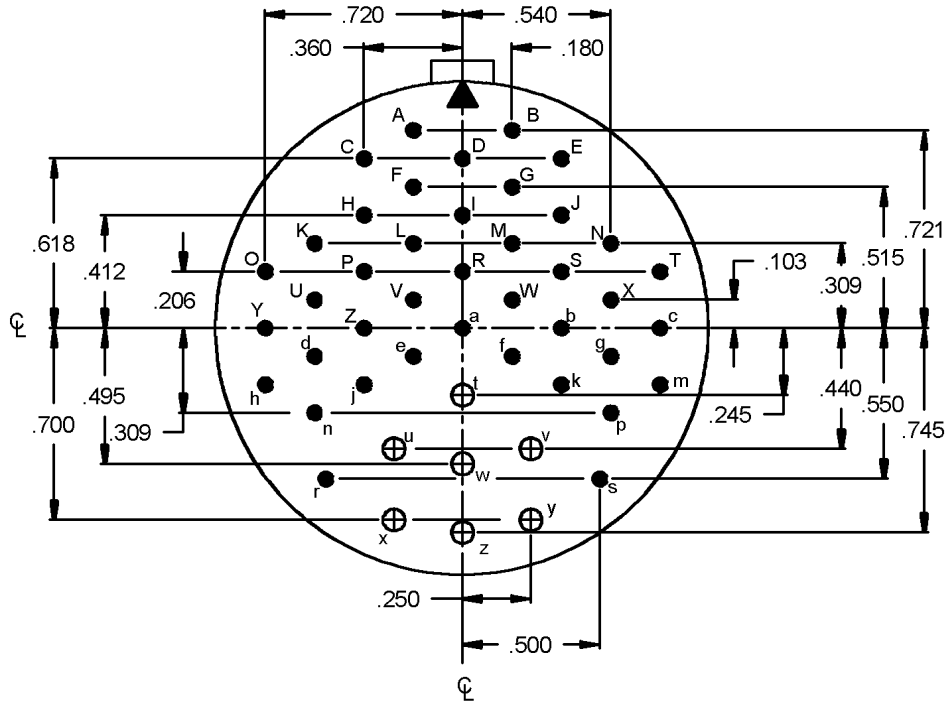
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -6 | 4 | 4 | A | All others | MS33690-6 |
| | | 2 | 0 | | A,D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 35 | 110 | 250 | 325 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-7

| Contacts (Insert arrangement 36-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.180 (4.57) | +.721 (18.31) |
| B | +.180 (4.57) | +.721 (18.31) |
| C | -.360 (9.14) | +.618 (15.70) |
| D | +.000 (0.00) | +.618 (15.70) |
| E | +.360 (9.14) | +.618 (15.70) |
| F | -.180 (4.57) | +.515 (13.08) |
| G | +.180 (4.57) | +.515 (13.08) |
| H | -.360 (9.14) | +.412 (10.46) |
| I | +.000 (0.00) | +.412 (10.46) |
| J | +.360 (9.14) | +.412 (10.46) |
| K | -.540 (13.72) | +.309 (7.85) |
| L | -.180 (4.57) | +.309 (7.85) |
| M | +.180 (4.57) | +.309 (7.85) |
| N | +.540 (13.72) | +.309 (7.85) |
| O | -.720 (18.29) | +.206 (5.23) |
| P | -.360 (9.14) | +.206 (5.23) |
| R | +.000 (0.00) | +.206 (5.23) |
| S | +.360 (9.14) | +.206 (5.23) |
| T | +.720 (18.29) | +.206 (5.23) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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| Contacts (Insert arrangement 36-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| U | -.540 (13.72) | +.103 (2.62) |
| V | -.180 (4.57) | +.103 (2.62) |
| W | +.180 (4.57) | +.103 (2.62) |
| X | +.540 (13.72) | +.103 (2.62) |
| Y | -.720 (18.29) | +.000 (0.00) |
| Z | -.360 (9.14) | +.000 (0.00) |
| a | +.000 (0.00) | +.000 (0.00) |
| b | +.360 (9.14) | +.000 (0.00) |
| c | +.720 (18.29) | +.000 (0.00) |
| d | -.540 (13.72) | -.103 (2.62) |
| e | -.180 (4.57) | -.103 (2.62) |
| f | +.180 (4.57) | -.103 (2.62) |
| g | +.540 (13.72) | -.103 (2.62) |
| h | -.720 (18.29) | -.206 (5.23) |
| j | -.360 (9.14) | -.206 (5.23) |
| k | +.360 (9.14) | -.206 (5.23) |
| m | +.720 (18.29) | -.206 (5.23) |
| n | -.540 (13.72) | -.309 (7.85) |
| p | +.540 (13.72) | -.309 (7.85) |
| r | -.500 (12.70) | -.550 (13.97) |
| s | +.500 (12.70) | -.550 (13.97) |
| t | +.000 (0.00) | -.245 (6.22) |
| u | -.250 (6.35) | -.440 (11.18) |
| v | +.250 (6.35) | -.440 (11.18) |
| w | +.000 (0.00) | -.495 (12.57) |
| x | -.250 (6.35) | -.700 (17.78) |
| y | +.250 (6.35) | -.700 (17.78) |
| z | +.000 (0.00) | -.745 (18.92) |

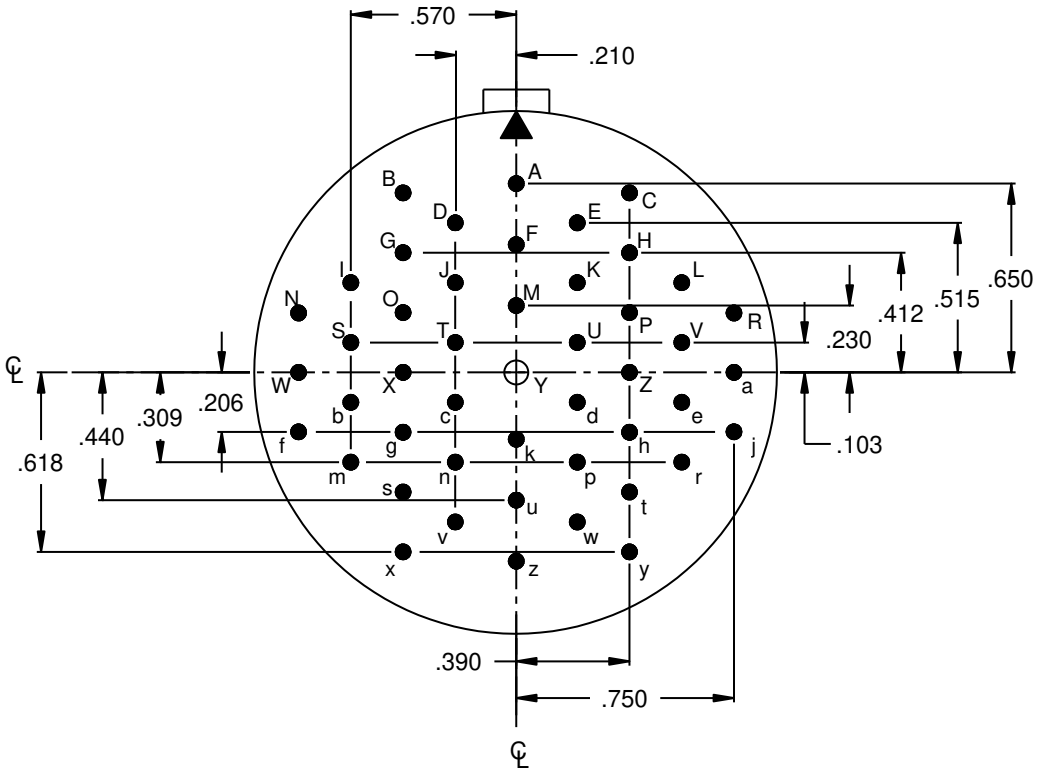
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -7 | 40 | 16 | A | All others | MS33690-7 |
| | | 7 | 12 | A | t,u,v,w,x,y,z | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-8

| Contacts (Insert arrangement 36-8) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.650 (16.51) |
| B | -.390 (9.91) | +.618 (15.70) |
| C | +.390 (9.91) | +.618 (15.70) |
| D | -.210 (5.33) | +.515 (13.08) |
| E | +.210 (5.33) | +.515 (13.08) |
| F | +.000 (0.00) | +.440 (11.18) |
| G | -.390 (9.91) | +.412 (10.46) |
| H | +.390 (9.91) | +.412 (10.46) |
| I | -.570 (14.48) | +.309 (7.85) |
| J | -.210 (5.33) | +.309 (7.85) |
| K | +.210 (5.33) | +.309 (7.85) |
| L | +.570 (14.48) | +.309 (7.85) |
| M | +.000 (0.00) | +.230 (5.84) |
| N | -.750 (19.05) | +.206 (5.23) |
| O | -.390 (9.91) | +.206 (5.23) |
| P | +.390 (9.91) | +.206 (5.23) |
| R | +.750 (19.05) | +.206 (5.23) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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| Contacts (Insert arrangement 36-8) | | |
|---|---|---|
| | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| S | -.570 (14.48) | +.103 (2.62) |
| T | -.210 (5.33) | +.103 (2.62) |
| U | +.210 (5.33) | +.103 (2.62) |
| V | +.570 (14.48) | +.103 (2.62) |
| W | -.750 (19.05) | +.000 (0.00) |
| X | -.390 (9.91) | +.000 (0.00) |
| Y | +.000 (0.00) | +.000 (0.00) |
| Z | +.390 (9.91) | +.000 (0.00) |
| a | +.750 (19.05) | +.000 (0.00) |
| b | -.570 (14.48) | -.103 (2.62) |
| c | -.210 (5.33) | -.103 (2.62) |
| d | +.210 (5.33) | -.103 (2.62) |
| e | +.570 (14.48) | -.103 (2.62) |
| f | -.750 (19.05) | -.206 (5.23) |
| g | -.390 (9.91) | -.206 (5.23) |
| h | +.390 (9.91) | -.206 (5.23) |
| j | +.750 (19.05) | -.206 (5.23) |
| k | +.000 (0.00) | -.230 (5.84) |
| m | -.570 (14.48) | -.309 (7.85) |
| n | -.210 (5.33) | -.309 (7.85) |
| p | +.210 (5.33) | -.309 (7.85) |
| r | +.570 (14.48) | -.309 (7.85) |
| s | -.390 (9.91) | -.412 (10.46) |
| t | +.390 (9.91) | -.412 (10.46) |
| u | +.000 (0.00) | -.440 (11.18) |
| v | -.210 (5.33) | -.515 (13.08) |
| w | +.210 (5.33) | -.515 (13.08) |
| x | -.390 (9.91) | -.618 (15.70) |
| y | +.390 (9.91) | -.618 (15.70) |
| z | +.000 (0.00) | -.650 (16.51) |

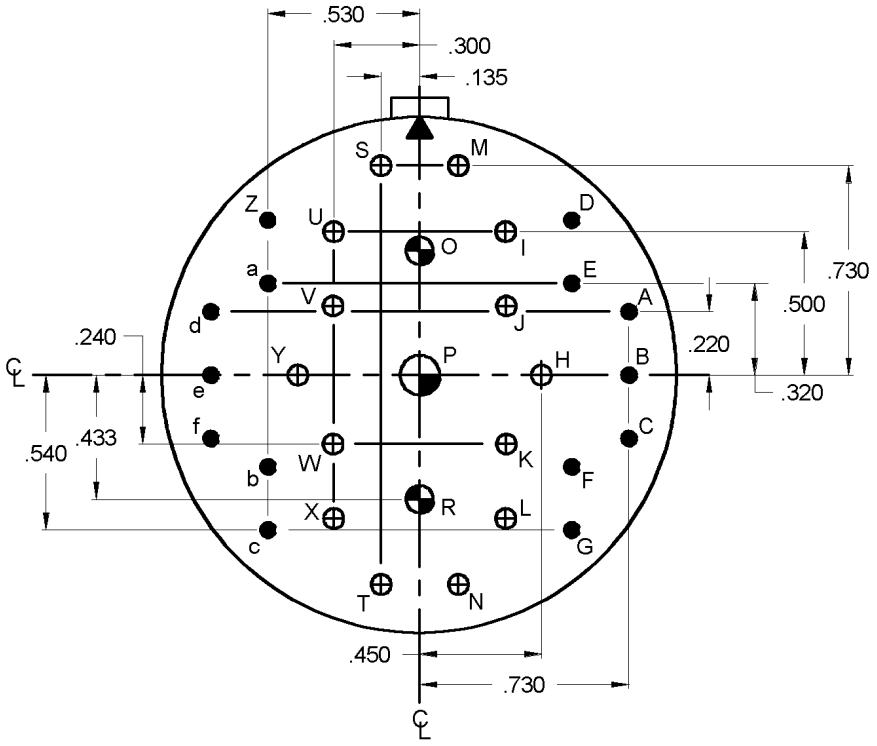
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -8 | 46 | 16 | A | All others | MS33690-8 |
| | | 1 | 12 | A | Y | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-9

| Contacts (Insert arrangement 36-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.730 (18.54) | +.220 (5.59) |
| B | +.730 (18.54) | +.000 (0.00) |
| C | +.730 (18.54) | -.220 (5.59) |
| D | +.530 (13.46) | +.540 (13.72) |
| E | +.530 (13.46) | +.320 (8.13) |
| F | +.530 (13.46) | -.320 (8.13) |
| G | +.530 (13.46) | -.540 (13.72) |
| H | +.450 (11.43) | +.000 (0.00) |
| I | +.300 (7.62) | +.500 (12.70) |
| J | +.300 (7.62) | +.240 (6.10) |
| K | +.300 (7.62) | -.240 (6.10) |
| L | +.300 (7.62) | -.500 (12.70) |
| M | +.135 (3.43) | +.730 (18.54) |
| N | +.135 (3.43) | -.730 (18.54) |
| O | +.000 (0.00) | +.433 (11.00) |
| P | +.000 (0.00) | +.000 (0.00) |
| R | +.000 (0.00) | -.433 (11.00) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 36-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| S | -.135 (3.43) | +.730 (18.54) |
| T | -.135 (3.43) | -.730 (18.54) |
| U | -.300 (7.62) | +.500 (12.70) |
| V | -.300 (7.62) | +.240 (6.10) |
| W | -.300 (7.62) | -.240 (6.10) |
| X | -.300 (7.62) | -.500 (12.70) |
| Y | -.450 (11.43) | +.000 (0.00) |
| Z | -.530 (13.46) | +.540 (13.72) |
| a | -.530 (13.46) | +.320 (8.13) |
| b | -.530 (13.46) | -.320 (8.13) |
| c | -.530 (13.46) | -.540 (13.72) |
| d | -.730 (18.54) | +.220 (5.59) |
| e | -.730 (18.54) | +.000 (0.00) |
| f | -.730 (18.54) | -.220 (5.59) |

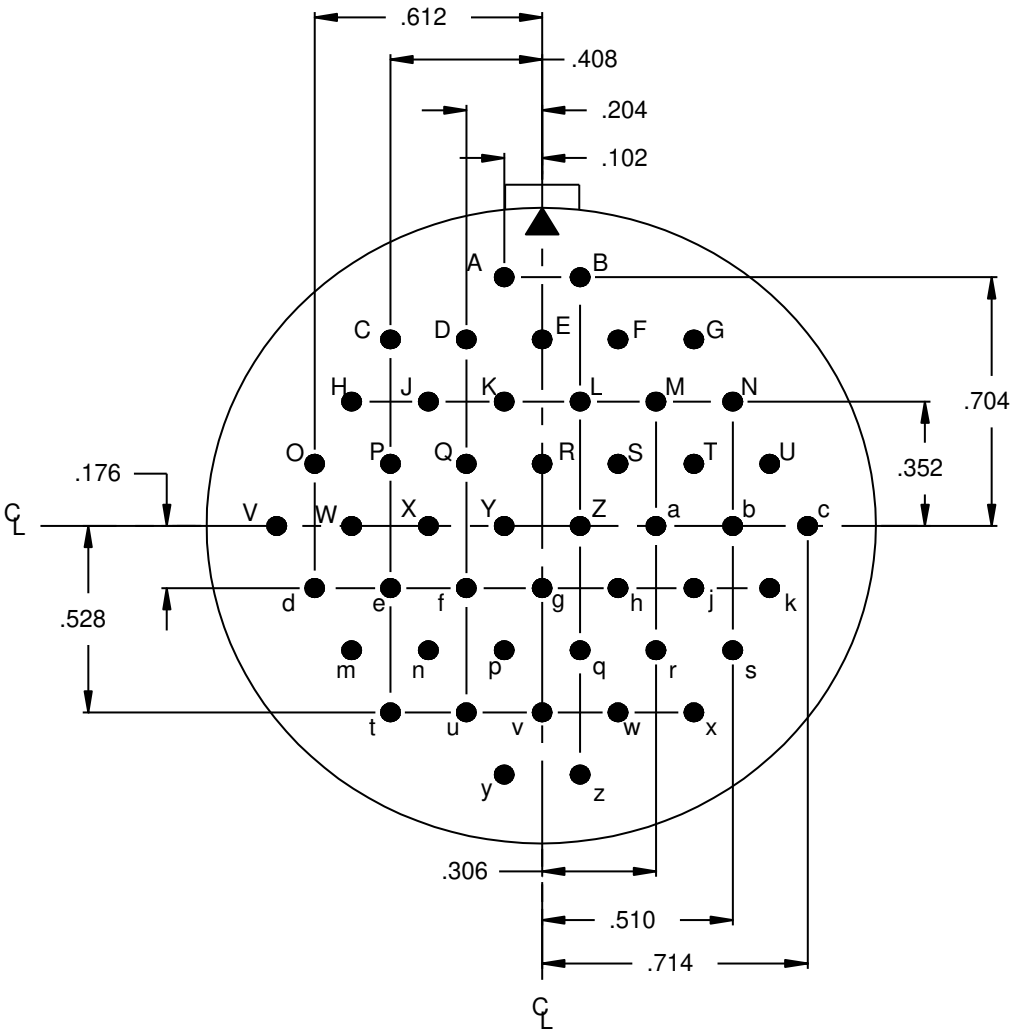
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -9 | 14 | 16 | A | All others | MS33690-9 |
| | | 14 | 12 | A | H,I,J,K,L,M,N,S,T,U,V,Y,W,X | |
| | | 2 | 8 | A | O,R | |
| | | 1 | 4 | A | P | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 125 | 235 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-10

| Contacts (Insert arrangement 36-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.102 (2.59) | +.704 (17.88) |
| B | +.102 (2.59) | +.704 (17.88) |
| C | -.408 (10.36) | +.528 (13.41) |
| D | -.204 (5.18) | +.528 (13.41) |
| E | +.000 (0.00) | +.528 (13.41) |
| F | +.204 (5.18) | +.528 (13.41) |
| G | +.408 (10.36) | +.528 (13.41) |
| H | -.510 (12.95) | +.352 (8.94) |
| J | -.306 (0.00) | +.352 (8.94) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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| Contacts (Insert arrangement 36-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| K | -.102 (2.59) | +.352 (8.94) |
| L | +.102 (2.59) | +.352 (8.94) |
| M | +.306 (0.00) | +.352 (8.94) |
| N | +.510 (12.95) | +.352 (8.94) |
| O | -.612 (15.54) | +.176 (4.47) |
| P | -.408 (10.36) | +.176 (4.47) |
| Q | -.204 (5.18) | +.176 (4.47) |
| R | +.000 (0.00) | +.176 (4.47) |
| S | +.204 (5.18) | +.176 (4.47) |
| T | +.408 (10.36) | +.176 (4.47) |
| U | +.612 (15.54) | +.176 (4.47) |
| V | -.714 (18.14) | +.000 (0.00) |
| W | -.510 (12.95) | +.000 (0.00) |
| X | -.306 (0.00) | +.000 (0.00) |
| Y | -.102 (2.59) | +.000 (0.00) |
| Z | +.102 (2.59) | +.000 (0.00) |
| a | +.306 (0.00) | +.000 (0.00) |
| b | +.510 (12.95) | +.000 (0.00) |
| c | +.714 (18.14) | +.000 (0.00) |
| d | -.612 (15.54) | -.176 (4.47) |
| e | -.408 (10.36) | -.176 (4.47) |
| f | -.204 (5.18) | -.176 (4.47) |
| g | +.000 (0.00) | -.176 (4.47) |
| h | +.204 (5.18) | -.176 (4.47) |
| j | +.408 (10.36) | -.176 (4.47) |
| k | +.612 (15.54) | -.176 (4.47) |
| m | -.510 (12.95) | -.352 (8.94) |
| n | -.306 (0.00) | -.352 (8.94) |
| p | -.102 (2.59) | -.352 (8.94) |
| q | +.102 (2.59) | -.352 (8.94) |
| r | +.306 (0.00) | -.352 (8.94) |
| s | +.510 (12.95) | -.352 (8.94) |
| t | -.408 (10.36) | -.528 (13.41) |
| u | -.204 (5.18) | -.528 (13.41) |
| v | +.000 (0.00) | -.528 (13.41) |
| w | +.204 (5.18) | -.528 (13.41) |
| x | +.408 (10.36) | -.528 (13.41) |
| y | -.102 (2.59) | -.704 (17.88) |
| z | +.102 (2.59) | -.704 (17.88) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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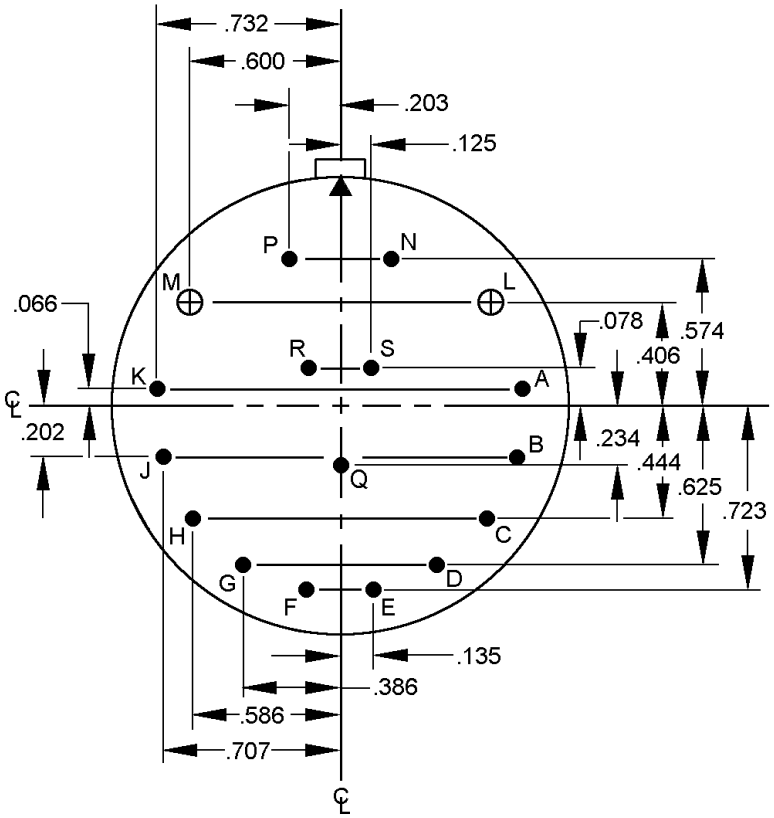
| Insert arrangement 36-10 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -10 | 48 | 16 | A | All | MS33690-10 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 125 | 235 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-13

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 36-13) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.732 (18.59) | +.066 (1.68) |
| B | +.707 (17.96) | -.202 (5.13) |
| C | +.586 (14.88) | -.444 (11.28) |
| D | +.386 (9.80) | -.625 (15.88) |
| E | +.135 (3.43) | -.723 (18.36) |
| F | -.135 (3.43) | -.723 (18.36) |
| G | -.386 (9.80) | -.625 (15.88) |
| H | -.586 (14.88) | -.444 (11.28) |
| J | -.707 (17.96) | -.202 (5.13) |

| Contacts (Insert arrangement 36-13) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| K | -.732 (18.59) | +.066 (1.68) |
| L | +.600 (15.24) | +.406 (10.31) |
| M | -.600 (14.24) | +.406 (10.31) |
| N | +.203 (5.16) | +.574 (14.58) |
| P | -.203 (5.16) | +.574 (14.58) |
| R | -.125 (3.18) | +.078 (1.98) |
| S | +.125 (3.18) | +.078 (1.98) |
| Q | +.000 (0.00) | -.234 (5.94) |
| --- | --- | --- |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued..



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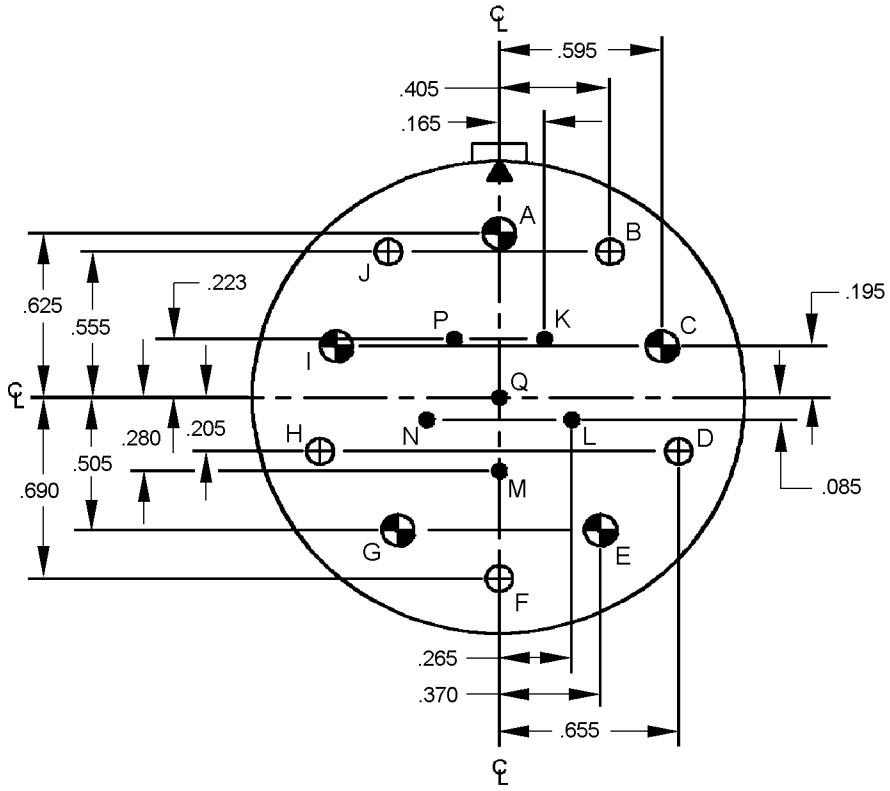
| Insert arrangement 36-13 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -13 | 3 | 16 | E | N,P,Q | MS33690-13 |
| | | 12 | | A | All others | |
| | | 2 | 12 | A | M,L | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued..



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36-14

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 36-14) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.625 (15.88) |
| B | +.405 (10.29) | +.555 (14.10) |
| C | +.595 (15.11) | +.195 (4.95) |
| D | +.655 (16.64) | -.205 (5.21) |
| E | +.370 (9.40) | -.505 (12.83) |
| F | +.000 (0.00) | -.690 (17.53) |
| G | -.370 (9.40) | -.505 (12.83) |
| H | -.655 (16.64) | -.205 (5.21) |

| Contacts (Insert arrangement 36-14) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| I | -.595 (15.11) | +.195 (4.95) |
| J | -.405 (10.29) | +.555 (14.10) |
| K | +.165 (4.19) | +.223 (5.66) |
| L | +.265 (6.73) | -.085 (2.16) |
| M | +.000 (0.00) | -.280 (7.11) |
| N | -.265 (6.73) | -.085 (2.16) |
| P | -.165 (4.19) | +.223 (5.66) |
| Q | +.000 (0.00) | +.000 (0.00) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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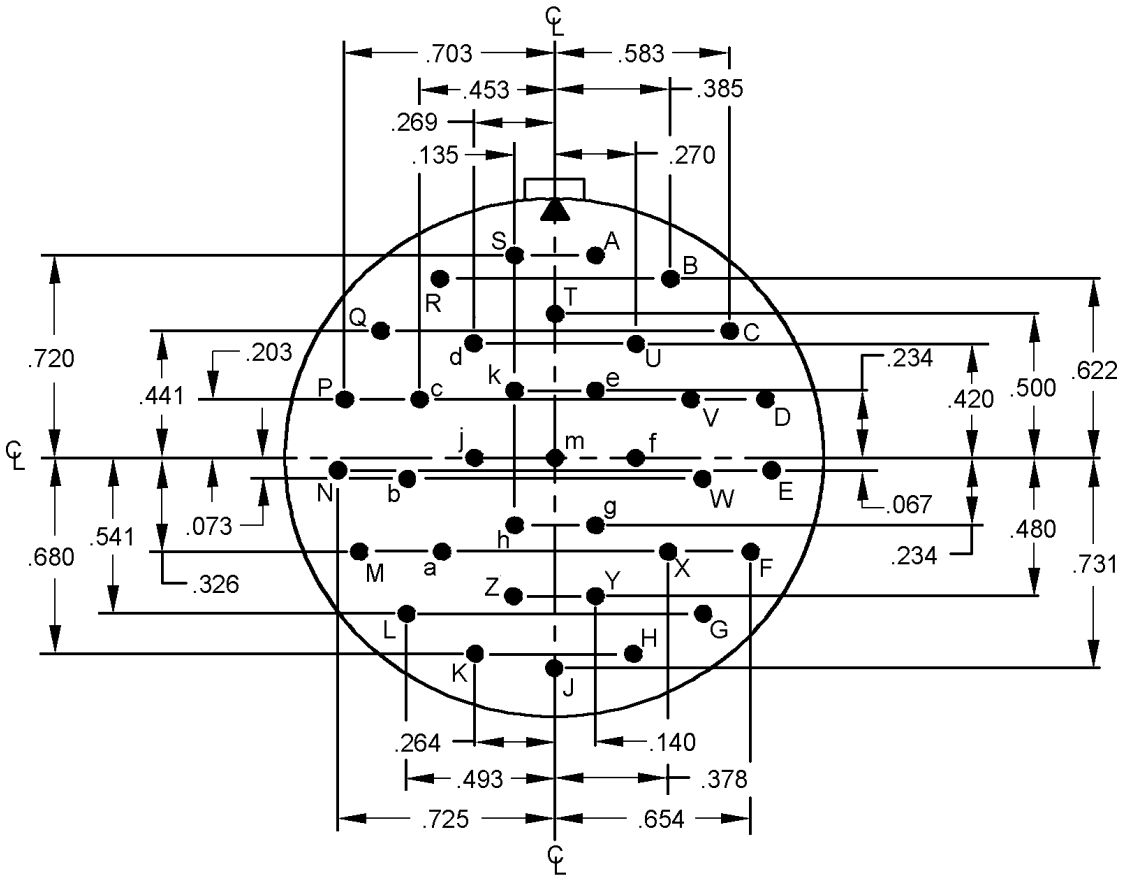
| Insert arrangement 36-14 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -14 | 6 | 16 | D | All others | MS33690-14 |
| | | 5 | 12 | D | B,D,F,H,J | |
| | | 5 | 8 | D | A,C,E,G,I | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 90 | 180 | 270 | --- |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-15

| Contacts (Insert arrangement 36-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.140 (3.56) | +.720 (18.29) |
| B | +.102 (2.59) | +.622 (15.80) |
| C | +.583 (14.81) | +.441 (11.20) |
| D | +.703 (17.86) | +.203 (5.16) |
| E | +.140 (3.56) | -.067 (1.70) |
| F | +.654 (16.61) | -.326 (8.28) |
| G | +.493 (12.52) | -.541 (13.74) |
| H | -.510 (12.95) | -.680 (17.27) |
| J | +.000 (0.00) | -.731 (18.57) |
| K | -.102 (2.59) | -.680 (17.27) |
| L | -.493 (12.52) | -.541 (13.74) |
| M | -.654 (16.61) | -.326 (8.28) |
| N | -.725 (18.42) | -.067 (1.70) |
| O | -.720 (18.29) | +.176 (4.47) |
| P | -.408 (10.36) | +.203 (5.16) |
| Q | -.204 (5.18) | +.176 (4.47) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 36-15) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| R | +.000 (0.00) | +.622 (15.80) |
| S | +.204 (5.18) | +.720 (18.29) |
| T | +.000 (0.00) | +.500 (12.70) |
| U | +.270 (6.86) | +.420 (10.67) |
| V | +.453 (11.51) | +.203 (5.16) |
| W | +.493 (12.52) | +.000 (0.00) |
| X | +.378 (9.60) | +.000 (0.00) |
| Y | +.140 (3.56) | +.000 (0.00) |
| Z | +.102 (2.59) | +.000 (0.00) |
| a | -.378 (9.60) | -.326 (8.28) |
| b | -.493 (12.52) | +.000 (0.00) |
| c | -.453 (11.51) | +.203 (5.16) |
| d | -.269 (6.83) | +.420 (10.67) |
| e | -.140 (3.56) | -.176 (4.47) |
| f | +.270 (6.86) | +.000 (0.00) |
| g | +.000 (0.00) | -.176 (4.47) |
| h | +.204 (5.18) | -.176 (4.47) |
| j | -.269 (6.83) | +.000 (0.00) |
| k | +.360 (9.14) | +.234 (5.94) |
| m | +.000 (0.00) | +.000 (0.00) |

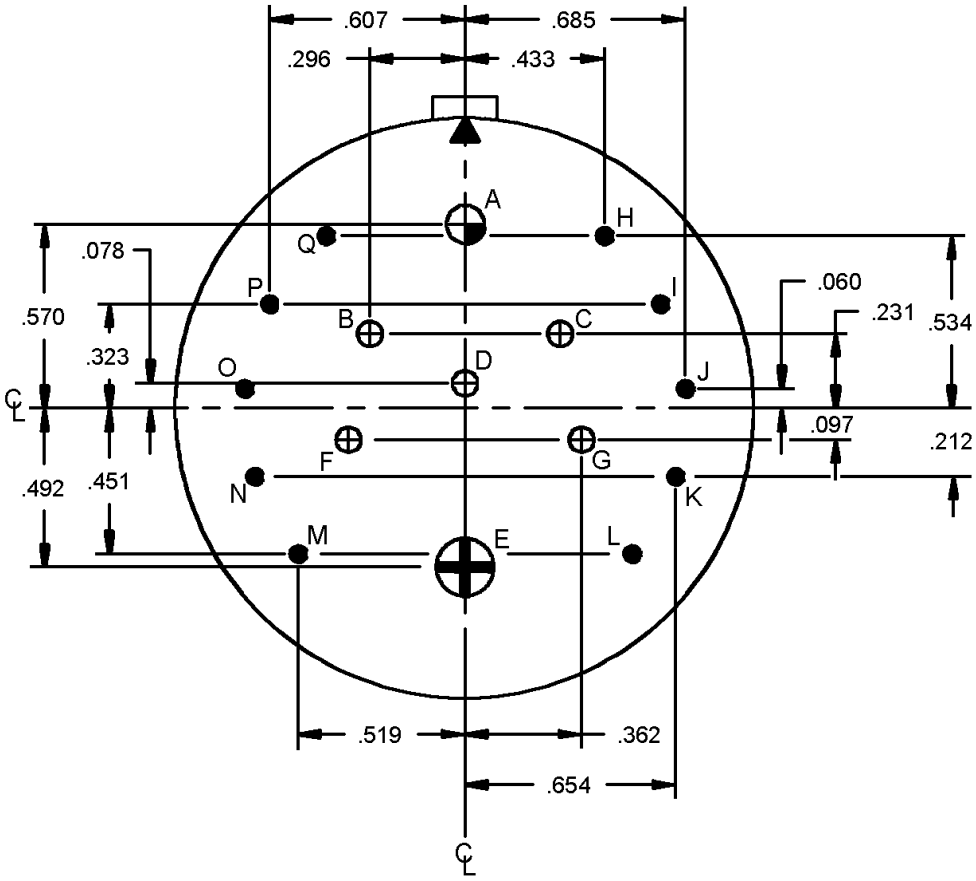
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -15 | 35 | 16 | D | M | MS33690-15 |
| | | | | A | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 60 | 125 | 245 | 305 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



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36-19

(Inactive for New Design after 17 Dec 1958 for SAE-AS50151 and MIL-DTL-83723 series II. Active for new design for SAE-AS95234.)

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 36-19) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.570 (14.48) |
| B | -.296 (7.52) | +.231 (5.87) |
| C | +.296 (7.52) | +.231 (5.87) |
| D | +.000 (0.00) | +.078 (1.98) |
| E | +.000 (0.00) | -.492 (12.50) |
| F | -.362 (9.19) | -.097 (2.46) |
| G | +.362 (9.19) | -.097 (2.46) |
| H | +.433 (11.00) | +.534 (13.56) |
| I | +.607 (15.42) | +.323 (8.20) |

| Contacts (Insert arrangement 36-19) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| J | +.685 (17.40) | +.060 (1.52) |
| K | +.654 (16.61) | -.212 (5.38) |
| L | +.519 (13.18) | -.451 (11.46) |
| M | -.519 (13.18) | -.451 (11.46) |
| N | -.654 (16.61) | -.212 (5.38) |
| O | -.685 (17.40) | +.060 (1.52) |
| P | -.607 (15.42) | +.323 (8.20) |
| Q | -.433 (11.00) | +.534 (13.56) |
| --- | --- | --- |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

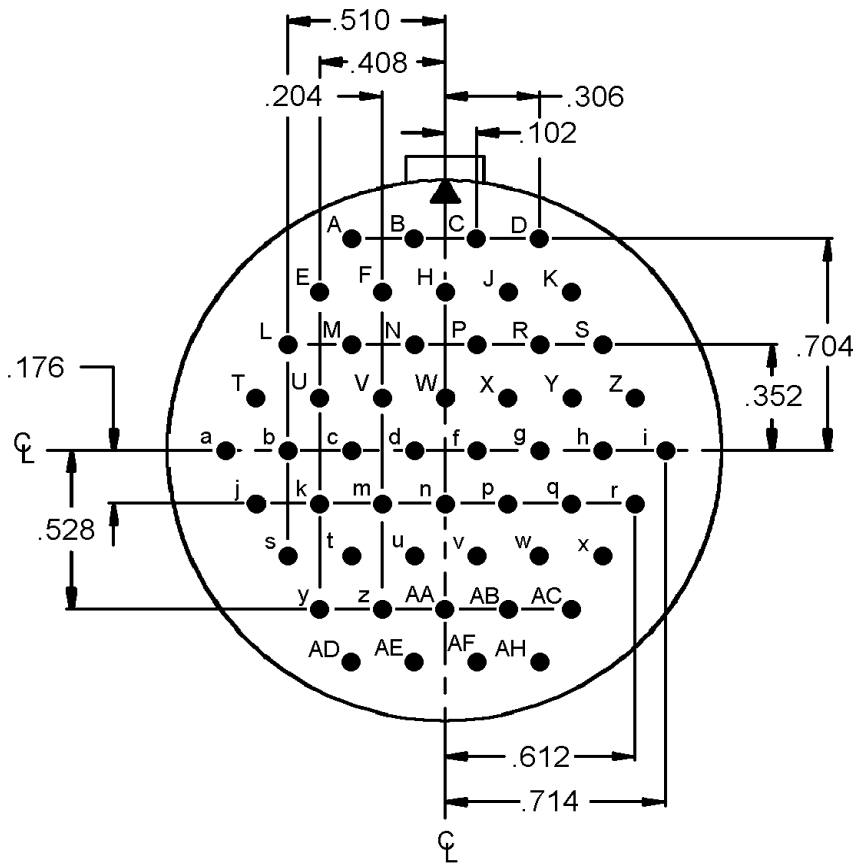
| Insert arrangement 36-19 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -19 | 10 | 16 | D | All others | MS33690-19 |
| | | 5 | 12 | D | B,C,D,F,G | |
| | | 1 | 4 | D | A | |
| | | 1 | 0 | D | E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




36-52

| Contacts (Insert arrangement 36-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.306 (7.77) | +.704 (17.88) |
| B | -.102 (2.59) | +.704 (17.88) |
| C | +.102 (2.59) | +.704 (17.88) |
| D | +.306 (7.77) | +.704 (17.88) |
| E | -.408 (10.36) | +.528 (13.41) |
| F | -.204 (5.18) | +.528 (13.41) |
| H | +.000 (0.00) | +.528 (13.41) |
| J | +.204 (5.18) | +.528 (13.41) |
| K | +.408 (10.36) | +.528 (13.41) |
| L | -.510 (12.95) | +.352 (8.94) |
| M | -.306 (7.77) | +.352 (8.94) |
| N | -.102 (2.59) | +.352 (8.94) |
| P | +.102 (2.59) | +.352 (8.94) |
| R | +.306 (7.77) | +.352 (8.94) |
| S | +.510 (12.95) | +.352 (8.94) |
| T | -.612 (15.54) | +.176 (4.47) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 36-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| U | -.408 (10.36) | +.176 (4.47) |
| V | -.204 (5.18) | +.176 (4.47) |
| W | +.000 (0.00) | +.176 (4.47) |
| X | +.204 (5.18) | +.176 (4.47) |
| Y | +.408 (10.36) | +.176 (4.47) |
| Z | +.612 (15.54) | +.176 (4.47) |
| a | -.714 (18.14) | +.000 (0.00) |
| b | -.510 (12.95) | +.000 (0.00) |
| c | -.306 (7.77) | +.000 (0.00) |
| d | -.102 (2.59) | +.000 (0.00) |
| f | +.102 (2.59) | +.000 (0.00) |
| g | +.306 (7.77) | +.000 (0.00) |
| h | +.510 (12.95) | +.000 (0.00) |
| i | +.714 (18.14) | +.000 (0.00) |
| j | -.612 (15.54) | -.176 (4.47) |
| k | -.408 (10.36) | -.176 (4.47) |
| m | -.204 (5.18) | -.176 (4.47) |
| n | +.000 (0.00) | -.176 (4.47) |
| p | +.204 (5.18) | -.176 (4.47) |
| q | +.408 (10.36) | -.176 (4.47) |
| r | +.612 (15.54) | -.176 (4.47) |
| s | -.510 (12.95) | -.352 (8.94) |
| t | -.306 (7.77) | -.352 (8.94) |
| u | -.102 (2.59) | -.352 (8.94) |
| v | +.102 (2.59) | -.352 (8.94) |
| w | +.306 (7.77) | -.352 (8.94) |
| x | +.510 (12.95) | -.352 (8.94) |
| y | -.408 (10.36) | -.528 (13.41) |
| z | -.204 (5.18) | -.528 (13.41) |
| AA | +.000 (0.00) | -.528 (13.41) |
| AB | +.204 (5.18) | -.528 (13.41) |
| AC | +.408 (10.36) | -.528 (13.41) |
| AD | -.306 (7.77) | -.704 (17.88) |
| AE | -.102 (2.59) | -.704 (17.88) |
| AF | +.102 (2.59) | -.704 (17.88) |
| AH | +.306 (7.77) | -.704 (17.88) |

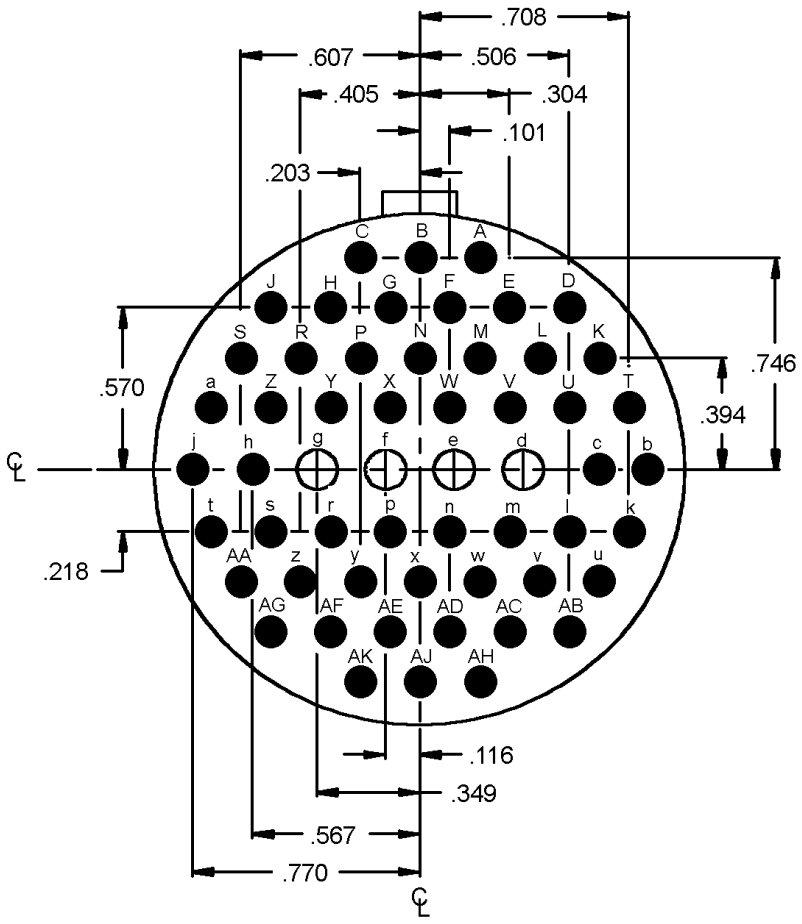
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 36 | -52 | 52 | 16 | A | All | MS33690-52 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 72 | 144 | 216 | 288 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




36-66

| Contacts (Insert arrangement 36-66) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.203 (5.16) | +.746 (18.95) |
| B | +.000 (0.00) | +.746 (18.95) |
| C | -.203 (5.16) | +.746 (18.95) |
| D | +.506 (12.85) | +.570 (14.48) |
| E | +.304 (7.72) | +.570 (14.48) |
| F | +.101 (2.57) | +.570 (14.48) |
| G | -.101 (2.57) | +.570 (14.48) |
| H | -.304 (7.72) | +.570 (14.48) |
| J | -.506 (12.85) | +.570 (14.48) |
| K | +.607 (15.42) | +.394 (10.01) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 36-66) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | +.405 (10.29) | +.394 (10.01) |
| M | +.203 (5.16) | +.394 (10.01) |
| N | +.000 (0.00) | +.394 (10.01) |
| P | -.203 (5.16) | +.394 (10.01) |
| R | -.405 (10.29) | +.394 (10.01) |
| S | -.607 (15.42) | +.394 (10.01) |
| T | +.708 (17.98) | +.218 (5.54) |
| U | +.506 (12.85) | +.218 (5.54) |
| V | +.304 (7.72) | +.218 (5.54) |
| W | +.101 (2.57) | +.218 (5.54) |
| X | -.101 (2.57) | +.218 (5.54) |
| Y | -.304 (7.72) | +.218 (5.54) |
| Z | -.506 (12.85) | +.218 (5.54) |
| a | -.708 (17.98) | +.218 (5.54) |
| b | +.770 (19.56) | +.000 (0.00) |
| c | +.567 (14.40) | +.000 (0.00) |
| d | +.349 (8.86) | +.000 (0.00) |
| e | +.101 (2.57) | +.000 (0.00) |
| f | -.101 (2.57) | +.000 (0.00) |
| g | -.349 (8.86) | +.000 (0.00) |
| h | -.567 (14.40) | +.000 (0.00) |
| j | -.770 (19.56) | +.000 (0.00) |
| k | +.708 (17.98) | -.218 (5.54) |
| l | +.506 (12.85) | -.218 (5.54) |
| m | +.304 (7.72) | -.218 (5.54) |
| n | +.101 (2.57) | -.218 (5.54) |
| p | -.101 (2.57) | -.218 (5.54) |
| r | -.304 (7.72) | -.218 (5.54) |
| s | -.506 (12.85) | -.218 (5.54) |
| t | -.708 (17.98) | -.218 (5.54) |
| u | +.607 (15.42) | -.394 (10.01) |
| v | +.405 (10.29) | -.394 (10.01) |
| w | +.203 (5.16) | -.394 (10.01) |
| x | +.000 (0.00) | -.394 (10.01) |
| y | -.203 (5.16) | -.394 (10.01) |
| z | -.405 (10.29) | -.394 (10.01) |
| AA | -.607 (15.42) | -.394 (10.01) |
| AB | +.506 (12.85) | -.570 (14.48) |
| AC | +.304 (7.72) | -.570 (14.48) |
| AD | +.101 (2.57) | -.570 (14.48) |
| AE | -.101 (2.57) | -.570 (14.48) |
| AF | -.304 (7.72) | -.570 (14.48) |
| AG | -.506 (12.85) | -.570 (14.48) |
| AH | +.203 (5.16) | -.746 (18.95) |
| AJ | +.000 (0.00) | -.746 (18.95) |
| AK | -.203 (5.16) | -.746 (18.95) |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

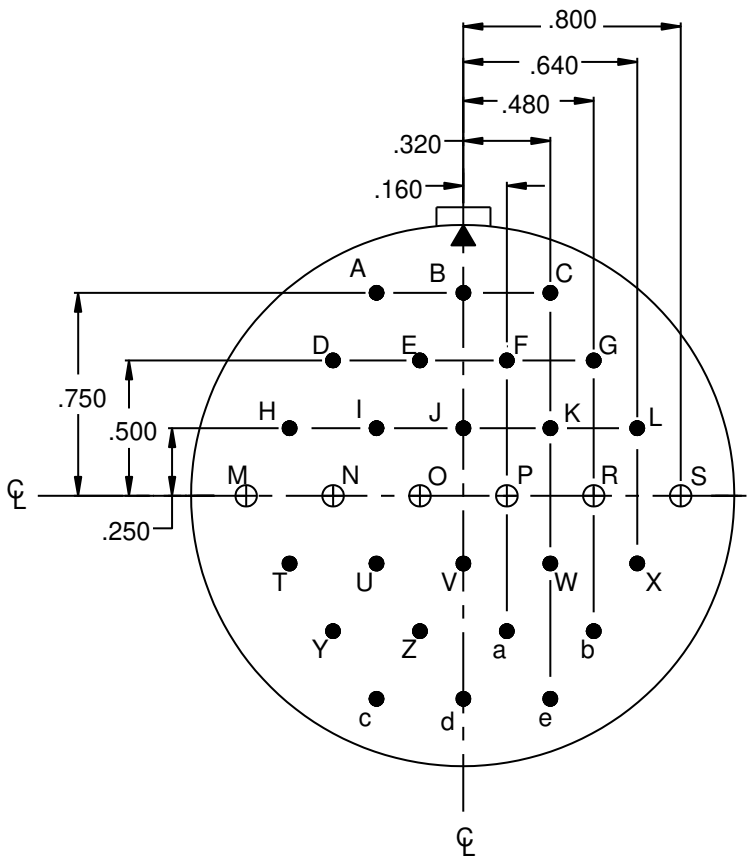
| Insert arrangement 36-66 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 36 | -66 | 52 | 16 | A | All others | MS33690-66 |
| | | 4 | 12 | | g,f,e,d | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 110 | 250 | 260 | 280 |

FIGURE 12. Insert arrangement, electrical connector, shell size 36, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS95234 - Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




40-1

| Contacts (Insert arrangement 40-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.320 (8.13) | +.750 (19.05) |
| B | +.000 (0.00) | +.750 (19.05) |
| C | +.320 (8.13) | +.750 (19.05) |
| D | -.480 (12.19) | +.500 (12.70) |
| E | -.160 (4.06) | +.500 (12.70) |
| F | +.160 (4.06) | +.500 (12.70) |
| G | +.480 (12.19) | +.500 (12.70) |
| H | -.640 (16.26) | +.250 (6.35) |
| I | -.320 (8.13) | +.250 (6.35) |
| J | +.000 (0.00) | +.250 (6.35) |
| K | +.320 (8.13) | +.250 (6.35) |
| L | +.640 (16.26) | +.250 (6.35) |
| M | -.800 (20.32) | +.000 (0.00) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | -.480 (12.19) | +.000 (0.00) |
| O | -.160 (4.06) | +.000 (0.00) |
| P | +.160 (4.06) | +.000 (0.00) |
| R | +.480 (12.19) | +.000 (0.00) |
| S | +.800 (20.32) | +.000 (0.00) |
| T | -.640 (16.26) | -.250 (6.35) |
| U | -.320 (8.13) | -.250 (6.35) |
| V | +.000 (0.00) | -.250 (6.35) |
| W | +.320 (8.13) | -.250 (6.35) |
| X | +.640 (16.26) | -.250 (6.35) |
| Y | -.480 (12.19) | -.500 (12.70) |
| Z | -.160 (4.06) | -.500 (12.70) |
| a | +.160 (4.06) | -.500 (12.70) |
| b | +.480 (12.19) | -.500 (12.70) |
| c | -.320 (8.13) | -.750 (19.05) |
| d | +.000 (0.00) | -.750 (19.05) |
| e | +.320 (8.13) | -.750 (19.05) |

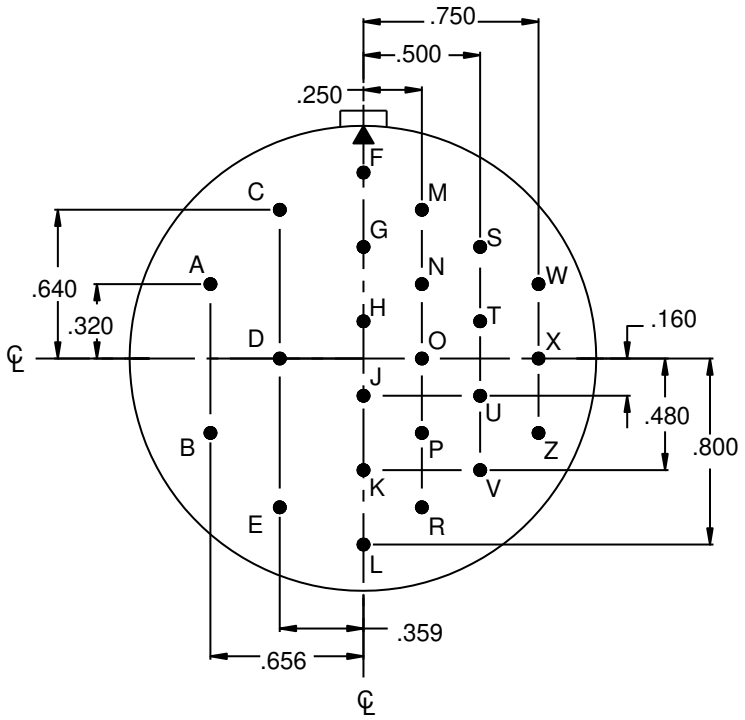
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -1 | 24 | 16 | D | All others | MS33691-1 |
| | | 6 | 12 | | M,N,O,P,R,S | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 65 | 130 | 235 | 300 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




40-2

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-2) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.656 (16.66) | +.320 (8.13) |
| B | -.656 (16.66) | -.320 (8.13) |
| C | -.359 (9.12) | +.640 (16.26) |
| D | -.359 (9.12) | +.000 (0.00) |
| E | -.359 (9.12) | -.640 (16.26) |
| F | +.000 (0.00) | +.800 (20.32) |
| G | +.000 (0.00) | +.480 (12.19) |
| H | +.000 (0.00) | +.160 (4.06) |
| J | +.000 (0.00) | -.160 (4.06) |
| K | +.000 (0.00) | -.480 (12.19) |
| L | +.000 (0.00) | -.800 (20.32) |
| M | +.250 (6.35) | +.640 (16.26) |

| Contacts (Insert arrangement 40-2) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | +.250 (6.35) | +.320 (8.13) |
| O | +.250 (6.35) | +.000 (0.00) |
| P | +.250 (6.35) | -.320 (8.13) |
| R | +.250 (6.35) | -.640 (16.26) |
| S | +.500 (12.70) | +.480 (12.19) |
| T | +.500 (12.70) | +.160 (4.06) |
| U | +.500 (12.70) | -.160 (4.06) |
| V | +.500 (12.70) | -.480 (12.19) |
| W | +.750 (19.05) | +.320 (8.13) |
| X | +.750 (19.05) | +.000 (0.00) |
| Z | +.750 (19.05) | -.320 (8.13) |
| --- | --- | --- |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

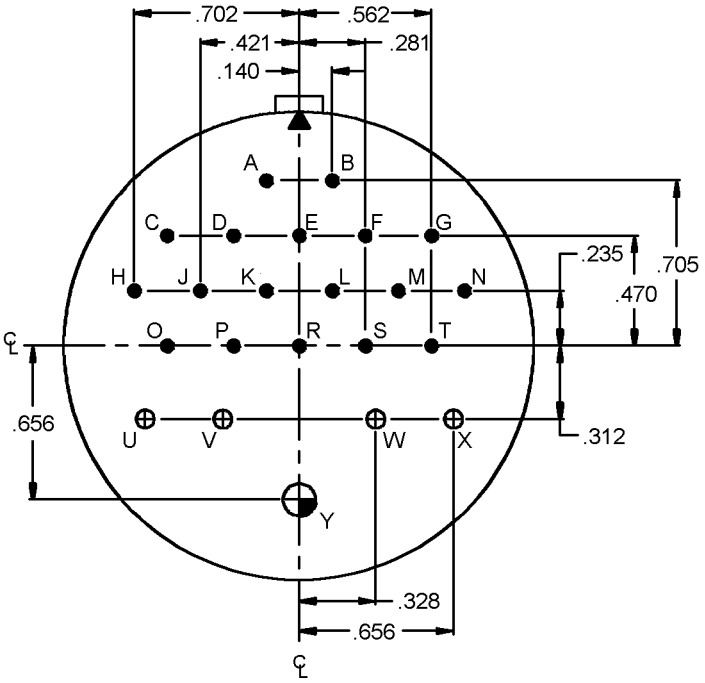
| Insert arrangement 40-2 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -2 | 5 | 16 | B | A,B,C,D,E | MS33691-2 |
| | | 18 | 16 | D | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




40-3

(Not applicable to MIL-DTL-22992)

| Contacts (Insert arrangement 40-3) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.140 (3.56) | +.705 (17.91) |
| B | +.140 (3.56) | +.705 (17.91) |
| C | -.562 (14.27) | +.470 (11.94) |
| D | -.281 (7.14) | +.470 (11.94) |
| E | +.000 (0.00) | +.470 (11.94) |
| F | +.281 (7.14) | +.470 (11.94) |
| G | +.562 (14.27) | +.470 (11.94) |
| H | -.702 (17.83) | +.235 (5.97) |
| J | -.421 (10.69) | +.235 (5.97) |
| K | -.140 (3.56) | +.235 (5.97) |
| L | +.140 (3.56) | +.235 (5.97) |
| M | +.421 (10.69) | +.235 (5.97) |

| Contacts (Insert arrangement 40-3) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | +.702 (17.83) | +.235 (5.97) |
| O | -.562 (14.27) | +.000 (0.00) |
| P | -.281 (7.14) | +.000 (0.00) |
| R | +.000 (0.00) | +.000 (0.00) |
| S | +.281 (7.14) | +.000 (0.00) |
| T | +.562 (14.27) | +.000 (0.00) |
| U | -.656 (16.66) | -.312 (7.92) |
| V | -.328 (8.33) | -.312 (7.92) |
| W | +.328 (8.33) | -.312 (7.92) |
| X | +.656 (16.66) | -.312 (7.92) |
| Y | +.000 (0.00) | -.656 (16.66) |
| --- | --- | --- |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

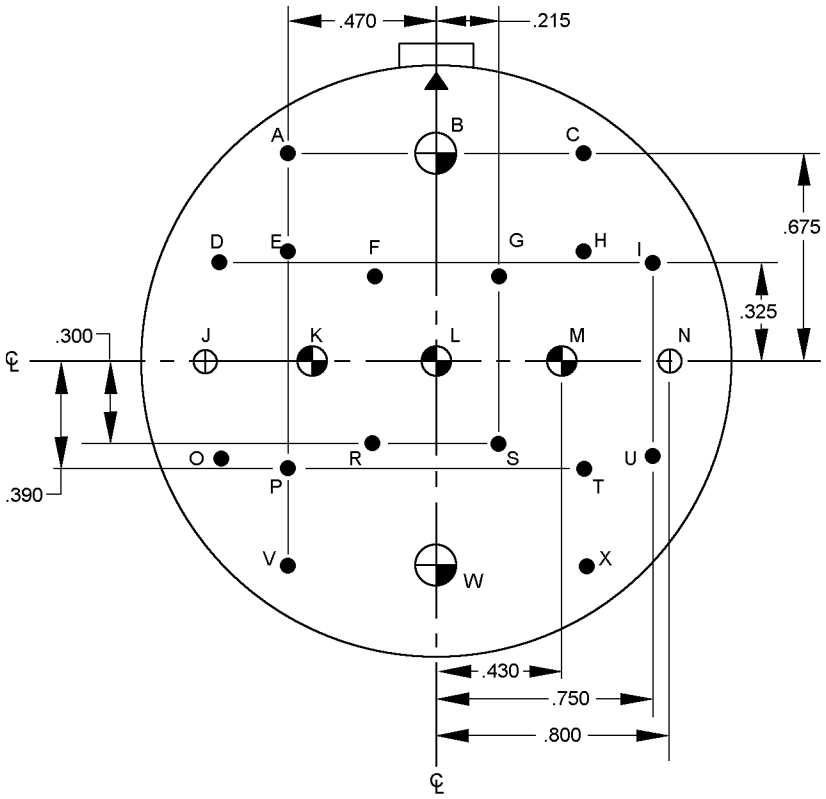
| Insert arrangement 40-3 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -3 | 18 | 16 | D | All others | MS33691-3 |
| | | 4 | 12 | D | U,V,W,X | |
| | | 1 | 4 | D | Y | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




40-4

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-4) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.470 (11.94) | +.675 (17.15) |
| B | +.000 (0.00) | +.675 (17.15) |
| C | +.470 (11.94) | +.675 (17.15) |
| D | -.750 (19.05) | +.325 (8.26) |
| E | -.470 (11.94) | +.390 (9.91) |
| F | -.215 (5.46) | +.300 (7.62) |
| G | +.215 (5.46) | +.300 (7.62) |
| H | +.470 (11.94) | +.390 (9.91) |
| I | +.750 (19.05) | +.325 (8.26) |
| J | -.800 (20.32) | +.000 (0.00) |
| K | -.430 (10.92) | +.000 (0.00) |
| L | +.000 (0.00) | +.000 (0.00) |

| Contacts (Insert arrangement 40-4) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| M | +.430 (10.92) | +.000 (0.00) |
| N | +.800 (20.32) | +.000 (0.00) |
| O | -.750 (19.05) | -.325 (8.26) |
| P | -.470 (11.94) | -.390 (9.91) |
| R | -.215 (5.46) | -.300 (7.62) |
| S | +.215 (5.46) | -.300 (7.62) |
| T | +.470 (11.94) | -.390 (9.91) |
| U | +.750 (19.05) | -.325 (8.26) |
| V | -.470 (11.94) | -.675 (17.15) |
| W | +.000 (0.00) | -.675 (17.15) |
| X | +.470 (11.94) | -.675 (17.15) |
| --- | --- | --- |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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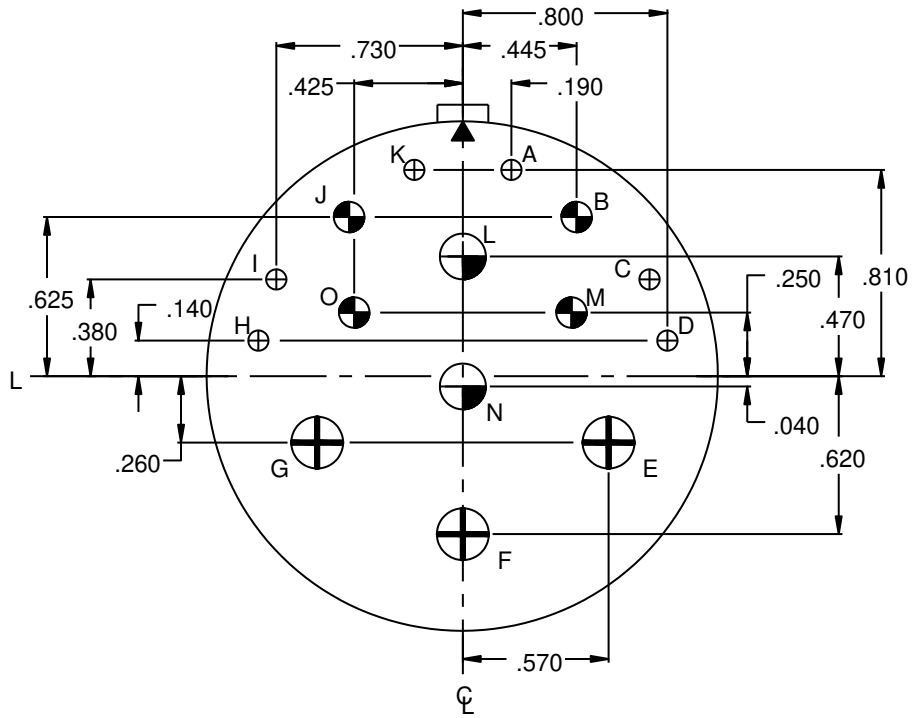
| Insert arrangement 40-4 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -4 | 16 | 16 | D | All others | MS33691-4 |
| | | 2 | 12 | D | J,N | |
| | | 3 | 8 | D | K,L,M | |
| | | 2 | 4 | D | B,W | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
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40-5

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-5) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.190 (4.83) | +.810 (20.57) |
| B | +.445 (11.30) | +.625 (15.88) |
| C | +.730 (18.54) | +.380 (9.65) |
| D | +.800 (20.32) | +.140 (3.56) |
| E | +.570 (14.48) | -.260 (6.60) |
| F | +.000 (0.00) | -.620 (15.75) |
| G | -.570 (14.48) | -.260 (6.60) |
| H | -.800 (20.32) | +.140 (3.56) |

| Contacts (Insert arrangement 40-5) | | |
|---|---|---|
| Contact position (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| I | -.730 (18.54) | +.380 (9.65) |
| J | -.445 (11.30) | +.625 (15.88) |
| K | -.190 (4.83) | +.810 (20.57) |
| L | +.000 (0.00) | +.470 (11.94) |
| M | +.425 (10.80) | +.250 (6.35) |
| N | +.000 (0.00) | -.040 (1.02) |
| O | -.425 (10.80) | +.250 (6.35) |
| --- | --- | --- |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

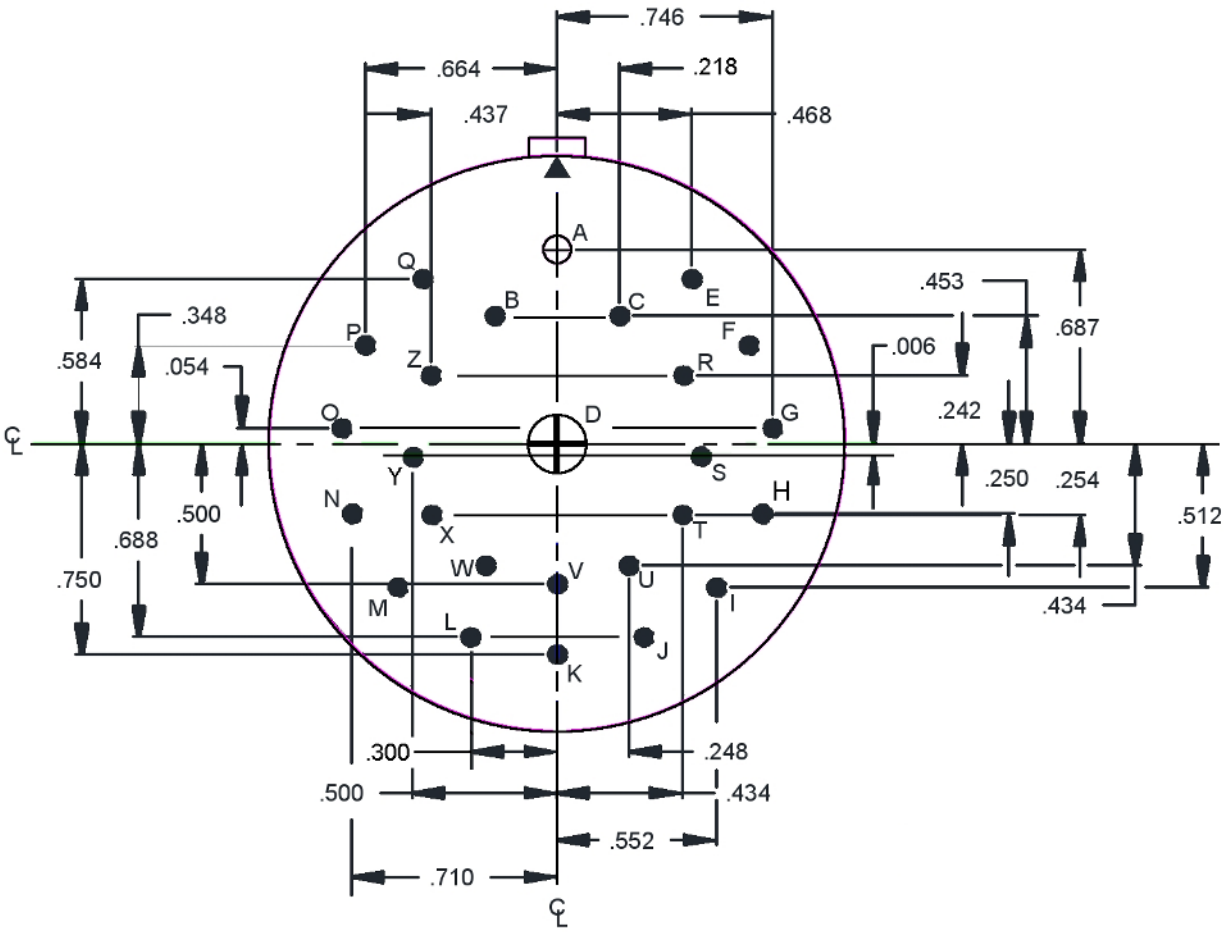
| Insert arrangement 40-5 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -5 | 6 | 12 | A | A,C,D,H,I,K | MS33691-5 |
| | | 4 | 8 | A | B,M,O,J | |
| | | 2 | 4 | A | L,N | |
| | | 3 | 0 | A | G,F,E | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



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40-6

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.687 (17.45) |
| B | -.218 (5.54) | +.453 (11.51) |
| C | +.218 (5.54) | +.453 (11.51) |
| D | +.000 (0.00) | +.000 (0.00) |
| E | +.468 (11.89) | +.584 (14.83) |
| F | +.664 (16.87) | +.348 (8.84) |
| G | +.746 (18.95) | +.054 (1.37) |
| H | +.710 (18.03) | -.250 (6.35) |
| I | +.552 (14.02) | -.512 (13.00) |
| J | +.300 (7.62) | -.688 (17.45) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-6) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| K | +.000 (0.00) | -.750 (19.05) |
| L | -.300 (7.62) | -.688 (17.45) |
| M | -.552 (14.02) | -.512 (13.00) |
| N | -.710 (18.03) | -.250 (6.35) |
| O | -.746 (18.95) | +.054 (1.37) |
| P | -.664 (16.87) | +.348 (8.84) |
| Q | -.468 (11.89) | +.584 (14.84) |
| R | +.437 (11.10) | +.242 (6.15) |
| S | +.500 (12.70) | -.006 (0.15) |
| T | +.434 (11.02) | -.254 (6.15) |
| U | +.248 (6.30) | -.434 (11.02) |
| V | +.000 (0.00) | -.500 (12.70) |
| W | -.248 (6.30) | -.434 (11.02) |
| X | -.434 (11.02) | -.254 (6.15) |
| Y | -.500 (12.70) | -.006 (0.15) |
| Z | -.437 (11.10) | +.242 (6.15) |

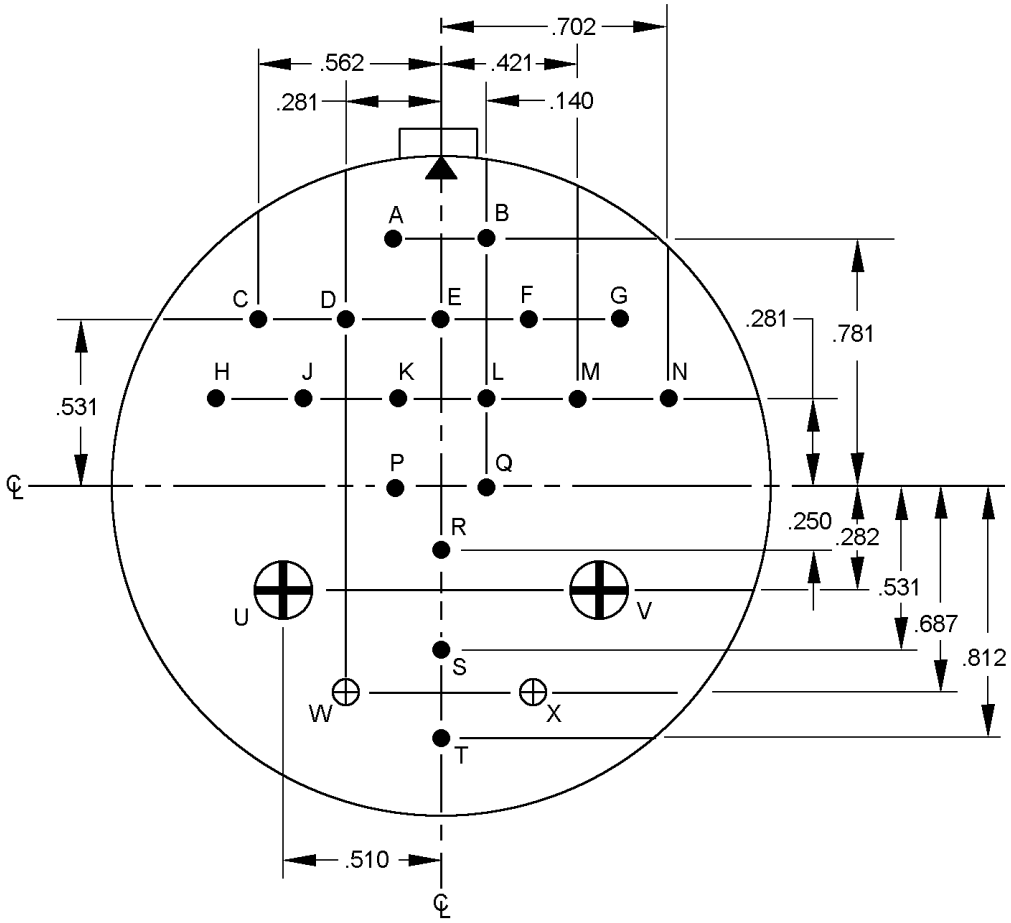
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -6 | 24 | 16 | D | All others | MS33691-6 |
| | | 1 | 12 | D | A | |
| | | 1 | 0 | D | D | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



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40-7

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.140 (3.56) | +.781 (19.84) |
| B | +.140 (3.56) | +.781 (19.84) |
| C | -.562 (14.27) | +.531 (13.49) |
| D | -.281 (7.14) | +.531 (13.49) |
| E | +.000 (0.00) | +.531 (13.49) |
| F | +.281 (7.14) | +.531 (13.49) |
| G | +.562 (14.27) | +.531 (13.49) |
| H | -.702 (17.83) | +.281 (7.14) |
| J | -.421 (10.69) | +.281 (7.14) |
| K | -.140 (3.56) | +.281 (7.14) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-7) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| L | +.140 (3.56) | +.281 (7.14) |
| M | +.421 (10.69) | +.281 (7.14) |
| N | +.702 (17.83) | +.281 (7.14) |
| P | -.140 (3.56) | +.000 (0.00) |
| Q | +.140 (3.56) | +.000 (0.00) |
| R | +.000 (0.00) | -.250 (6.35) |
| S | +.000 (0.00) | -.531 (13.49) |
| T | +.000 (0.00) | -.812 (20.62) |
| U | -.510 (12.95) | -.282 (7.16) |
| V | +.510 (12.95) | -.282 (7.16) |
| W | -.281 (7.14) | -.687 (17.45) |
| X | +.281 (7.14) | -.687 (17.45) |

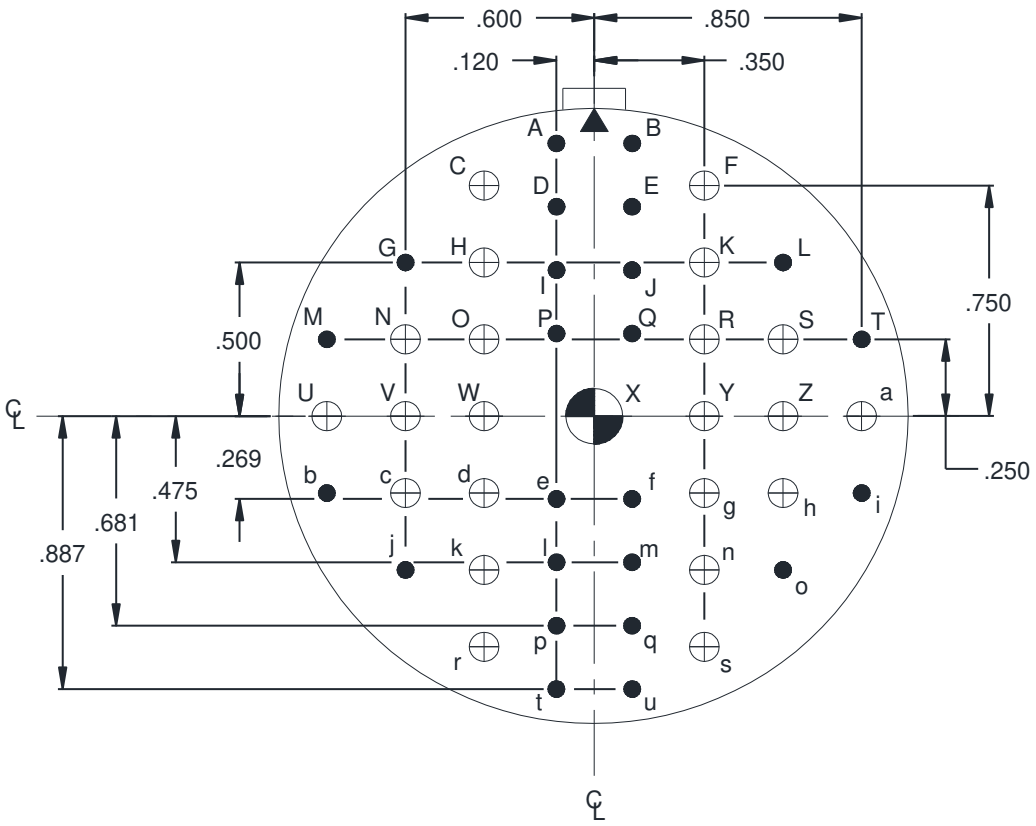
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -7 | 2 | 12 | A | W,X | MS33691-7 |
| | | 2 | 0 | A | U,V | |
| | | 2 | 16 | A | P,Q | |
| | | 16 | 16 | D | All others | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



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40-9

| Contacts (Insert arrangement 40-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.120 (3.05) | +.887 (22.53) |
| B | +.120 (3.05) | +.887 (22.53) |
| C | -.350 (8.89) | +.750 (19.05) |
| D | -.120 (3.05) | +.681 (17.30) |
| E | +.120 (3.05) | +.681 (17.30) |
| F | +.350 (8.89) | +.750 (19.05) |
| G | -.600 (15.24) | +.500 (12.70) |
| H | -.350 (8.89) | +.500 (12.70) |
| I | -.120 (3.05) | +.475 (12.07) |
| J | +.120 (3.05) | +.475 (12.07) |
| K | +.350 (8.89) | +.500 (12.70) |
| L | +.600 (15.24) | +.500 (12.70) |
| M | -.850 (21.59) | +.250 (6.35) |
| N | -.600 (15.24) | +.250 (6.35) |
| O | -.350 (8.89) | +.250 (6.35) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-9) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | -.120 (3.05) | +.269 (6.83) |
| Q | +.120 (3.05) | +.269 (6.83) |
| R | +.350 (8.89) | +.250 (6.35) |
| S | +.600 (15.24) | +.250 (6.35) |
| T | +.850 (21.59) | +.250 (6.35) |
| U | -.850 (21.59) | +.000 (0.00) |
| V | -.600 (15.24) | +.000 (0.00) |
| W | -.350 (8.89) | +.000 (0.00) |
| X | +.000 (0.00) | +.000 (0.00) |
| Y | +.350 (8.89) | +.000 (0.00) |
| Z | +.600 (15.24) | +.000 (0.00) |
| a | +.850 (21.59) | +.000 (0.00) |
| b | -.850 (21.59) | -.250 (6.35) |
| c | -.600 (15.24) | -.250 (6.35) |
| d | -.350 (8.89) | -.250 (6.35) |
| e | -.120 (3.05) | -.269 (6.83) |
| f | +.120 (3.05) | -.269 (6.83) |
| g | +.350 (8.89) | -.250 (6.35) |
| h | +.600 (15.24) | -.250 (6.35) |
| i | +.850 (21.59) | -.250 (6.35) |
| j | -.600 (15.24) | -.500 (12.70) |
| k | -.350 (8.89) | -.500 (12.70) |
| l | -.120 (3.05) | -.475 (12.07) |
| m | +.120 (3.05) | -.475 (12.07) |
| n | +.350 (8.89) | -.500 (12.70) |
| o | +.600 (15.24) | -.500 (12.70) |
| p | -.120 (3.05) | -.681 (17.30) |
| q | +.120 (3.05) | -.681 (17.30) |
| r | -.350 (8.89) | -.750 (19.05) |
| s | +.350 (8.89) | -.750 (19.05) |
| t | -.120 (3.05) | -.887 (22.53) |
| u | +.120 (3.05) | -.887 (22.53) |

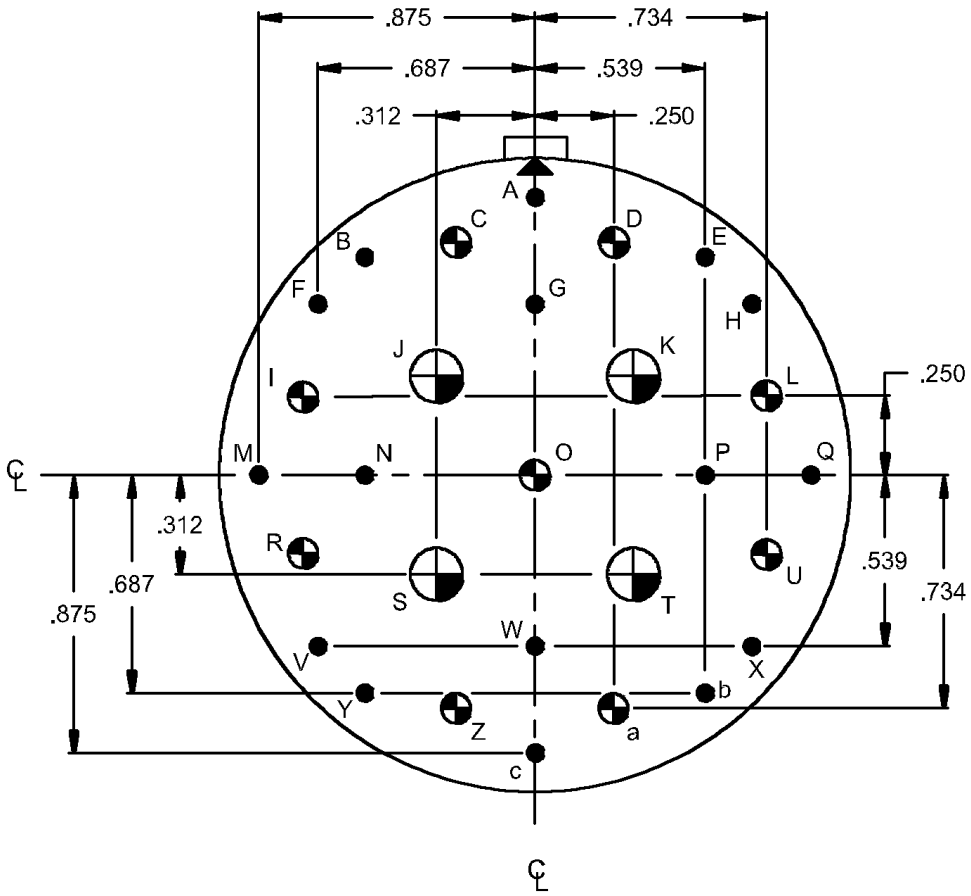
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -9 | 24 | 16 | A | All others | MS33691-9 |
| | | 22 | 12 | | C,F,H,K,N,O,R,S,U,V,W,Y,Z, a,c,d,g,h,k,n,r,s | |
| | | 1 | 8 | | X | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 65 | 125 | 225 | 310 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



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40-10

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.000 (0.00) | +.875 (22.23) |
| B | -.539 (13.69) | +.687 (17.45) |
| C | -.250 (6.35) | +.734 (18.64) |
| D | +.250 (6.35) | +.734 (18.64) |
| E | +.539 (13.69) | +.687 (17.45) |
| F | -.687 (17.45) | +.539 (13.69) |
| G | +.000 (0.00) | +.539 (13.69) |
| H | +.687 (17.45) | +.539 (13.69) |
| I | -.734 (18.64) | +.250 (6.35) |
| J | -.312 (7.92) | +.312 (7.92) |
| K | +.312 (7.92) | +.312 (7.92) |
| L | +.734 (18.64) | +.250 (6.35) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-10) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| M | -.875 (22.23) | +.000 (0.00) |
| N | -.539 (13.69) | +.000 (0.00) |
| O | +.000 (0.00) | +.000 (0.00) |
| P | +.539 (13.69) | +.000 (0.00) |
| Q | +.875 (22.23) | +.000 (0.00) |
| R | -.734 (18.64) | -.250 (6.35) |
| S | -.312 (7.92) | -.312 (7.92) |
| T | +.312 (7.92) | -.312 (7.92) |
| U | +.734 (18.64) | -.250 (6.35) |
| V | -.687 (17.45) | -.539 (13.69) |
| W | +.000 (0.00) | -.539 (13.69) |
| X | +.687 (17.45) | -.539 (13.69) |
| Y | -.539 (13.69) | -.687 (17.45) |
| Z | -.250 (6.35) | -.734 (18.64) |
| a | +.250 (6.35) | -.734 (18.64) |
| b | +.539 (13.69) | -.687 (17.45) |
| c | +.000 (0.00) | -.875 (22.23) |

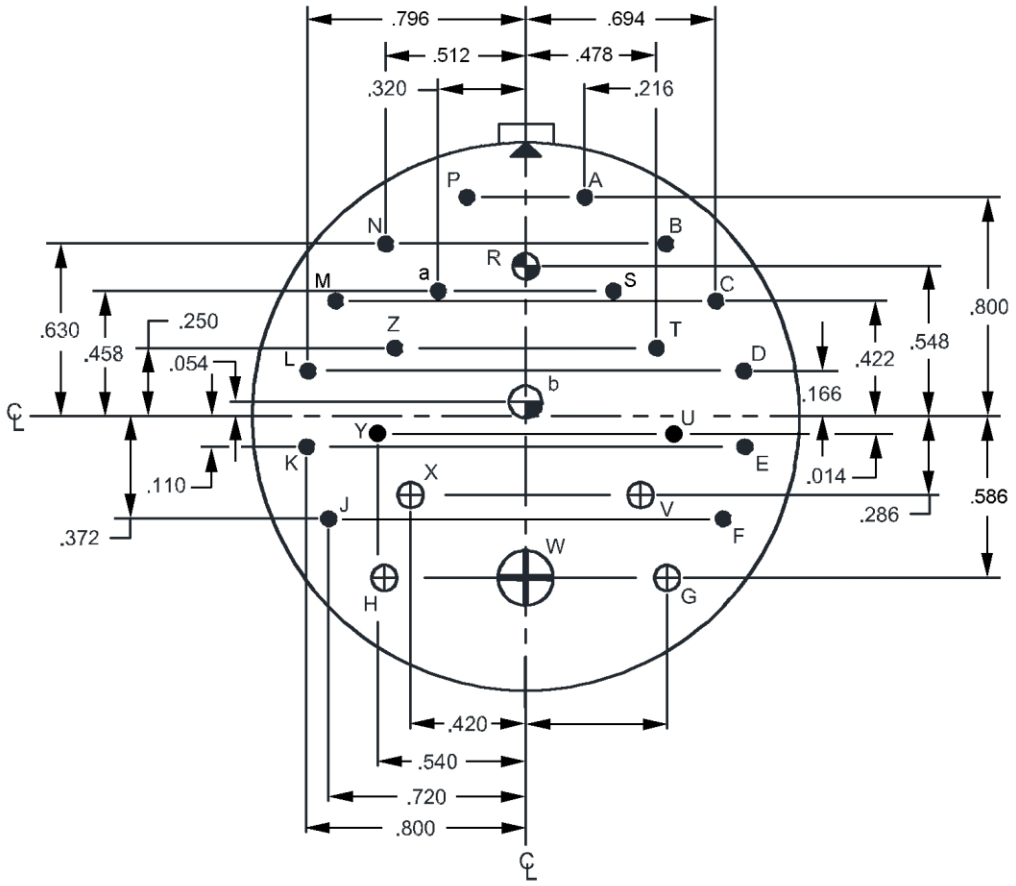
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -10 | 16 | 16 | A | All others | MS33691-10 |
| | | 9 | 8 | A | C,D,I,L,O,R,U,Z,a | |
| | | 4 | 4 | A | J,K,S,T | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 65 | 125 | 225 | 310 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



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40-11

(Not applicable to MIL-DTL-22992.)

| Contacts (Insert arrangement 40-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | +.216 (5.49) | +.800 (20.32) |
| B | +.512 (13.00) | +.630 (16.00) |
| C | +.694 (17.63) | +.422 (10.72) |
| D | +.796 (20.22) | +.166 (4.22) |
| E | +.800 (20.32) | -.110 (2.79) |
| F | +.720 (18.29) | -.372 (9.45) |
| G | +.516 (13.11) | -.586 (14.88) |
| H | -.516 (13.11) | -.586 (14.88) |
| J | -.720 (18.29) | -.372 (9.45) |
| K | -.800 (20.32) | -.110 (2.79) |
| L | -.796 (20.22) | +.166 (4.22) |
| M | -.694 (17.63) | +.422 (10.72) |
| N | -.512 (13.00) | +.630 (16.00) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-11) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | -.216 (5.49) | +.800 (20.32) |
| R | +.000 (0.00) | +.548 (13.92) |
| S | +.320 (8.13) | +.458 (11.63) |
| T | +.478 (12.14) | +.250 (6.35) |
| U | +.540 (13.72) | -.014 (0.36) |
| V | +.420 (10.67) | -.286 (7.26) |
| W | +.000 (0.00) | -.586 (14.88) |
| X | -.420 (10.67) | -.286 (7.26) |
| Y | -.540 (13.72) | -.014 (0.36) |
| Z | -.478 (12.14) | +.250 (6.35) |
| a | -.320 (8.13) | +.458 (11.63) |
| b | +.000 (0.00) | +.054 (1.37) |

| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 40 | -11 | 18 | 16 | D | All others | MS33691-11 |
| | | 4 | 12 | D | G,H,X,V | |
| | | 1 | 8 | D | R | |
| | | 1 | 4 | D | b | |
| | | 1 | 0 | D | W | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 80 | 110 | 250 | 280 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

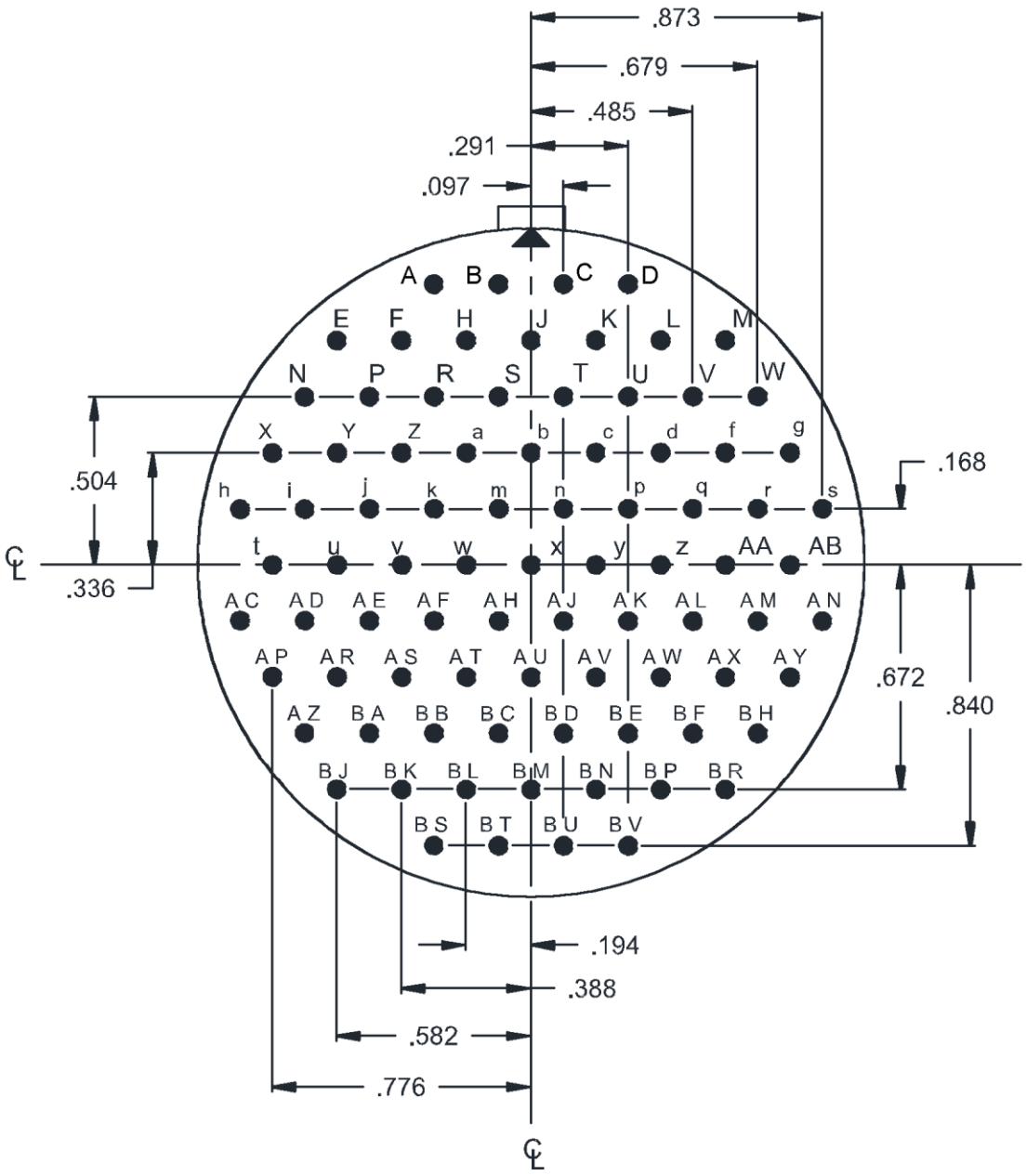



FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert Arrangement 40-56) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.291 (7.39) | +.840 (21.34) |
| B | -.097 (2.46) | +.840 (21.34) |
| C | +.097 (2.46) | +.840 (21.34) |
| D | +.291 (7.39) | +.840 (21.34) |
| E | -.582 (14.78) | +.672 (17.07) |
| F | -.388 (9.86) | +.672 (17.07) |
| H | -.194 (4.93) | +.672 (17.07) |
| J | +.000 (0.00) | +.672 (17.07) |
| K | +.194 (4.93) | +.672 (17.07) |
| L | +.388 (9.86) | +.672 (17.07) |
| M | +.582 (14.78) | +.672 (17.07) |
| N | -.679 (17.25) | +.504 (12.80) |
| P | -.485 (12.32) | +.504 (12.80) |
| R | -.291 (7.39) | +.504 (12.80) |
| S | -.097 (2.46) | +.504 (12.80) |
| T | +.097 (2.46) | +.504 (12.80) |
| U | +.291 (7.39) | +.504 (12.80) |
| V | +.485 (12.32) | +.504 (12.80) |
| W | +.679 (17.25) | +.504 (12.80) |
| X | -.776 (19.71) | +.336 (8.53) |
| Y | -.582 (14.78) | +.336 (8.53) |
| Z | -.388 (9.86) | +.336 (8.53) |
| a | -.194 (4.93) | +.336 (8.53) |
| b | +.000 (0.00) | +.336 (8.53) |
| c | +.194 (4.93) | +.336 (8.53) |
| d | +.388 (9.86) | +.336 (8.53) |
| f | +.582 (14.78) | +.336 (8.53) |
| g | +.776 (19.71) | +.336 (8.53) |
| h | -.873 (22.17) | +.168 (4.27) |
| i | -.679 (17.25) | +.168 (4.27) |
| j | -.485 (12.32) | +.168 (4.27) |
| k | -.291 (7.39) | +.168 (4.27) |
| m | -.097 (2.46) | +.168 (4.27) |
| n | +.097 (2.46) | +.168 (4.27) |
| p | +.291 (7.39) | +.168 (4.27) |
| q | +.485 (12.32) | +.168 (4.27) |
| r | +.679 (17.25) | +.168 (4.27) |
| s | +.873 (22.17) | +.168 (4.27) |
| t | -.776 (19.71) | +.000 (0.00) |
| u | -.582 (14.78) | +.000 (0.00) |
| v | -.388 (9.86) | +.000 (0.00) |
| w | -.194 (4.93) | +.000 (0.00) |
| x | +.000 (0.00) | +.000 (0.00) |
| y | +.194 (4.93) | +.000 (0.00) |
| z | +.388 (9.86) | +.000 (0.00) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert Arrangement 40-56) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| AA | +.582 (14.78) | +.000 (0.00) |
| AB | +.776 (19.71) | +.000 (0.00) |
| AC | -.873 (22.17) | -.168 (4.27) |
| AD | -.679 (17.25) | -.168 (4.27) |
| AE | -.485 (12.32) | -.168 (4.27) |
| AF | -.291 (7.39) | -.168 (4.27) |
| AH | -.097 (2.46) | -.168 (4.27) |
| AJ | +.097 (2.46) | -.168 (4.27) |
| AK | +.291 (7.39) | -.168 (4.27) |
| AL | +.485 (12.32) | -.168 (4.27) |
| AM | +.679 (17.25) | -.168 (4.27) |
| AN | +.873 (22.17) | -.168 (4.27) |
| AP | -.776 (19.71) | -.336 (8.53) |
| AR | -.582 (14.78) | -.336 (8.53) |
| AS | -.388 (9.86) | -.336 (8.53) |
| AT | -.194 (4.93) | -.336 (8.53) |
| AU | +.000 (0.00) | -.336 (8.53) |
| AV | +.194 (4.93) | -.336 (8.53) |
| AW | +.388 (9.86) | -.336 (8.53) |
| AX | +.582 (14.78) | -.336 (8.53) |
| AY | +.776 (19.71) | -.336 (8.53) |
| AZ | -.679 (17.25) | -.504 (12.80) |
| BA | -.485 (12.32) | -.504 (12.80) |
| BB | -.291 (7.39) | -.504 (12.80) |
| BC | -.097 (2.46) | -.504 (12.80) |
| BD | +.097 (2.46) | -.504 (12.80) |
| BE | +.291 (7.39) | -.504 (12.80) |
| BF | +.485 (12.32) | -.504 (12.80) |
| BH | +.679 (17.25) | -.504 (12.80) |
| BJ | -.582 (14.78) | -.672 (17.07) |
| BK | -.388 (9.86) | -.672 (17.07) |
| BL | -.194 (4.93) | -.672 (17.07) |
| BM | +.000 (0.00) | -.672 (17.07) |
| BN | +.194 (4.93) | -.672 (17.07) |
| BP | +.388 (9.86) | -.672 (17.07) |
| BR | +.582 (14.78) | -.672 (17.07) |
| BS | -.291 (7.39) | -.840 (21.34) |
| BT | -.097 (2.46) | -.840 (21.34) |
| BU | +.097 (2.46) | -.840 (21.34) |
| BV | +.291 (7.39) | -.840 (21.34) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

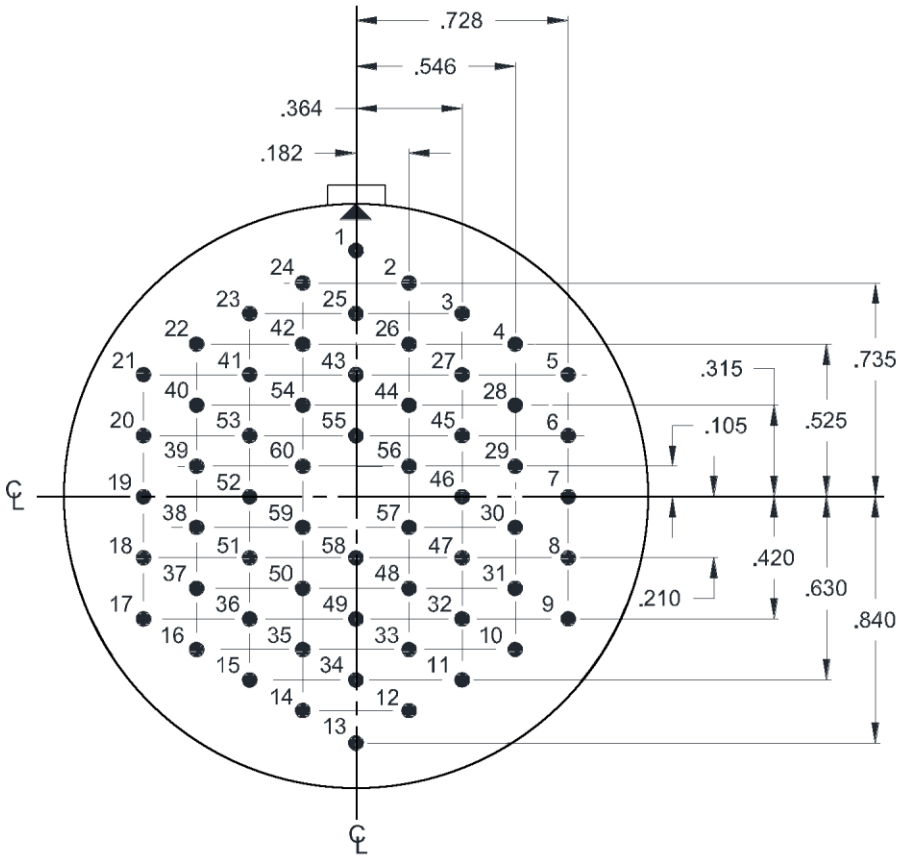
| Insert Arrangement 40-56 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -56 | 85 | 16 | A | All | MS33691-56 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 72 | 144 | 216 | 288 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




40-62

| Contacts (Insert arrangement 40-62) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 1 | +.000 (0.00) | +.840 (21.34) |
| 2 | +.182 (4.62) | +.735 (18.67) |
| 3 | +.364 (9.25) | +.630 (16.00) |
| 4 | +.546 (13.87) | +.525 (13.34) |
| 5 | +.728 (18.49) | +.420 (10.67) |
| 6 | +.728 (18.49) | +.210 (5.33) |
| 7 | +.728 (18.49) | +.000 (0.00) |
| 8 | +.728 (18.49) | -.210 (5.33) |
| 9 | +.728 (18.49) | -.420 (10.67) |
| 10 | +.546 (13.87) | -.525 (13.34) |
| 11 | +.364 (9.25) | -.630 (16.00) |
| 12 | +.182 (4.62) | -.735 (18.67) |
| 13 | +.000 (0.00) | -.840 (21.34) |
| 14 | -.182 (4.62) | -.735 (18.67) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 40-62) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 15 | -.364 (9.25) | -.630 (16.00) |
| 16 | -.546 (13.87) | -.525 (13.34) |
| 17 | -.728 (18.49) | -.420 (10.67) |
| 18 | -.728 (18.49) | -.210 (5.33) |
| 19 | -.728 (18.49) | +.000 (0.00) |
| 20 | -.728 (18.49) | +.210 (5.33) |
| 21 | -.728 (18.49) | +.420 (10.67) |
| 22 | -.546 (13.87) | +.525 (13.34) |
| 23 | -.364 9.25) | +.630 (16.00) |
| 24 | -.182 (4.62) | +.735 (18.67) |
| 25 | +.000 (0.00) | +.630 (16.00) |
| 26 | +.182 (4.62) | +.525 (13.34) |
| 27 | +.364 (9.25) | +.420 (10.67) |
| 28 | +.546 (13.87) | +.315 (8.00) |
| 29 | +.546 (13.87) | +.105 (2.67) |
| 30 | +.546 (13.87) | -.105 (2.67) |
| 31 | +.546 (13.87) | -.315 (8.00) |
| 32 | +.364 (9.25) | -.420 (10.67) |
| 33 | +.182 (4.62) | -.525 (13.34) |
| 34 | +.000 (0.00) | -.630 (16.00) |
| 35 | -.182 (4.62) | -.525 (13.34) |
| 36 | -.364 (9.25) | -.420 (10.67) |
| 37 | -.546 (13.87) | -.315 (8.00) |
| 38 | -.546 (13.87) | -.105 (2.67) |
| 39 | -.546 (13.87) | +.105 (2.67) |
| 40 | -.546 (13.87) | +.315 (8.00) |
| 41 | -.364 (9.25) | +.420 (10.67) |
| 42 | -.182 (4.62) | +.525 (13.34) |
| 43 | +.000 (0.00) | +.420 (10.67) |
| 44 | +.182 (4.62) | +.315 (8.00) |
| 45 | +.364 (9.25) | +.210 (5.33) |
| 46 | +.364 (9.25) | +.000 (0.00) |
| 47 | +.364 (9.25) | -.210 (5.33) |
| 48 | +.182 (4.62) | -.315 (8.00) |
| 49 | +.000 (0.00) | -.420 (10.67) |
| 50 | -.182 (4.62) | -.315 (8.00) |
| 51 | -.364 (9.25) | -.210 (5.33) |
| 52 | -.364 (9.25) | +.000 (0.00) |
| 53 | -.364 (9.25) | +.210 (5.33) |
| 54 | -.182 (4.62) | +.315 (8.00) |
| 55 | +.000 (0.00) | +.210 (5.33) |
| 56 | +.182 (4.62) | +.105 (2.67) |
| 57 | +.182 (4.62) | -.105 (2.67) |
| 58 | +.000 (0.00) | -.210 (5.33) |
| 59 | -.182 (4.62) | -.105 (2.67) |
| 60 | -.182 (4.62) | +.105 (2.67) |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

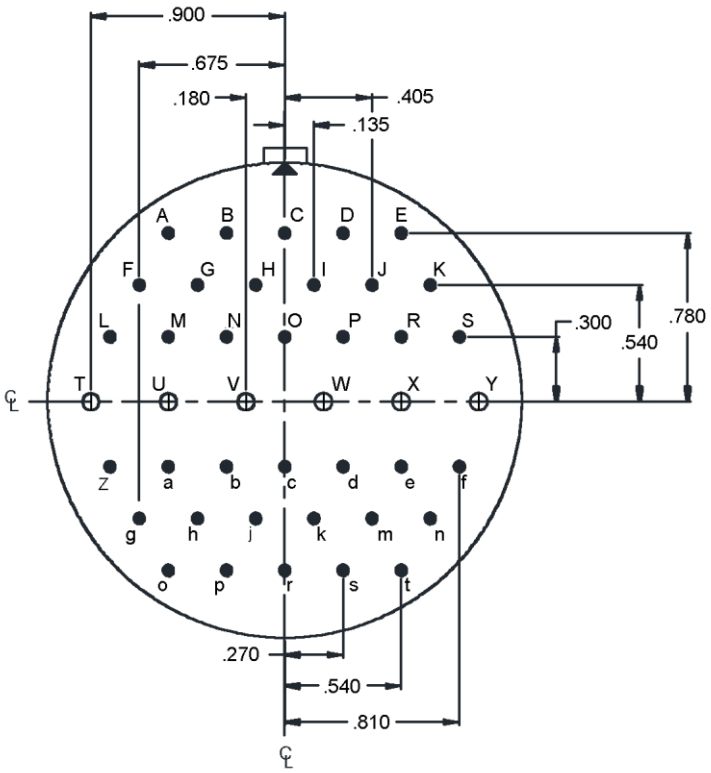
| Insert arrangement 40-62 | | | | | | |
|---|---|---|---|---|---|---|
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
| 40 | -62 | 60 | 16 | A | All | MS33691-62 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 30 | 130 | 220 | 290 |

FIGURE 13. Insert arrangement, electrical connector, shell size 40, MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




44-1

| Contacts (Insert arrangement 44-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.540 (13.72) | +.780 (19.81) |
| B | -.270 (6.86) | +.780 (19.81) |
| C | +.000 (0.00) | +.780 (19.81) |
| D | +.270 (6.86) | +.780 (19.81) |
| E | +.540 (13.72) | +.780 (19.81) |
| F | -.675 (17.15) | +.540 (13.72) |
| G | -.405 (10.29) | +.540 (13.72) |
| H | -.135 (3.43) | +.540 (13.72) |
| I | +.135 (3.43) | +.540 (13.72) |
| J | +.405 (10.29) | +.540 (13.72) |
| K | +.675 (17.15) | +.540 (13.72) |
| L | -.810 (20.57) | +.300 (7.62) |
| M | -.540 (13.72) | +.300 (7.62) |
| N | -.270 (6.86) | +.300 (7.62) |
| O | +.000 (0.00) | +.300 (7.62) |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert arrangement 44-1) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| P | +.270 (6.86) | +.300 (7.62) |
| R | +.540 (13.72) | +.300 (7.62) |
| S | +.810 (20.57) | +.300 (7.62) |
| T | -.900 (22.86) | +.000 (0.00) |
| U | -.540 (13.72) | +.000 (0.00) |
| V | -.180 (4.57) | +.000 (0.00) |
| W | +.180 (4.57) | +.000 (0.00) |
| X | +.540 (13.72) | +.000 (0.00) |
| Y | +.900 (22.86) | +.000 (0.00) |
| Z | -.810 (20.57) | -.300 (7.62) |
| a | -.540 (13.72) | -.300 (7.62) |
| b | -.270 (6.86) | -.300 (7.62) |
| c | +.000 (0.00) | -.300 (7.62) |
| d | +.270 (6.86) | -.300 (7.62) |
| e | +.540 (13.72) | -.300 (7.62) |
| f | +.810 (20.57) | -.300 (7.62) |
| g | -.675 (17.15) | -.540 (13.72) |
| h | -.405 (10.29) | -.540 (13.72) |
| j | -.135 (3.43) | -.540 (13.72) |
| k | +.135 (3.43) | -.540 (13.72) |
| m | +.405 (10.29) | -.540 (13.72) |
| n | +.675 (17.15) | -.540 (13.72) |
| o | -.540 (13.72) | -.780 (19.81) |
| p | -.270 (6.86) | -.780 (19.81) |
| r | +.000 (0.00) | -.780 (19.81) |
| s | +.270 (6.86) | -.780 (19.81) |
| t | +.540 (13.72) | -.780 (19.81) |

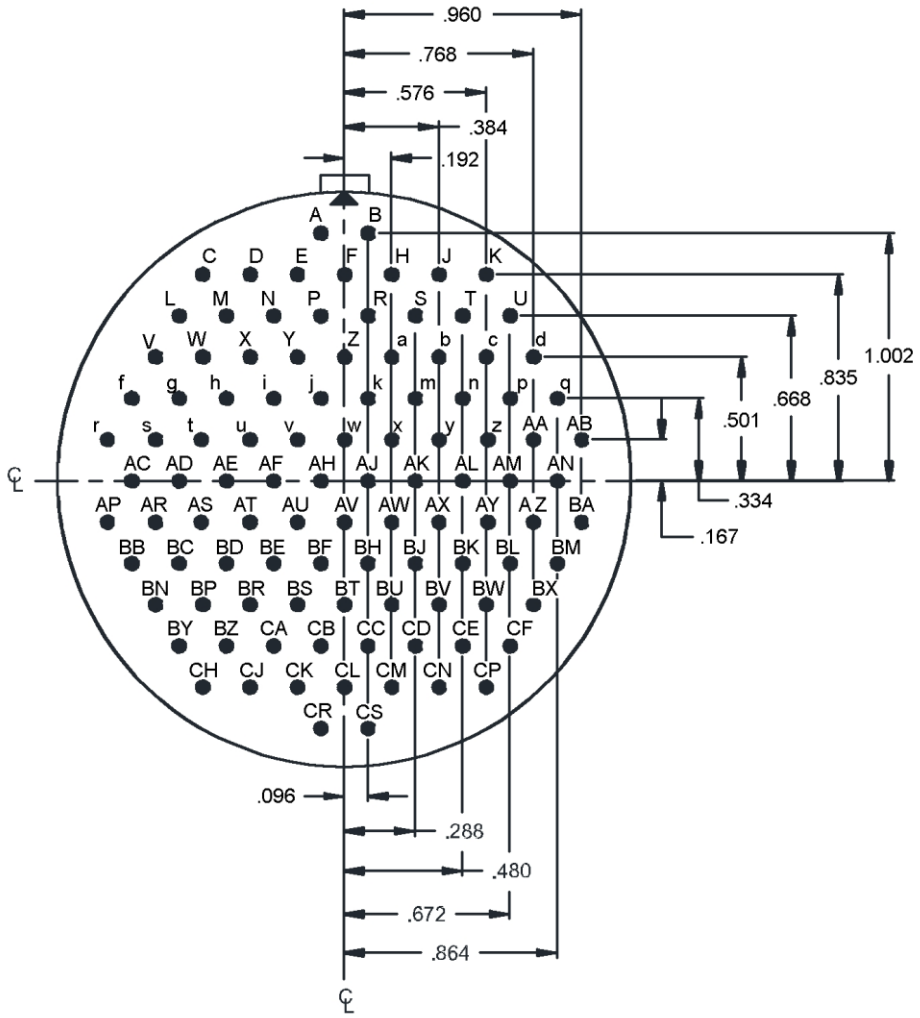
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 44 | -1 | 36 | 16 | D | All others | MS33692-1 |
| | | 6 | 12 | | T,U,V,W,X,Y | |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 65 | 125 | 225 | 310 |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.




44-52

| Contacts (Insert Arrangement 44-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| A | -.096 (2.44) | +1.002 (25.45) |
| B | +.096 (2.44) | +1.002 (25.45) |
| C | -.576 (14.63) | +.835 (21.21) |
| D | -.384 (9.75) | +.835 (21.21) |
| E | -.192 (4.88) | +.835 (21.21) |
| F | +.000 (0.00) | +.835 (21.21) |
| H | +.192 (4.88) | +.835 (21.21) |
| J | +.384 (9.75) | +.835 (21.21) |
| K | +.576 (14.63) | +.835 (21.21) |
| L | -.672 (17.07) | +.668 (16.97) |
| M | -.480 (12.19) | +.668 (16.97) |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert Arrangement 44-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| N | -.288 (7.32) | +.668 (16.97) |
| P | -.096 (2.44) | +.668 (16.97) |
| R | +.096 (2.44) | +.668 (16.97) |
| S | +.288 (7.32) | +.668 (16.97) |
| T | +.480 (12.19) | +.668 (16.97) |
| U | +.672 (17.07) | +.668 (16.97) |
| V | -.768 (19.51) | +.501 (12.73) |
| W | -.576 (14.63) | +.501 (12.73) |
| X | -.384 (9.75) | +.501 (12.73) |
| Y | -.192 (4.88) | +.501 (12.73) |
| Z | +.000 (0.00) | +.501 (12.73) |
| a | +.192 (4.88) | +.501 (12.73) |
| b | +.384 (9.75) | +.501 (12.73) |
| c | +.576 (14.63) | +.501 (12.73) |
| d | +.768 (19.51) | +.501 (12.73) |
| f | -.864 (21.95) | +.334 (8.48) |
| g | -.672 (17.07) | +.334 (8.48) |
| h | -.480 (12.19) | +.334 (8.48) |
| i | -.288 (7.32) | +.334 (8.48) |
| j | -.096 (2.44) | +.334 (8.48) |
| k | +.096 (2.44) | +.334 (8.48) |
| m | +.288 (7.32) | +.334 (8.48) |
| n | +.480 (12.19) | +.334 (8.48) |
| p | +.672 (17.07) | +.334 (8.48) |
| q | +.864 (21.95) | +.334 (8.48) |
| r | -.960 (24.38) | +.167 (4.24) |
| s | -.768 (19.51) | +.167 (4.24) |
| t | -.576 (14.63) | +.167 (4.24) |
| u | -.384 (9.75) | +.167 (4.24) |
| v | -.192 (4.88) | +.167 (4.24) |
| w | +.000 (0.00) | +.167 (4.24) |
| x | +.192 (4.88) | +.167 (4.24) |
| y | +.384 (9.75) | +.167 (4.24) |
| z | +.576 (14.63) | +.167 (4.24) |
| AA | +.768 (19.51) | +.167 (4.24) |
| AB | +.960 (24.38) | +.167 (4.24) |
| AC | -.864 (21.95) | +.000 (0.00) |
| AD | -.672 (17.07) | +.000 (0.00) |
| AE | -.480 (12.19) | +.000 (0.00) |
| AF | -.288 (7.32) | +.000 (0.00) |
| AH | -.096 (2.44) | +.000 (0.00) |
| AJ | +.096 (2.44) | +.000 (0.00) |
| AK | +.288 (7.32) | +.000 (0.00) |
| AL | +.480 (12.19) | +.000 (0.00) |
| AM | +.672 (17.07) | +.000 (0.00) |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:50Z
Check the source to verify that this is the current version before use.

| Contacts (Insert Arrangement 44-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| AN | +.864 (21.95) | +.000 (0.00) |
| AP | -.960 (24.38) | -.167 (4.24) |
| AR | -.768 (19.51) | -.167 (4.24) |
| AS | -.576 (14.63) | -.167 (4.24) |
| AT | -.384 (9.75) | -.167 (4.24) |
| AU | -.192 (4.88) | -.167 (4.24) |
| AV | +.000 (0.00) | -.167 (4.24) |
| AW | +.192 (4.88) | -.167 (4.24) |
| AX | +.384 (9.75) | -.167 (4.24) |
| AY | +.576 (14.63) | -.167 (4.24) |
| AZ | +.768 (19.51) | -.167 (4.24) |
| BA | +.960 (24.38) | -.167 (4.24) |
| BB | -.864 (21.95) | -.334 (8.46) |
| BC | -.672 (17.07) | -.334 (8.46) |
| BD | -.480 (12.19) | -.334 (8.46) |
| BE | -.288 (7.32) | -.334 (8.46) |
| BF | -.096 (2.44) | -.334 (8.46) |
| BH | +.096 (2.44) | -.334 (8.46) |
| BJ | +.288 (7.32) | -.334 (8.46) |
| BK | +.480 (12.19) | -.334 (8.46) |
| BL | +.672 (17.07) | -.334 (8.46) |
| BM | +.864 (21.95) | -.334 (8.46) |
| BN | -.768 (19.51) | -.501 (12.73) |
| BP | -.576 (14.63) | -.501 (12.73) |
| BR | -.384 (9.75) | -.501 (12.73) |
| BS | -.192 (4.88) | -.501 (12.73) |
| BT | +.000 (0.00) | -.501 (12.73) |
| BU | +.192 (4.88) | -.501 (12.73) |
| BV | +.384 (9.75) | -.501 (12.73) |
| BW | +.576 (14.63) | -.501 (12.73) |
| BX | +.768 (19.51) | -.501 (12.73) |
| BY | -.672 (17.07) | -.668 (16.97) |
| BZ | -.480 (12.19) | -.668 (16.97) |
| CA | -.288 (7.32) | -.668 (16.97) |
| CB | -.096 (2.44) | -.668 (16.97) |
| CC | +.096 (2.44) | -.668 (16.97) |
| CD | +.288 (7.32) | -.668 (16.97) |
| CE | +.480 (12.19) | -.668 (16.97) |
| CF | +.672 (17.07) | -.668 (16.97) |
| CH | -.576 (14.63) | -.835 (21.21) |
| CJ | -.384 (9.75) | -.835 (21.21) |
| CK | -.192 (4.88) | -.835 (21.21) |
| CL | +.000 (0.00) | -.835 (21.21) |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



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| Contacts (Insert Arrangement 44-52) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| CM | +.192 (4.88) | -.835 (21.21) |
| CN | +.384 (9.75) | -.835 (21.21) |
| CP | +.576 (14.63) | -.835 (21.21) |
| CR | -.096 (2.44) | -1.002 (25.45) |
| CS | +.096 (2.44) | -1.002 (25.45) |

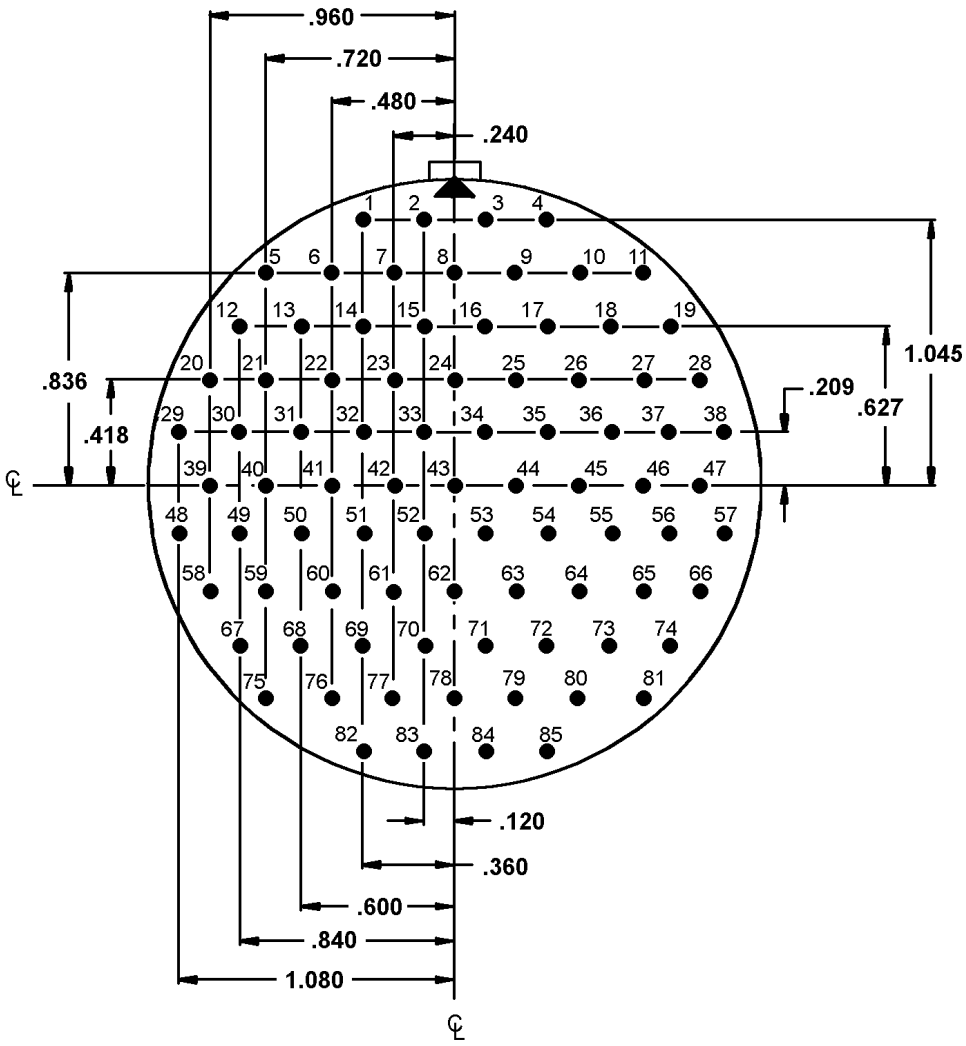
| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 44 | -52 | 104 | 16 | A | All | MS33692-52 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 72 | 135 | 225 | 288 |

FIGURE 14. Insert arrangement, electrical connector, shell size 44 (MIL-DTL-22992, classes C, J and R, MIL-DTL-83723, series II, and SAE-AS50151 and SAE-AS50151).



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48-62

(Shell size 48 applies to SAE-AS5015 only.)

| Contacts (Insert arrangement 48-62) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 1 | -.360 (9.14) | +1.045 (26.54) |
| 2 | -.120 (3.05) | +1.045 (26.54) |
| 3 | +.120 (3.05) | +1.045 (26.54) |
| 4 | +.360 (9.14) | +1.045 (26.54) |
| 5 | -.720 (18.29) | +.836 (21.23) |
| 6 | -.480 (12.19) | +.836 (21.23) |

FIGURE 15. Insert arrangement, electrical connector, shell size 48, SAE-AS50151.



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| Contacts (Insert arrangement 48-62) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | Y-axis (mm) |
| 7 | -.240 (6.10) | +.836 (21.23) |
| 8 | +.000 (0.00) | +.836 (21.23) |
| 9 | +.240 (6.10) | +.836 (21.23) |
| 10 | +.480 (12.19) | +.836 (21.23) |
| 11 | +.720 (18.29) | +.836 (21.23) |
| 12 | -.840 (21.34) | +.627 (15.93) |
| 13 | -.600 (15.24) | +.627 (15.93) |
| 14 | -.360 (9.14) | +.627 (15.93) |
| 15 | -.120 (3.05) | +.627 (15.93) |
| 16 | +.120 (3.05) | +.627 (15.93) |
| 17 | +.360 (9.14) | +.627 (15.93) |
| 18 | +.600 (15.24) | +.627 (15.93) |
| 19 | +.840 (21.34) | +.627 (15.93) |
| 20 | -.960 (24.38) | +.418 (10.62) |
| 21 | -.720 (18.29) | +.418 (10.62) |
| 22 | -.480 (12.19) | +.418 (10.62) |
| 23 | -.240 (6.10) | +.418 (10.62) |
| 24 | +.000 (0.00) | +.418 (10.62) |
| 25 | +.240 (6.10) | +.418 (10.62) |
| 26 | +.480 (12.19) | +.418 (10.62) |
| 27 | +.720 (18.29) | +.418 (10.62) |
| 28 | +.960 (24.38) | +.418 (10.62) |
| 29 | -1.080 (27.43) | +.209 (5.31) |
| 30 | -.840 (21.34) | +.209 (5.31) |
| 31 | -.600 (15.24) | +.209 (5.31) |
| 32 | -.360 (9.14) | +.209 (5.31) |
| 33 | -.120 (3.05) | +.209 (5.31) |
| 34 | +.120 (3.05) | +.209 (5.31) |
| 35 | +.360 (9.14) | +.209 (5.31) |
| 36 | +.600 (15.24) | +.209 (5.31) |
| 37 | +.840 (21.34) | +.209 (5.31) |
| 38 | +1.080 (27.43) | +.209 (5.31) |
| 39 | -.960 (24.38) | +.000 (0.00) |
| 40 | -.720 (18.29) | +.000 (0.00) |
| 41 | -.480 (12.19) | +.000 (0.00) |
| 42 | -.240 (6.10) | +.000 (0.00) |
| 43 | +.000 (0.00) | +.000 (0.00) |
| 44 | +.240 (6.10) | +.000 (0.00) |
| 45 | +.480 (12.19) | +.000 (0.00) |
| 46 | +.720 (18.29) | +.000 (0.00) |
| 47 | +.960 (24.38) | +.000 (0.00) |
| 48 | -1.080 (27.43) | -.209 (5.31) |
| 49 | -.840 (21.34) | -.209 (5.31) |
| 50 | -.600 (15.24) | -.209 (5.31) |
| 51 | -.360 (9.14) | -.209 (5.31) |

FIGURE 15. Insert arrangement, electrical connector, shell size 48, SAE-AS50151.



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| Contacts (Insert arrangement 48-62) | | |
|---|---|---|
| Contact position identification (ID) | Contact location | |
| | X-axis (mm) | X-axis (mm) |
| 52 | -.120 (3.05) | -.209 (5.31) |
| 53 | +.120 (3.05) | -.209 (5.31) |
| 54 | +.360 (9.14) | -.209 (5.31) |
| 55 | +.600 (15.24) | -.209 (5.31) |
| 56 | +.840 (21.34) | -.209 (5.31) |
| 57 | +1.080 (27.43) | -.209 (5.31) |
| 58 | -.960 (24.38) | -.418 (10.62) |
| 59 | -.720 (18.29) | -.418 (10.62) |
| 60 | -.480 (12.19) | -.418 (10.62) |
| 61 | -.240 (6.10) | -.418 (10.62) |
| 62 | +.000 (0.00) | -.418 (10.62) |
| 63 | +.240 (6.10) | -.418 (10.62) |
| 64 | +.480 (12.19) | -.418 (10.62) |
| 65 | +.720 (18.29) | -.418 (10.62) |
| 66 | +.960 (24.38) | -.418 (10.62) |
| 67 | -.840 (21.34) | -.627 (15.93) |
| 68 | -.600 (15.24) | -.627 (15.93) |
| 69 | -.360 (9.14) | -.627 (15.93) |
| 70 | -.120 (3.05) | -.627 (15.93) |
| 71 | +.120 (3.05) | -.627 (15.93) |
| 72 | +.360 (9.14) | -.627 (15.93) |
| 73 | +.600 (15.24) | -.627 (15.93) |
| 74 | +.840 (21.34) | -.627 (15.93) |
| 75 | -.720 (18.29) | -.836 (21.23) |
| 76 | -.480 (12.19) | -.836 (21.23) |
| 77 | -.240 (6.10) | -.836 (21.23) |
| 78 | +.000 (0.00) | -.836 (21.23) |
| 79 | +.240 (6.10) | -.836 (21.23) |
| 80 | +.480 (12.19) | -.836 (21.23) |
| 81 | +.720 (18.29) | -.836 (21.23) |
| 82 | -.360 (9.14) | -1.045 (26.54) |
| 83 | -.120 (3.05) | -1.045 (26.54) |
| 84 | +.120 (3.05) | -1.045 (26.54) |
| 85 | +.360 (9.14) | -1.045 (26.54) |

| Shell size | Arrangement number | Number of contacts | Size contacts | Service rating | Contact location | Supersedes |
|---|---|---|---|---|---|---|
| 48 | -62 | 85 | 16 | D | All | MS33693-62 |

| Alternate insert positions | | | |
|---|---|---|---|
| A degrees (A°) | | | |
| W | X | Y | Z |
| 72 | 144 | 216 | 288 |

FIGURE 15. Insert arrangement, electrical connector, shell size 48, SAE-AS50151.



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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2016-146)

Review activities:
Army – AR
Navy – EC
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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